Subject: **Call Center Protocol Specification for Switch / Reporting Adjunct Interface - Overview and Syntax of SPI Messages Issue: 0.7**

COMPAS ID: 160965

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ABSTRACT

Ancestor versions of this document are:

- **COMPAS ID 4585DR (SPI languages 1-7)**
- **COMPAS ID 96484 (SPI languages 8-9)**
- **COMPAS ID 109193 (SPI languages 20-23)**

This document contains an overview of the protocol specification for the switch interface to Call Center reporting adjuncts starting with SPI language 24. Reporting adjuncts are management information systems that reside on an adjunct processor. They track ACD activity and collect data provided by the switch, and use this data to generate various real-time and historical reports. The data collected includes information about agents, splits, trunks, trunk groups, VDNs, and vectors. Reporting adjuncts also have some limited switch administration capabilities.

The entire definition of the protocol specification has been split in three documents. This document contains an overview of the communications protocol and the layout and rules for each message. The syntax for messages to the reporting adjuncts is covered. The semantics to the protocol (call trees) can be found in COMPAS ID 21003. The semantics document gives message sequences for various call scenarios. The database definitions for the R3V8 database can be found in COMPAS ID 19974.

This document is written using generic product terms as much as possible. “Switch” can refer to releases of G3, DEFINITY®, MultiVantage® or Communication Manager®. “Reporting Adjunct” can refer to releases of CMS®, R3Vx, IQ®, WAE (Work Assignment Engine), etc. All SPI messages apply to all reporting adjunct products with the following exceptions:

- When the switch receives CHGVDNSK, MVAGSTFD, MVVDN, SKILLADD or CHGNAM from IQ (SPI language 21+) and WAE, it will wait for USERID message
- The switch will only accept REQNAM, RCODENAM, CHGNAM or USERID from IQ (SPI language 21+) and WAE
- If the switch receives RLTB from IQ (SPI language 21+) or WAE, it will abort an ADMNAM
- download in progress
- WAE does not do pumpup
- The only audits that the switch sends to IQ are EWTAUDIT or VAUDIT.
- IQ only supports an EAS environment. Any non-EAS messages are not sent to IQ.
- IQ does not support vector admin (VECTOR, CHGVEC).

Issue 1.0 has the following changes from COMPAS ID 109193:

- "Quick Links to SPI Messages" was added.
- The Cross-reference table in section 3.1 was updated
- COMPAS MRs (79738, 80260) from the prior version of this document (COMPAS ID 109193) were resolved
- References to specific reporting products were minimized
- General cleanup and minor reorganization. Obsolete information was removed (see ancestor documents for this information).
- Add SPI language 24 (based on SRAD Compas ID 166079), including,

sorted by requirements:

- Overhead for new SPI L24 language
  - New XSTAT24 for support of SPI language 24 (CMS R3V18) and Release 3 (CM7.0)
- Support increased agent-skill pairs
  - New RTCS24 and XFAIL24 for support of 360,000 agent/skill pairs
- Support CM increased locations to 2000
  - New XTRUNK24, ARTRUNK24 and ALOGON24 for support of 2,000 location IDs
- Send quantity of licensed logged in agents allowed
  - New XSTAT24 to send quantity of licensed logged in agents allowed
- Expand DADFWD24 for support of 8000 skills
  - New DADFWD24 for support of 8000 skills (outage from SPI language 23)
- Add specific RCODE for tenant number mismatch for Match Agent Skills feature
- Add "b" beginning bit to ADMNAM24 for agent object type along with spare bits
  - Noted partial support of the "b" (beginning) bit in ADMNAM20 starting with FP4
  - New ADMNAM24 for full support of "b" (beginning) bit
- Support increased extension size (to 16 digits)
    - support increased dnumber (from 4 to 5 bits) AND increase digits from d1-d15 to d1-d16: XVDN24, XAGENT24, ARVDN24, ARAGENT24, MVAGSTFD24, APARMS24, AGENTSK24, SKILLADD24, VECTOR24, CHGVEC24, ALOGON24, DFWD24
    - support increased pnumber (from 4 to 5 bits) AND increase digits from p1-p15 to p1-p16: ADMNAM24, CHGNAM24, ALOGON24, ALOGOFF24, AVAIL24, ACD24, ACW24, AUX24, OTHER24, DIGITS24, AEVENT24, DAQUEUED24, DCON24, DABN24, DADFWD24, HOLD24, RECON24, AUDAVAIL24, AUDACD24, AUDACW24, AUDAUX24, AUDOTHER24, AUDUNSTF24, XFER24, CPERROR24
- Support increased trunk capacity
  - support of 24,000 measured trunks (64,000 future): RTCS24, XFAIL24
  - allow sending 64,000 trunk messages: XTRUNK24
  - allow sending 9999 XTRUNK messages per trunk group: XCLOSE20
  - Increase ITN from 15 to 16 bits: XTRUNK24, ARTRUNK24, DIGITS24, IDLE24, SETUP24, QUEUED24, DAQUEUED24, DCON24, DABN24, DADFWD24, DFWD24, MBUSY24, FBUSY24, VSTART24, VSTOP24, FDISC24, HOLD24, RECON24, COMP24, LOOKAH24, AROUTE24, DNEVENT24, RAUDIT24, TAUDIT24, XFER24, CPERROR24
- Support Tenant related info to reporting adjunct
  - send Tenant Number: XVDN24, XSPLIT24, XTKGRP24, XTGVDN24, ARVDN24, ARSPLIT24, ARTKGRP24, ARTGVDN24, VDNMOVE24, ALOGON24

- Add why sent info to SETUP
    - Add a 4 bit reason code to SETUP24 indicating why the SETUP was sent
- Change trunk eqloc format to match virtual trunk locations
    - Use a spare bit to indicate the location fields are for virtual IP/SIP trunk number: XTRUNK24, ARTRUNK24
    - Allow the location fields to be a virtual IP/SIP trunk number (range 0001-64,000): XTRUNK24, ARTRUNK24
- Send Attribute reporting attribute in ALOGON message

sorted by message:

- RTCS
    - support of 360,000 agent/skill pairs
    - support of 24,000 measured trunks (64,000 future)
- XFAIL
    - support of 360,000 agent/skill pairs
    - support of 24,000 measured trunks (64,000 future)
- XSTAT
    - support of SPI language 24 (CMS R3V18) and Release 3 (CM7.0)
    - send quantity of licensed logged in agents allowed
- XCLOSE
    - allow sending 9999 XTRUNK messages per trunk group
- XVDN
    - support increased extension size (dnumber from 4 to 5 bits, digits from d1-d15 to d1-d16)
    - send tenant number (new field, 10 bits)
- XSPLIT
    - send tenant number (new field, 10 bits)
- XAGENT
    - support increased extension size (dnumber from 4 to 5 bits, digits from d1-d15 to d1-d16)
- XTKGRP
    - send tenant number (new field, 10 bits)
- XTGVDN
    - send tenant number (new field, 10 bits)
- XTRUNK
    - Support CM increased locations to 2000
    - Increase ITN from 15 to 16 bits
    - Use a spare bit to indicate the location fields are for virtual IP/SIP trunk number
    - Allow the location fields to be a virtual IP/SIP trunk number (range 00001-64,000)
    - allow sending 24,000 trunk messages (64,000 future)
- ARVDN
    - support increased extension size (dnumber from 4 to 5 bits, digits from d1-d15 to d1-d16)
    - send tenant number (new field, 10 bits)
- ARAGENT
    - support increased extension size (dnumber from 4 to 5 bits, digits from d1-d15 to d1-d16)
- ARSPLIT
    - send tenant number (new field, 10 bits)
- ARTKGRP
    - send tenant number (new field, 10 bits)
- ARTGVDN
    - send tenant number (new field, 10 bits)
- ARTRUNK

    - Support CM increased locations to 2000
    - Move “r” bit to make room for ITN increase.
    - Increase ITN from 15 to 16 bits
    - Use a spare bit to indicate the location fields are for virtual IP/SIP trunk number
    - Allow the location fields to be a virtual IP/SIP trunk number (range 00001-64,000)
- MVAGSTFD
    - support increased extension size (pos from 13 nybles to 16 nybles)
    - receive up to 2 messages instead of 1. Add an end bit.
- APARMS
    - support increased extension size (pnumber from 4 to 5 bits, digits from p1-p15 to p1-p16)
- AGENTSK
    - support increased extension size (pnumber from 4 to 5 bits, digits from p1-p15 to p1-p16)
- SKILLADD
    - support increased extension size (pnumber from 4 to 5 bits, digits from p1-p15 to p1-p16)
- VDNMOVE
    - send tenant number (new field, 10 bits)
- VECTOR/CHGVEC
    - support increased ANI
        - Goto format D (ANI), new condition 24 supports 24 digits in ANI
        - Route-to number
    - support increased extension size (go to d1-d16)
        - announcement step
        - wait step
        - goto step (format D)
        - disconnect step
        - collect step
        - messaging step
- ADMNAM
    - Add “b” beginning bit to ADMNAM24 for agent object type along with spare bits
    - support increased extension size (pnumber from 4 to 5 bits, digits from p1-p15 to p1-p16)
- CHGNAM
    - support increased extension size (pnumber from 4 to 5 bits, digits from p1-p15 to p1-p16)
- RCODE20
    - Add specific RCODE for tenant number mismatch for Match Agent Skills feature
- ALOGON
    - Support CM increased locations to 2000
    - support increased extension size (pnumber from 4 to 5 bits, digits from p1-p15 to p1-p16)
    - support increased extension size (dnumber from 4 to 5 bits, digits from d1-d15 to d1-d16)
    - send tenant number (new field, 10 bits)
    - Send Attribute reporting attribute in ALOGON message
    - Send up to 3 messages instead of 2.
- ALOGOFF
    - support increased extension size (pnumber from 4 to 5 bits, digits from p1-p15 to p1-p16)
- AVAIL
    - support increased extension size (pnumber from 4 to 5 bits, digits from p1-p15 to p1-p16)
- ACD
    - support increased extension size (pnumber from 4 to 5 bits, digits from p1-p15 to p1-p16)
- ACW
    - support increased extension size (pnumber from 4 to 5 bits, digits from p1-p15 to p1-p16)
- AUX

    - support increased extension size (pnumber from 4 to 5 bits, digits from p1-p15 to p1-p16)
- OTHER
    - support increased extension size (pnumber from 4 to 5 bits, digits from p1-p15 to p1-p16)
- DIGITS
    - support increased extension size (pnumber from 4 to 5 bits, digits from p1-p15 to p1-p16)
        - used to hold both agent position and ITN
- AEVENT
    - support increased extension size (pnumber from 4 to 5 bits, digits from p1-p15 to p1-p16)
- IDLE
    - Increase ITN from 15 to 16 bits
- SETUP
    - Increase ITN from 15 to 16 bits
    - Add a 4 bit reason code to SETUP24 indicating why the SETUP was sent
- QUEUED
    - Increase ITN from 15 to 16 bits
- DAQUEUED
    - support increased extension size (pnumber from 4 to 5 bits, digits from p1-p15 to p1-p16)
    - Increase ITN from 15 to 16 bits
- DCON
    - support increased extension size (pnumber from 4 to 5 bits, digits from p1-p15 to p1-p16)
    - Increase ITN from 15 to 16 bits
- DABN
    - support increased extension size (pnumber from 4 to 5 bits, digits from p1-p15 to p1-p16)
    - Increase ITN from 15 to 16 bits
- DADFWD
    - Expand DADFWD24 for support of 8000 skills
    - support increased extension size (pnumber from 4 to 5 bits, digits from p1-p15 to p1-p16)
    - Increase ITN from 15 to 16 bits
- DFWD
    - support increased extension size (dnumber from 4 to 5 bits, digits from d1-d15 to d1-d16)
    - Increase ITN from 15 to 16 bits
- MBUSY
    - Increase ITN from 15 to 16 bits
- FBUSY
    - Increase ITN from 15 to 16 bits
- VSTART
    - Increase ITN from 15 to 16 bits
- VSTOP
    - Increase ITN from 15 to 16 bits
- FDISC
    - Increase ITN from 15 to 16 bits
- HOLD
    - support increased extension size (pnumber from 4 to 5 bits, digits from p1-p15 to p1-p16)
    - Increase ITN from 15 to 16 bits
- RECON
    - support increased extension size (pnumber from 4 to 5 bits, digits from p1-p15 to p1-p16)
    - Increase ITN from 15 to 16 bits
- COMP
    - Increase ITN from 15 to 16 bits
- LOOKAH
    - Increase ITN from 15 to 16 bits

- AROUTE
  - Increase ITN from 15 to 16 bits
- ASA
  - Increase ITN from 15 to 16 bits
- DNEVENT
  - Increase ITN from 15 to 16 bits
- RAUDIT
  - Increase ITN from 15 to 16 bits
- TAUDIT
  - Increase ITN from 15 to 16 bits
- AUDAVAIL
  - support increased extension size (pnumber from 4 to 5 bits, digits from p1-p15 to p1-p16)
- AUDACD
  - support increased extension size (pnumber from 4 to 5 bits, digits from p1-p15 to p1-p16)
- AUDACW
  - support increased extension size (pnumber from 4 to 5 bits, digits from p1-p15 to p1-p16)
- AUDAUX
  - support increased extension size (pnumber from 4 to 5 bits, digits from p1-p15 to p1-p16)
- AUDOTHER
  - support increased extension size (pnumber from 4 to 5 bits, digits from p1-p15 to p1-p16)
- AUDUNSTF
  - support increased extension size (pnumber from 4 to 5 bits, digits from p1-p15 to p1-p16)
- XFER
  - support increased extension size (pnumber from 4 to 5 bits, digits from p1-p15 to p1-p16)
  - Increase ITN from 15 to 16 bits
- CPERROR
  - support increased extension size (pnumber from 4 to 5 bits, digits from p1-p15 to p1-p16)
  - Increase ITN from 15 to 16 bits

LINKS TO SPI MESSAGE DESIGNS

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Subject: **Call Center Protocol Specification for Switch / Reporting Adjunct Interface - Overview and Syntax of SPI Messages**
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REQUIREMENTS

1. Introduction

This document contains the protocol specification for the switch interface to Call Center reporting adjuncts. Reporting adjuncts are management information systems that reside on an adjunct processor. They track ACD activity and collect data provided by the switch, and use this data to generate various real-time and historical reports that allow the customer to monitor and manage their call centers. The data collected includes information about agents, splits, trunks, trunk groups, VDNs, and vectors. Reporting adjuncts also have some limited switch administration capabilities.

First an overview of communication protocols is presented. This is followed by the message syntax; these define the Application protocol that is used in the switch communications with reporting adjuncts. The syntax defines the message layouts used in the protocol. The semantics document[1] (call trees) provides protocol rules and example call trees, the message sequences for the call trees, and how the reporting adjuncts interpret each message sequence including what data items get updated. The syntax section and semantics document are separated into subsections on Translations Pumpup, Switch Administration, and Call Processing. The database tables document[2] lists all of the database tables and defines each of the items in the tables.

1.1 Purpose

This paper was written to document and baseline the switch/reporting adjunct interface protocol. It will be used as the requirements and design specification for the Switch Protocol Interpreter (SPI). It is also intended to serve as an interface specification and reference for switch development.

It is hoped that this will also prove useful as a reference for System Test in verifying the switch-reporting adjunct protocol, Field Support and Services in isolating link and data collection problems, Technical Writers in providing training and customer documentation, and Product Management and Product Marketing in answering questions about the how reporting adjuncts interpret call data and what information gets recorded.

1.2 Overview

This section provides an overview of the functions of a data communications protocol. Included is some background and definitions that will be helpful in understanding the protocol specified in this document.

1.2.1 OSI Reference Model

The Open Systems Interconnection (OSI) Reference Model separates the data communication functions into seven layers (or levels) as shown in Figure 1. The OSI Model is included to compare what is provided by the TCP/IP protocol to what is needed for process communications.

Each layer of the OSI Reference Model defines both the services that that layer provides to the upper layers and the functions it must perform to provide these services. Each layer needs the services of the layers beneath it to accomplish its task. The text, taken from [3] only provides an introductory description to each layer.

<table>
<tr><td>APPLICATION</td><td rowspan="4"></td><td rowspan="2">USER
DEFINED</td><td rowspan="2">USER
DEFINED</td></tr>
<tr><td>PRESENTATION</td></tr>
<tr><td>SESSION</td><td rowspan="2">SESSION</td><td rowspan="2">SESSION</td></tr>
<tr><td>TRANSPORT</td></tr>
<tr><td>NETWORK</td><td rowspan="2">PACKET</td><td rowspan="2">PACKET</td><td>TCP</td></tr>
<tr><td>DATA LINK</td><td>IP</td></tr>
<tr><td>PHYSICAL</td><td>LINK</td><td>LINK</td><td>LINK</td></tr>
<tr><td>OSI MODEL</td><td>X.25</td><td>BX.25</td><td>TCP/IP</td></tr>
</table>

Figure 1 Data Communication Protocol Layers

1.2.1.1 Physical Layer

The Physical Layer provides for the transmission of data bits between two data link entities. To do this it must provide the mechanical, electrical, functional, and procedural mechanisms needed to activate, maintain, and deactivate physical connections between Data Link entities.

1.2.1.2 Data Link Layer

The Data Link Layer is responsible for providing error-free delivery of Network Layer data over the data circuit provided by the Physical Layer. It can also provide sequential delivery of data and flow control to prevent a fast transmitter from drowning a slow receiver.

1.2.1.3 Network Layer

The Network Layer establishes, maintains, and terminates connections between communicating end systems. It is responsible for providing a path between the end systems specified by the Transport Layer.

1.2.1.4 Transport Layer

The Transport Layer provides all transport services to the Session Layer, either directly or through the lower layers. The Transport Layer also provides end-to-end error detection and recovery, and flow control on each transport connection so that users receiving data can temporarily stop the flow of data if they cannot process it as fast as it is arriving.

1.2.1.5 Session Layer

The Session Layer provides the means for cooperating Presentation entities to organize and synchronize their dialogues and manage their data exchanges.

1.2.1.6 Presentation Layer

It is the responsibility of the Presentation Layer to request that the Session Layer establish and later terminate a session. The Presentation Layer deals with the internal attributes of the communicating resources. It can on request modify the syntax of data without changing its meaning (e.g., change data characters from ASCII to EBCDIC). The Presentation Layer can also provide other data transformation services such as data compression.

1.2.1.7 Application Layer

The Application Layer dictates how communicating users will manage their conversations.

1.2.2 X.25 Protocol

X.25[4] is a layered data communications protocol patterned after the OSI Reference Model but it only covers the first three layers. If services at other layers are required, other protocols must be used in addition to X.25 (e.g., BX.25).

The X.25 Physical Layer and Link Layer closely resemble their OSI peers with only a few exceptions, the most notable being that X.25 only deals with single, full-duplex, point-to-point links.

The X.25 Packet Layer provides most of the services of the OSI Network Layer and a portion of the Transport Layer services. The main OSI Network Layer functions not included in the X.25 Packet Layer are routing and switching, since X.25 is a network interface specification and not an internal network specification. The main OSI Transport Layer feature provided in the X.25 Packet Layer is the end-to-end data acknowledgement.

1.2.3 BX.25 Protocol

BX.25[5] goes beyond X.25 and defines five levels of protocol. The first three correspond to the Physical, Link, and Packet Layers of X.25. Since only a few of the features from the OSI Presentation Layer are incorporated in BX.25, they are included in the definition of the Session Layer rather than in a separate layer.

At the highest level is the application, which must be defined by the user. The Application Protocol must specify the syntax and semantics of the information that will flow across the connection.

1.2.4 TCP/IP Protocol

The TCP/IP connection requires DEFINITY ECS 7.1 or later software and hardware including the C-LAN card. This connection is only available on the Sun SPARCservers and requires CentreVu CMS R3V6 or later. TCP/IP will be provided over both an Ethernet (for a LAN/WAN) and a PPP (for a modem) connection.

The TCP/IP connection will be transparent to the R3V6 or later CMS application except for link administration. The differences between the TCP/IP connection and the X.25 connection will be at the session layer and below.

One 10BaseT Ethernet connection will support the message traffic equivalent to a fully loaded 19.2 Kbps X.25 connection. The TCP/IP over Ethernet connection and the TCP/IP over PPP connection will continue to support the advertised 40,000 Call Center busy hour calls per CMS.

1.2.5 Reporting Adjunct Protocol

The reporting adjunct will use the AT&T X.25 Network Interface [Error: Reference source not found] software to provide the services of the Physical, Link, and Packet levels of the BX.25 protocol. The reporting adjunct will package the X.25 p-command interface library supplied with this software into a Transport Connection library which will provide for the transparent transfer of data between the communicating Session Layers. It relieves the Session Layer of any concern about the way in which the data transfer is achieved, including the case where the reporting adjunct is communicating with a switch simulator that may reside on the same or a different computer as the reporting adjunct.
CMS also supports a TCP/IP connection to the switch.

The CMS Session Layer will comply with the standard BX.25 Session Layer protocol [Error: Reference source not found].

This document specifies the syntax and semantics of the user-defined Application Protocol for the CMS application.

Switches will send buffers of data according to the following table.

Switch	CMS Buffer Size	Interval
S75 R1V3	n/a	n/a
G1.1	200	1 second
G3i	244	1 second
G3r	252 bytes	1 second
CM6.2 SP4 CM6.3	252 bytes	0.1 seconds

The switch also sends the packet if the buffer is full (which is the case with CHGVEC and VECTOR messages).

The overhead on the link is 4 bytes for the session layer for each buffer, 3 bytes for the packet layer for each 128 byte packet, and 5 bytes for the link layer for each 128 byte packet.

1.3 **Terminology**

This section provides definitions for some commonly used but often misunderstood terms.

ATM	Asynchronous Transfer Mode - Definity® support of ATM falls into two areas. The first is Port Network Connectivity (PNC), primarily remote PNC over an ATM Wide Area Network (WAN). The second is a set of new ATM trunk circuit packs which can support 256 circuits each.
BSR	Best service routing. Provides call vectoring functionality which builds upon Lookahead Interflow to route a call to the "best" split or skill on a single switch or to the "best" split/skill in a network of Definity® switches.
CentreVu Advocate	The external product name for a set of features which were originally called Agent Allocation.
CentreVu Virtual Routing	The external product name for a set of features which were originally called Best Service Routing (BSR).
iln	The switch assigns an internal line number to all extensions. The *iln*-VDN extension association is reported to reporting adjuncts during the translation pumpup and then the *iln* is used to identify VDN extensions in all subsequent transactions between CMS and the switch. The *iln* is only used by reporting adjuncts to track VDNs (see *itn* for an explanation of extension call tracking).
itn	The switch assigns an internal trunk number to all trunks (including software facilities used to place station-to-station calls or used to transfer a call to another destination). The *itn*-trunk association is reported to reporting adjuncts during the translation pumpup and then the *itn* is used to identify trunks in all subsequent transactions between the reporting adjuncts and the switch. When software facilities are used to process a call, the switch will "create" an *itn* value and use it for notifying the reporting adjuncts of the events associated with a call.
location-id	A number, 1-44 (G3V8-G3V11), 1-250 (MV2-CM6.2) and 1-2000 (CM6.3-, but not supported until SPI language 24), that is assigned to one or more port networks on Communication Manager. All hardware residing on a port network will be associated with that port network's location-id (via the "change cabinet X" form). Location-id is supported by the optional 'Multiple Locations' feature.
loginID	Each EAS agent is given a login number called a loginID. This means that when the agent logs into a physical extension, the properties such as assigned skills, class of restriction,

	coverage path, etc., are associated with the agent's loginID rather than with the physical extension.
measured agent	See *measured extension*.
measured agent position	See *measured extension*.
measured domain	The measured domain of a reporting adjunct is the collection of all measured splits, measured extensions, measured trunk groups, measured trunks, measured VDNs, and measured vectors on the switch.
measured extension	A measured extension is one that is administered as measured on the switch. All extensions that are members of a measured split are automatically measured extensions. The switch will give the reporting adjunct information about measured extensions during the translation pumpup, and will then notify the reporting adjunct of any events and translation changes associated with the measured extensions, and the reporting adjunct will collect agent data for the agent that has staffed that extension.
measured split	A measured split is one that is administered as measured on the switch. The switch tells the reporting adjunct which splits are measured. The switch will give the reporting adjunct information about measured splits during the translation pumpup, and will then notify the reporting adjunct of any events and translation changes associated with the measured splits, and the reporting adjunct will collect split data.
measured trunk	A measured trunk is a trunk that is a member of a measured trunk group. The switch will give the reporting adjunct information about measured trunks during the translation pumpup, and will then notify the reporting adjunct of any events associated with measured trunks, and the reporting adjunct will collect trunk and trunk group data.
measured trunk group	A measured trunk group is one that is administered as measured on the switch. The switch will give the reporting adjunct information about measured trunk groups during the translation pumpup, and will then notify the reporting adjunct of any events and translation changes associated with the measured trunk groups, and the reporting adjunct will collect trunk group data.
measured VDN	A measured VDN is one that is administered as measured on the switch. VDNs must be administered as measured. The switch also sends messages for unmeasured VDNs. In CM6, the SPI messages for unmeasured VDNs can be stopped by setting “Used for BSR Polling?”. The switch will give the reporting adjunct information about measured VDNs during the translation pumpup, and will then notify the reporting adjunct of any events and translation changes associated with the measured VDNs, and the reporting adjunct will collect VDN data.
measured vector	All vectors are automatically measured by the reporting adjunct. The switch will notify the reporting adjunct of all events associated with vectors.
modification number	A modification number associated with a particular entity on the switch is used by the reporting adjunct and its clients to identify whether cached data for the entity is valid. Entities to which modification numbers apply include VDNs and vectors. The modification number for an entity is incremented when the administration for the entity is changed.
reason code	The code agents enter when changing to the AUX work state or logging out. This code indicates the reason for the state change.
skill	In an EAS environment, a skill can be defined as a specific customer need/requirement that can be assigned to an agent.
terminal number	The switch assigns a terminal number (also referred to as an agent position number) to each measured extension in a non-EAS environment. The *terminal number*-extension association is reported to the reporting adjunct during the translation pumpup, and then the *terminal number* is used to identify extensions (agent positions) in all subsequent transactions between the reporting adjunct and the switch.

	In an EAS environment the *terminal number* is the agent's login ID in packed decimal format.
unmeasured extension	An unmeasured extension is one that is not administered as measured on the switch. The switch will notify the reporting adjunct of certain events associated with unmeasured extensions, for example, if a measured trunk is forwarded to an unmeasured extension. In this case, the trunk is in the *measured domain* so event notification is required, and the reporting adjunct will collect trunk (and VDN/vector and split) data, but does not collect agent data. The agent position number is reported to the reporting adjunct as terminal "0", which the reporting adjunct interprets as an unmeasured extension.
unmeasured split	An unmeasured split is one that is not administered as measured on the switch. The switch will notify the reporting adjunct of certain events associated with unmeasured splits, for example, if a measured trunk is queued to an unmeasured split. In this case, the trunk is in the *measured domain* so event notification is required, and the reporting adjunct will collect trunk (and VDN/vector) data but does not collect split data. The split number is reported to the reporting adjunct as split "0", which the reporting adjunct interprets as an unmeasured split.
unmeasured trunk	An unmeasured trunk is one that is not a member of a measured trunk group. This includes, for example, software facilities used to place station-to-station calls (see *itn* for further details). The switch will notify the reporting adjunct of certain events associated with unmeasured trunks, for example, if an unmeasured trunk is used to carry a measured VDN call. In this case, the VDN (and vector) is in the *measured domain* so event notification is required, and the reporting adjunct will collect VDN/vector (and split/agent) data but does not collect trunk or trunk group data. The trunk number is reported to the reporting adjunct as an *itn*, but the reporting adjunct will not recognize it as a measured trunk from the translation pumpup so it interprets it as an unmeasured trunk, and tracks the events on this trunk for the duration of the call so that is can collect the call data.
unmeasured trunk group	An unmeasured trunk group is one that is not administered as measured on the switch. The switch does not tell the reporting adjunct about these trunk groups (but may tell the reporting adjunct about its members - see *unmeasured trunk*).
unmeasured VDN	An unmeasured VDN is one that is not administered as measured on the switch. The switch will notify the reporting adjunct of all events associated with unmeasured VDNs. The VDN is reported to the reporting adjunct as *iln* "0", which the reporting adjunct interprets as an unmeasured VDN.

2. PROTOCOL SPECIFICATION

This section defines the reporting adjunct Application Protocol. The reporting adjunct Application (a.k.a. the reporting adjunct Session) is responsible for establishing communications between the reporting adjunct and switch applications. The reporting adjunct Application can be in any of the following states (on the reporting adjunct side of the communications path).

QUIESCENT	The reporting adjunct Application is idle. Data Collection is turned off. The reporting adjunct is waiting for the user to request that a reporting adjunct Application be initiated.
WAITING SESSION ACCEPT (WSA)	The reporting adjunct Application is waiting for notification from the session layer that the session has been established. The reporting adjunct Application will be in this state whenever the session layer does not have a session currently established.
REQUEST TRANSLATION CHANGE STATUS (**RTCS**)	When the session layer has notified the reporting adjunct Application that the session has been established, the reporting adjunct Application begins its connection by requesting translation change status. The reporting adjunct Application enters this state and remains here until a valid answer (XSTAT) has been received. The status of translations and the state of the reporting adjunct database determine the form of pumpup used. This state may also be entered when the user requests new translations or when the Application timer AT3 expires. When the **RTCS** state is entered, the Application timer AT1 is started (2 minutes). If the timer expires before the switch response is received, a new **RTCS** message is sent and the timer is set again.
PUMPUP	Once the answer to the request for translation change status has been received, the application enters the pumpup state. During this state the switch sends the requested translations, depending on whether the request was for full translations (**RFTB**) or logons (**RLTB**). When the final message of the pumpup is received and processed successfully, the reporting adjunct Application enters the *DATA TRANSFER* state. If for some reason the pumpup fails (e.g., the reporting adjunct gets more data than it expects), the reporting adjunct Application will return to the **RTCS** state. During pumpup, the Application timer AT3 is set (2 minutes) and is reset (2 minutes) each time a **TIME** message is received. If the timer ever is allowed to expire, the pumpup is aborted, the Application returns to the **RTCS** state, and a new reporting adjunct Application connection will be established.
DATA TRANSFER	The reporting adjunct Application is in data transfer state. The reporting adjunct database is up-to-date with the switch database and the reporting adjunct is actively tracking the status of the ACD facilities and collecting real-time data. Switch administration commands can now communicate with the switch application. While the reporting adjunct Application is in *DATA TRANSFER* state, the reporting adjunct will set the AT3 timer (2 minutes). The reporting adjunct will reset the AT3 timer whenever a **TIME** message is received (the switch will send a **TIME** message every minute). If the timer expires (i.e., data transfer timed-out) the reporting adjunct will initiate another reporting adjunct Application connection.
BUSIED-OUT	The reporting adjunct Application has been busied-out for administration purposes at the switch. This is usually done by the switch administrator when making changes that will affect the reporting adjunct database configuration. These changes typically result in the "translation-change" flag being set at the switch so that when the reporting adjunct initiates another reporting adjunct Application connection, full translations will be requested and the reporting adjunct real-time database will be updated to reflect the changes that were made at the switch.

	While the reporting adjunct Application is *BUSIED-OUT*, the reporting adjunct will set the AT2 timer (2 minutes). When the timer expires, the reporting adjunct will initiate another reporting adjunct Application connection to see if the session has been released from busy-out at the switch. If the reporting adjunct Application is still busied-out the timer will be reset. The switch will busy-out the link if the Session layer queue in the switch overflows.

3. PROTOCOL SYNTAX

3.1 Compatibility

This section indicates the switch releases that support each of the messages. Each message layout is also preceded with a compatibility table. Most message types (functions) have more than one version of the message used for different switch releases. Only one version is used in each release. Typically a new version was created to accommodate larger capacities. The message layouts for a single message type are grouped under a single third-level heading.

The table indicates which releases of the switch use each of the messages. In the syntax section that follows the table, a left arrow "←" indicates messages from the reporting adjunct and a right arrow "→" indicates messages from the switch.

Cross-reference tables in the appendix list the messages alphabetically and by opcode.

Message	Op Code	Language 22	Language 23	Language 24
Pumpup				
RTCS21	x34	y	n	n
RTCS23	x37	n	y	n
RTCS24	**x3D**	**n**	**n**	**y**
XFAIL21	xf9	y	n	n
XFAIL23	x0d	n	y	n
XFAIL24	**x29**	**n**	**n**	**y**
XSTAT20	xb1	y	y	n
XSTAT24	**x2c**	**n**	**n**	**y**
RFTB	x06	y	y	y
RLTB	x0c	y	y	y
XOPEN20	xb2	y	y	y
XCLOSE20	xb3	y	y	y
XVDN20	xb4	y	n	n
XVDN23	x11	n	y	n
XVDN24	**x2d**	**n**	**n**	**y**
XSPLIT20	xb5	y	y	n
XSPLIT24	**x2e**	**n**	**n**	**y**
XAGENT20	xb6	y	y	n
XAGENT24	**x2f**	**n**	**n**	**y**
XTKGRP20	xb7	y	y	n
XTKGRP24	**x30**	**n**	**n**	**y**
XTGVDN20	xb8	y	y	n
XTGVDN24	**x31**	**n**	**n**	**y**
XTRUNK20	xb9	y	y	n
XTRUNK24	**x25**	**n**	**n**	**y**
Switch Administration				
BUSYOUT20	xba	y	y	y
RELBUSY20	xbb	y	y	y
ARVDN20	xbc	y	n	n
ARVDN23	x12	n	y	n
ARVDN24	**x32**	**n**	**n**	**y**
ARAGENT20	xbd	y	y	n

Message	Op Code	Language 22	Language 23	Language 24
ARAGENT24	**x33**	**n**	**n**	**y**
ARSPLIT20	xbe	y	n	n
ARSPLIT23	x13	n	y	n
ARSPLIT24	**x34**	**n**	**n**	**y**
ARTKGRP20	xbf	y	n	n
ARTKGRP23	x14	n	y	n
ARTKGRP24	**x35**	**n**	**n**	**y**
ARTGVDN20	xc0	y	y	n
ARTGVDN24	**x36**	**n**	**n**	**y**
ARTRUNK20	xc1	y	y	n
ARTRUNK24	**x26**	**n**	**n**	**y**
AGMOVE20	xf8	y	y	y
MVAGSTFD21	x35	y	n	n
MVAGSTFD23	x38	n	y	n
MVAGSTFD24	**x3e**	**n**	**n**	**y**
APARMS7	x28	y	y	n
APARMS24	**x3f**	**n**	**n**	**y**
AGENTSK20	xc2	y	n	n
AGENTSK23	x04	n	y	n
AGENTSK24	**x37**	**n**	**n**	**y**
SKILLADD8	x2d	y	n	n
SKILLADD23	x39	n	y	n
SKILLADD24	**x40**	**n**	**n**	**y**
TGMOVE20	xc3	y	n	n
TGMOVE23	x16	n	y	y
VTGMOVE20	xc4	y	y	y
VDNMOVE20	xc5	y	n	n
VDNMOVE23	x18	n	y	n
VDNMOVE24	**x38**	**n**	**n**	**y**
MVVDN9	x2f	y	n	n
MVVDN23	x3c	n	y	y
CHGVDNSK8	x2e	y	n	n
CHGVDNSK23	x3a	n	y	y
VPARMS	x15	y	y	y
VECTOR20	xc6	y	y	n
VECTOR24	**x52**	**n**	**n**	**y**
CHGVEC20	x33	y	y	n
CHGVEC24	**x44**	**n**	**n**	**y**
REQLIST5	x23	y	y	y
ADMLIST20	xc8	y	y	y
REQNAM20	x30	y	y	y
ADMNAM20	xc9	y	y	n
ADMNAM24	**X2b**	**n**	**n**	**y**
RCODENAM20	0x31	y	y	y
CHGNAM20	0x32	y	y	n
CHGNAM24	**x41**	**n**	**n**	**y**
MVRCODE20	xca	y	y	y
RCODE20	xcb	y	y	y
CRCODE20	xcc	y	y	y

Message	Op Code	Language 22	Language 23	Language 24
USERID21	x36	y	y	y
Call Processing				
TIME20	xcd	y	y	y
ALOGON20	xce	y	n	n
ALOGON23	x05	n	y	n
ALOGON24	**x28**	**n**	**n**	**y**
ALOGOFF20	xcf	y	y	n
ALOGOFF24	**x39**	**n**	**n**	**y**
AVAIL22	x01	y	n	n
AVAIL23	x0e	n	y	n
AVAIL24	**x3a**	**n**	**n**	**y**
ACD20	xd1	y	y	n
ACD24	**x3b**	**n**	**n**	**y**
ACW22	xff	y	y	n
ACW24	**x3c**	**n**	**n**	**y**
AUX22	x03	y	n	n
AUX23	x0f	n	y	n
AUX24	**x3d**	**n**	**n**	**y**
OTHER22	xfa	y	y	y
OTHER24	**x3e**	**n**	**n**	**y**
DIGITS20	xd5	y	y	n
DIGITS24	**x3f**	**n**	**n**	**y**
AEVENT20	xd6	y	y	n
AEVENT24	**x40**	**n**	**n**	**y**
IDLE20	xd7	y	n	n
IDLE23	x1b	n	y	n
IDLE24	**x41**	**n**	**n**	**y**
SETUP22	xfe	y	y	n
SETUP24	**x42**	**n**	**n**	**y**
QUEUED20	xd9	y	n	n
QUEUED23	x1c	n	y	n
QUEUED24	**x43**	**n**	**n**	**y**
DAQUEUED20	xda	y	n	n
DAQUEUED23	x1d	n	y	n
DAQUEUED24	**x44**	**n**	**n**	**y**
DCON22	xfd	y	n	n
DCON23	x0b	n	y	y
DCON24	**x45**	**n**	**n**	**y**
DABN22	xfc	y	y	n
DABN24	**x46**	**n**	**n**	**y**
DADFWD20	xdd	y	y	n
DADFWD24	**x2a**	**n**	**n**	**y**
DFWD20	xde	y	n	n
DFWD23	x0c	n	y	n
DFWD24	**x47**	**n**	**n**	**y**
MBUSY20	xdf	y	y	n
MBUSY24	**x48**	**n**	**n**	**y**
FBUSY20	xe0	y	y	n
FBUSY24	**x49**	**n**	**n**	**y**

Message	Op Code	Language 22	Language 23	Language 24
VSTART20	xe1	y	y	n
VSTART24	**x4a**	**n**	**n**	**y**
VSTOP20	xe2	y	y	n
VSTOP24	**x4b**	**n**	**n**	**y**
FDISC20	xe3	y	y	n
FDISC24	**x4c**	**n**	**n**	**y**
HOLD20	xe4	y	y	n
HOLD24	**x4d**	**n**	**n**	**y**
RECON20	xe5	y	n	n
RECON23	x1e	n	y	n
RECON24	**x4e**	**n**	**n**	**y**
COMP20	xe6	y	y	n
COMP24	**x4f**	**n**	**n**	**y**
LOOKAH20	xe7	y	y	n
LOOKAH24	**x50**	**n**	**n**	**y**
AROUTE20	xe8	y	y	n
AROUTE24	**x51**	**n**	**n**	**y**
SKSTATE20	xe9	y	n	n
SKSTATE23	x1f	n	y	y
ASA20	xea	y	n	n
ASA23	x20	n	y	y
DNEVENT20	xeb	y	y	n
DNEVENT24	**x53**	**n**	**n**	**y**
MODEVENT20	xec	y	y	y
RAUDIT	x08	y	y	n
RAUDIT24	**x42**	**n**	**n**	**y**
TAUDIT20	xee	y	y	n
TAUDIT24	**x54**	**n**	**n**	**y**
AUDAVAIL22	x02	y	n	n
AUDAVAIL23	x08	n	y	n
AUDAVAIL24	**x55**	**n**	**n**	**y**
AUDACD20	xf0	y	n	n
AUDACD23	x06	n	y	n
AUDACD24	**x56**	**n**	**n**	**y**
AUDACW20	xf1	y	n	n
AUDACW23	x07	n	y	n
AUDACW24	**x57**	**n**	**n**	**y**
AUDAUX20	xf2	y	n	n
AUDAUX23	x10	n	y	n
AUDAUX24	**x58**	**n**	**n**	**y**
AUDOTHER22	xfb	y	y	n
AUDOTHER24	**x59**	**n**	**n**	**y**
AUDUNSTF20	xf4	y	y	n
AUDUNSTF24	**x5a**	**n**	**n**	**y**
EWTAUDIT20	xf5	y	n	n
EWTAUDIT23	x22	n	y	y
VAUDIT20	xf6	y	y	y
XFER20	xf7	y	y	n
XFER24	**x5b**	**n**	**n**	**y**

Message	Op Code	Language 22	Language 23	Language 24
CPERROR7	x2a	y	y	n
CPERROR24	**x43**	**n**	**n**	**y**
Miscellaneous				
NOP	x0b	y	y	y

3.2 Translation Pumpup

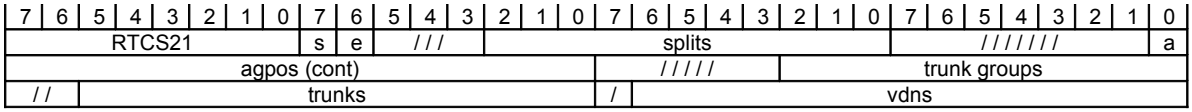
3.2.1 ←RTCS21, RTCS23 and RTCS24

The **Request Translation Change Status** message is sent by the reporting adjunct as part of the translation pumpup for Communication Manager® servers. It is the initial message used to request a session and tells the switch the maximum number of measured entities configured on the reporting adjunct. If the switch's current measured capacities are within the reporting adjuncts maximum capacities, the expected response is an **XSTAT** message and a **TIME** message. If the switch has more capacity then the reporting adjunct can measure, the switch will reject the session with a **XFAIL** message. The switch may also reject the session with a **BUSYOUT** or an **RCODE** (reason code = 1) message.

After a successful pumpup the switch will not allow more than the maximum number to be later administered as measured for each entity. With EAS, the switch will restrict an agent from logging on if the maximum number of measured agents/skills are already logged on.

3.2.1.1 ←RTCS21 (New CM4)

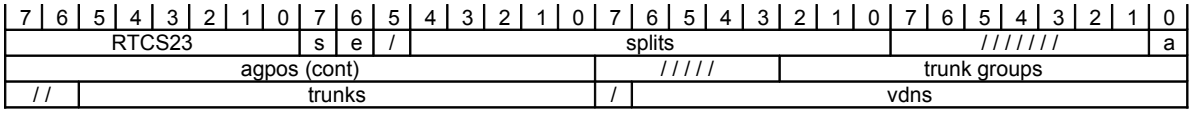
	Switch	Vect	Skill
First Release	CM4	y/n	y/n
Last Release	CM5.2		



opcode	RTCS21 = 52 (0x34)
s	system flag;0 = CMS, 1 = IQ
e	enhanced dial-plan (EDP) support flag; 0 = no longer used, 1 = always
splits	number of splits that can be measured, 1-2000
agpos	number of agent/skill pairs that can be measured, 1-60000
trunk groups	number of trunk groups that can be measured, 1-2000
trunks	number of trunks that can be measured, 1-12000
vdns	number of VDNs that can be measured, 1-20000

3.2.1.2 ←RTCS23 (New CM6.0)

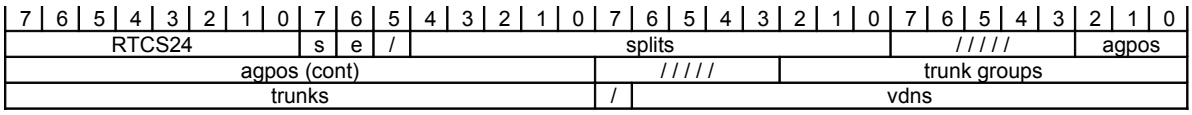
	Switch	Vect	Skill
First Release	CM6.0	y/n	y/n
Last Release	CM6.3		



opcode	RTCS23 = 55 (0x37)
s	system flag;0 = CMS, 1 = IQ or WAE
e	enhanced dial-plan (EDP) support flag; 0 = no longer used, 1 = always
splits	number of splits/skills that can be measured, 1-8000
agpos	number of agent/skill pairs that can be measured, 1-100000
trunk groups	number of trunk groups that can be measured, 1-2000
trunks	number of trunks that can be measured, 1-12000
vdns	number of VDNs that can be measured, 1-30000

3.2.1.3 **←RTCS24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y/n	y/n
Last Release			



opcode	RTCS24 = 61 (0x3D)
s	system flag;0 = CMS, 1 = IQ or WAE
e	enhanced dial-plan (EDP) support flag; 0 = no longer used, 1 = always
splits	number of splits/skills that can be measured, 1-8000
agpos	**number of agent/skill pairs that can be measured, 1-360000**
trunk groups	number of trunk groups that can be measured, 1-2000
trunks	**number of trunks that can be measured, 1-24,000 (64,000 future)**
vdns	number of VDNs that can be measured, 1-30000

3.2.2 → RCODE20

The switch may reject the request to establish communications with the rcode 1 (illegal request).

Refer to **RCODE20** in Switch Administration (section 3.3.30) for the layout.

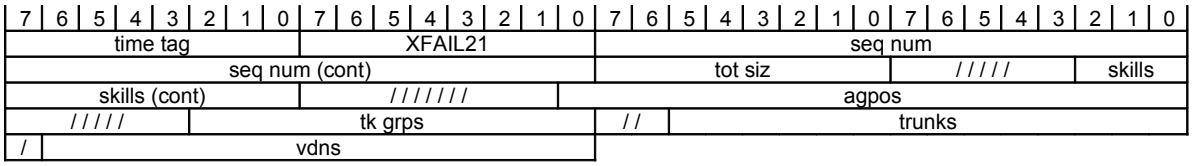
3.2.3 → XFAIL21, XFAIL23 and XFAIL24

The **Translations Fail** message is sent by the switch in response to an **RTCS** that does not have large enough capacities for the current switch configuration. The reporting adjunct will send the maximum number of measured entities configured on the reporting adjunct in the **RTCS**. If the switch has more measured entities than the reporting adjunct capacity, the switch will send back its current configuration for each measured entity. With EAS the *agpos* field will contain the current number of agent/skill pairs logged in.

When an **XFAIL** is received, the reporting adjunct will inform the user of the switch's capacities, giving the user the choice of increasing the reporting adjunct capacities or working with the switch administrator to reduce the switch capacities before the reporting adjunct Application connection can be established.

3.2.3.1 → XFAIL21 (New CM4)

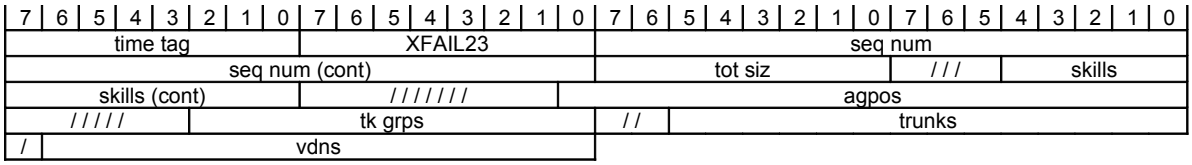
	Switch	Vect	Skill
First Release	CM4	y/n	y/n
Last Release	CM5.2		



time_tag	seconds past minute when message was formed, 0-59
opcode	XFAIL21 = 249 (0xf9)
seq num	sequence number, 0-0xffff ffff
tot siz	total size of message in bytes, 1-255
skills	number of splits that are currently measured, 1-2000
agpos	number of agent/skill pairs that are currently measured, 1-100000
tk grps	number of trunk groups that are currently measured, 1-2000
trunks	number of trunks that are currently measured, 1-12000
vdns	number of VDNs that are currently measured, 1-20000

3.2.3.2 → XFAIL23 (New CM6.0)

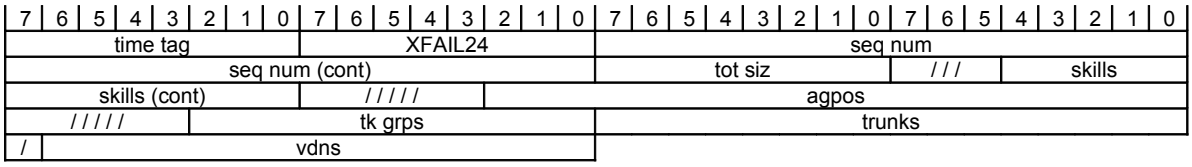
	Switch	Vect	Skill
First Release	CM6.0	y/n	y/n
Last Release	CM6.3		



time_tag	seconds past minute when message was formed, 0-59
opcode	XFAIL23 = 13 (0x0d)
seq num	sequence number, 0-0xffff ffff
tot siz	total size of message in bytes, 1-255
skills	number of splits that are currently measured, 1-8,000 (CM6.0 and later)
agpos	number of agent/skill pairs that are currently measured, 1-100,000
tk grps	number of trunk groups that are currently measured, 1-2,000
trunks	number of trunks that are currently measured, 1-12,000
vdns	number of VDNs that are currently measured, 1-30,000 (CM6.0 and later)

3.2.3.3 → XFAIL24 (New CM7.0)

	Switch	Vect	Skill
First Release	CM7.0	y/n	y/n
Last Release			



time_tag	seconds past minute when message was formed, 0-59
opcode	XFAIL24 = 41 (0x29)
seq num	sequence number, 0-0xffff ffff
tot siz	total size of message in bytes, 1-255
skills	number of splits that are currently measured, 1-8,000 (CM6.0 and later)
agpos	**number of agent/skill pairs that are currently measured, 1-360,000**
tk grps	number of trunk groups that are currently measured, 1-2,000
trunks	**number of trunks that are currently measured, 1-24,000 (64,000 future)**
vdns	number of VDNs that are currently measured, 1-30,000 (CM6.0 and later)

3.2.4 → BUSYOUT20 (New CM3.1)

The switch may reject the request to establish communications **RTCS** by sending a **BUSYOUT20** message. Refer to **BUSYOUT20** in Switch Administration (section 3.3.1) for the layout.

3.2.5 →XSTAT20 and XSTAT24

The **Translation Status** message is sent by the switch when the request (**RTCS**) for a reporting adjunct Application connection is accepted (the *splits* and *agpos* values are valid). If the "translations-changed" flag in the **XSTAT** is set (*c*=1), the reporting adjunct will request full translations by sending the **RFTB** message. If the "translations-changed" flag is not set (*c*=0), the reporting adjunct will request only logon translations by sending the **RLTB** message. However, if the reporting adjunct must (re)configure the real-time database because shared memory was cleared (e.g., initial session or adjunct rebooted), the reporting adjunct will send the **RFTB** message regardless of the value of the "translations-changed" flag.

The **XSTAT** message also indicates that status (enabled/disabled) of the Call Vectoring, Call Prompting, and EAS (skills) features on the switch.

The **XSTAT** message contains the reason for the **XSTAT** request in *fault*.

The **XSTAT** is sent by the switch, and is the last message sent in the buffer. The message set SPI language must change when capacities or supported features change.

3.2.5.1 →XSTAT20 (New CM3.1)

	Switch	Vect	Skill
First Release	CM3.1	y/n	y/n
Last Release	CM6.3		

7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
time_tag								XSTAT20								seq num															
seq num (cont)																tot siz								c	v	p	s	fault			
switch				release				/ /				language				l		/ / / /				dot		switch name							
switch name (cont)																															
switch name (cont)																															
switch name (cont)																															
switch name (cont)																															
switch name (cont)																														switch id	
switch id (cont)								time zone																							
time zone (cont)																															
time zone (cont)																															
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time zone (cont)																															
time zone (cont)																															
time zone (cont)																															
time zone (cont)																															
time zone (cont)																														country	
display																															

XSTAT20 (continued)

time_tag	seconds past minute when message was formed, 0-59
opcode	XSTAT20 = 177 (0xb1)
seq num	sequence number, 1-0xffff ffff
tot siz	total size of message in bytes; 1-255
c	change status; 1 = translations changed
v	vectoring; 1 = call vectoring is enabled
p	prompting; 1 = call prompting is enabled
s	EAS; 1 = EAS (skills) is enabled
fault	fault type; 0 = data link down, 1 = short power hit, 2 = buffer blocking, 3 = bad message, 4 = switch reload, 5 = modification
switch	switch identifier; 3 = G3
release	switch release; 1 = (not used), 2 = CM6.0, 3 = G3V3 (obsolete), 4 = G3V4 (obsolete), 5 = G3V5 (obsolete), 6 = G3V6/G3V7 (obsolete), 8 = G3V8 (obsolete), 9 = G3V9/G3V10 (obsolete), 11 = G3V11 (obsolete), 12 = MV2, 13 = CM3, 14 = CM4/CM5.0, 15 = CM5.2
language	message set SPI language - must change when capacities or supported features change. 0 = not used (=> XSTAT) 1 = R3.0 message set w/FSTATE (obsolete) 2 = G3V2/R3V2 message set w/multiple trunk state messages (obsolete) 3 = G3V4/R3V4 message set w/ vectoring enhancements (obsolete) 4 = G3V5/R3V5 message set w/ multiple agent states and audit messages (obsolete) 5 = G3V6/R3V6 message set w/ BSR and Agent Allocation features (obsolete) 6 = G3V8/R3V8 message set w/ ATM enhancements and R8.3/G3V9/R3V9 w/ NCR and Holiday Vectoring (obsolete) 7 = G3V11/R3V11 message set w/ Enhanced Dialing Plan (6 and 7 digit extensions) and Trunk Group Capacity Increases and BSR Application Plan Capacity Increase (obsolete) 8 = MV2/R12 message set w/ Capacity Increases and Maximizer and Variables in Vectors and Media Gateway and ASAI UU 9 = CM3/R13 message set w/ Variables in Vectors Enhancement 20 = CM3.1/R13.1 and IQ message set w/ IQ Names Admin, Sequence Numbers, Message Sizes, and UCID in FSTATE/ASTATE message 21 = CM4.0/CM5.0/CM5.1/CMS R14/CMS R14.1/IQ 4.0 message set 22 = CM5.2/CMS R15/IQ 5.0 message set 23 = CM6.0/CMS R16/IQ 5.1
l	LSP flag; 0 = Primary, 1 = LSP/ESS
dot	switch dot release number extracted from switch release string; 0-7
switch name	switch name (ASCII)
switch id	switch UCID node, 1-32767
time zone	time zone (ASCII)
country	country code; 1-25
display	display character set; 0 = roman (default), 1 = cyrillic, 2 = ukranian, 3 = katakana. 0, 1, 2 map to Euro font character set

3.2.5.2 **→XSTAT24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y/n	y/n
Last Release			

7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
time_tag								XSTAT24								seq num															
seq num (cont)																tot siz								c		v		p		s	fault
switch				release				/ /				language				l		/ / / /				dot		switch name							
switch name (cont)																															
switch name (cont)																															
switch name (cont)																															
switch name (cont)																															
switch name (cont)																															
switch id																															
switch id (cont)								time zone																							
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time zone (cont)																															
time zone (cont)																															
time zone (cont)																															
time zone (cont)																															
time zone (cont)																														country	
display								/ /				max_acd																			

time_tag	seconds past minute when message was formed, 0-59
opcode	XSTAT24 = 44 (0x2c)
seq num	sequence number, 1-0xffff ffff
tot siz	total size of message in bytes; 1-255
c	change status; 1 = translations changed
v	vectoring; 1 = call vectoring is enabled
p	prompting; 1 = call prompting is enabled
s	EAS; 1 = EAS (skills) is enabled
fault	fault type; 0 = data link down, 1 = short power hit, 2 = buffer blocking, 3 = bad message, 4 = switch reload, 5 = modification
switch	switch identifier; 3 = G3
release	switch release; 1 = (not used), 2 = CM6.0, **3 = CM7.0,** 4-13 = (not used), 14 = CM4/CM5.0, 15 = CM5.2
language	message set SPI language - must change when capacities or supported features change. 0 = not used (→ XSTAT) 21 = CM4.0/CM5.0/CM5.1/CMS R14/CMS R14.1/IQ 4.0 message set 22 = CM5.2/CMS R15/IQ 5.0 message set 23 = CM6.0/CMS R16/IQ 5.1 **24 = CM7.0/CMS R18**
l	LSP flag; 0 = Primary, 1 = LSP/ESS
dot	switch dot release number extracted from switch release string; 0-7
switch name	switch name (ASCII)
switch id	switch UCID node, 1-32767
time zone	time zone (ASCII)
country	country code; 1-25
display	display character set; 0 = roman (default), 1 = cyrillic, 2 = ukranian, 3 = katakana. 0, 1, 2 map to Euro font character set
max_acd	**maximum logged in ACD agents, 1-10,000**

3.2.6 → TIME20 (New CM3.1)

The **TIME** message is sent by the switch immediately after the **XSTAT** message in response to the **RTCS** message. The reporting adjunct first verifies that the switch clock is set (all parameters are valid) and if so, the reporting adjunct clock is set. If the switch clock is not set, the reporting adjunct will reinitiate the reporting adjunct Application connection. The switch administrator must set the clock to enable the connection to be established.

Once the reporting adjunct clock is set, the reporting adjunct sets an AT3 timer (2 minutes). If this timer expires, the reporting adjunct will report that the translation pumpup timed-out and then initiate another reporting adjunct session (i.e., the reporting adjunct sends another **RTCS** message). The reporting adjunct will reset the AT3 timer each time a **TIME** message is received. If the switch expects the translation pumpup to take longer than the time-out period, another **TIME** message will be sent. Additionally, **TIME** messages can be inserted anywhere in the translation blocks - preferably every minute, as is done during Call Processing.

Refer to **TIME** in Call Processing (section 3.4.1) for the layout.

3.2.7 ←RFTB

	Switch	Vect	Skill
First Release	G3V1	y/n	y/n
Last Release			

The **Request Full Translation Block** message is sent by the reporting adjunct to request a full translation pumpup. This is required if the reporting adjunct user requests that additional or fewer splits be measured, the switch indicates that translations have changed since the last session, or when the reporting adjunct shared memory has been cleared (e.g., initial session or after an adjunct reboot). The expected response is shown in the semantics document (depending on switch release and active features) and includes all messages from the **XOPEN** *block* = 0 to the **XCLOSE** *block* = 0.

7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
RFTB								NOP								NOP								NOP							

opcode RFTB = 6 (0x06)

3.2.8 ←RLTB

	Switch	Vect	Skill
First Release	G3V1	y/n	y/n
Last Release			

The **Request Logon/Status Translations** message is sent by the reporting adjunct as part of the translation pumpup when the real-time database does not have to be reconfigured since the real-time database is in synch with the switch translations. The reporting adjunct only needs an update of the agents that are currently logged in, and the current state of each skill.

The expected response is a logon/status translation block. **XOPEN** *block* = 5 surrounds the messages of the logon/status pumpup. An **ALOGON** message is sent for each staffed agent position. If there are no staffed agents, then an empty block (**XOPEN** to **XCLOSE**) is sent. Within the **XOPEN** *block* = 5 and **XCLOSE** *block* = 5 messages, the **XOPEN** *block* = 6 and **XCLOSE** *block* = 6 surrounds the **SKSTATE** for all measured skills.

7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
RLTB								NOP								NOP								NOP							

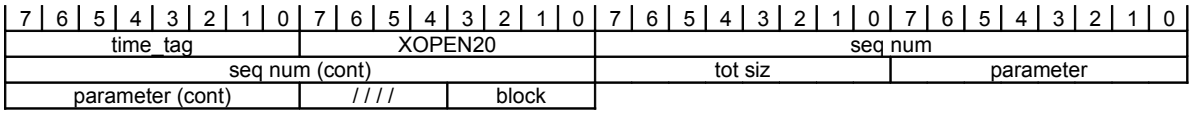
opcode RLTB = 12 (0x0c)

3.2.9 → XOPEN20 (New CM3.1)

	Switch	Vect	Skill
First Release	CM3.1	y/n	y/n
Last Release			

The **Open Translation Block** message is sent by the switch during the translation pumpup. It indicates the start of a new block of translation data.

Following the **XOPEN** is a group of messages for that block type indication the members of that block. For example, following the **XOPEN** for a trunk group (block type 2), an **XTRUNK** would be sent for each member of the trunk group. It is up to the switch to ensure that a single **XTRUNK** is sent for each trunk group member. If the reporting adjunct receives two **XTRUNK** messages for the same trunk group member, two rows of data will be created for the same trunk group member with the possibility of incorrect reporting for that trunk group member. This is true for all of the translations pumpup messages. The switch will not send duplicates.



time_tag	seconds past minute when message was formatted, 0-59
opcode	XOPEN20 = 178 (0xb2)
seq mi,	sequence number, 1-0xffff ffff
tot siz	total size of message in bytes, 1-255
parameter	value (split or trunk group number) for items in the translation block.
block	translation block:

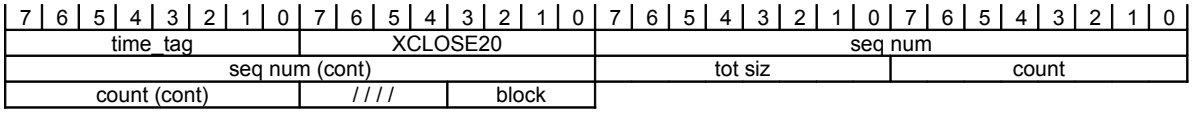
0 open full translation block
1 open agent position translation block for split
(*parameter* is split number 0-2000 (up to CM5.2), 0-8000 (CM6.0 and later)
2 open trunk translation block for trunk group
(*parameter* is trunk group number 0-2000
3 open agent logon ID translation block
4 open VDN translation block
5 open logon/status block
6 open skill state block

3.2.10 → XCLOSE20 (New CM3.1)

	Switch	Vect	Skill
First Release	CM3.1	y/n	y/n
Last Release			

The **Close Translation Block** message is sent by the switch during the translation pumpup. It indicates the end of a block of translation data. The reporting adjunct will validate the *count* value with the actual number of items received and if there is a discrepancy, the reporting adjunct Application connection will not be established until the switch administrator resolves the problem.

An **XOPEN/XCLOSE** will be sent for each skill with a count of 0 agents. Logged in agents will also be sent at the end of the pumpup.



time_tag seconds past minute when event occurred, 0-59
opcode XCLOSE20 = 179 (0xb3)
seq num sequence number, 1-0xffff ffff
tot siz total size of message in bytes, 1-255
Count number of items sent in this translation block,

agent positions	block 1, agents per split 0-1500
trunks	**block 2, trunks per trunk group 0-255 (< SPI language 24), 0-9999 (>= SPI language 24)**
logid IDs	block 3, total login IDs currently logged in 0-5200 (w/ EAS), 0-10000 (non-EAS)
VDNs	block 4, all VDNs 0-20000 (up to CM5.2), 1-30000 (CM6.0 and later)
Skill state	block 6, number measured skills 0-2000 (up to CM5.2), 0-8000 (CM6.0 and later)

Block translation block:

0 close full translation block (count field will be 0)
1 close agent position translation block
2 close trunk translation block
3 close agent logon ID translation block
4 close VDN translation block
5 open logon/status block (count field will be 0)
6 open skill state block

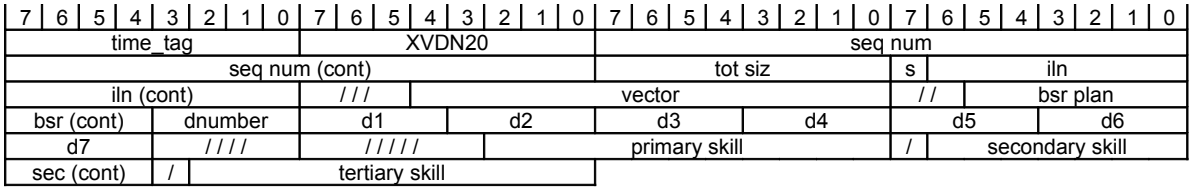
3.2.11 → XVDN20, XVDN23 and XVDN24

The **Vector Translations** messages will be sent by the switch as part of a VDN translation block, if Call Vectoring is enabled. One of these messages is sent for each measured VDN. The reporting adjunct will record the extension and vector termination for each VDN in the database. For messages which contain additional information such as VDN skill preference or BSR plan, the reporting adjunct will also record this information in the database. In subsequent messages the reporting adjunct and the switch will refer to the VDNs by the internal line number (*iln*) specified in the message. Note that the extension is sent in packed decimal format, where hex a is the digit 0 and 0 in the field is null.

This is a variable length message based on the "s" bit and "dnumber". If the "s" bit is set then 5 additional bytes are sent for the "primary-skill", "secondary-skill" and "tertiary-skill".

3.2.11.1 → XVDN20 (New CM3.1)

	Switch	Vect	Skill
First Release	CM3.1	y	y/n
Last Release	CM5.2		



time_tag	seconds past minute when message was formatted, 0-59
opcode	XVDN20 = 180 (0xb4)
seq num	sequence number, 1-0xffff ffff
tot siz	total size of message in bytes, 1-255
S	skill flag, 0 = no skills in message, 1 = skills in message
Iln	internal line number, 1-32766
vector	vector on which this VDN terminates, 1-2000
bsr plan	bsr application plan number, 1-512
dnumber	number of digits in VDN extension, 1-15
d1-d15	digits of VDN extension (packed decimal format, where hex a is digit 0 and 0 in this field is null, zero fill extension digits to a byte boundary)
primary skill	when s = 1, primary VDN agent skill 1-2000, 0 = no skill used
secondary skill	when s = 1, secondary VDN agent skill 1-2000, 0 = no skill used
tertiary skill	when s = 1, tertiary VDN agent skill 1-2000, 0 = no skill used

3.2.11.2 **→XVDN23 (New CM6.0)**

	Switch	Vect	Skill
First Release	CM6.0	y	y/n
Last Release	CM6.3		

7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
time_tag								XVDN23								seq num															
seq num (cont)																tot siz								s	iln						
iln (cont)								/ / /								vector								/ /	bsr plan						
bsr (cont)				dnumber				d1				d2				d3				d4				d5				d6			
d7				d8				d9				d10				d11				d12				d13				d14			
d15				/ / / /				/ / /				primary skill												/ / /				sec skill			
sec skill (cont)								/ / /				tertiary skill																			

time_tag	seconds past minute when message was formatted, 0-59
opcode	XVDN23 = 17 (0x11)
seq num	sequence number, 1-0xffff ffff
tot siz	total size of message in bytes, 1-255
s	skill flag, 0 = no skills in message, 1 = skills in message
iln	internal line number, 1-32766
vector	vector on which this VDN terminates, 1-8000 (CM6.0 and later)
bsr plan	bsr application plan number, 1-512
dnumber	number of digits in VDN extension, 1-15
d1-d15	digits of VDN extension (packed decimal format, where hex a is digit 0 and 0 in this field is null, zero fill extension digits to a byte boundary)
primary skill	when s = 1, primary VDN agent skill 1-8000 (CM6.0 and later), 0 = no skill used
secondary skill	when s = 1, secondary VDN agent skill 1-8000 (CM6.0 and later), 0 = no skill used
tertiary skill	when s = 1, tertiary VDN agent skill 1-8000 (CM6.0 and later), 0 = no skill used

3.2.11.3 **→XVDN24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y	y/n
Last Release			

7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
time_tag								XVDN24								seq num															
seq num (cont)																tot siz								s				iln			
iln (cont)								/ / /				vector												/ /				bsr plan			
bsr (cont)				/ / / /				/ / /				dnumber				d1				d2				d3				d4			
d5				d6				d7				d8				d9				d10				d11				d12			
d13				d14				d15				d16				/ / / / / /								tenant							
/ / /				primary skill												/ / /				secondary skill											
/ / /				tertiary skill																											

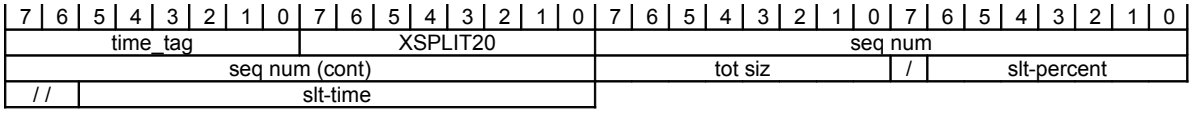
time_tag	seconds past minute when message was formatted, 0-59
opcode	XVDN24 = 45 (0x2d)
seq num	sequence number, 1-0xffff ffff
tot siz	total size of message in bytes, 1-255
s	skill flag, 0 = no skills in message, 1 = skills in message
iln	internal line number, 1-32766
vector	vector on which this VDN terminates, 1-8000 (CM6.0 and later)
bsr plan	bsr application plan number, 1-512
dnumber	**number of digits in VDN extension, 1-16**
d1-d16	**digits of VDN extension (packed decimal format, where hex a is digit 0 and 0 in this field is null, zero fill extension digits to a byte boundary).**
tenant	**VDN tenant number, 1-250**
primary skill	when s = 1, primary VDN agent skill 1-8000 (CM6.0 and later), 0 = no skill used
secondary skill	when s = 1, secondary VDN agent skill 1-8000 (CM6.0 and later), 0 = no skill used
tertiary skill	when s = 1, tertiary VDN agent skill 1-8000 (CM6.0 and later), 0 = no skill used

3.2.12 →XSPLIT20 and XSPLIT24

The **Split Translations** messages is sent by the switch and is the first message of the agent position translation block (**XOPEN** message **block** type 2).

3.2.12.1 →XSPLIT20 (New CM3.1)

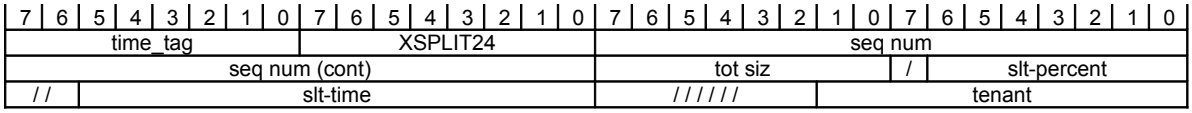
	Switch	Vect	Skill
First Release	CM3.1	y/n	y/n
Last Release	CM6.3		



time_tag	seconds past minute when message was formatted, 0-59
opcode	XSPLIT20 = 181 (0xb5)
seq num	sequence number, 0-4,294-967,295
tot siz	total size in bytes, 1-255
slt-percent	service level target percentage 1-99
slt-time	service level target time 1-9999 seconds

3.2.12.2 →XSPLIT24 (New CM7.0)

	Switch	Vect	Skill
First Release	CM7.0	y/n	y/n
Last Release			



time_tag	seconds past minute when message was formatted, 0-59
opcode	XSPLIT24 = 46 (0x2e)
seq num	sequence number, 0-4,294-967,295
tot siz	total size in bytes, 1-255
slt-percent	service level target percentage 1-99
slt-time	service level target time 1-9999 seconds
tenant	**Split tenant number, 1-250**

3.2.13 → XAGENT20 and XAGENT24

This message is for non-EAS only.

The **Agent Position Translations** message is sent by the switch as part of a split translation block. One of these messages is sent for each agent position assigned to the split. The reporting adjunct will record the extension and split assignment for each agent position in the database. In subsequent messages the reporting adjunct and the switch will refer to the agent positions by the terminal number (*pos*) specified in the message. Note that the extension is sent in packed decimal format, where hex a is the digit 0 and 0 in the field is null.

Without EAS, a single extension may be assigned to multiple splits. In this case, the switch will send a separate **XAGENT** message for each extension appearance in a split. In other words, each agent/split pair has a unique terminal number and is tracked as a separate agent position by the reporting adjunct. With EAS, only agents logged in are reported in translation pumpup therefore this message is not used.

3.2.13.1 → XAGENT20 (New CM3.1)

	Switch	Vect	Skill
First Release	CM3.1	y/n	n
Last Release	CM6.3		

7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
time_tag								XAGENT20								seq num															
seq num (cont)																tot siz								pos							
pos (cont)								dnumber				d1				d2				d3				d4				d5			
d6								d7				d8				d9				d10				d11				d12			
d13								d14				d15																			

time_tag	seconds past minute when message was formatted, 0-59
opcode	XAGENT20 = 182 (0xb6)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
pos	agent position - without EAS (1-10,000),
dnumber	number of digits in agent extension, 1-15
d1-d15	digits of agent extension, packed decimal format, where hex a is digit 0 and 0 in this field is null, zero fill extension digits to a byte boundary

3.2.13.2 **→XAGENT24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y/n	n
Last Release			

7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
time_tag								XAGENT24								seq num															
seq num (cont)																tot siz								pos							
pos (cont)								/ / /				dnumber				d1				d2				d3				d4			
d5				d6				d7				d8				d9				d10				d11				d12			
d13				d14				d15				d16																			

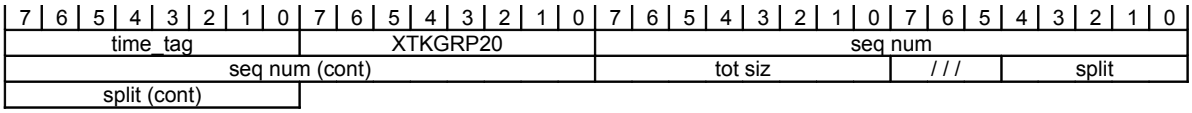
time_tag seconds past minute when message was formatted, 0-59
opcode XAGENT24 = 47 (0x2f)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
pos agent position - without EAS, 1-10,000
dnumber number of digits in agent extension, 1-16
d1-d16 digits of agent extension, packed decimal format, where hex a is digit 0 and 0 in this field is null, zero fill extension digits to a byte boundary.

3.2.14 →XTKGRP20 and XTKGRP24

The **Trunk Group-Split Translations** messages is sent by the switch and is the first message of a trunk group translation block. One message is sent for each trunk group that does not terminate to a VDN. The reporting adjunct records the split assignment for each trunk group in the database.

3.2.14.1 →XTKGRP20 (New CM3.1)

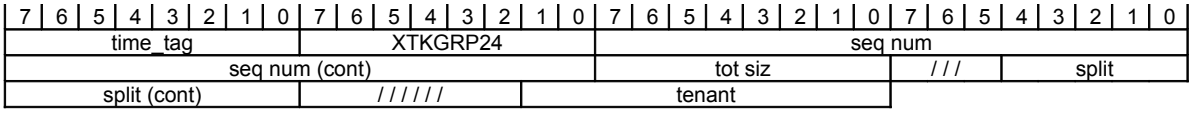
	Switch	Vect	Skill
First Release	CM3.1	y/n	y/n
Last Release	CM6.3		



time_tag seconds past minute when message was formatted, 0-59
opcode XTKGRP20 = 183 (0xb7)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
split split on which trunk group terminates, 0-2000 (up to CM5.2), 0-8000 (CM6.0 and later)
(0 indicates that the trunk group does not terminate on a split or terminates on an unmeasured split)

3.2.14.2 →XTKGRP24 (New CM7.0)

	Switch	Vect	Skill
First Release	CM7.0	y/n	y/n
Last Release			



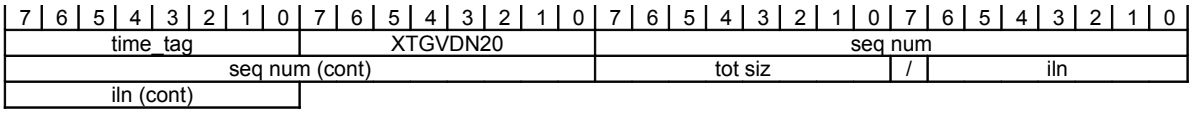
time_tag seconds past minute when message was formatted, 0-59
opcode XTKGRP24 = 48 (0x30)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
split split on which trunk group terminates, 0-2000 (up to CM5.2), 0-8000 (CM6.0 and later)
(0 indicates that the trunk group does not terminate on a split or terminates on an unmeasured split)
tenant **Trunk Group tenant number, 1-250.**

3.2.15 → XTGVDN20 and XTGVDN24

3.2.15.1 → XTGVDN20 (New CM3.1)

	Switch	Vect	Skill
First Release	CM3.1	y	y/n
Last Release	CM6.3		

The **Trunk Group-VDN Translations** message is sent by the switch and is the first message of a trunk group translation block. It is used if Call Vectoring is enabled and/or the trunk group terminates on a VDN. One message is sent for each measured trunk group. The reporting adjunct records the VDN termination for each trunk group in the database.

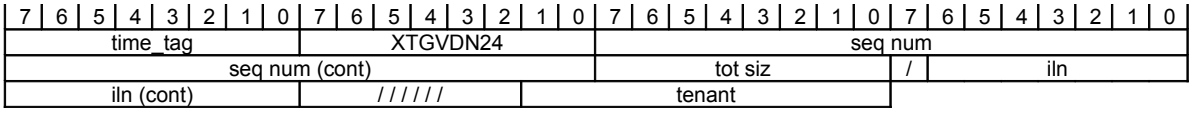


time_tag seconds past minute when message was formatted, 0-59
opcode XTGVDN20 = 184 (0xb8)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
iln internal line number, 0-32766 (0 indicates that the trunk group does not terminate on a VDN or terminates on an unmeasured VDN)

3.2.15.2 → XTGVDN24 (New CM7.0)

	Switch	Vect	Skill
First Release	CM7.0	y	y/n
Last Release			

The **Trunk Group-VDN Translations** message is sent by the switch and is the first message of a trunk group translation block. It is used if Call Vectoring is enabled and/or the trunk group terminates on a VDN. One message is sent for each measured trunk group. The reporting adjunct records the VDN termination for each trunk group in the database.



time_tag seconds past minute when message was formatted, 0-59
opcode XTGVDN24 = 49 (0x31)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
iln internal line number, 0-32766 (0 indicates that the trunk group does not terminate on a VDN or terminates on an unmeasured VDN)
tenant **Trunk Group tenant number, 1-250**

3.2.16 → XTRUNK20 and XTRUNK24

The **Trunk Translations** messages are sent by the switch as part of a trunk group translation block. One message is sent for each trunk that is a member of the opened trunk group. The reporting adjunct will record the equipment location, trunk group assignment and location-id for each trunk in the database. In subsequent messages the reporting adjunct and the switch will refer to the trunk by the internal trunk number (*itn*) specified in the message.

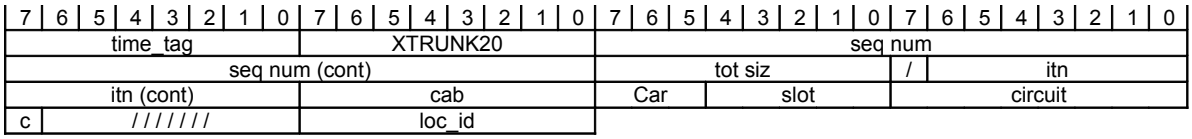
SPI pumpup supports sending up to 24,000 XTRUNK20 messages.
SPI pumpup supports sending up to 24,000 XTRUNK24 messages (64,000 future), with 9,999 per trunk group. (Currently only sending up to 999 per trunk group.)

For **XTRUNK20** and non-H.323/SIP trunks in XTRUNK24, the mapping into the cab, carrier and slot are defined in the table in section 5.

For **XTRUNK24**, H.323/SIP trunks set the "t" flag to indicate that the cabinet, carrier and slot fields contain the binary number for a virtual IP or SIP trunk number (range 00001 to 64,000).

3.2.16.1 → XTRUNK20 (New CM3.1)

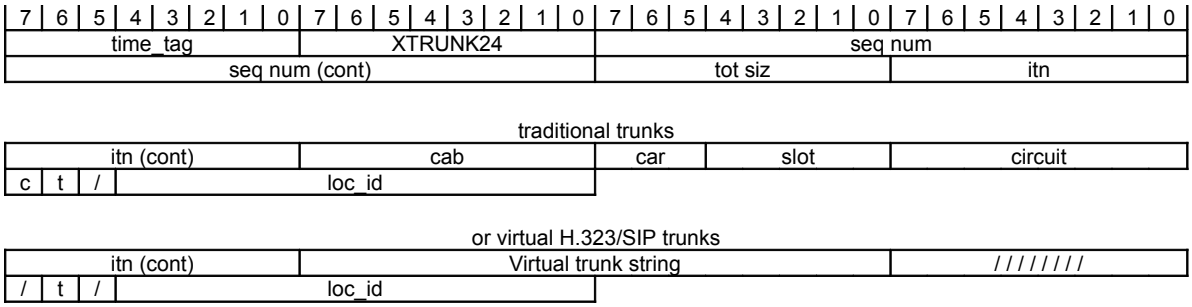
	Switch	Vect	Skill
First Release	CM3.1	y/n	y/n
Last Release	CM6.3		



time_tag	seconds past minute when message was formatted, 0-59
opcode	XTRUNK20 = 185 (0xb9)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
itn	internal trunk number, 1-32766
cab	cabinet, 1-44 (G3V8-G3V11), 1-250 (MV2 and later)
car	carrier, 1-5 representing letters A-E, 7 representing letter V
slot	slot, 1-25
circuit	circuit, 1-256 (9 bits, continued in second byte)
loc_id	location id number, 0-250, 0 is default for no location id

3.2.16.2 **→XTRUNK24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y/n	y/n
Last Release			



time_tag	seconds past minute when message was formatted, 0-59
opcode	XTRUNK24 = 37 (0x25)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
itn	**internal trunk number, 1-65,535**
cab	cabinet, 1-250
car	carrier, 1-5 representing letters A-E, 7 representing letter V
slot	slot, 1-25
circuit	circuit, 1-256 (9 bits, continued in second byte)
Virtual trunk string	**The Txxxxx virtual H.323 or SIP trunk string, 1-24,000 (1-64,000 in the future)**
t	**trunk location format. 0=use format with cab, car, slot, circuit 1=use format with Virtual trunk string**
loc_id	**location id number, 0-2000. 0 is default for no location id (13 bits for future expansion)**

3.2.17 **→SKSTATE20 and SKSTATE23**

Refer to **SKSTATE** in Call Processing (section 3.4.29) for the layout.

3.2.18 **→ALOGON20, ALOGON23 and ALOGON24**

Refer to **ALOGON** in the Call Processing (section 3.4.2) for message layout.

3.3 **Switch Administration**

Switch Administration request messages will only be sent by the reporting adjunct to the switch when the reporting adjunct Session is in the *DATA TRANSFER* mode. The switch response messages will be intermixed with the Call Processing messages and will be time-stamped and sent (or buffered) in the order that they were processed.

Every request message that is sent by the reporting adjunct will be tagged (*tag*) and the switch will return this *tag*, unchanged, in each response message that is associated with that particular request. The reporting adjunct uses the *tag* to correlate the switch response with the reporting adjunct switch Administration process that sent the request.

If the switch does not recognize a request message, or can not interpret one or more of the message parameters, the switch will return the **RCODE** message with *rcode*=1 (illegal request). Otherwise the normal response message(s) will be returned.

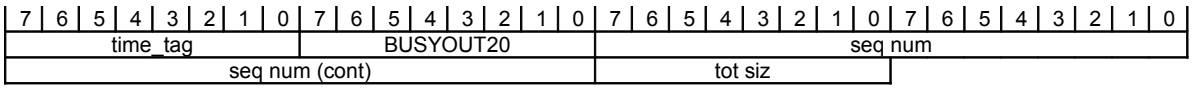
3.3.1 → BUSYOUT20 (New CM3.1)

	Switch	Vect	Skill
First Release	CM3.1	y/n	y/n
Last Release			

The **Busy-Out Link** message is sent by the switch when the reporting adjunct Session is busied-out for administration purposes. This is usually done by the switch administrator when making changes that will affect the reporting adjunct database configuration (e.g., adding another extension to a measured split or administering another trunk group as measured). These changes will typically result in the "translations changed" flag being set so that when the reporting adjunct reinitiates the session (**RTCS**) the switch will report that translations changed (**XSTAT20** *c*=1) and the reporting adjunct will request full translations (**RFTB**) in order to update the real-time database to reflect the changes that were made. This message is also sent in response to an **RTCS** if the reporting adjunct Session is busied-out at the time of the request.

The reporting adjunct will mark the reporting adjunct Session *BUSIED-OUT*, initialize all measured agent positions and trunks, and wait for another reporting adjunct Session to be established.

The switch supports busying out the link, but this does not need to be done for database configuration changes.

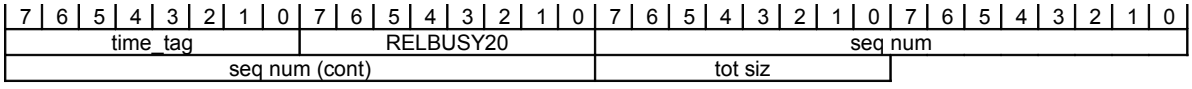


time_tag	seconds past minute when event occurred, 0-59
opcode	BUSYOUT20 = 186 (0xba)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255

3.3.2 → RELBUSY20 (New CM3.1)

	Switch	Vect	Skill
First Release	CM3.1	y/n	y/n
Last Release			

The **Release Busy Link** message is sent by the switch when the reporting adjunct session is released from busy-out. The reporting adjunct will attempt to initiate another reporting adjunct session by sending the **RTCS** message.



time_tag	seconds past minute when event occurred, 0-59
opcode	RELBUSY20 = 187 (0xbb)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255

3.3.3 → ARVDN20, ARVDN23 and ARVDN24

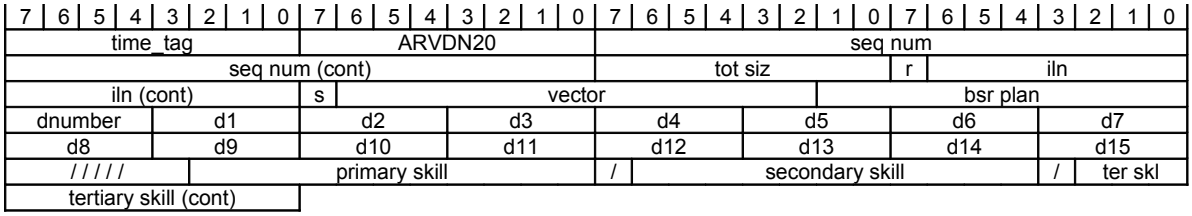
The **Add/Remove Vector Translations** message is sent by the switch when a VDN is added/removed as a measured VDN if Call Vectoring is enabled. This allows for dynamic administration without bringing down the reporting adjunct link. One of these messages is sent for each measured VDN. VDNs are referenced by internal line number (*iln*).

VDNs can be made measured or unmeasured or removed using switch administration while calls are in progress. VDN statistics will not be accumulated for these calls in progress (i.e, calls in progress will look like calls to an unmeasured VDN).

This is a variable length message based on the "s" bit and "dnumber". If the "s" bit is set then 5 additional bytes are sent for the "primary-skill", "secondary-skill" and "tertiary-skill".

3.3.3.1 → ARVDN20 (New CM3.1)

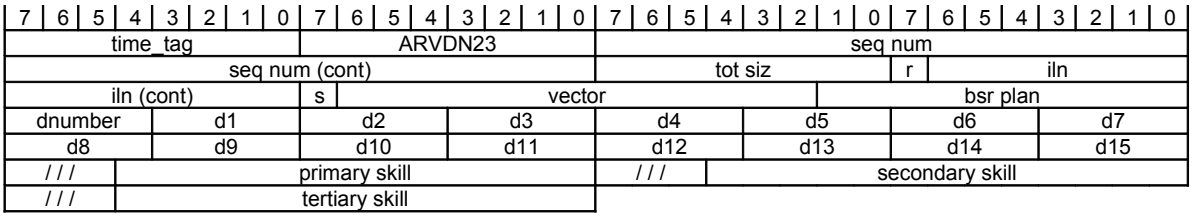
	Switch	Vect	Skill
First Release	CM3.1	y	y/n
Last Release	CM5.2		



time_tag	seconds past minute when message was formed, 0-59
opcode	ARVDN20 = 188 (0xbc)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
r	remove, 0 = add, 1 = remove
iln	internal line number, 1-32766
s	skill flag, 0 = no skills in message, 1 = skills in message
vector	vector on which this VDN terminates, 1-2000
bsr plan	bsr application plan number, 1-512
dnumber	number of digits in VDN extension, 1-15
d1-d15	digits of VDN extension (packed decimal format, where hex a is digit 0 and 0 in this field is null, zero fill extension digits to a byte boundary)
primary skill	when s = 1, primary VDN agent skill 1-2000, 0 = no skill used
secondary skill	when s = 1, secondary VDN agent skill 1-2000, 0 = no skill used
tertiary skill	when s = 1, tertiary VDN agent skill 1-2000, 0 = no skill used

3.3.3.2 **→ARVDN23 (New CM6.0)**

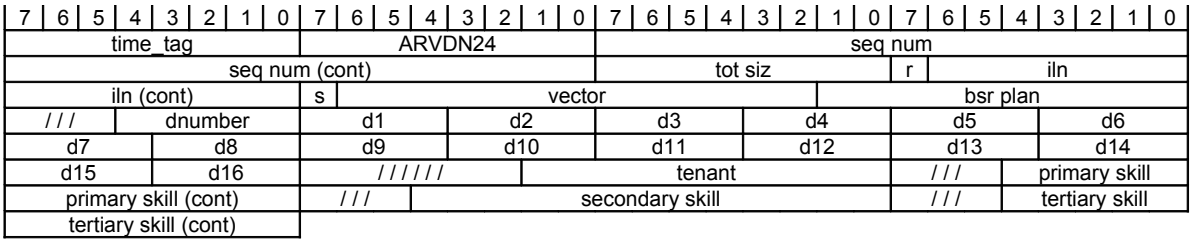
	Switch	Vect	Skill
First Release	CM6.0	y	y/n
Last Release	CM6.3		



time_tag	seconds past minute when message was formed, 0-59
opcode	ARVDN23 = 18 (0x12)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
r	remove, 0 = add, 1 = remove
iln	internal line number, 1-32766
s	skill flag, 0 = no skills in message, 1 = skills in message
vector	vector on which this VDN terminates, 1-8000 (CM6.0 and later)
bsr plan	bsr application plan number, 1-512
dnumber	number of digits in VDN extension, 1-15
d1-d15	digits of VDN extension (packed decimal format, where hex a is digit 0 and 0 in this field is null, zero fill extension digits to a byte boundary)
primary skill	when s = 1, primary VDN agent skill 1-8000 (CM6.0 and later), 0 = no skill used
secondary skill	when s = 1, secondary VDN agent skill 1-8000 (CM6.0 and later), 0 = no skill used
tertiary skill	when s = 1, tertiary VDN agent skill 1-8000 (CM6.0 and later), 0 = no skill used

3.3.3.3 **→ARVDN24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y	y/n
Last Release			



time_tag	seconds past minute when message was formed, 0-59
opcode	ARVDN24 = 50 (0x32)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
r	remove, 0 = add, 1 = remove
iln	internal line number, 1-32766
s	skill flag, 0 = no skills in message, 1 = skills in message
vector	vector on which this VDN terminates, 1-8000 (CM6.0 and later)
bsr plan	bsr application plan number, 1-512
dnumber	**number of digits in VDN extension, 1-16**
d1-d16	**digits of VDN extension (packed decimal format, where hex a is digit 0 and 0 in this field is null, zero fill extension digits to a byte boundary).**
tenant	**VDN tenant number, 1-250**
primary skill	when s = 1, primary VDN agent skill 1-8000 (CM6.0 and later), 0 = no skill used
secondary skill	when s = 1, secondary VDN agent skill 1-8000 (CM6.0 and later), 0 = no skill used
tertiary skill	when s = 1, tertiary VDN agent skill 1-8000 (CM6.0 and later), 0 = no skill used

3.3.4 → ARAGENT20 and ARAGENT24

This message is for non-EAS only.

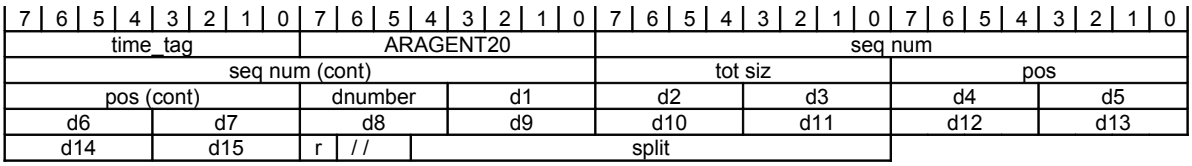
The **Add/Remove Agent Position Translations** message is sent by the switch when an agent is added/removed from a measured split. This allows for dynamic administration without bringing down the -adjunct reporting link. One of these messages is sent for each measured agent position that belongs to the measured split. With EAS, only agents logged in are reported in translation pumpup therefore there is no need to add/remove agents with this message.

When a split is added, the **ARSPLIT** message is sent first followed by the **ARAGENT** messages. When a split is removed, the **ARAGENT** messages are sent before the **ARSPLIT** message.

The agent cannot be changed to measured or unmeasured or removed while the agent is active on a call. The request to change the "measured" field on the switch hunt group form will be blocked by the switch.

3.3.4.1 → ARAGENT20 (New CM3.1)

	Switch	Vect	Skill
First Release	CM3.1	y/n	n
Last Release	CM6.3		



time_tag	seconds past minute when message was formatted, 0-59
opcode	ARAGENT20 = 189 (0xbd)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
pos	agent position - 1-10000
dnumber	number of digits in agent extension, 1-15,
d1-d15	digits of agent extension, packed decimal format, where hex a is digit 0 and 0 in this field is null, zero fill extension digits to a byte boundary
r	remove; 0 = add, 1 = remove
split	split number 0-8000 (CM6.0 and later)

3.3.4.2 **→ARAGENT24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y/n	n
Last Release			

7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
time_tag								ARAGENT24								seq num															
seq num (cont)																tot siz								pos							
pos (cont)								/ / /				dnumber				d1				d2				d3				d4			
d5				d6				d7				d8				d9				d10				d11				d12			
d13				d14				d15				d16				r	/ /			split											

time_tag seconds past minute when message was formatted, 0-59
opcode ARAGENT24 = 51 (0x33)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
pos agent position - 1-10000
dnumber number of digits in agent extension, 1-16
d1-d16 digits of agent extension, packed decimal format, where hex a is digit 0 and 0 in this field is null, zero fill extension digits to a byte boundary.
r remove; 0 = add, 1 = remove
split split number 0-8000 (CM6.0 and later)

3.3.5 → ARSPLIT20, ARSPLIT23 and ARSPLIT24

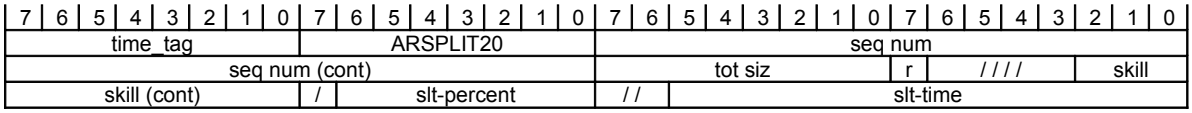
The **Add/Remove Split Translations** message is sent by the switch when a skill is added/removed as a measured skill. A skill can not be removed if agents are logged in to that skill. With EAS, after the **ARSPLIT** message is sent, the **SKSTATE** message will follow.

This allows for dynamic administration without bringing down the reporting adjunct link. One of these messages is sent for each measured skill.

If calls are active in the skill, a request to change the "measured" field on the switch hunt group form will be blocked by the switch.

3.3.5.1 → ARSPLIT20 (New CM3.1)

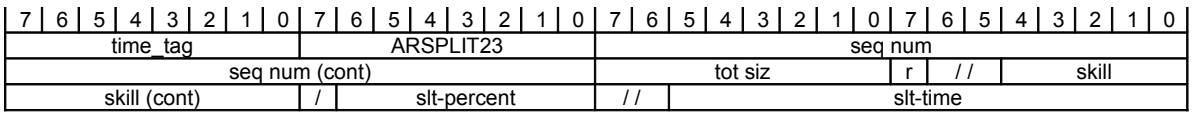
	Switch	Vect	Skill
First Release	CM3.1	y/n	y/n
Last Release	CM5.2		



time_tag seconds past minute when message was formatted, 0-59
opcode ARSPLIT20 = 190 (0xbe)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
r remove; 0 = add; 1 = remove
skill skill number 0-2000
slt-percent service level target percentage 1-99
slt-time service level target time 1-9999 seconds

3.3.5.2 → ARSPLIT23 (New CM6.0)

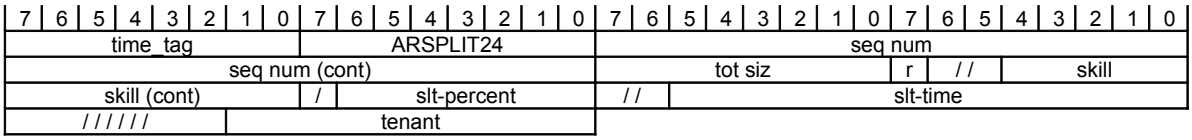
	Switch	Vect	Skill
First Release	CM6.0	y/n	y/n
Last Release	CM6.3		



time_tag seconds past minute when message was formatted, 0-59
opcode ARSPLIT23 = 19 (0x13)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
r remove; 0 = add; 1 = remove
skill skill number 0-8000 (CM6.0 and later)
slt-percent service level target percentage 1-99
slt-time service level target time 1-9999 seconds

3.3.5.3 **→ARSPLIT24 (New CM7.0**

	Switch	Vect	Skill
First Release	CM7.0	y/n	y/n
Last Release			



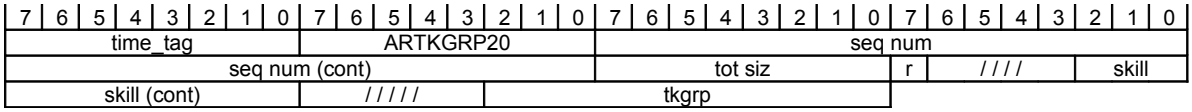
time_tag	seconds past minute when message was formatted, 0-59
opcode	ARSPLIT24 = 52 (0x34)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
r	remove; 0 = add; 1 = remove
skill	skill number 0-8000 (CM6.0 and later)
slt-percent	service level target percentage 1-99
slt-time	service level target time 1-9999 seconds
tenant	**Split tenant number, 1-250**

3.3.6 → ARTKGRP20, ARTKGRP23 and ARTKGRP24

The **Add/Remove Trunk Group-Split Translations** message is sent by the switch when a trunk group is added/removed as a measured trunk group and terminates on a skill. This allows for dynamic administration without bringing down the reporting adjunct link. One of these messages is sent for each measured trunk group. Typically **ARTRUNK** messages containing the trunk members follow an **ARTKGRP** message. Measured trunks cannot be changed to unmeasured or added or removed using switch administration with calls active. This is blocked by the switch. If the trunk group is changed to measured with calls active, the active calls will be treated as an unmeasured trunk group call.

3.3.6.1 → ARTKGRP20 (New CM3.1)

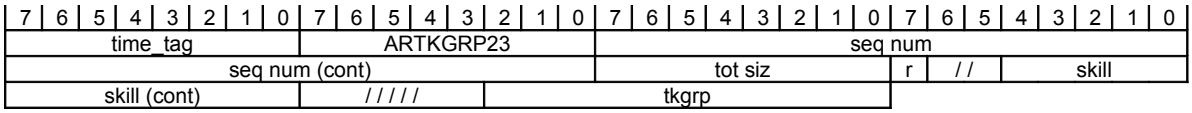
	Switch	Vect	Skill
First Release	CM3.1	y/n	y/n
Last Release	CM5.2		



time_tag seconds past minute when message was formatted, 0-59
opcode ARTKGRP20 = 191 (0xbf)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
r remove; 0 = add; 1 = remove
skill skill on which trunk group terminates, 0-2000 (0 indicates that the trunk group does not terminate on a split or terminates on an unmeasured split)
tkgrp trunk group number, 0-2000

3.3.6.2 → ARTKGRP23 (New CM6.0)

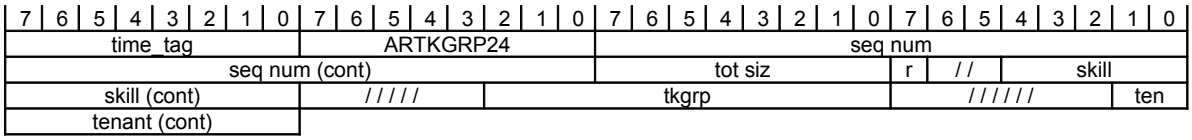
	Switch	Vect	Skill
First Release	CM6.0	y/n	y/n
Last Release	CM6.3		



time_tag seconds past minute when message was formatted, 0-59
opcode ARTKGRP23 = 20 (0x14)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
r remove; 0 = add; 1 = remove
skill skill on which trunk group terminates, 0-8000 (CM6.0 and later)
(0 indicates that the trunk group does not terminate on a split or terminates on an unmeasured split)
tkgrp trunk group number, 0-2000

3.3.6.3 **→ARTKGRP24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y/n	y/n
Last Release			



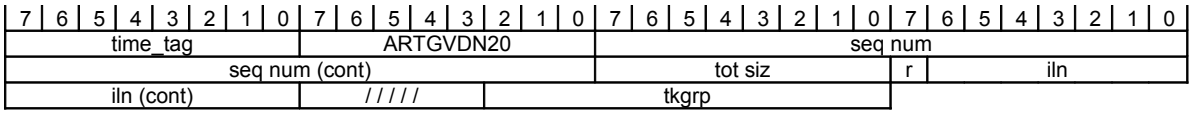
time_tag seconds past minute when message was formatted, 0-59
opcode ARTKGRP24 = 53 (0x35)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
r remove; 0 = add; 1 = remove
skill skill on which trunk group terminates, 0-8000 (CM6.0 and later)
(0 indicates that the trunk group does not terminate on a split or terminates on an unmeasured split)
tkgrp trunk group number, 0-2000
tenant Trunk Group tenant number, 1-250

3.3.7 → ARTGVDN20 and ARTGVDN24

The **Add/Remove Trunk Group-VDN Translations** message is sent by the switch when a trunk group that terminates on a VDN is add/removed as a measured trunk group. This allows for dynamic administration without bringing down the reporting adjunct link. One of these messages is sent for each measured trunk group.

3.3.7.1 → ARTGVDN20 (New CM3.1)

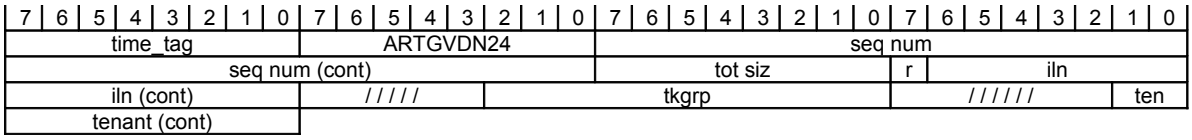
	Switch	Vect	Skill
First Release	CM3.1	y	y/n
Last Release	CM6.3		



time_tag seconds past minute when message was formatted, 0-59
opcode ARTGVDN20 = 192 (0xc0)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
r remove; 0 = add, 1 = remove
iln internal line number of the VDN, 0-32766
(0 indicates that the trunk group does not terminate on a VDN or terminates on an unmeasured VDN)
tkgrp trunk group number, 0-666 (G3V8-G3V10), 0-2000 (G3V11 and later)

3.3.7.2 → ARTGVDN24 (New CM7.0)

	Switch	Vect	Skill
First Release	CM7.0	y	y/n
Last Release			



time_tag seconds past minute when message was formatted, 0-59
opcode ARTGVDN24 = 54 (0x36)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
r remove; 0 = add, 1 = remove
iln internal line number of the VDN, 0-32766
(0 indicates that the trunk group does not terminate on a VDN or terminates on an unmeasured VDN)
tkgrp trunk group number, 0-2000
tenant **Trunk group tenant number, 1-250**

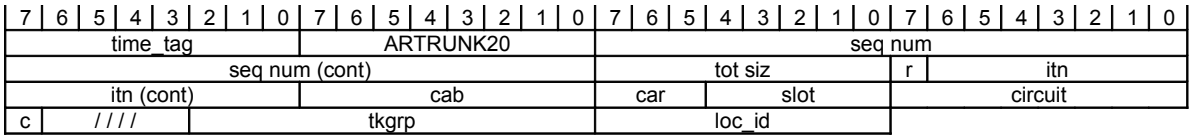
3.3.8 → ARTRUNK20 and ARTRUNK24

The **Add/Remove Trunk Translations** message is sent by the switch when a trunk is added or removed from a measured trunk group. This allows for dynamic administration without bringing down the reporting adjunct link. One of these messages is sent for each trunk that is added/removed. Trunks are referenced by internal trunk number (*itn*). For virtual IP/SIP trunk ports, the mapping into the cab, slot and circuit are defined in the table in section 5.

Measured trunks cannot be changed to unmeasured or added or removed using switch administration with calls active. This is blocked by the switch. If the trunk group is changed to measured with calls active, the active calls will be treated as an unmeasured trunk group call.

3.3.8.1 → ARTRUNK20 (New CM3.1)

	Switch	Vect	Skill
First Release	CM3.1	y/n	y/n
Last Release	CM6.3		



time_tag	seconds past minute when message was formatted, 0-59
opcode	ARTRUNK20 = 193 (0xc1)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
r	remove; 0 = add; 1 = remove
itn	internal trunk number, 1-32766
cab	cabinet, 1-44 (G3V8-G3V11), 1-250 (MV2 and later)
car	carrier, 1-5 representing letters A-E, 7 representing letter V (G3V11 and later)
slot	slot, 1-25
circuit	circuit, 1-256 (9 bits, continued in second byte)
tkgrp	trunk group number, 0-666 (G3V8-G3V10), 0-2000 (G3V11 and later)
loc id	location id number, 0-44 (G3V8-G3V11), 0-250 (MV2 and later), 0 is default for no location id

3.3.8.2 **→ARTRUNK24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y/n	y/n
Last Release			

7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
time_tag								ARTRUNK24								seq num															
seq num (cont)																tot siz								itn							

traditional trunks

itn (cont)												cab						car			slot						circuit			
c		r		t		/	/	tkgrp										/	/	/	loc_id										

or Virtual H.323/SIP trunks

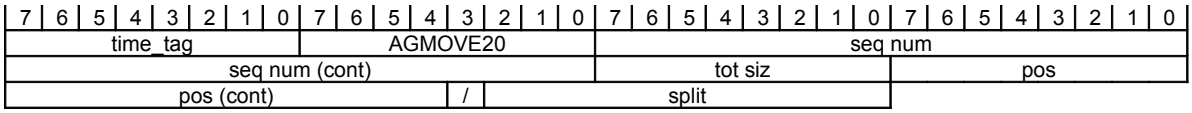
itn (cont)												Virtual trunk string									/	/	/	/	/	/	/	/	/	/	/
/		r		t		/	/	tkgrp										/	/	/	loc_id										

time_tag	seconds past minute when message was formatted, 0-59
opcode	ARTRUNK24 = 38 (0x26)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
itn	**internal trunk number, 1-65535**
cab	cabinet, 1-44 (G3V8-G3V11), 1-250 (MV2 and later)
car	carrier, 1-5 representing letters A-E, 7 representing letter V (G3V11 and later)
slot	slot, 1-25
circuit	circuit, 1-256 (9 bits, continued in second byte)
r	remove; 0 = add; 1 = remove
Virtual trunk string	**The Txxxxx virtual H.323 or SIP trunk string, 1-24,000 (1-64,000 in the future)**
t	**trunk location format. 0=use format with cab, car, slot, circuit 1=use format with Virtual trunk string**
tkgrp	trunk group number, 0-666 (G3V8-G3V10), 0-2000 (G3V11 and later)
loc_id	**location id number, 0-2000, 0 is default for no location id (13 bits for future expansion)**

3.3.9 → AGMOVE20 (New CM3.1)

	Switch	Vect	Skill
First Release	CM3.1	y/n	n
Last Release			

The **Agent Position Move** message is sent by the switch when an agent position (extension) is removed from and/or moved to a new split. When an agent position is removed from a split or added to an unmeasured split, the switch will send an **Agent Position Move** message with *split*=0. When an agent position is added to a measured split, the switch will send an **Agent Position Move** message with appropriate split number. When an agent position is moved from one split to another split the switch may send both the "remove" and "add" messages or simply the "add" ("move") message. The reporting adjunct will update the split assignment for the specified agent position. The **AGMOVE** message is sent by switch to the reporting adjunct in response to a MVAGSTFD from the reporting adjunct (only when EAS is not optioned).



time_tag	seconds past minute when event occurred, 0-59
opcode	AGMOVE20 = 248 (0xF8)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
pos	agent position - 1-10000 (G3V8 and later)
split	split number, 1-999 (G3V8-G3V11), 1-2000 (MV2 and later)

3.3.10 ← MVAGSTFD21, MVAGSTFD23 and MVAGSTFD24

The **Move Agent While Staffed** message is a variable length message sent by the reporting adjunct to the switch. In an EAS environment, the **MVAGSTFD** message is a request to add, remove, or change the skill assignments for up to 32 agent positions (LoginID).

An agent with one skill cannot have that skill removed, and an agent with the maximum number of skills cannot have a skill added. The remaining skills are not shifted when a skill is removed, so gaps can occur in the administered positions for an agent. For example, if the first of four skills is removed, the agent will have three remaining skills, in positions 2, 3, and 4. Skills are added in the first available position.

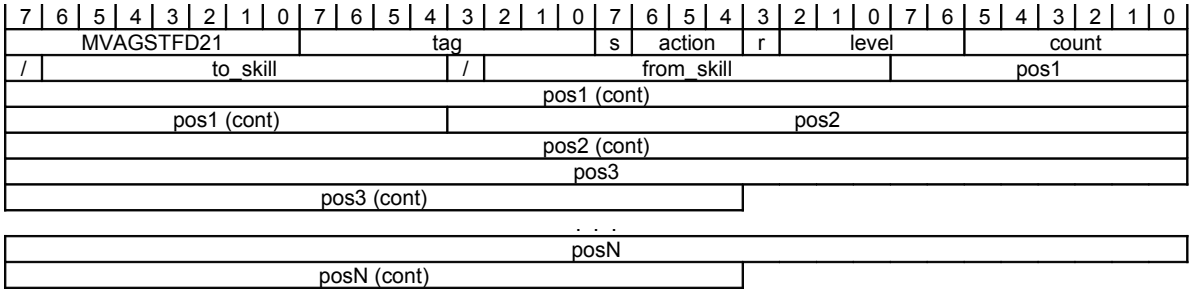
The expected response is one **RCODE** message *or* one **CRCODE** and up to 32 sets of messages to effect the move. With EAS, this is **ALOGON**.

Starting with **MVAGSTFD24**, with the increased extension size, this request may not fit into a single **MVAGSTFD** message. we are changing skills on more than 16 agents, a second message will be sent. The 1st message in series covers agents 1-16 and the 2nd message covers agents 17-32, mapping to pos1-pos16. The end bit is set for the last message in a series.

3.3.10.1 **←MVAGSTFD21 (New CM4)**

	Switch	Vect	Skill
First Release	CM4	y/n	y/n
Last Release	CM5.2		

The first 14 bytes are always sent.
This message is specific to the 13 Digit Dial Plan Expansion enhancement.

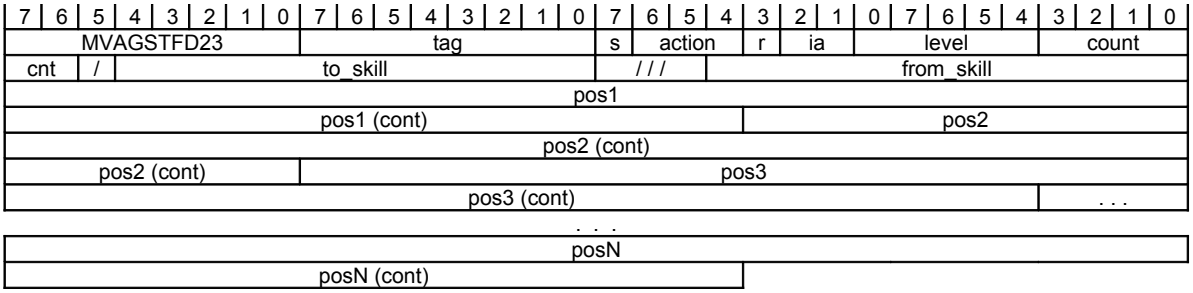


opcode	MVAGSTFD21 = 53 (0x35)
tag	message tag, 0-255 (it will be returned in the switch response message(s))
s	skills, 1=skills are used in this message, 0=skills not used. If s=0, the only valid operation is the "move" action.
action	1 = add to_skill to agent (from_skill is unused). 2 = remove from_skill from agent (to_skill is unused). 3 = move agents between splits/skills
r	flag indicating level is reserved, 0=skill level, 1=reserve level.
level	skill level (1-16) for to_skill (used only if s = 1). Reserve level (1-2) if r = 1. 0 = level is unspecified (no change requested).
count	the number of agent positions included in the message (1-32).
to_skill, from_skill	skill or split number, 0-999 (G3V8-G3V11), 0-2000 (MV2 and later), measured (EAS or non-EAS) or unmeasured (EAS only). The "from_skill" is not used for `add' (action = 1); therefore it will contain "0". The "to_skill is not used for `delete' (action = 2); therefore it will contain "0". When action = 3, to_skill is the 'to' split or skill and from_skill is the 'from' split or skill.
pos1 – pos32	agent position - without EAS (1-10000), with EAS (0-9999999999999) login ID in BCD format, left-justified where hex a is digit 0 and 0 in this field is null

3.3.10.2 **←MVAGSTFD23 (New CM6.0)**

	Switch	Vect	Skill
First Release	CM6.0	y/n	y/n
Last Release	CM6.3		

The first 15 bytes are always sent.
This message is specific to the 13 Digit Dial Plan Expansion enhancement.
If the dial plan is expanded beyond 13 digits, a new protocol message would be needed.



opcode — MVAGSTFD23 = 56 (0x38)

tag — message tag, 0-255 (it will be returned in the switch response message(s))

s — skills, 1=skills are used in this message, 0=skills not used. If s=0, the only valid operation is the "move" action.

action — 1 = add to_skill to agent (from_skill is unused).
2 = remove from_skill from agent (to_skill is unused).
3 = move agents between splits/skills

r — flag indicating level is reserved, 0=skill level, 1=reserve level.

ia — field that indicates whether the skill is interruptible. This field is only valid when the “r” (reserve) field is 0.

- 0 not an interruptible skill
- 1 auto-in-interrupt
- 2 manual-in-interrupt
- 3 notify-interrupt

level — skill level (1-16) for to_skill (used only if s = 1). Reserve level (1-2) if r = 1. 0 = level is unspecified (no change requested).

count — the number of agent positions included in the message (1-32).

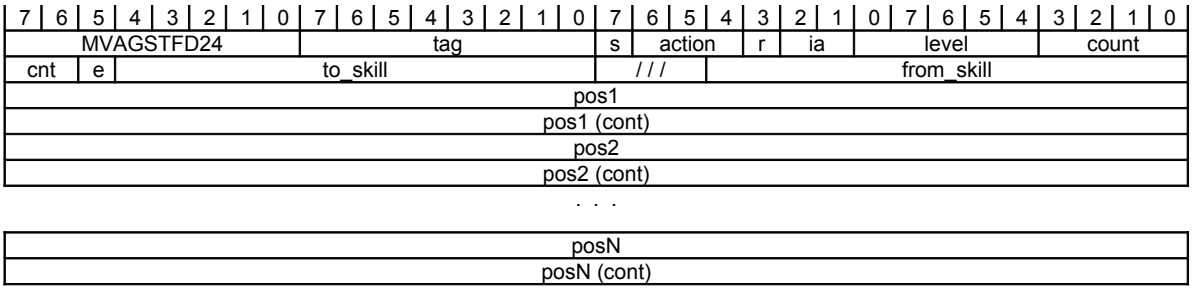
to_skill, from_skill — skill or split number, 0-8000 measured (EAS or non-EAS) or unmeasured (EAS only). The "from_skill" is not used for `add' (action = 1); therefore it will contain "0". The "to_skill is not used for `delete' (action = 2); therefore it will contain "0". When action = 3, to_skill is the 'to' split or skill and from_skill is the 'from' split or skill.

pos1 – pos32 — agent position - without EAS (1-10000), with EAS login ID in BCD format, left-justified where hex a is digit 0 and 0 in this field is null

3.3.10.3 **←MVAGSTFD24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y/n	y/n
Last Release			

The first 20 bytes are always sent.



opcode — MVAGSTFD24 = 62 (0x3e)

tag — message tag, 0-255 (it will be returned in the switch response message(s))

s — skills, 1=skills are used in this message, 0=skills not used. If s=0, the only valid operation is the "move" action.

action — 1 = add to_skill to agent (from_skill is unused).
2 = remove from_skill from agent (to_skill is unused).
3 = move agents between splits/skills

r — flag indicating level is reserved, 0=skill level, 1=reserve level.

ia — field that indicates whether the skill is interruptible. This field is only valid when the “r” (reserve) field is 0.

0 not an interruptible skill
1 auto-in-interrupt
2 manual-in-interrupt
3 notify-interrupt

level — skill level (1-16) for to_skill (used only if s = 1). Reserve level (1-2) if r = 1. 0 = level is unspecified (no change requested).

count — the number of agent positions included in the message (1-16).

e — end bit, set when this is the last message of the set needed to satisfy the request

to_skill, from_skill — skill or split number, 0-8000 measured (EAS or non-EAS) or unmeasured (EAS only). The "from_skill" is not used for `add' (action = 1); therefore it will contain "0". The "to_skill is not used for `delete' (action = 2); therefore it will contain "0". When action = 3, to_skill is the 'to' split or skill and from_skill is the 'from' split or skill.

pos1 – pos16 — agent position - without EAS (1-10000), with EAS login ID in BCD format, left-justified where hex a is digit 0 and 0 in this field is null

3.3.11 ←APARMS7 and APARMS24

The **Agent Parameters** message is sent by the reporting adjunct to the switch to request the skills for an agent. The expected response is an **AGENTSK** with the agent's skills or an **RCODE** 1 or 31.This message is only used in EAS configurations.

3.3.11.1 ←APARMS7 (New G3V11)

	Switch	Vect	Skill
First Release	G3V11	y	y
Last Release	CM6.3		

7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
APARMS7								tag								pnumber				p1				p2				p3			
p4				p5				p6				p7				p8				p9				p10				p11			
p12				p13				p14				p15																			

opcode APARMS7 = 40 (0x28)
tag message tag, 0-255 (it will be returned in the switch response message)
pnumber number of digits in agent position, 1-15
p1 – p15 agent position - with EAS (0-9999999) login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill position to a byte boundary

3.3.11.2 ←APARMS24 (New CM7.0)

	Switch	Vect	Skill
First Release	CM7.0	y	y
Last Release			

7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
APARMS24								tag								/ / /				pnumber				p1				p2			
p3				p4				p5				p6				p7				p8				p9				p10			
p11				p12				p13				p14				p15				p16											

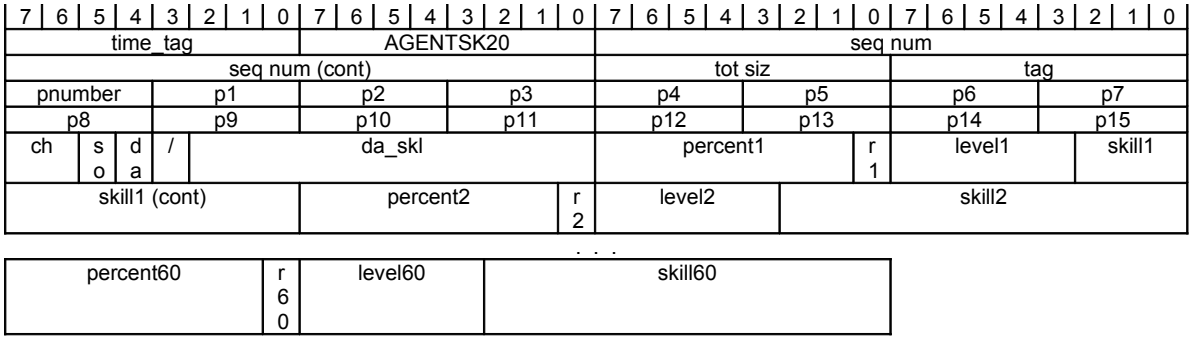
opcode APARMS24 = 63 (0x3f)
tag message tag, 0-255 (it will be returned in the switch response message)
pnumber number of digits in agent position, 1-16
p1 – p16 agent position - with EAS (0-9999999) login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill position to a byte boundary.

3.3.12 → AGENTSK20, AGENTSK23, AGENTSK24

The **Agent Skills** message is sent by switch to the reporting adjunct in response to an **APARMS** message to indicate the agent's skills. Blanks (or holes) in the administration will be represented by skill=0, level=0, reserve=0, percent=0.

3.3.12.1 → AGENTSK20 (New CM3.1)

	Switch	Vect	Skill
First Release	CM3.1	y	y
Last Release	CM5.2		

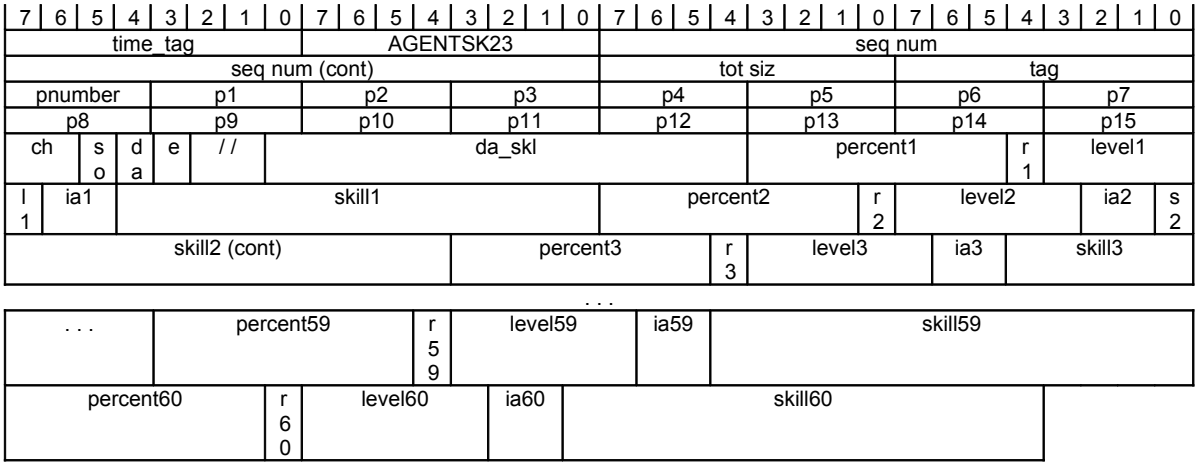


time_tag	seconds past minute when message was formed, 0-59
opcode	AGENTSK20 = 194 (0xc2)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
tag	message tag, 0-255 (it will be returned in the switch response message).
pnumber	number of digits in agent position, 1-15
p1 - p15	agent position - (0-9999999) login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary
ch	agent's call handling preference, 0 = skill level, 1 = greatest need, 2 = percent allocation
so	service objective flag: 0 = not in use, 1 = in use. This option is only used when agent call handling preference is skill level or greatest need.
da	direct agent calls handled first: 0 = no, 1 = yes. This option is only used when agent call handling preference is percent allocation
da_skill	agent's direct-agent skill (must be one of agent's 60 skills or 0 if no direct agent skill is assigned)
percent1-60	target percent of calls(0-100) handled by this agent which are for this skill. Used only when call handling preference is percent allocation.
r1-60	flag indicating if level for this skill is reserve: 0 = not in reserve (regular agent), 1 = reserve
level1-60	skill level (1-16) or reserve level (1-2)
skill1-60	skill number, 0-999 (G3V8-G3V11), 0-2000 (MV2-CM5.2)

3.3.12.2 **→AGENTSK23 (New CM6.0)**

	Switch	Vect	Skill
First Release	CM6.0	y	y
Last Release	CM6.3		

With expansion to 120 skills this request may not fit into a single **AGENTSK23** message. If the agent has more than 60 skills, a second message will be sent. The 1st message in the series covers skills 1 through 60 and the 2nd message covers skills 61 through 120, mapping to skills 1-60. The end bit is set for the last message in a series.



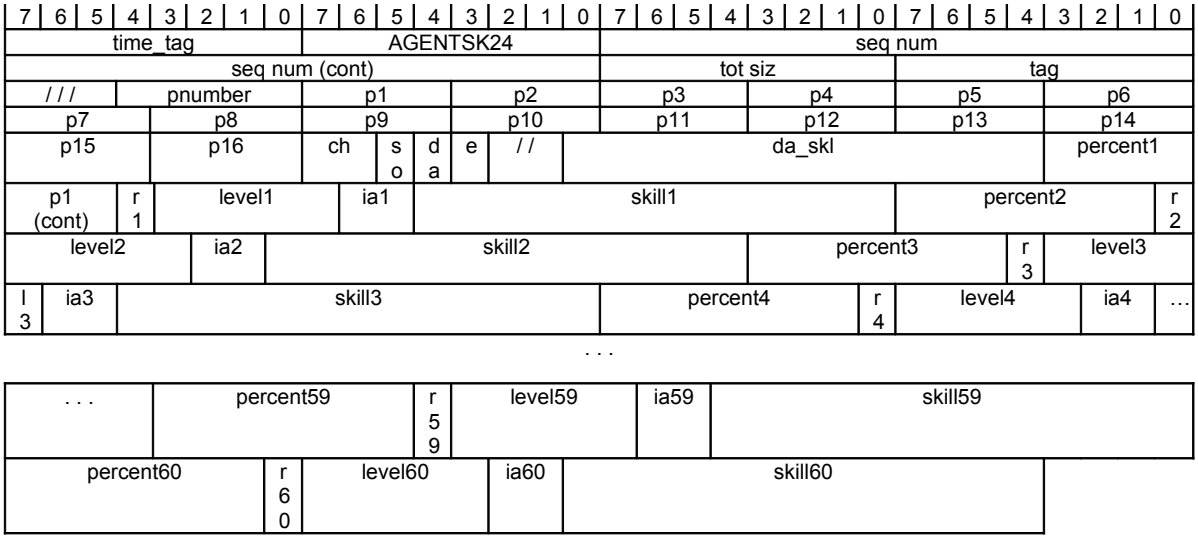
AGENTSK23 (continued)

time_tag	seconds past minute when message was formed, 0-59
opcode	AGENTSK23 = 4 (0x04)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
tag	message tag, 0-255 (it will be returned in the switch response message).
pnumber	number of digits in agent position, 1-15
p1 - p15	agent position - EAS login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary
ch	agent's call handling preference, 0 = skill level, 1 = greatest need, 2 = percent allocation
so	service objective flag: 0 = not in use, 1 = in use. This option is only used when agent call handling preference is skill level or greatest need.
da	direct agent calls handled first: 0 = no, 1 = yes. This option is only used when agent call handling preference is percent allocation
e	end bit, set when this is the last message of the set needed to satisfy the request
da_skl	agent's direct-agent skill (must be one of agent's 120 skills or 0 if no direct agent skill is assigned)
percent1-60	target percent of calls(0-100) handled by this agent which are for this skill. Used only when call handling preference is percent allocation.
r1-60	flag indicating if level for this skill is reserve: 0 = not in reserve (regular agent), 1 = reserve
level1-60	skill level (1-16) or reserve level (1-2)
ia1-60	field that indicates whether the corresponding skill is interruptible. This field is only valid when the corresponding “rx” (reserve) field is 0. 0 not an interruptible skill 1 auto-in-interrupt 2 manual-in-interrupt 3 notify-interrupt
skill1-60	skill number, 0-8000

3.3.12.3 **→AGENTSK24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y	y
Last Release			

If the agent has more than 60 skills, a second message will be sent. The 1st message in the series covers skills 1 through 60 and the 2nd message covers skills 61 through 120, mapping to skills 1-60. The end bit is set for the last message in a series.



AGENTSK24 (continued)

time_tag	seconds past minute when message was formed, 0-59
opcode	AGENTSK24 = 55 (0x37)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
tag	message tag, 0-255 (it will be returned in the switch response message).
pnumber	**number of digits in agent position, 1-16**
p1 – p16	**agent position - EAS login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary.**
ch	agent's call handling preference, 0 = skill level, 1 = greatest need, 2 = percent allocation
so	service objective flag: 0 = not in use, 1 = in use. This option is only used when agent call handling preference is skill level or greatest need.
da	direct agent calls handled first: 0 = no, 1 = yes. This option is only used when agent call handling preference is percent allocation
e	end bit, set when this is the last message of the set needed to satisfy the request
da_skl	agent's direct-agent skill (must be one of agent's 120 skills or 0 if no direct agent skill is assigned)
percent1-60	target percent of calls(0-100) handled by this agent which are for this skill. Used only when call handling preference is percent allocation.
r1-60	flag indicating if level for this skill is reserve: 0 = not in reserve (regular agent), 1 = reserve
level1-60	skill level (1-16) or reserve level (1-2)
ia1-60	field that indicates whether the corresponding skill is interruptible. This field is only valid when the corresponding “rx” (reserve) field is 0. 0 not an interruptible skill 1 auto-in-interrupt 2 manual-in-interrupt 3 notify-interrupt
skill1-60	skill number, 0-8000

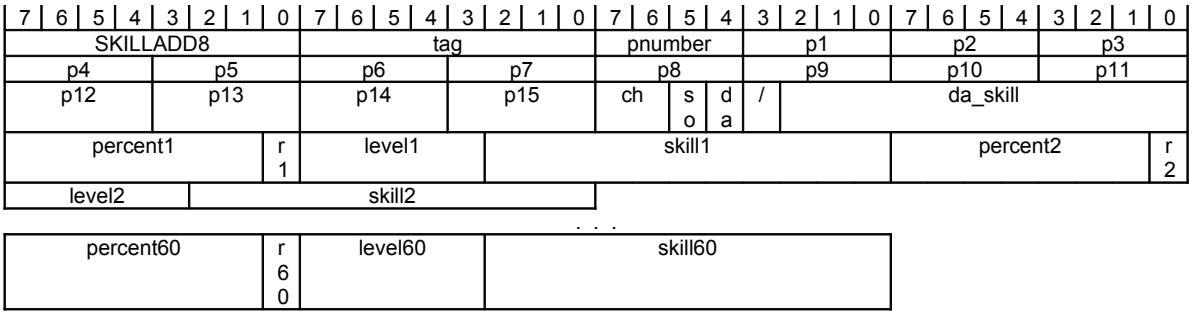
3.3.13 ←SKILLADD8, SKILLADD23 and SKILLADD24

The **Skill Add** message is sent by the reporting adjunct to the switch. It is a request to change the skill assignments for an agent position (LoginID). At least one skill must be present. All skills will be replaced. Blanks (or holes) in the administration will be represented by skill=0. The expected response is an **RCODE** 0, 1, 38, 45, 31 or 60 message. The expected response also includes **RCODE** 62 and 63 messages. An agent's skill assignment changes can take place while the agent is staffed.

If the agent is using the percent allocation call handling preference, the administration change for a logged in agent means that accumulated metrics for the percent allocation on the switch will be zeroed out and restarted with the new login.

3.3.13.1 ←SKILLADD8 (New MV2)

	Switch	Vect	Skill
First Release	MV2	y	y
Last Release	CM5.2		

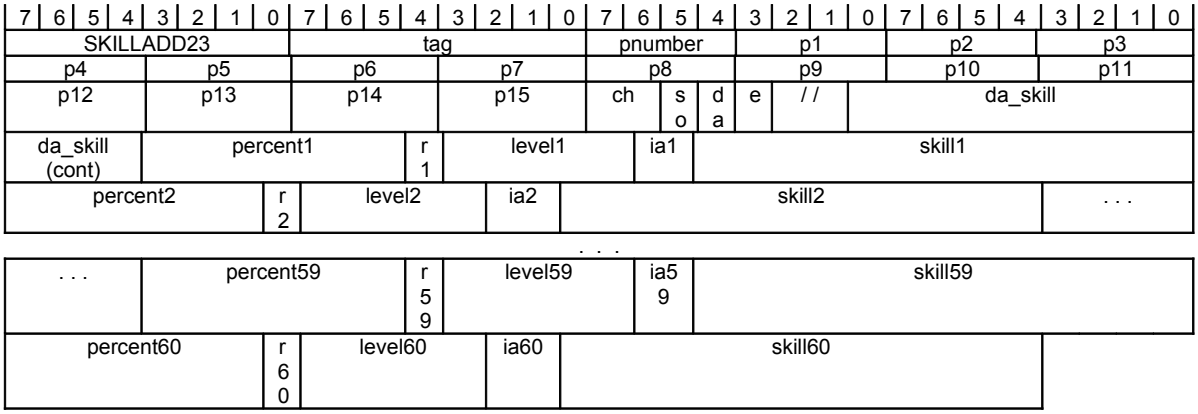


opcode	SKILLADD8 = 45 (0x2d)
tag	message tag, 0-255 (it will be returned in the switch response message)
pnumber	number of digits in agent position, 1-15
p1-p15	agent position - (0-9999999) login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary
ch	agent's call handling preference, 0 = skill level, 1 = greatest need, 2 = percent allocation
so	service objective flag: 0 = not in use, 1 = in use. This option is only used when agent call handling preference is skill level or greatest need.
da	direct agent calls handled first: 0 = no, 1 = yes. This option is only used when agent call handling preference is percent allocation.
da_skill	agent's direct-agent skill (must be one of agent's 60 skills or zero which indicates no DA skill)
percent1-60	Percent allocation to apply to this skill, (0-100). Used only when the call handling preference is percent allocation and the level is not reserve (r bit is not set for the skill).
r1-60	flag indicating if the level for this skill is reserve: 0 = not in reserve (regular agent), 1 = reserve
level1-60	skill level (1-16) or reserve level (1-2)
skill1-60	skill number, 0-999 (G3V8-G3V11), 0-2000 (MV2-CM5.2)

3.3.13.2 **←SKILLADD23 (New CM6.0)**

	Switch	Vect	Skill
First Release	CM6.0	y	y
Last Release	CM6.3		

With expansion to 120 skills this request may not fit into a single **SKILLADD23** message. If the agent has more than 60 skills, a second message will be sent. The 1st message in the series covers skills 1 through 60 and the 2nd message covers skills 61 through 120, mapping to skills 1-60. The end bit is set for the last message in a series.



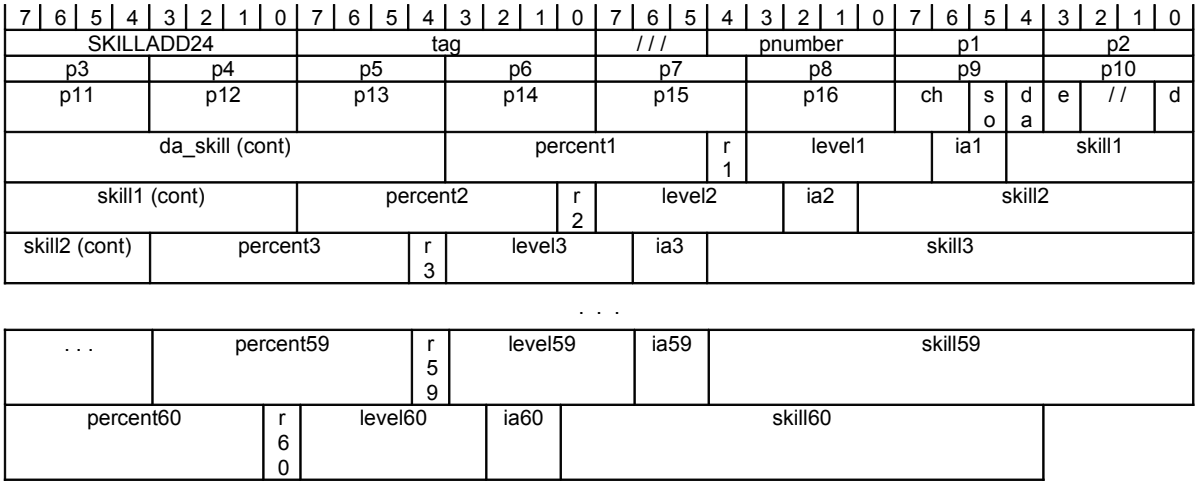
opcode	SKILLADD23 = 57(0x39)
tag	message tag, 0-255 (it will be returned in the switch response message)
pnumber	number of digits in agent position, 1-15
p1-p15	agent position - EAS login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary
ch	agent's call handling preference, 0 = skill level, 1 = greatest need, 2 = percent allocation
so	service objective flag: 0 = not in use, 1 = in use. This option is only used when agent call handling preference is skill level or greatest need.
da	direct agent calls handled first: 0 = no, 1 = yes. This option is only used when agent call handling preference is percent allocation.
e	end bit, set when this is the last message of the set needed to satisfy the request
da_skill	agent's direct-agent skill (must be one of agent's 120 skills or zero which indicates no DA skill)
percent1-60	Percent allocation to apply to this skill, (0-100). Used only when the call handling preference is percent allocation and the level is not reserve (r bit is not set for the skill).
r1-60	flag indicating if the level for this skill is reserve: 0 = not in reserve (regular agent), 1 = reserve
level1-60	skill level (1-16) or reserve level (1-2)
skill1-60	skill number, 0-8000
ia1-60	field that indicates whether the corresponding skill is interruptible. This field is only valid when the corresponding “rx” (reserve) field is 0.

0 not an interruptible skill
1 auto-in-interrupt
2 manual-in-interrupt
3 notify-interrupt

3.3.13.3 **←SKILLADD24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y	y
Last Release			

With expansion to 120 skills this request may not fit into a single **SKILLADD23** message. If the agent has more than 60 skills, a second message will be sent. The 1st message in the series covers skills 1 through 60 and the 2nd message covers skills 61 through 120, mapping to skills 1-60. The end bit is set for the last message in a series.



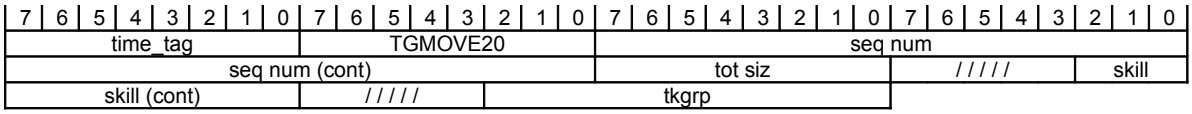
opcode	SKILLADD24 = 64 (0x40)
tag	message tag, 0-255 (it will be returned in the switch response message)
pnumber	**number of digits in agent position, 1-16**
p1-p16	**agent position - EAS login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary.**
ch	agent's call handling preference, 0 = skill level, 1 = greatest need, 2 = percent allocation
so	service objective flag: 0 = not in use, 1 = in use. This option is only used when agent call handling preference is skill level or greatest need.
da	direct agent calls handled first: 0 = no, 1 = yes. This option is only used when agent call handling preference is percent allocation.
e	end bit, set when this is the last message of the set needed to satisfy the request
da_skill	agent's direct-agent skill (must be one of agent's 120 skills or zero which indicates no DA skill)
percent1-60	Percent allocation to apply to this skill, (0-100). Used only when the call handling preference is percent allocation and the level is not reserve (r bit is not set for the skill).
r1-60	flag indicating if the level for this skill is reserve: 0 = not in reserve (regular agent), 1 = reserve
level1-60	skill level (1-16) or reserve level (1-2)
skill1-60	skill number, 0-8000
ia1-60	field that indicates whether the corresponding skill is interruptible. This field is only valid when the corresponding “rx” (reserve) field is 0. 0 not an interruptible skill 1 auto-in-interrupt 2 manual-in-interrupt 3 notify-interrupt

3.3.14 → TGMOVE20 and TGMOVE23

The **Trunk Group-Split Move** message is sent by the switch whenever a trunk group is removed from and/or added to another split. The move request will have originated from a switch administration procedure. When a trunk group is removed from a split or added to an unmeasured split, the switch will send a **TGMOVE** *split*=0 message. The reporting adjunct will update the split assignment for the specified trunk group.

3.3.14.1 → TGMOVE20 (New CM3.1)

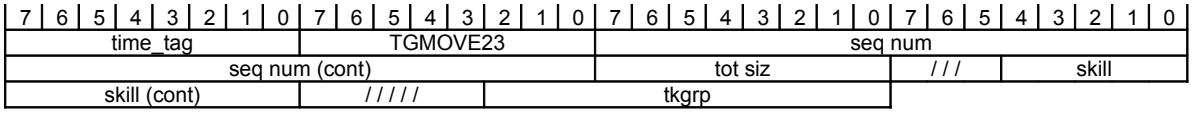
	Switch	Vect	Skill
First Release	CM3.1	y/n	y/n
Last Release	CM5.2		



time_tag seconds past minute when event occurred, 0-59
opcode TGMOVE20 = 195 (0xc3)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
skill skill number, 0-999 (G3V8-G3V11), 0-2000 (MV2 and later)
(0 indicates that the trunk group is not assigned to a skill or is assigned to an unmeasured skill)
tkgrp trunk group number, 1-666 (G3V8-G3V10), 0-2000 (G3V11 and later)

3.3.14.2 → TGMOVE23 (New CM6.0)

	Switch	Vect	Skill
First Release	CM6.0	y/n	y/n
Last Release			

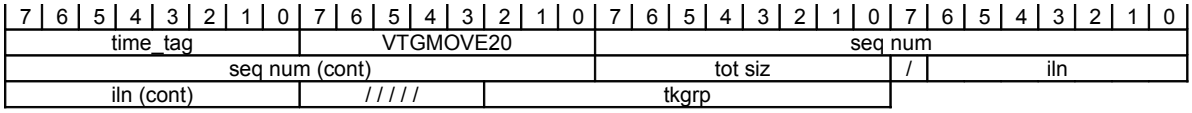


time_tag seconds past minute when event occurred, 0-59
opcode TGMOVE23 = 22 (0x16)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
skill skill number, 0-8000 (CM6.0 and later)
(0 indicates that the trunk group is not assigned to a skill or is assigned to an unmeasured skill)
tkgrp trunk group number, 0-2000 (G3V11 and later)

3.3.15 → VTGMOVE20 (New CM3.1)

	Switch	Vect	Skill
First Release	CM3.1	y	y/n
Last Release			

The **Trunk Group-VDN Move** message is sent by the switch when Call Vectoring is enabled and a trunk group is removed from and/or moved to a new VDN. The move request will have originated from a switch administration procedure. When a trunk group is removed from a VDN or added to an unmeasured VDN, the switch will send a **VTGMOVE** *iln*=0 message. The reporting adjunct will update the VDN assignment for the specified trunk group.



time_tag seconds past minute when event occurred, 0-59
opcode VTGMOVE20 = 196 (0xc4)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
iln internal line number, 0-32766 (0 indicates that the trunk group is not assigned to a VDN or is assigned to an unmeasured VDN)
tkgrp trunk group number, 1-666 (G3V8-G3V10), 0-2000 (G3V11 and later)

3.3.16 → VDNMOVE20, VDNMOVE23 and VDNMOVE24

A **VDN-Vector Move** message is sent by the switch when the administration information for a VDN is changed. This information will always include the vector. Other information may include, depending on switch type and feature set, the assigned vector, the VDN skill preferences (with EAS), and the BSR plan number associated with the VDN. The move request may have originated from the reporting adjunct (**MVVDN** or **CHGVDNSK** messages) or from a switch administration procedure.

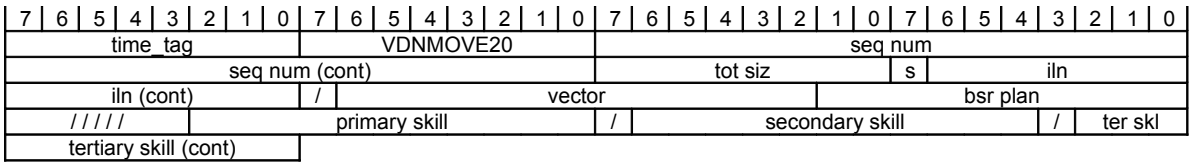
VDNs are referenced by internal line numbers.

When a VDN is removed from a vector, the switch will send a **VDNMOVE** message.

This is a variable-length message based on the "s" bit. The first 7 bytes are always sent. If the "s" bit is set then 5 additional bytes are sent for the "primary-skill", "secondary-skill" and "tertiary-skill".

3.3.16.1 → VDNMOVE20 (New CM3.1)

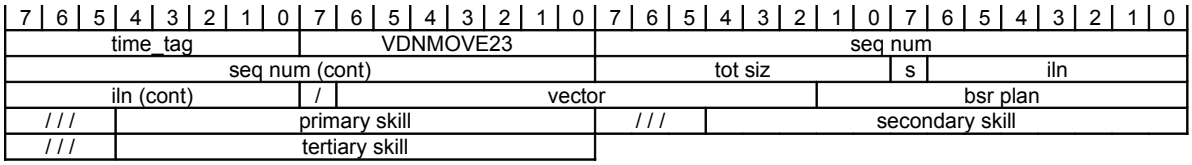
	Switch	Vect	Skill
First Release	CM3.1	y	y
Last Release	CM5.2		



time_tag	seconds past minute when event occurred, 0-59
opcode	VDNMOVE20 = 197 (0xc5)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
s	skill flag, 0 = no skills in message, 1 = skills in message
iln	internal line number (1-32766)
vector	vector number, 1-999 (G3V8-CM2), 1-2000 (CM3 and later)
bsr plan	bsr application plan number, 1-255 (G3V8-G3V10), 1-512 (G3V11 and later)
primary skill	when s = 1, primary VDN agent skill 1-999 (G3V8-G3V11), 1-2000 (MV2 and later), 0 = no skill used
secondary skill	when s = 1, secondary VDN agent skill 1-999 (G3V8-G3V11), 1-2000 (MV2 and later), 0 = no skill used
tertiary skill	when s = 1, tertiary VDN agent skill 1-999 (G3V8-G3V11), 1-2000 (MV2 and later), 0 = no skill used

3.3.16.2 **→VDNMOVE23 (New CM6.0)**

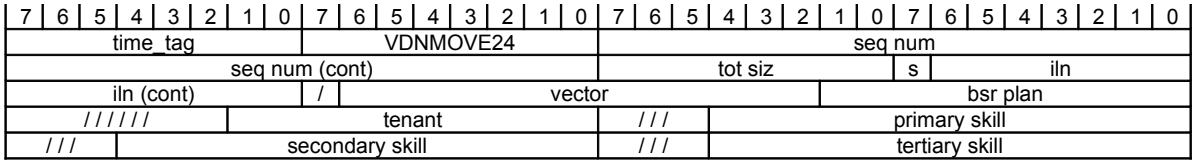
	Switch	Vect	Skill
First Release	CM6.0	y	y
Last Release	CM6.3		



time_tag seconds past minute when event occurred, 0-59
opcode VDNMOVE23 = 24 (0x18)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
s skill flag, 0 = no skills in message, 1 = skills in message
iln internal line number (1-32766)
vector vector number, 1-8000 (CM6.0 and later)
bsr plan bsr application plan number, 1-255 (G3V8-G3V10), 1-512 (G3V11 and later)
primary skill when s = 1, primary VDN agent skill 1-8000 (CM6.0 and later), 0 = no skill used
secondary skill when s = 1, secondary VDN agent skill 1-8000 (CM6.0 and later), 0 = no skill used
tertiary skill when s = 1, tertiary VDN agent skill 1-8000 (CM6.0 and later), 0 = no skill used

3.3.16.3 **→VDNMOVE24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y	y
Last Release			



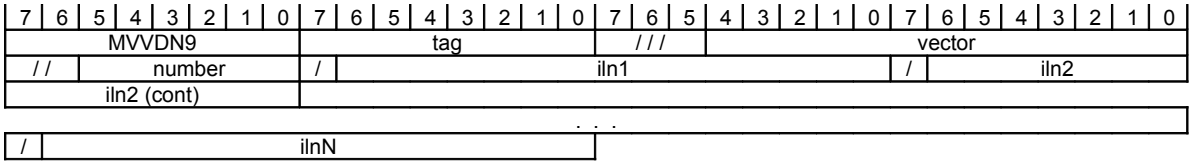
time_tag seconds past minute when event occurred, 0-59
opcode VDNMOVE24 = 56 (0x38)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
s skill flag, 0 = no skills in message, 1 = skills in message
iln internal line number (1-32766)
vector vector number, 1-8000 (CM6.0 and later)
bsr plan bsr application plan number, 1-512
tenant **VDN tenant number, 1-250**
primary skill when s = 1, primary VDN agent skill 1-8000 (CM6.0 and later), 0 = no skill used
secondary skill when s = 1, secondary VDN agent skill 1-8000 (CM6.0 and later), 0 = no skill used
tertiary skill when s = 1, tertiary VDN agent skill 1-8000 (CM6.0 and later), 0 = no skill used

3.3.17 ← MVVDN9 and MVVDN23

The **Move VDN** message is a variable-length message based on "number" sent by the reporting adjunct. The first seven bytes are always sent. It is a request to change the vector termination of one or more (up to 32) VDNs. VDNs are referenced by internal line numbers. The switch will attempt to reassign each VDN and report the results back to the reporting adjunct. For each VDN that is successfully moved, the switch will send one **VDNMOVE** (or equivalent) message so that the reporting adjunct can update the real-time database. Lastly, the switch sends a **MVRCODE** message that specifies which VDN(s) were successfully moved and which were not.

3.3.17.1 ←MVVDN9 (New CM3)

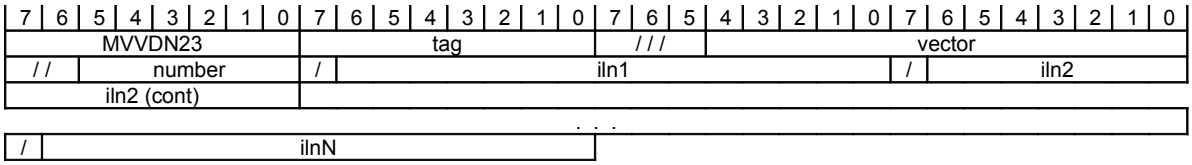
	Switch	Vect	Skill
First Release	CM3	y	y/n
Last Release	CM5.2		



opcode MVVDN9 = 47 (0x2f)
tag message tag, 0-255 (it will be returned in the switch response message)
vector vector number, 1-999 (G3V8-CM2), 1-2000 (CM3 and later)
number number of VDNs (N) in move request, 1-32
iln1-N list of N internal line numbers, 1-32766

3.3.17.2 ←MVVDN23 (New CM6.0)

	Switch	Vect	Skill
First Release	CM6.0	y	y/n
Last Release			



opcode MVVDN23 = 60 (0x3c)
tag message tag, 0-255 (it will be returned in the switch response message)
vector vector number, 1-8000 (CM6.0 and later)
number number of VDNs (N) in move request, 1-32
iln1-N list of N internal line numbers, 1-32766

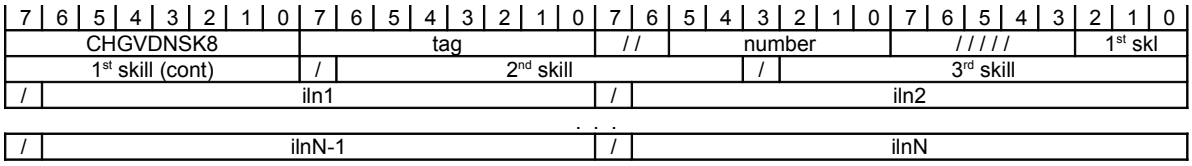
3.3.18 ←CHGVDNSK8 and CHGVDNSK23

The **Change VDN's skills** message is sent by the reporting adjunct to request a VDN's skills be changed for up to 32 VDNs. All three of the VDN's skills are always included, zero will indicate no skills are assigned. The switch sends **VDNMOVE** (or equivalent) messages and then a **MVRCODE** message that specifies which VDNs were successfully changed.

This is a variable length message based on the "number" of VDNs sent.

3.3.18.1 ←CHGVDNSK8 (New MV2)

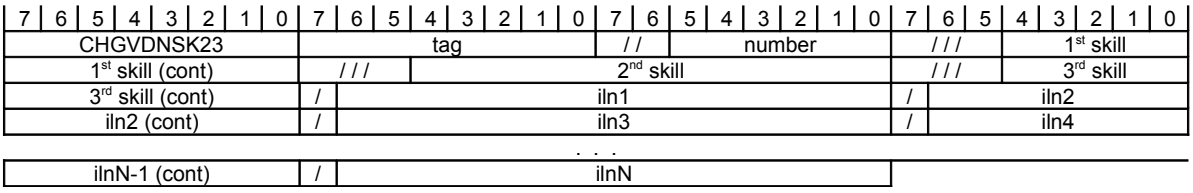
	Switch	Vect	Skill
First Release	MV2	y	y
Last Release	CM5.2		



opcode	CHGVDNSK8 = 46 (0x2e)
tag	message tag, 0-255 (it will be returned in the switch response message)
number	number of VDNs (N) in change request, (1-32)
1st, 2nd, 3rd skill	skill associated with the VDN skill, 1-999 (G3V8-G3V11), 1-2000 (MV2 and later), measured or unmeasured
iln1-N	list of N internal line number (1-32767)

3.3.18.2 ←CHGVDNSK23 (New CM6.0)

	Switch	Vect	Skill
First Release	CM6.0	y	y
Last Release			



opcode	CHGVDNSK23 = 58 (0x3a)
tag	message tag, 0-255 (it will be returned in the switch response message)
number	number of VDNs (N) in change request, (1-32)
1st, 2nd, 3rd skill	skill associated with the VDN skill, 1-8000, measured or unmeasured
iln1-N	list of N internal line number (1-32767)

3.3.19 ←VPARMS

The **Request Vector Parameters** message is sent by the reporting adjunct. It is a request for the vector steps of the specified vector. The expected response is a **VECTOR** message.

For CM6.0 and later, if the requested vector contains a check step with a preference level other than “all-levels” and the SPI language is prior to 23, the request will be denied (rcode=43).

	Switch	Vect	Skill
First Release	G3V1	y	y/n
Last Release			

7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
VPARMS								tag								/ / /				vector											

opcode VPARMS = 21 (0x15)
tag message tag, 0-255 (it will be returned in the switch response message)
vector vector number, 1-999 (G3V8-CM2), 1-2000 (CM3-CM5.2), 1-8000 (CM6.0 and later)

3.3.20 **VECTOR20 and VECTOR24**

The **VECTOR** message is sent by the switch in response to a **VPARMS** message. This is a variable-length message based on the "steps" field. The first 9 bytes are always sent and will be the same for all messages sent for a single vector. Multiple messages will be sent for large vectors.

Vectors can have 99 steps per vector. The number of steps sent per message varies based on the message.

3.3.20.1 **→VECTOR20 (New CM3.1)**

Each vector step in VECTOR20 must be 24 bytes. The switch will send 9 steps per message, with up to 11 messages to send all 99 steps.

	Switch	Vect	Skill
First Release	CM3.1	y	y/n
Last Release	CM6.3		

7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	**0**
time_tag								VECTOR20								seq num															
seq num (cont)																tot siz								tag							
m	steps																														

. . .

stype1			pri/cov	c	-					cond				threshold or table			
s	sday or comp		shour			smin			/	eday		ehour or type			emin or subtype		
dest										step-num					var_t		
v_t	var_c					-								-			
d1-d3 or s_num or type_val						d4			d5			d6			d7		d8
d9			d10			d11			d12			d13			d14		d15	d16

. . .

stypeN			pri/cov	c	-					cond				threshold or table			
s	sday or comp		shour			smin			/	eday		ehour or type			emin or subtype		
dest										step-num					var_t		
v_t	var_c					-								-			
d1-d3 or s_num or type_val						*d4*			*d5*			*d6*			*d7*		*d8*
d9			*d10*			*d11*			*d12*			*d13*			*d14*		*d15*	*d16*

time_tag seconds past minute when event occurred, 0-59
opcode VECTOR20 = 198 (0xc6)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
tag message tag, 0-255 (it will be returned in the switch response message)
m m=1, more steps for this vector will be sent in a subsequent vector, m=0, this is the last message for this vector
steps number of steps (N) 0-9. This is the number of steps in this message.
stype1-N step type for steps 1-N (see section 3.3.21)

stype	description	stype	description	stype	description
1	queue-to	11	collect digits	21	return (CM 3.0)
2	check	12	route-to digits	22	set (CM3.0)
3	route-to number	13	messaging skill	23	unused
4	announcement	14	adjunct route	24	unused
5	wait	15	converse-on split	25	unused
6	goto step (obsolete)	16	send digits (not supported)	26	unused
7	disconnect	17	goto step	27	unused
8	busy	18	goto vector	28	unused
9	stop	19	consider	29	unused
10	goto vector (obsolete)	20	reply-best	30	unused
				31	unused

3.3.20.2 **→VECTOR24 (New CM7.0)**

Each vector step in VECTOR24 must be 28 bytes. The switch will send 8 steps per message, with up to 11 messages to send all 99 steps.

	Switch	Vect	Skill
First Release	CM7.0	y	y/n
Last Release			

7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	**0**
time_tag								VECTOR24								seq num															
seq num (cont)																tot siz								tag							
m	steps																														

. . .

stype1			pri/cov	c	-				cond				threshold or table			
s	sday or comp		shour		smin		/	eday		ehour or type			emin or subtype			
dest								step-num					var_t			
v_t	var_c				-								-			
d1-d3 or s_num or type_val					d4			d5			d6			d7		d8
d9			d10			d11		d12			d13			d14	d15	d16
d17			d18			d19		d20			d21			d22	d23	d24

. . .

stypeN			pri/cov	c	-				cond				threshold or table			
s	sday or comp		shour		smin		/	eday		ehour or type			emin or subtype			
dest								step-num					var_t			
v_t	var_c				-								-			
d1-d3 or s_num or type_val					d4			d5			d6			d7		d8
d9			d10			d11		d12			d13			d14	d15	d16
d17			d18			d19		d20			d21			d22	d23	d24

time_tag seconds past minute when event occurred, 0-59
opcode VECTOR24 = 82 (0x52)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
tag message tag, 0-255 (it will be returned in the switch response message)
m m=1, more steps for this vector will be sent in a subsequent vector, m=0, this is the last message for this vector
steps number of steps (N) 0-8. This is the number of steps in this message.
stype1-N step type for steps 1-N (see section 3.3.21)

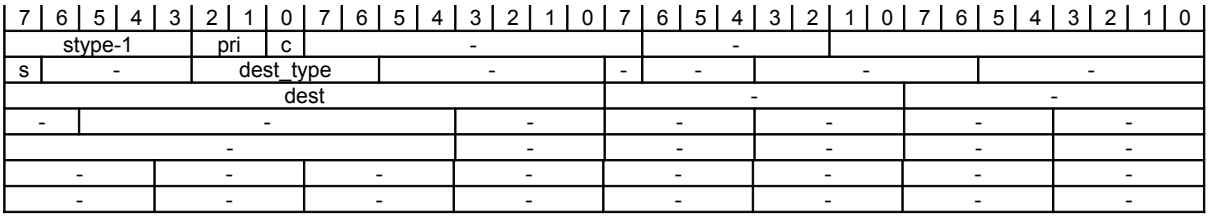
stype	description	stype	description	stype	description
1	queue-to	11	collect digits	21	return (CM 3.0)
2	check	12	route-to digits	22	set (CM3.0)
3	route-to number	13	messaging skill	23	unused
4	announcement	14	adjunct route	24	unused
5	wait	15	converse-on split	25	unused
6	goto step (obsolete)	16	send digits (not supported)	26	unused
7	disconnect	17	goto step	27	unused
8	busy	18	goto vector	28	unused
9	stop	19	consider	29	unused
10	goto vector (obsolete)	20	reply-best	30	unused
				31	unused

3.3.21 **Vector Steps**

The following steps are common to both VECTOR (section 3.3.20) and CHGVEC (section 3.3.22) messages.

In all of the following descriptions, 28 bytes are shown.
For SPI language < 24, the last 4 bytes are NOT present.

3.3.21.1 **Queue to**

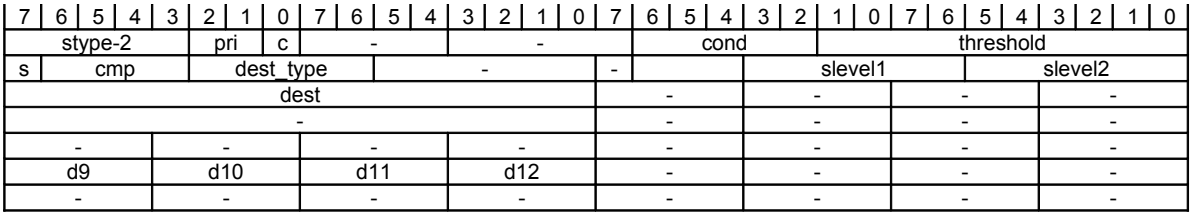


stype-1 queue-to
pri 0=low, 1=medium, 2=high, 3=top
c 1=this step is commented out
dest destination

dest_type	s	destination value
0	0	1-999 (G3V8-G3V11), 1-2000 (MV2- CM5.2), 1-8000 (CM6.0 and later)
0	1	VDN skill preference 1, 2, or 3 (valid only for EAS)
1	N/A	N/A

s 0=specific skill (if dest_type=0), 1=VDN skill preference (valid only for EAS, and if dest_type=0).
dest_type 0=skill number or VDN skill preference, 1=best.

3.3.21.2 **Check**



7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
stype-2					pri		c	-				-				cond								threshold							
s	cmp				dest_type			-								-				slevel1				slevel2							
dest																-				-				-							
-																-				-				-							
-					-				-				-			-				-				-							
d9					d10				d11				d12			-				-				-							
-					-				-				-			-				-				-							

stype-2 check

pri 0=low, 1=medium, 2=high, 3=top

c 1=this step is commented out

slevel1 the preferred level (for pref-level) or the start of the range (for pref-range). Only used in Call Center 6.0 or later with SPI language 23 or greater.

Skill Level Preference	Skill Level 1 (slevel1)
all-levels	0
pref-level	1-16
pref-range	1-16

slevel2 the end of the range (for pref-range). Only used in Call Center 6.0 or later with SPI language 23 or greater.

Skill Level Preference	Skill Level 2 (slevel2)
all-levels	0
pref-level	0
pref-range	1-16

dest destination value. split or skill 1-999 (G3V8-G3V11), 1-2000 (MV2-CM5.2), 1-8000 (CM6.0 and later). For s=1, skill = 1, 2, or 3

dest_type	s	destination value
0	0	1-999 (G3V8-G3V11), 1-2000 (MV2-CM5.2), 1-8000 (CM6.0 and later)
0	1	N/A
1	N/A	N/A

cond/threshold, and cmp

Conditions	Comparator	Threshold
0- unconditional	N/A	N/A
1- avail agents	1 >	0-998(G3V6,G3V7) [for G3V8+, see d9-d12]
3- staffed agents	1 >	0-998(G3V6,G3V7) [for G3V8+, see d9-d12]
6- calls queued	2 <	1-999
8- oldest call	2 <	1-999
11- ASA	2 <	1-999
12- EWT	2 <	[see d9-d12]
16- wait-improved	1 >	[see d9-d12]

s 0=specific skill, 1=VDN skill preference (EAS only, and if dest_type=0)

cmp

Comparators	
1	>
2	<

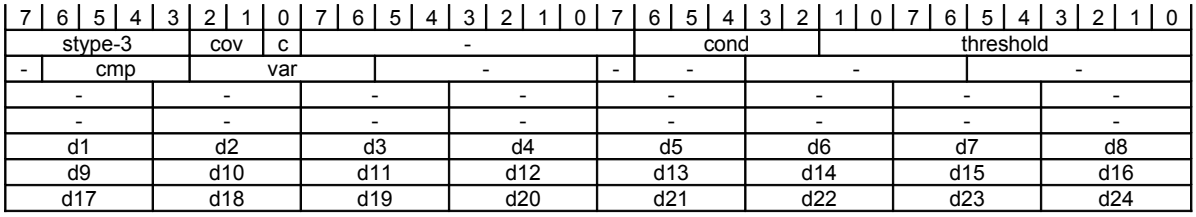
dest_type 0=split, 1=best.

d9-d12

Condition	Threshold
1-avail agents	Agent threshold (0-1500) G3V8 and later
3-staffed agents	Agent threshold (0-1500) G3V8 and later
12-EWT	Expected wait time, in seconds (1-9999)
16-wait-improved	Wait-improved (0-9999)

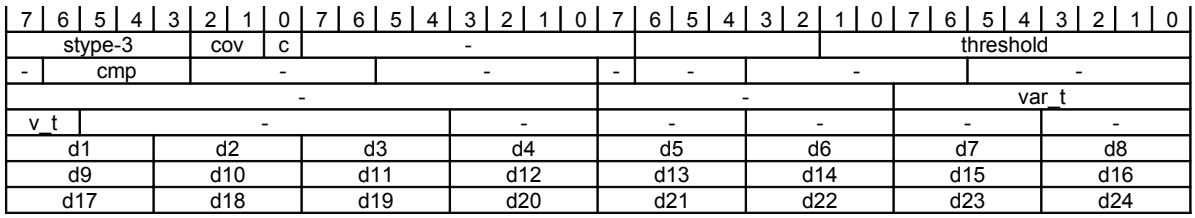
3.3.21.3 **Route To Number**

Variant 1:



7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
stype-3					cov	c		-								cond								threshold							
-	cmp				var			-								-	-							-							
-					-			-				-				-				-				-							
-					-			-				-				-				-				-							
d1					d2			d3				d4				d5				d6				d7				d8			
d9					d10			d11				d12				d13				d14				d15				d16			
d17					d18			d19				d20				d21				d22				d23				d24			

Variant 2:



7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
stype-3					cov	c		-																threshold							
-	cmp				-			-								-	-							-							
-																-								var_t							
v_t					-			-								-				-				-							
d1					d2			d3				d4				d5				d6				d7				d8			
d9					d10			d11				d12				d13				d14				d15				d16			
d17					d18			d19				d20				d21				d22				d23				d24			

Variant 1 is used when the route to number:

- does not contain a variable (< SPI language 20)
- does contain a variable
    - IF the route to number is ~r (< SPI language 20)

Variant 2 is used when:

- does not contain a variable (>= SPI language 20)
- does contain a variable (>= SPI language 23)
- does contain a variable (> SPI language 20 AND < SPI language 23) IF
    - the route to number is ~r (>= SPI language 20)

stype-3 route-to number

cov with coverage, 0=no, 1=yes

c 1=this step is commented out

cond/threshold

Condition		Threshold
0	unconditional	N/A
10	digits	0-9, hex c = #
17	interflow-qpos	1-9

cmp for condition 10 or 17

Comparators	Comparators
1 >	4 <>
2 <	5 >=
3 =	6 <=

var Variable: 1-26 for variable A-Z, 27-31 for VDN variable V1-V5
Used in message variant 1 (prior to SPI language 20).

var_t/v_t Variable: 1-26 for variable A-Z, 27-31 for VDN variable V1-V5, 34-37 for VDN variables V6-V9, 101-776 for variables AA-ZZ
Used in message variant 2 (starting with SPI language 20).

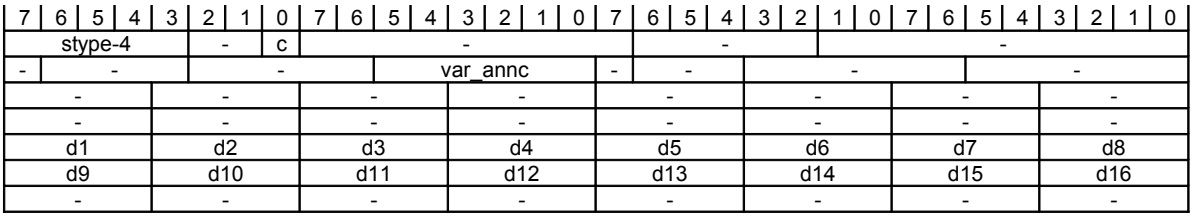
d1-d16 (< language24) Digits to route to (encoded in hex).
(if have ~r in d1-d2 then may optionally have "var")

d1-d24 (>= language24) The hex 0, representing null, is used when digits = "none".

hex	char	hex	char
0	null	7	7
1	1	8	8
2	2	9	9
3	3	a	0
4	4	b	*
5	5	c	#
6	6		

hex	char	meaning
e0	+	used with ~r to indicate E.164 phone number for SIP NCR
e1	~p	pause 1.5 seconds
e2	~w	wait for dial tone
e3	~m	mark
e4	~s	suppress display
e5	~W	wait indefinitely
e6	~r	redirection via NCR (allowed in d1,d2 only) [for G3V8.3 or later]

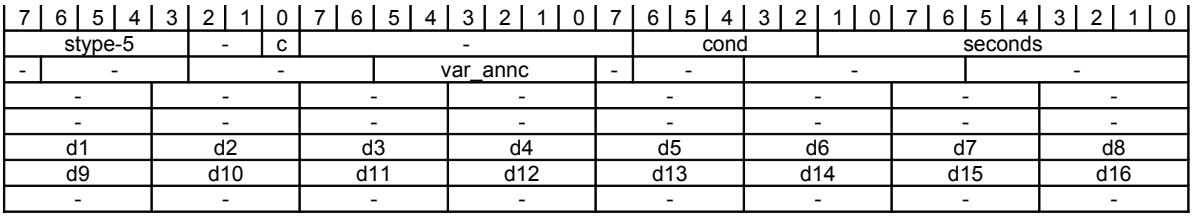
3.3.21.4 **Announcement**



7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
stype-4					-		c	-								-								-							
-	-				-			var_annc								-	-			-				-							
-					-			-				-				-				-				-							
-					-			-				-				-				-				-							
d1					d2			d3				d4				d5				d6				d7				d8			
d9					d10			d11				d12				d13				d14				d15				d16			
-					-			-				-				-				-				-				-			

stype-4 announcement
c 1=this step is commented out
var_annc Variable announcement: 1-26 for variable A-Z, 27-31 for VDN variable V1-V5 (if "var" is set then d1-d15 must all be hex 0) (CM 3.0 and later)
d1-d15 (< language 24) **Announcement extension (packed decimal where 0= null and hex a is digit 0)**
d1-d16 (>= language 24) **If present then var must be 0.**

3.3.21.5 **Wait**



7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
stype-5					-		c	-								cond								seconds							
-	-				-			var_annc								-	-			-				-							
-					-			-				-				-				-				-							
-					-			-				-				-				-				-							
d1					d2			d3				d4				d5				d6				d7				d8			
d9					d10			d11				d12				d13				d14				d15				d16			
-					-			-				-				-				-				-				-			

stype-5 wait
c 1=this step is commented out
cond 0=silence, 1=ringback, 2=music, 3=continue, 4=i-silent
seconds 0-998 even numbers
var_annc Variable announcement: 1-26 for variable A-Z, 27-31 for VDN variable V1-V5 (CM 3.0 and later)
d1-d15 (< language 24) **Audio/music source extension.**

d1-d16 (>= language 24)

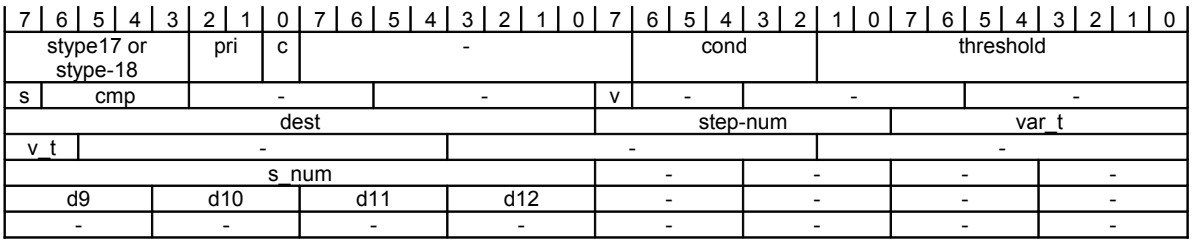
3.3.21.6 **Goto Step and Goto Vector**

For the Go To Step, the field names depend on the value in the "cond" field (the condition of the goto). The possible conditions are as follows:

Conditions	Thresholds	Format
0-unconditional	N/A	A
1-avail agents	0-999 (obsolete)	A
3-staffed agents	0-999 (obsolete)	A
6-calls queued	0-999	A
8-oldest call	0-999	A
9-TOD	none	B
10-digits	none	C
11-ASA	0-999	D or E
12-EWT	1-9999	E
13-ANI	none	D or E
14-counted-calls	0-999	D or E
15-ii-digits	none	F
16-wait-improved	0-999	E
17-interflow-qpos	1-9	A
18-holiday	1-99	G
19-A-Z (variable)	none	C
20-pn	none	C
21-mg	none	C
22-server	none	C
23-svhour	1-99	G

Format A:

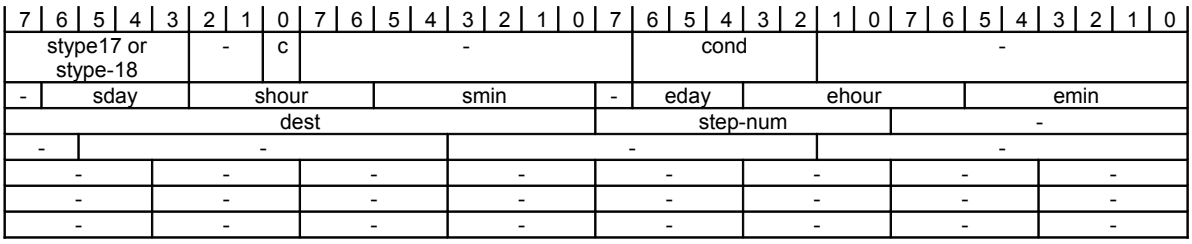
For conditions 0, 1, 3, 6, 8, and 17 the message will contain the following fields.



stype-17 or stype-18	goto step or goto vector
pri	0=low, 1=medium, 2=high, 3=top, for conditions 6 and 8
c	1=this step is commented out
dest	Step number to go to (1-32) if stype-17. Vector to go to (1-999 (G3V8-CM2), 1-2000 (CM3-CM5.2), 1-8000 (CM6.0 and later) if stype-18
threshold	see ranges for thresholds above (paired with conditions) and "v" is zero
s	0=specific split/skill in s_num, 1=VDN skill preference in s_num (EAS only)
cmp	Comparator s 1 >= 2 < 3 = 4 <> 5 >= 6 <=
var_t	Variable threshold: 1-26 for variable A-Z (MV2 and later), 27-31 for VDN variable V1-V5 (CM 3.0 and later)
v	Variable threshold flag: 0=no variable in var_t, 1=variable is in var_t (MV2 and later)
s_num	Number of the split/skill: 1-999 (G3V8-G3V11), 1-2000 (MV2-CM5.2), 1-8000 (CM6.0 and later), or VDN skill preference (1-3),
step_num	Vector Step number to go to: 0-31 (stype-18 only CM 3.0 and later)
d9-d12	Agent threshold (0-1500), for conditions 1 and 3 and "v" is 0, (G3V8 and later)

Format B:

For condition 9 (Time Of Day), the following fields are contained in the message.



stype-17 or stype-18	goto step or goto vector
c	1=this step is commented out
dest	Step number to go to (1-32) if stype-17. Vector to go to (1-999 (G3V8-CM2), 1-2000 (CM3-CM5.2), 1-8000 (CM6.0 and later) if stype-18
sday	start day for TOD condition; 0-7, 0 = all, 1 = Mon, ...
shour	start hour for TOD condition; 0-23
smin	start minutes for TOD condition; 0-59
eday	end day for TOD condition; 0-7, 0 = all, 1 = Mon, ...
ehour	end hour for TOD condition; 0-23
emin	end minutes for TOD condition; 0-59
step_num	Vector Step number to go to: 0-31 (stype-18 only CM 3.0 and later)

Format C:

For condition 10 (if digits), the following fields are contained in the message.
For condition 19 (A-Z, V1-V5 (variable)), var_c applies.
For condition 20, 21 and 22 (CM 3.0 and later), the following fields are contained in the message.

7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
stype17 or stype-18					-		c	-								cond								table							
-	cmp				-			-								v	-							-							
dest																step-num								var_t							
v_t	var_c							-								-															
d1					d2				d3				d4				d5				d6				d7				d8			
d9					d10				d11				d12				d13				d14				d15				d16			
-					-				-				-				-				-				-				-			

stype-17 or stype-18 — goto step or goto vector

c — 1=this step is commented out

dest — Step number to go to (1-32) if stype-17.
Vector to go to (1-999 (G3V8-CM2), 1-2000 (CM3-CM5.2), 1-8000 (CM6.0 and later) if stype-18

table — Vector routing table number in which to check digits: 1-100 (through CM5.2), 1-999 (CM6.0 or later). If comparator is 7 or 8, if table=0 then v=1.

cmp — For condition 20 and 21 only comparators 3 and 4 are valid.
For condition 22 comparators 1-6 are valid.

Comparators	Comparators	Comparators
1 >	4 <>	7 in (table)
2 <	5 >=	8 not-in (table)
3 =	6 <=	

var_t — Variable threshold: 1-26 for variable A-Z (if comparator is 1-6) (MV2 and later),
1-26 for variable A-Z (if comparator is 7-8) (CM 3.0 and later),
27-31 for VDN variable V1-V5 (if comparator is 1-8) (CM 3.0 and later)

var_c — Variable condition:
1-26 for variable A-Z, for condition 19 only (if comparator is 1-6) (MV2 and later),
27-31 for VDN variable V1-V5, for condition 19 only (if comparator is 1-8) (CM 3.0 and later)

v — Variable threshold flag: 0=no variable in var_t, 1=variable is in var_t (MV2 and later)

step_num — Vector Step number to go to: 0-31 (stype-18 only CM 3.0 and later)

d1-d16 — Digits for condition 10, 19, 20, 21 and 22 (if comparator is 1-6 and "v" is 0) are as follows:
The hex 0, representing null, is used when digits = "none".

hex	char	hex	char
0	null	a	0
1	1	b	*
2	2	c	# (only valid in a single-character digit string)
3	3	d	?
4	4	e	any (only condition 20 and 21 in a single-character digit string)
5	5	f	all (only condition 20 and 21 in a single-character digit string)
6	6		
7	7	d	main (only condition 22 in a single-character digit string)
8	8	e	ess (only condition 22 in a single-character digit string)
9	9	f	lsp (only condition 22 in a single-character digit string)

hex	char	meaning
e0	+	(used in wildcarding)

Format D:

For conditions 11, 13 and 14 (ASA, ANI, and counted-calls) and condition 16 (wait-improved), one of the two following field layouts for the message will apply. If the condition, type, and subtype match the table below, the first message layout applies. Otherwise, a match should be found in the second message layout (Format E).

Condition	Type	Subtype	# of Digits	Threshold	var_t Allowed
11-ASA	VDN	ext	1-15 (< language 24) **1-16 (>= language 24)**	0-999	yes
13-ANI	digits	N/A	1-16 (< language 24) **1-24 (>= language 24)**	N/A	yes
14-counted-calls	VDN	ext	1-15 (< language 24) **1-16 (>= language 24)**	0-999	yes

7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
stype17 or stype-18					-		c	-								cond								table							
-	cmp				-			-							v	-			type					subtype							
dest																step-num								var_t							
v_t	-							-								-															
d1					d2				d3				d4				d5				d6				d7				d8			
d9					d10				d11				d12				d13				d14				d15				d16			
d17					d18				d19				d20				d21				d22				d23				d24			

Format D (continued)

stype-17 or stype-18	goto step or goto vector
c	1=this step is commented out
dest	destination. Step number to go to (1-32) if stype-17. Vector to go to: 1-999 (G3V8-CM2), 1-2000 (CM3-CM5.2), 1-8000 (CM6.0 and later) if stype-18
cond	see the condition/threshold above
threshold	see the condition/threshold above (if "v" is 0)

cmp — Comparator

s	
1	>
2	<
3	=
4	<>
5	>=
6	<=

step_num	Vector Step number to go to: 0-31 (stype-18 only CM 3.0 and later)
var_t	Variable threshold: 1-26 for variable A-Z, condition 11 and 14 (MV2 and later), 27-31 for VDN variable V1-V5, condition 11 and 14 (CM 3.0 and later)
v	Variable threshold flag: 0=no variable in var_t, 1=variable is in var_t, condition 11 and 14 (MV2 and later)

type

2	VDN extension
5	digits

subtype

0	N/A
1	extension

d1-d16 (< language 24) — digits for condition 11, 13 and 14 as follows:
The hex 0, representing null, is used when digits = "none"

d1-d24 (>= language 24)

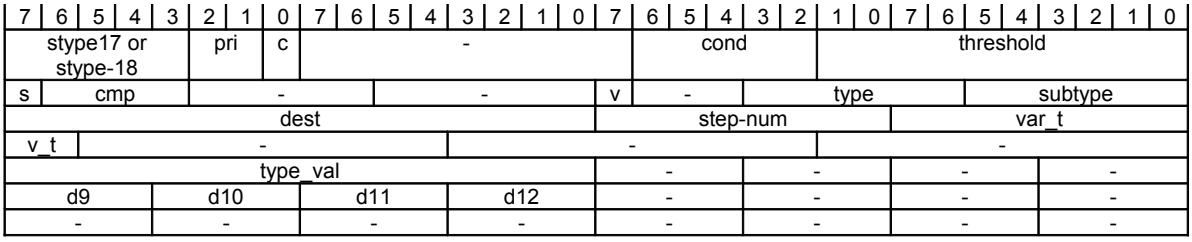
hex	char	hex	char
0	nul	7	7
1	1	8	8
2	2	9	9
3	3	a	0
4	4	b	* (not valid)
5	5	c	# (not valid)
6	6	d	? (condition 13 only)

hex	char	interpretation
e0	+	(used in wildcarding - condition 13 only))

Format E:

For conditions 11, 12, 13 and 14 (ASA, EWT, ANI, and counted-calls) and condition 16 (wait-improved) which do not have the message layout specified above, the message will contain the fields identified in the following table, and the message will have the following layout.

Condition	Type	Subtype	Type_val	Threshold	d9-d12 (wait time)	var_t
11-ASA	split/skill	N/A	split/skill number	0-999	N/A	yes
11-ASA	VDN	active or latest	N/A	0-999	N/A	yes
12-EWT	split/skill	N/A	split/skill number	N/A	1-9999	yes
12-EWT	Call	N/A	N/A	N/A	1-9999	yes
12-EWT	Best	N/A	N/A	N/A	1-9999	yes
13-ANI	Table	N/A	table number	N/A	N/A	yes
14-counted-calls	VDN	active or latest	N/A	0-999	N/A	yes
16-wait-improved	split/skill	N/A	split/skill number	N/A	0-9999	yes
16-wait-improved	Best	N/A	N/A	N/A	0-9999	yes

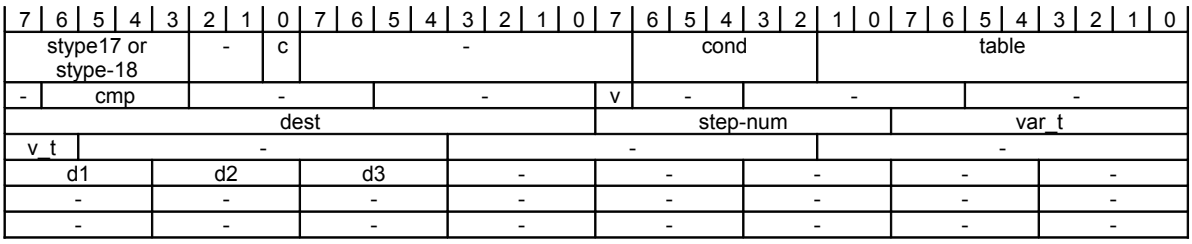


Format E (continued)

stype-17 or stype-18	goto step or goto vector
pri	0 = low, 1 = medium, 2 = high, 3 = top (for conditions 6, 8, 12)
c	1=this step is commented out
dest	Destination. Step number to go to (1-32) if stype-17. Vector to go to (1-999 (G3V8-CM2), 1-2000 (CM3-CM5.2), 1-8000 (CM6.0 and later) if stype-18
cond	see the condition/threshold above
threshold	see the condition/threshold above (if "v" is 0)
s	0=specific split/skill in type_val, 1=VDN skill preference in type_val
cmp	Comparators 1 >, 2 <, 3 =, 4 <>, 5 >=, 6 <=, 7 in (applies only to conditions 10 and 13), 8 not-in (applies only to conditions 10 and 13)
var_t	Variable threshold: 1-26 for variable A-Z, conditions 11 and 14 (for comparator 1-6) (MV2 and later), 1-26 for variable A-Z, conditions 11 and 14 (for comparator 7-8) (CM 3.0 and later), 27-31 for VDN variable V1-V5, conditions 11 and 14 (for comparator 1-8) (CM 3.0 and later)
v	Variable threshold flag: 0=no variable in var_t, 1=variable is in var_t, conditions 11 and 14 (MV2 and later)
type	1 split/skill 2 VDN 3 call 4 table 7 best
subtype	(applies only to conditions 11 and 14) 0 N/A 2 active 3 latest
type_val	split/skill for type=1, 1-999 (G3V8-G3V11), 1-2000 (MV2-CM5.2), 1-8000 (CM6.0 and later), or table number if type=4: 1-100 (through CM5.2), 1-999 (CM6.0 and later)
step_num	Vector Step number to go to: 0-31 (stype-18 only CM 3.0 and later)
d9-d12	Wait time, in seconds (0-9999)

Format F:

For condition 15 (if ii-digits), the following fields are contained in the message.



stype-17 or stype-18	goto step or goto vector
c	1=this step is commented out
dest	destination. Step number to go to (1-32) if stype-17. Vector to go to (1-999 (G3V8-CM2), 1-2000 (CM3-CM5.2), 1-8000 (CM6.0 and later) if stype-18
table	table number in which to check digits: 1-100 (through CM5.2), 1-999 (CM6.0 or later) if comparator is 7 or 8
cmp	Comparators 1 >, 2 <, 3 =, 4 <>, 5 >=, 6 <=, 7 in (table), 8 not-in (table)
var_t	Variable threshold: 1-26 for variable A-Z (for comparator 1-6) (MV2 and later), 1-26 for variable A-Z (for comparator 7-8) (CM 3.0 and later), 27-31 for VDN variable V1-V5 (for comparator 1-8) (CM 3.0 and later)
v	Variable threshold flag: 0=no variable in var_t, 1=variable is in var_t (MV2 and later)
d1-d2	digits for condition 15 (if comparator is 1-6 and "v" is 0) are as follows: The hex 0, representing null, is used when digits = "none".
d1-d3	d1-d2 as above for most cases, d3 also needed when one digit is "+" represented by hex e0.

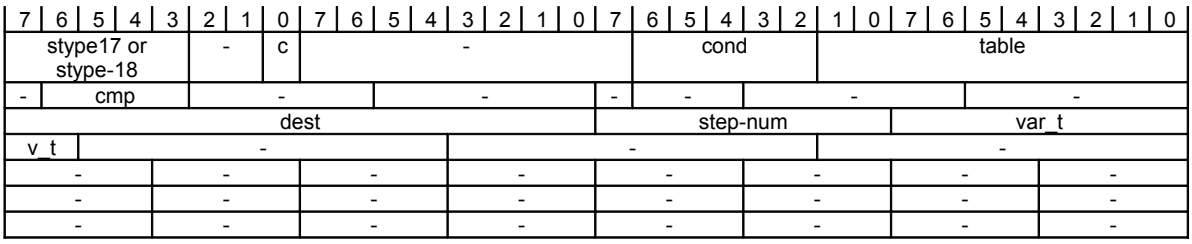
hex	char	hex	char
0	null	7	7
1	1	8	8
2	2	9	9
3	3	a	0
4	4	b	* (not valid)
5	5	c	# (not valid)
6	6	d	?

hex	char	interpretation
e0	+	(used in wildcarding)

step_num	Vector Step number to go to: 0-31 (stype-18 only CM 3.0 and later)

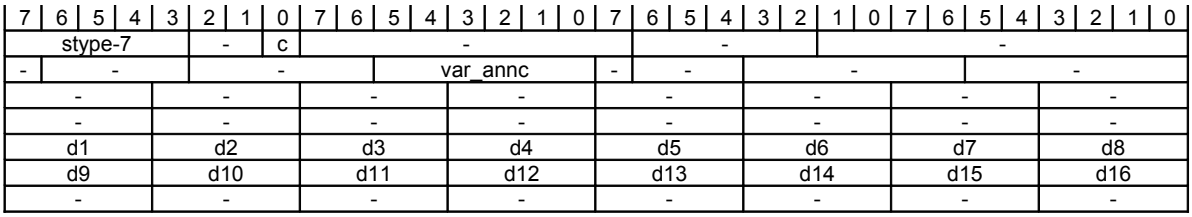
Format G:

For condition 18 (if holiday) and 23 (service hours), the following fields are contained in the message.



stype-17 or stype-18	goto step or goto vector
c	1=this step is commented out
dest	destination. Step number to go to (1-32) if stype-17. Vector to go to (1-999 (G3V8-CM2), 1-2000 (CM3-CM5.2), 1-8000 (CM6.0 and later) if stype-18
table	holiday table number in which to check holiday: 1-10 (pre-CM 3.0), 1-99 (CM 3.0-CM5.2), 1-999 (CM6.0 and later); comparator is 7 or 8 only
cmp	Comparators 1 > (not valid) 2 < (not valid) 3 = (not valid) 4 <> (not valid) 5 >= (not valid) 6 <= (not valid) 7 in (table) 8 not-in (table)
var_t	Variable threshold: 1-26 for variable A-Z, 27-31 for VDN variable V1-V5 (for comparator 1-8) (CM 3.0 and later)
step_num	Vector Step number to go to: 0-31 (stype-18 only CM 3.0 and later)

3.3.21.7 **Disconnect**

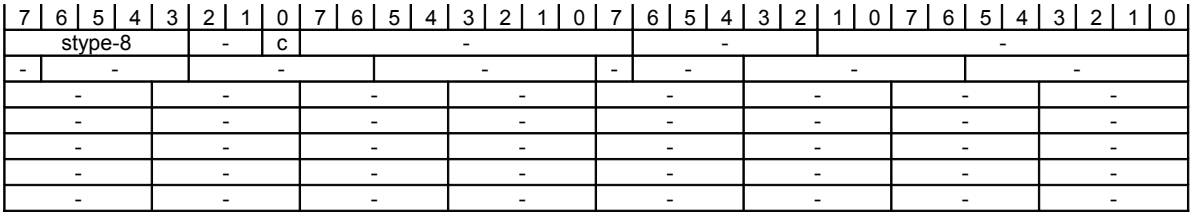


7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
stype-7					-		c	-								-								-							
-	-				-			var_annc								-	-							-							
-					-			-				-				-								-							
-					-			-				-				-								-							
d1					d2			d3				d4				d5								d6							
d9					d10			d11				d12				d13								d14							
-					-			-				-				-								-							

stype-7 disconnect
c 1=this step is commented out
var_annc Variable announcement: 1-26 for variable A-Z, 27-31 for VDN variable V1-V5 (CM 3.0 and later)
d1-d15 (< language 24) **announcement extension (packed decimal where 0= null or "none" and hex a is digit 0).**

d1-d16 (>= language 24)

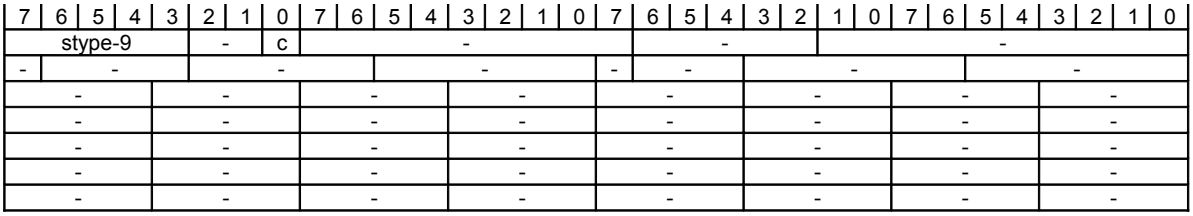
3.3.21.8 **Busy**



7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
stype-8					-		c	-								-								-							
-	-				-			-								-	-							-							
-					-			-				-				-								-							
-					-			-				-				-								-							
-					-			-				-				-								-							
-					-			-				-				-								-							
-					-			-				-				-								-							

stype-8 busy
c 1=this step is commented out

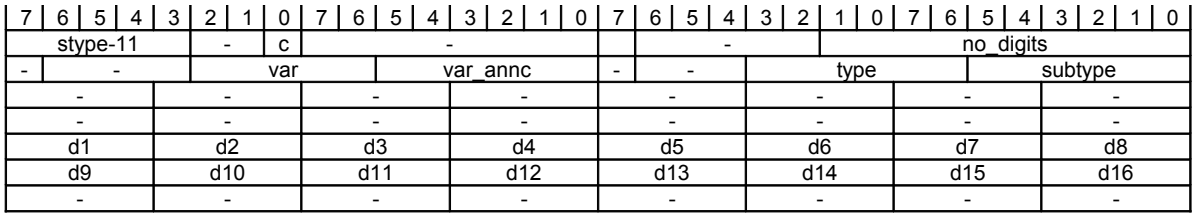
3.3.21.9 **Stop**



7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
stype-9					-		c	-								-								-							
-	-				-			-								-	-							-							
-					-			-				-				-								-							
-					-			-				-				-								-							
-					-			-				-				-								-							
-					-			-				-				-								-							
-					-			-				-				-								-							

stype-9 stop
c 1=this step is commented out

3.3.21.10 **Collect Digits**

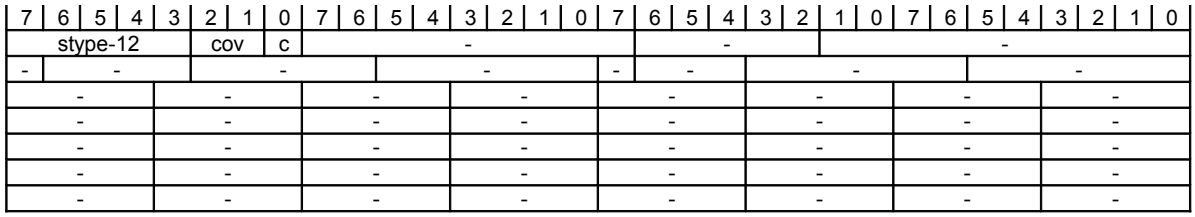


7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
stype-11					-	c		-									-							no_digits							
-	-				var			var_annc								-	-			type				subtype							
-					-			-				-				-				-				-							
-					-			-				-				-				-				-							
d1					d2			d3				d4				d5				d6				d7				d8			
d9					d10			d11				d12				d13				d14				d15				d16			
-					-			-				-				-				-				-							

stype-11	collect digits
c	1=this step is commented out
no_digits	number of digits to collect (1-16). It is used if the type is 5 (digits)
var	variable: 1-26 for variable A-Z, where 0="none",
var_annc	Variable announcement: 1-26 for variable A-Z, 27-31 for VDN variable V1-V5 (CM 3.0 and later)
type	5=digits, 6=special digits
subtype	4=ced, 5=cdpd; only valid with type=6
d1-d15 (< language 24)	**announcement extension (packed decimal where 0= "none" and hex a is digit 0); only valid with type=5.**

d1-d16 (>= language 24)

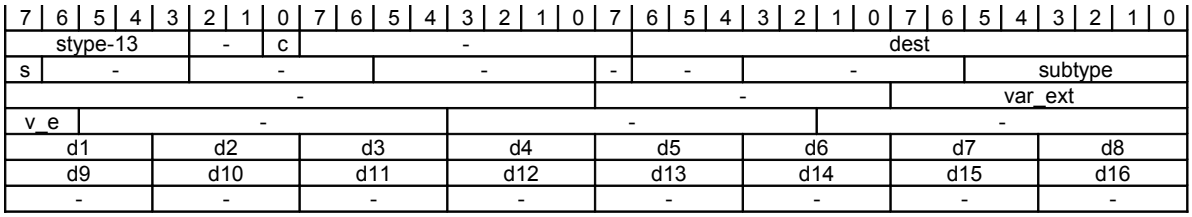
3.3.21.11 **Route To Digits**



7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
stype-12					cov	c		-									-							-							
-	-				-			-				-				-	-			-				-							
-					-			-				-				-				-				-							
-					-			-				-				-				-				-							
-					-			-				-				-				-				-							
-					-			-				-				-				-				-							
-					-			-				-				-				-				-							

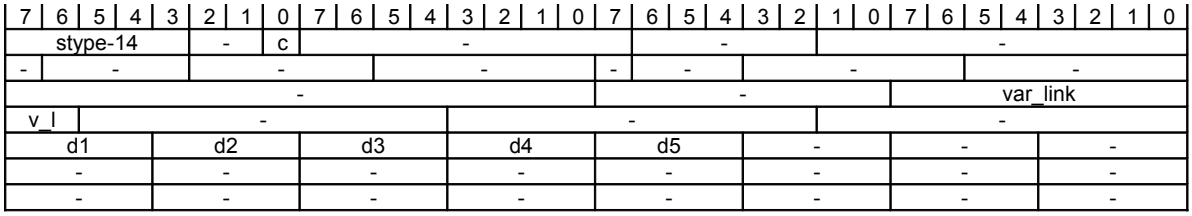
stype-12	route-to digits
cov	with coverage, 0=no, 1=yes
c	1=this step is commented out

3.3.21.12 **Messaging Split**



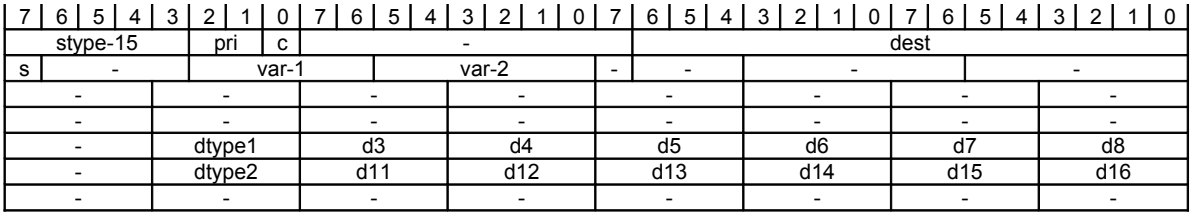
stype-13	messaging split/skill
c	1=this step is commented out
dest	destination. Split or skill 1-999 (G3V8-G3V11), 1-2000 (MV2-CM5.2), 1-8000 (CM6.0 and later). For s=1 (EAS only), skill = 1, 2, or 3.
s	0=specific skill, 1=VDN skill preference (EAS only)
subtype	1 extension 2 active 3 latest
var_ext	Variable Extension: 1-26 for variable A-Z (CM 3.1 and later), 27-31 for VDN variable V1-V5 (CM 3.1 and later), 0 when "d1-d8" contains value.
d1-d15 (< language 24)	**the "for extension" parameter (packed decimal where hex a is digit 0). Will be empty unless subtype is 1 (extension).**
d1-d16 (>= language 24)	

3.3.21.13 **Adjunct Routing**



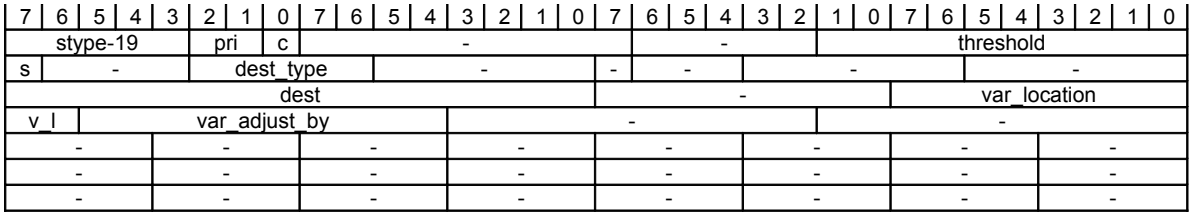
stype-14	adjunct route
c	1=this step is commented out
var_link	Variable Link: 1-26 for variable A-Z (CM 3.1 and later), 27-31 for VDN variable V1-V5 (CM 3.1 and later), 0 when "d1-d5" contains value.
d1-d5	the extension (G3V8-G3V10) or 1-digit to 2-digit link ID (G3V11 and later) of the adjunct link (packed decimal where 0= null and hex a is digit 0)

3.3.21.14 **Converse**



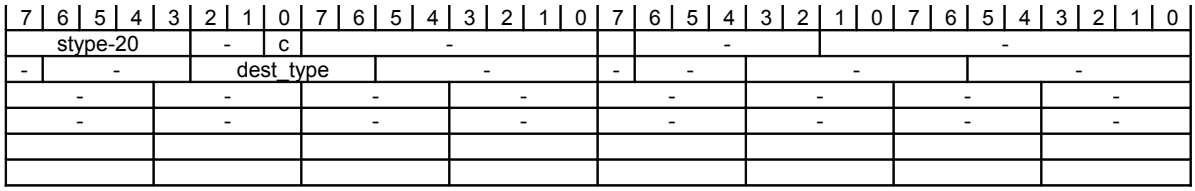
stype-15	converse-on split/skill
pri	0=low, 1=medium, 2=high, 3=top
c	1=this step is commented out
dest	Destination. Split or skill 1-999 (G3V8-G3V11), 1-2000 (MV2-CM5.2), 1-8000 (CM6.0 and later). For s=1 (EAS only), skill = 1, 2, or 3.
s	0=specific skill, 1=VDN skill preference (EAS only)
var-1	Variable: 1-26 for variable A-Z, 27-31 for VDN variable V1-V5 (dtype1 = 7)
var-2	Variable: 1-26 for variable A-Z, 27-31 for VDN variable V1-V5 (dtype2 = 7)
dtype1	The first passing digits type, 0=none, 1=string of digits, 2=vdn, 3=ani, 4=collected digits, 5=qpos, 6=ewt, 7=var-1
d3-d8	The first digits to pass (packed decimal where 0 is null, hex a is digit 0, hex b is *, hex c, #, is allowed as the first and only digit)
dtype2	The second passing digits type, 0=none, 1=string of digits, 2=vdn, 3=ani, 4=collected digits, 5=qpos, 6=ewt, 7=var-2
d11-d16	The second digits to pass (packed decimal where 0 is null, hex a is digit 0, hex b is *, hex c, #, is allowed as the first and only digit)

3.3.21.15 **Consider**



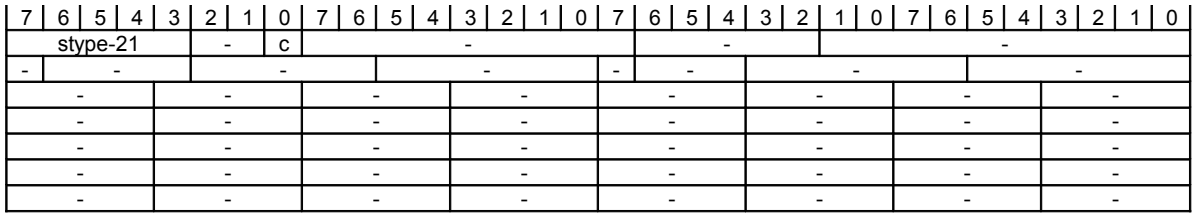
stype-19	consider
pri	Priority 0=low, 1=med, 2=high, 3=top
c	1=this step is commented out
threshold	Amount to adjust by, 0-100
s	0=specific split/skill (if dest_type=0), 1=VDN skill preference (EAS only, and if dest_type=0)
dest_type	0=split/skill or VDN preference, 2=location
dest	If dest_type=0, then split/skill 1-999 (G3V8-G3V11), 1-2000 (MV2-CM5.2), 1-8000 (CM6.0 and later) or VDN skill preference (1, 2, or 3). If dest_type=2, then location number, 1-255.
var_location	Variable Location: 1-26 for variable A-Z (CM 3.1 and later), 27-31 for VDN variable V1-V5 (CM 3.1 and later), 0 when "dest" contains value.
var_adjust_by	. Variable Adjust By: 1-26 for variable A-Z (CM 3.1 and later), 27-31 for VDN variable V1-V5 (CM 3.1 and later), 0 when "threshold" contains value.

3.3.21.16 **Reply**



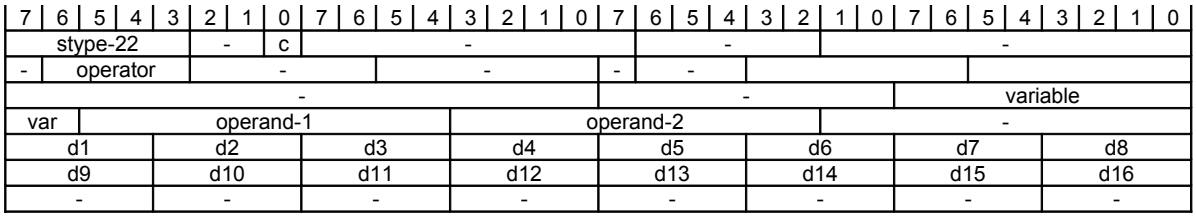
stype-20 reply-best
c 1=this step is commented out
dest_type 1 (best)

3.3.21.17 **Return**



stype-21 return
c 1=this step is commented out

3.3.21.18 **Set**



stype-22 set
c 1=this step is commented out
variable 0 for digits, 1-26 for variable A-Z
operand-1 0 for digits, 1-26 for variable A-Z, 27-31 for VDN variable V1-V5, 32 for none
operand-2 0 for digits, 1-26 for variable A-Z, 27-31 for VDN variable V1-V5, 32 for none, 33 for string of digits
d1-d16 digits to pass as follows:

hex	char	hex	char
0	null	7	7
1	1	8	8
2	2	9	9
3	3	a	0
4	4	b	*
5	5	c	+
6	6	d	-

operator
1 ADD
2 SUB
3 MUL
4 DIV
5 CATR
6 CATL
7 MOD70
8 SEL

3.3.22 ← CHGVEC20 AND CHGVEC24

The **CHGVEC20 - Change Vector Parameters** message is a variable-length message sent by the reporting adjunct based on the "steps" field. It is a request to change the vector steps of the specified vector. The expected response is an **RCODE20** message. The first 5 bytes are always sent and will be the same for all messages sent for a single vector. Multiple messages will be sent for large vectors.

Vectors can have 99 steps per vector. The number of steps sent per message varies based on the message.

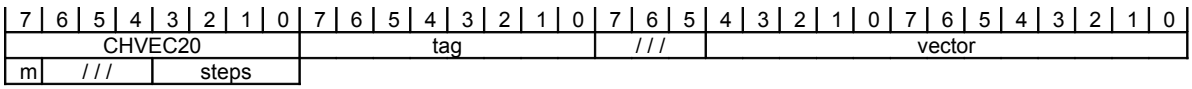
The **CHGVEC** message has the same steps as the **VECTOR** message. See section 3.3.21 for the layout of each step type.

3.3.22.1 ←CHGVEC20 (New CM3.1)

	Switch	Vect	Skill
First Release	CM3.1	y	y/n
Last Release	CM6.3		

The reporting adjunct will send 9 steps per message, with up to 11 messages to send all 99 steps.

For CM6.0 and later, if a **CHGVEC20** is sent with a check step with a preference level other than “all-levels” and the SPI language is prior to 23, the request will be denied (rcode=43).

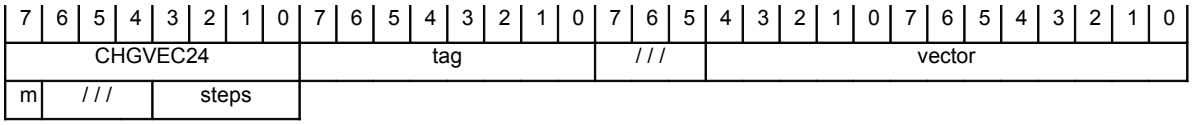


opcode CHGVEC20 = 51 (0x33)
tag message tag, 0-255 (it will be returned in the switch response message)
vector vector number, 1-999 (G3V8-CM2), 1-2000 (CM3.1-CM5.2), 1-8000 (CM6.0 and later)
m m=1, more steps for this vector will be sent in a subsequent vector, m=0, this is the last message for this vector
steps number of steps (N) 0-9. This is the number of steps in this message.

3.3.22.2 **←CHGVEC24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y	y/n
Last Release			

The reporting adjunct will send 8 steps per message, with up to 13 messages to send all 99 steps.



opcode	CHGVEC24 = 68 (0x44)
tag	message tag, 0-255 (it will be returned in the switch response message)
vector	vector number, 1-999 (G3V8-CM2), 1-2000 (CM3.1-CM5.2), 1-8000 (CM6.0 and later)
m	m=1, more steps for this vector will be sent in a subsequent vector, m=0, this is the last message for this vector
steps	number of steps (N) 0-8. This is the number of steps in this message.

3.3.23 ← REQLIST5 (New G3V6)

	Switch	Vect	Skill
First Release	G3V6	y	y/n
Last Release			

The **Request List** message is sent by the reporting adjunct to the switch to request a list of non-empty entities of a certain type and their modification numbers. The expected response is an **ADMLIST20** with the entities or an **RCODE20** 1.

7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
REQLIST5								tag								/ / / /				obj type			

opcode REQLIST5 = 35 (0x23)
tag message tag, 0-255 (it will be returned in the switch response message)
obj type type of object requested: 0=vectors, 1=VDNs (measured by CMS)

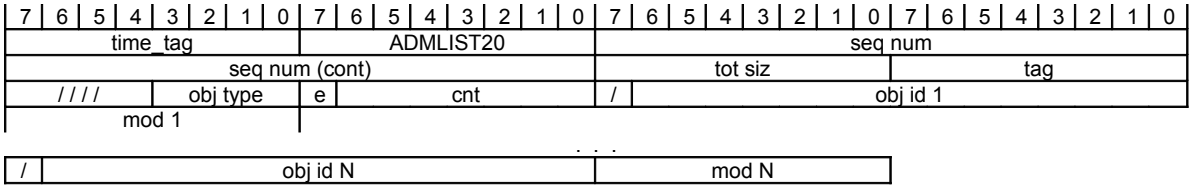
3.3.24 → ADMLIST20 (New CM3.1)

	Switch	Vect	Skill
First Release	CM3.1	y	y/n
Last Release			

The **Administration list** message is sent by switch to the reporting adjunct in response to a **REQLIST5** message to indicate the non-empty entities on the switch of the requested type. For the VDN type, these are the VDNs which are measured by the reporting adjunct only. For vectors, these are the vectors which are non-empty (i.e., there is a name or one or more steps).

The results of this request may not fit into a single **ADMLIST20** message. In this case, the cnt field indicates the number of items in the current message and the end bit is set for the last message in a series. This is a variable length message based on the cnt field. The first ten bytes are always sent.

These messages may form very long chains. G3V6/R3V6 supports 8000 CMS measured VDNs. A list of 20000 VDNs will require 254 messages to be sent. A list of 999 vectors will require 13 messages. These calculations are based on a message buffer with a size of 244 bytes.



time_tag	seconds past minute when messages was formed, 0-59
opcode	ADMLIST20 = 200 (0xc8)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
tag	message tag, 0-255 (as contained in the request message).
obj type	type of object which is listed: 0=vectors, 1=VDNs
e	end bit, set when this is the last message of the set needed to satisfy the request
cnt	number of objects listed in this message, 1-77
obj id 1-N	the identifying number for the object

obj type	entity	obj id values
0	vector	1-999 (G3V8-CM2), 1-2000 (CM3-CM5.2), 1-8000 (CM6.0 and later)
1	VDN (iln)	1-32767

mod 1-N modification number for the object

3.3.25 ←REQNAM20 (New CM3.1)

	Switch	Vect	Skill
First Release	CM3.1	y	y/n
Last Release			

The **Request NAME** message is sent by the reporting adjunct to CM to request a list of objects. The expected response is an **ADMNAM**.

7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
REQNAM20								tag								/ / / / / / /								m	/ / /				v	t	h	d	a

opcode	REQNAM20 = 48 (0x30)
tag	message tag, 1-255 (it will be returned in the switch **ADMNAM** response messages)
mode	mode; 0 = all names; 1 = changed names
v	vector names; 1 = report vector names
t	trunk group names; 1 = report trunk group names
h	hunt group names; 1 = report hunt group names
d	VDN names; 1 = report VDN names
a	agent names; 1 = report agent names

3.3.26 → ADMNAM20 and ADMNAM24

The **Administration NAME** message is sent by switch to the reporting adjunct in response to a **REQNAM20** message to provide switch data for the requested object type. The information for each returned object contains: object-id, name1-object-data-length, name1-object-data, name2-object-data-length and name2-object-data.

The end bit is set for the last message in a series. This is a variable length message based on the data-len-name1 and data-len-name2 fields. The first eight bytes are always sent.

3.3.26.1 → ADMNAM20 (New CM3.1)

	Switch	Vect	Skill
First Release	CM3.1	y	y/n
Last Release	CM6.3		

7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
time_tag								ADMNAM20								seq num															
seq num (cont)																tot siz								tag							
e	r	n 1	n 2	obj type																											

for vector, trunk, hunt and VDN

b	obj id			name 1 len M			name 1 byte 1
name 1 byte 2			name 1 byte 3	* * *			name 1 byte M
name 2 len N			name 2 script tag	name 2 byte 1			name 2 byte 2
name 2 byte 3			* * *	name 2 byte N			

for agent

pnumber		p1		p2		p3		p4		p5		p6		p7	
p8		p9		p10		p11		p12		p13		p14		p15	
name 1 len M				name 1 byte 1				name 1 byte 2				name 1 byte 3			
* * *				name 1 byte M				name 2 len N				name 2 script tag			
name 2 byte 1				name 2 byte 2				name 2 byte 3				* * *			
name 2 byte N															

ADMNAM20 (continued)

time_tag	seconds past minute when messages was formed, 0-59
opcode	ADMNAM20 = 201 (0xc9)
seq num	sequence number, 0-0xffff ffff
tot siz	total size of message in bytes; 1-255
tag	message tag, 0-255 from REQNAM20 message or 0 if admin change update
e	end bit, set when this is the last message of the set needed to satisfy the request
r	reason bit, 0 = response to sync request, 1 = admin change update
n1	name-1 flag, 0 = name-1 NOT in message, 1 = name-1 IS in message
n2	name-2 flag, 0 = name-2 NOT in message, 1 = name-2 IS in message
obj type	type of object which is listed:

obj type	entity
0	Vector Names
1	Trunk Group Names
2	Hunt Group Names
3	VDN (iln) Names
4	Agent Loginid Names

b	Beginning bit, set when this is the first message of the set needed to satisfy the request. This bit is only set for CM FP4 or later. (obj types 0-3 only)
obj id	the identifying number for the object (obj types 0-3 only)

obj type	entity	obj id values
0	Vector Names	1-2000 (CM3.1-CM5.2) 1-8000 (CM6.0 and later)
1	Trunk Group Names	1-2000
2	Hunt Group Names	1-2000 (CM3.1-CM5.2) 1-8000 (CM6.0 and later)
3	VDN (iln) Names	1-20000 (CM3.1-CM5.2) 1-30000 (CM6.0 and later)

pnumber	number of digits in agent position, 1-15 (obj type 4 only)
p1-p15	agent position - with EAS login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill position to a byte boundary (obj type 4 only)
name 1 len M	number of bytes of object data, (0-27)
name 1 byte X	object data bytes (ASCII encoded name1 data bytes)
name 2 len N	number of bytes of object data, (0-81)
name 2 script tag	UTF-8 script tag for name2 data bytes. The script tag represents a writing script. See the following table for the actual values, 1-32.
name 2 byte X	object data bytes (UTF-8 encoded name2 data bytes)

Script tag	Writing script	
1	Basic Latin	
2	Latin-1 Supplement	
3	Latin Extended-A	
4	Latin Extended-B	
5	Greek and Coptic	
6	Cyrillic and Cyrillic Supplementary	
7	Armenian	
8	Hebrew	
9	Arabic	
10	Devanagari	
11	Bengali	
12	Gurmukhi	
13	Gujarati	
14	Oriya	
15	Tamil	
16	Telugu	
17	Kannada	
18	Malayalam	
19	Sinhala	
20	Thai	
21	Lao	
22	Myanmar	
23	Georgian	
24	Tagalog	
25	Khmer	
26	Halfwidth and Fullwidth Forms	
27	Hiragana, Katakana, Katakana Phonetic Extensions	CJKV Radicals Supplement Kangxi Radicals CJKV Symbols and Punctuation Enclosed CJK Letters and Months CJKV Compatibility CJKV Unified Ideographs Extension A JCKV Unified Ideographs CJK Compatibility Ideographs CJK Compatibility Forms
28		
29	Bopomofo, Bopomofo Extended	
30		
31		
32	Hangul Jamo, Hangul Compatibilty Jamo, Hangul Syllables	

3.3.26.2 **→ADMNAM24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y	y/n
Last Release			

7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
time_tag								ADMNAM24								seq num															
seq num (cont)																tot siz								tag							
e	r	n 1	n 2	obj type																											

for vector, trunk, hunt and VDN

b	/ / / / / / /	/	obj_id		name 1 len M
name 1 byte 1		name 1 byte 2		name 1 byte 3	* * *
name 1 byte M		name 2 len N		name 2 script tag	name 2 byte 1
name 2 byte 2		name 2 byte 3		* * *	name 2 byte N

for agent

b	/ / / / / /	pnumber	p1	p2	p3	p4	p5
p6	p7	p8	p9	p10	p11	p12	p13
p14	p15	p16	/ / / /	Name 1 len M		Name 1 byte 1	
Name 1 bytes 2		Name 1 byte 3		* * *		Name 1 byte M	
Name 2 len N		Name 2 script tag		Name 2 byte 1		Name 2 byte 2	
Name 2 byte 3		* * *		Name 2 byte N			

ADMNAM24 (continued)

time_tag	seconds past minute when messages was formed, 0-59
opcode	ADMNAM24 = 43 (0x2b)
seq num	sequence number, 0-0xffff ffff
tot siz	total size of message in bytes; 1-255
tag	message tag, 0-255 from REQNAM20 message or 0 if admin change update
e	end bit, set when this is the last message of the set needed to satisfy the request
r	reason bit, 0 = response to sync request, 1 = admin change update
n1	name-1 flag, 0 = name-1 NOT in message, 1 = name-1 IS in message
n2	name-2 flag, 0 = name-2 NOT in message, 1 = name-2 IS in message
obj type	type of object which is listed:

obj type	entity
0	Vector Names
1	Trunk Group Names
2	Hunt Group Names
3	VDN (iln) Names
4	Agent Loginid Names

b	Beginning bit, set when this is the first message of the set needed to satisfy the request. **(all object types)**
obj id	the identifying number for the object (obj types 0-3 only)

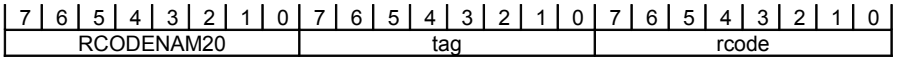
obj type	entity	obj id values
0	Vector Names	1-2000 (CM3.1-CM5.2) 1-8000 (CM6.0 and later)
1	Trunk Group Names	1-2000
2	Hunt Group Names	1-2000 (CM3.1-CM5.2) 1-8000 (CM6.0 and later)
3	VDN (iln) Names	1-20000 (CM3.1-CM5.2) 1-30000 (CM6.0 and later)

pnumber	**number of digits in agent position, 1-16 (obj type 4 only)**
p1-p16	**agent position - with EAS login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill position to a byte boundary (obj type 4 only)**
name 1 len M	number of bytes of object data, (0-27)
name 1 byte X	object data bytes (ASCII encoded name1 data bytes)
name 2 len N	number of bytes of object data, (0-81)
name 2 script tag	UTF-8 script tag for name2 data bytes. The script tag represents a writing script. See the table under ADMNAM20 for the actual values, 1-32.
name 2 byte X	object data bytes (UTF-8 encoded name2 data bytes)

3.3.27 ←RCODENAM20 (New CM3.1)

	Switch	Vect	Skill
First Release	CM3.1	y	y/n
Last Release			

The **Return Code Name** message is sent by the reporting adjunct to CM.



opcode RCODENAM20 = 49 (0x31)
tag message tag, 1-255
rcode

rcode	description
0	success
1	resend
2	skip

3.3.28 ← CHGNAM20 and CHGNAM24

The **Change Name** message is sent by the reporting adjunct to the switch to change the administered name of an entity. Entities included in this protocol are: vectors, trunk groups, hunt groups, VDNs and Agent LoginIDs. The expected response to each message is an **RCODE20** [rcode = 0].

3.3.28.1 ←CHGNAM20 (New CM3.1)

	Switch	Vect	Skill
First Release	CM3.1	y	y/n
Last Release	CM6.3		

7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
CHGNAM20								tag								/	r	n 1	n 2	obj type											

for vector, trunk, hunt and VDN

/	obj id		name 1 len M	name 1 byte 1
name 1 byte 2		name 1 byte 3	* * *	name 1 byte M
name 2 len N		name 2 script tag	name 2 byte 1	name 2 byte 2
name 2 byte 3		* * *	name 2 byte N	

for agent

pnumber	p1	p2	p3	p4	p5	p6	p7
p8	p9	p10	p11	p12	p13	p14	p15
name 1 len M		name 1 byte 1		name 1 byte 2		name 1 byte 3	
* * *		name 1 byte M		name 2 len N		name 2 script tag	
name 2 byte 1		name 2 byte 2		name 2 byte 3		* * *	
name 2 byte N							

CHGNAM20 (continued)

opcode	CHGNAM20 = 50 (0x32)
tag	message tag, 0-255 (it will be returned in the switch response message RCODE20)
r	reason bit, 0 = sync data, 1 = admin change update
n1	name-1 flag, 0 = name-1 NOT in message, 1 = name-1 IS in message
n2	name-2 flag, 0 = name-2 NOT in message, 1 = name-2 IS in message
obj type	type of object which is listed:

obj type	entity
0	Vector Names
1	Trunk Group Names
2	Hunt Group Names
3	VDN (iln) Names
4	Agent Loginid Names

obj id — the identifying number for the object

obj type	entity	obj id values (obj types 0-3 only)
0	Vector Names	1-2000 (CM3.1-CM5.2) 1-8000 (CM6.0 and later)
1	Trunk Group Names	1-2000
2	Hunt Group Names	1-2000 (CM3.1-CM5.2) 1-8000 (CM6.0 and later)
3	VDN (iln) Names	1-20000 (CM3.1-CM5.2) 1-30000 (CM6.0 and later)

pnumber	number of digits in agent position, 1-15 (obj type 4 only)
p1-p15	agent position - with EAS login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill position to a byte boundary (obj type 4 only)
name 1 len M	number of bytes of object data, (0-27)
name 1 byte X	object data bytes (ASCII encoded name1 for obj type 0 thru 4)
name 2 len N	number of bytes of object data, (0-81)
name 2 script tag	UTF-8 script tag for name2 data bytes. The script tag represents a writing script. See the table under ADMNAM20 for the actual values, 1-32.
name 2 byte X	object data bytes (UTF-8 encoded name2 for obj type 0 thru 4)

3.3.28.2 **← CHGNAM24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y	y/n
Last Release			

7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
CHGNAM24								tag								/	r	n 1	n 2	obj type											

for vector, trunk, hunt and VDN

b	/ / / / / / /	/	obj_id	name 1 len M
name 1 byte 1		name 1 byte 2	name 1 byte 3	* * *
name 1 byte M		name 2 len N	name 2 script tag	name 2 byte 1
name 2 byte 2		name 2 byte 3	* * *	name 2 byte N

for agent

b	/ / / / / /	pnumber	p1	p2	p3	p4	p5
p6	p7	p8	p9	p10	p11	p12	p13
p14	p15	p16	/ / / /	Name 1 len M		Name 1 byte 1	
Name 1 bytes 2		Name 1 byte 3		* * *		Name 1 byte M	
Name 2 len N		Name 2 script tag		Name 2 byte 1		Name 2 byte 2	
Name 2 byte 3		* * *		Name 2 byte N			

CHGNAM24 (continued)

opcode	CHGNAM24 = 65 (0x41)
tag	message tag, 0-255 (it will be returned in the switch response message RCODE20)
r	reason bit, 0 = sync data, 1 = admin change update
n1	name-1 flag, 0 = name-1 NOT in message, 1 = name-1 IS in message
n2	name-2 flag, 0 = name-2 NOT in message, 1 = name-2 IS in message
obj type	type of object which is listed:

obj type	entity
0	Vector Names
1	Trunk Group Names
2	Hunt Group Names
3	VDN (iln) Names
4	Agent Loginid Names

obj id — the identifying number for the object

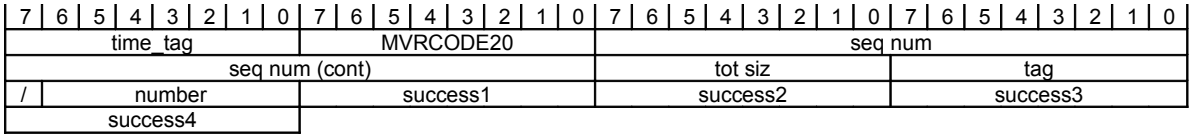
obj type	entity	obj id values (obj types 0-3 only)
0	Vector Names	1-2000 (CM3.1-CM5.2) 1-8000 (CM6.0 and later)
1	Trunk Group Names	1-2000
2	Hunt Group Names	1-2000 (CM3.1-CM5.2) 1-8000 (CM6.0 and later)
3	VDN (iln) Names	1-20000 (CM3.1-CM5.2) 1-30000 (CM6.0 and later)

pnumber	**number of digits in agent position, 1-16 (obj type 4 only)**
p1-p16	**agent position - with EAS login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill position to a byte boundary (obj type 4 only)**
name 1 len M	number of bytes of object data, (0-27)
name 1 byte X	object data bytes (ASCII encoded name1 for obj type 0 thru 4)
name 2 len N	number of bytes of object data, (0-81)
name 2 script tag	UTF-8 script tag for name2 data bytes. The script tag represents a writing script. See the table under ADMNAM20 for the actual values, 1-32.
name 2 byte X	object data bytes (UTF-8 encoded name2 for obj type 0 thru 4)

3.3.29 → MVRCODE20 (New CM3.1)

	Switch	Vect	Skill
First Release	CM3.1	y/n	y/n
Last Release			

The **Return Code from Move Request** message is sent by the switch in response to a move request: **CHGVDNSK** or **MVVDN**. A bit map is used to indicate the status of each (up to 32) move requests. The bit map is divided into four bytes, and the bytes are used from left to right. Within each byte, the bits are used from right to left. For example, if two extensions were moved, bits 0 and 1 of the first byte (*success1*) will be used and bit 0 will represent the status of the first item in the move request message and bit 1 will represent the status of the second item in the message.

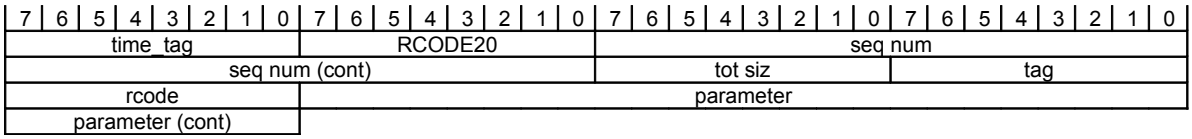


time_tag	seconds past minute when message was formatted, 0-59
opcode	MVRCODE20 = 202 (0xca)
seq_num	sequence number, 0-0xffff ffff
tot_siz	total size in bytes, 1-255
tag	message tag, 0-255 (it will be returned in the switch response message)
number	number of items in the request, 1-32
success1-4	one bit for each item in the move request, 0 = failure, 1 = success. The bits are right-justified within each byte, i.e., success1 represents the success of items 8-1, success2 represents items 16-9, and so on.

3.3.30 → RCODE20 (New CM3.1)

	Switch	Vect	Skill
First Release	CM3.1	y/n	y/n
Last Release			

The **Return Code from Request** message is sent by the switch in response to a request from the reporting adjunct. In general, it is used whenever the switch does not completely understand a request message (*rcode*=1) and specifically, to indicate the result of the following requests: **VPARMS**, **CHGVEC** and **SKILLADD**.



time_tag	seconds past minute when event occurred, 0-59
opcode	RCODE20 = 203 (0xcb)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
tag	message tag, 0-255 (it will be returned in the switch response message)
rcode	return code
parameter	value of item that error occurred on

RCODE20 (continued)

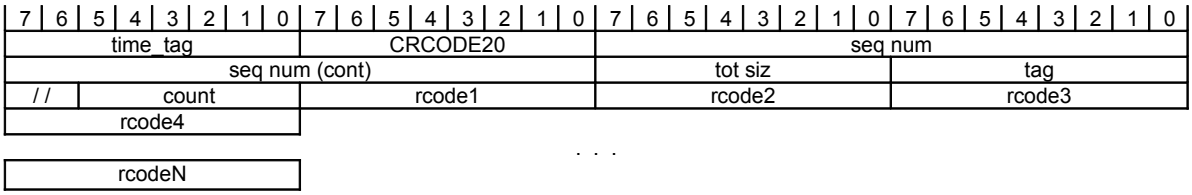
Rcode	Description	Parameter	In Response To
0	request successfully completed		CHGVEC,SKILLADD
1	illegal request		any
20	vector number out of range	vector no.	CHGVEC,VPARMS, MVVDN
21	field in vector step out of range	step no.	CHGVEC
22	bad vector 'goto' destination	step no.	CHGVEC
24	split/skill number out of range		MVAGSTFD
25	bad step type in vector specification	step no.	CHGVEC
26	undefined split number in CVP request	step no.	CHGVEC
27	vector 'goto' cannot go to same vector	step no.	CHGVEC
28	too many steps		CHGVEC
29	invalid step type	step no.	CHGVEC
31	extension not assigned	step no. none none	CHGVEC APARMS SKILLADD
32	extension not a BRI extension	step no.	CHGVEC
33	announcement not assigned	step no.	CHGVEC
34	illegal packed decimal digit	step no.	CHGVEC
35	invalid *i* condition	step no.	CHGVEC
36	invalid *wait* condition	step no.	CHGVEC
37	split is not vector controlled	step no.	CHGVEC
38	request pending until agent logs out		SKILLADD
39	switch run tape in progress, try later		NOT USED
42	split not messaging type split	step no.	CHGVEC
43	unsupported step type for this CMS release	step no.	VPARMS
44	cannot move to unmeasured split		NMVAGENT
45	request pending until agent is in AUX or AVAIL state		SKILLADD
46	specified split/skill not assigned to agent		(used in CRCODE)
47	specified split/skill already assigned to agent		(used in CRCODE)
48	agent already assigned to maximum split/skill pairs		(used in CRCODE)
49	agent cannot be assigned to both AUDIX and non-AUDIX splits/skills		(used in CRCODE)
50	maximum agents active in this split/skill		(used in CRCODE)
51	maximum agents assigned in system		(used in CRCODE)
52	cannot remove only skill for an agent		(used in CRCODE)
53	request denied - a change is currently pending for this agent		(used in CRCODE)
54	agent cannot be assigned to both AAS and non-AAS splits/skills		(used in CRCODE)
55	agent is adjunct controlled		(used in CRCODE)
56	skill type must be specified		MVAGSTFD
57	maximum number of agents allowed by B/CMS already administered		(used in CRCODE)
58	agent must have at least one administered skill		SKILLADD
59	feature not enabled on switch		CHGVEC
60	administration contention. Try again later		any
61	cannot remove direct agent skill		(used in CRCODE)
62	cannot add/remove standard skill		(used in CRCODE)

63	direct agent skill cannot be a reserve skill	SKILLADD
64	cannot change reserve skill to standard or vice versa	(used in CRCODE)
65	entity locked on switch	CHGVEC, VPARMS
66	**Skill tenant number doesn't match agent tenant number**	**SKILLADD**

3.3.31 → CRCODE20 (New CM3.1)

	Switch	Vect	Skill
First Release	CM3.1	y/n	y/n
Last Release			

This is a variable length message sent by the switch in response to a **MVAGSTFD** message. It contains up to 32 individual RCODEs, one for each agent position specified in the **MVAGSTFD** message. The number of RCODEs contained in the message is indicated by the "count" field. This message must be sent within 2 minutes of the time of the **MVAGSTFD** message.



time_tag	seconds past minute when event occurred, 0-59
opcode	CRCODE20 = 204 (0xcc)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
tag	message tag, 0-255, set to the tag value of the IQ message that generated the **CRCODE20**
count	the number of RCODES, 1-32, included in the message
rcode 1-N	return code for specific request. (See table below for valid codes.)

Rcode	Description	In Response To
0	request successfully completed	MVAGSTFD
1	illegal request	MVAGSTFD
20	vector number out of range	(used in RCODE)
21	field in vector step out of range	(used in RCODE)
22	bad vector 'goto' destination	(used in RCODE)
24	split/skill number out of range	MVAGSTFD
25	bad step type in vector specification	(used in RCODE)
26	undefined split number in CVP request	(used in RCODE)
27	vector 'goto' cannot go to same vector	(used in RCODE)
28	too many steps	(used in RCODE)
29	invalid step type	(used in RCODE)
31	extension not assigned	(used in RCODE)
32	extension not a BRI extension	(used in RCODE)
33	announcement not assigned	(used in RCODE)
34	illegal packed decimal digit	(used in RCODE)
35	invalid *if* condition	(used in RCODE)
36	invalid *wait* condition	(used in RCODE)
37	split is not vector controlled	(used in RCODE)
38	request pending until agent logs out	(used in RCODE)
39	switch run tape in progress, try later	NOT USED
42	split not messaging type split	(used in RCODE)
43	unsupported step type for this CMS release	(used in RCODE)
44	cannot move to unmeasured split	NMVAGENT,MVAGSTFD
45	request pending until agent is in AUX or AVAIL state	MVAGSTFD

46	specified split/skill not assigned to agent	MVAGSTFD
47	specified split/skill already assigned to agent	MVAGSTFD
48	agent already assigned to maximum skills	MVAGSTFD
49	agent cannot be in both AUDIX and non-AUDIX skills	MVAGSTFD
50	maximum agents active in this skill	MVAGSTFD
51	maximum agents logged into system	MVAGSTFD
52	cannot remove only skill for an agent	MVAGSTFD
53	request denied - change currently pending for agent	MVAGSTFD
54	agent cannot be in both AAS and non-AAS skills	MVAGSTFD
55	adjunct controlled agents limited to one skill	MVAGSTFD
56	skill type must be specified	MVAGSTFD
57	max agents allowed by B/CMS already administered	MVAGSTFD
58	agent must have at least one administered skill	MVAGSTFD
60	switch administration contention	any
61	cannot remove direct agent skill	MVAGSTFD
62	cannot add/remove standard skill	MVAGSTFD
63	direct agent skill cannot be a reserve skill	MVAGSTFD
64	cannot change reserve skill to standard or vice versa	MVAGSTFD
65	entity locked on switch	(used in RCODE)
66	**Skill tenant number doesn't match agent tenant number**	**MVAGSTFD**

3.3.32 ←**USERID21 (New CM4)**

	Switch	Vect	Skill
First Release	CM4	y/n	y/n
Last Release			

The **user Identity 21** format is to be an ASCII string of up to 32 characters. The population of user identity changes made by the reporting adjunct will be supported by the reporting adjunct only.

This is a variable length message based on the "id-length" field.

The first five bytes are always sent.

This message is to follow an ACD admin change message (**MVAGSTFD**, **SKILLADD**, **MVVDN**, **CHGVDNSK** or **CHGNAM**). The expected response is one **RCODE20** message.

7 6 5 4 3 2 1 0	7 6 5 4 3 2 1 0	7 6 5 4 3 2 1 0	7 6 5 4 3 2 1 0
USERID21	tag	admin-opcode	admin-tag
id-length	id_byte-1	id-byte-2	id-byte-3

. . .

id-byte-N

opcode	USERID21 = 54 (0x36)
tag	message tag, 0-255 (it will be returned in the switch response message(s))
admin-opcode	opcode of the admin message identity applies to
admin-tag	tag of the admin message identity applies to
id-length	number of bytes in id (0-32)
id-byte-N	object data bytes ASCII encoded - user identity for admin change (N ranges from 1 to 32)

3.4 Call Processing

Call Processing messages are sent continuously while the reporting adjunct Session is in the *DATA TRANSFER* mode. Each message will be time-stamped and sent (or buffered) in the order they are processed.

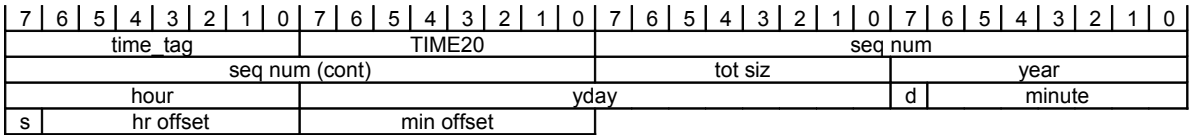
3.4.1 → TIME20 (New CM3.1)

	Switch	Vect	Skill
First Release	CM3.1	y/n	y/n
Last Release			

The **Time of Day** message is the first Call Processing message sent by the switch after the translation pumpup is completed. Thereafter, it is sent every minute, on the minute. the reporting adjunct uses this message to reset the reporting adjunct clock. In addition, every message sent by the switch to the reporting adjunct is time-stamped (*time_tag*) with the seconds past the minute when the event happened. The reporting adjunct sets the seconds of the clock with every message, therefore it is important that all messages between **TIME20** messages are sent in the order created so that time (*time_tag*) never goes backwards.

Since a **TIME20** message is required every minute, the reporting adjunct will set the AT2 timer (2 minutes) and reset this timer every time a **TIME20** message is received. If the timer expires (i.e., data transfer timeout) the reporting adjunct will initiate another reporting adjunct Session.

For the time zone offset fields (s, hr offset & min offset), the switch and the reporting adjunct use different formats. The switch uses Linux format and the reporting adjunct uses Java format. Therefore, the reporting adjunct will need to convert the time zone offset received into Java format before using it. For example, Java expects the Mountain Standard Time Zone to be GMT -7, but Linux configures it as GMT +7. So the *s* field in the **TIME20** would be zero. The reporting adjunct needs to convert the *s* field to 1 prior to using it.



time_tag	seconds past minute, 0-59
opcode	TIME20 = 205 (0xcd)
seq num	sequence number, 0-0xffff ffff
tot siz	total size of message in bytes, 1-255
year	current year on switch, 0-99
hour	current hour on switch, 0-23
yday	current day of the year on switch, 1-366
d	daylight savings time flag; 0 = standard time, 1 = daylight savings time
minute	current minute on switch, 0-59
s	time zone offset flag; 0 = +, 1 = -
hr offset	time zone offset hours, 0-23
min offset	time zone offset minutes, 0-59

3.4.2 → ALOGON20, ALOGON23 and ALOGON24

An **Agent Logon** message is sent by the switch during the translation pumpup and whenever an agent staffs a position. When sent as part of an agent logon translation block, one of these messages is sent for each extension that is staffed. The reporting adjunct will update the status of the agent position (record who is logged on) in the database. Note that the *logid* is sent in packed decimal format, where hex a is the digit 0 and 0 in the field is null.

With EAS, a single agent has one "agent position" independent of how may skills that agent has.

The switch will require at least one login digit if the reporting adjunct is optioned.
This is a variable length message sent from the switch to the reporting adjunct. The length is based on the "snumber", "pnumber" and "dnumber" fields. For the EAS version, field "f1" always starts on a byte boundary. Four bits will be zero filled for a digit if needed.

The skills are in administered order with any administered blanks eliminated (however, blanks may exist because a skill is not measured by the reporting adjunct, i.e., skill=0, level=0). The order and position of the skills in the message must correspond to the bit-map in the **AVAIL** and **AUDAVAIL** messages.

The **ALOGON** message is sent when an agent logs in, when an agent performs add/remove skills from a phone, and at pumpup. The **ALOGON** is also sent in response to the **MVAGSTFD** or **SKILLADD** message when a move is completed.

If the entire login attempt fails, no **ALOGON** message is sent; an exception to this rule is with the **MVAGSTFD** message where the **ALOGON** message may be sent with all fail bits set.

Starting with **ALOGON23**, the expansion to 120 skills this request may not fit into a single **ALOGON** message. If the agent has more than 60 skills, a second message will be sent. The 1st message in the series covers skills 1 through 60 and the 2nd message covers skills 61 through 120, mapping to skills 1-60. The end bit is set for the last message in a series.

Starting with **ALOGON24**, with the increases to number of digits and additional fields, 60 skills may not fit into one **ALOGON** message. Instead of splitting **ALOGON** into 2 messages that each send 60 skills, it will now be 3 messages that each send 40 skills.

3.4.2.1 **→ALOGON20 (New CM3.1)**

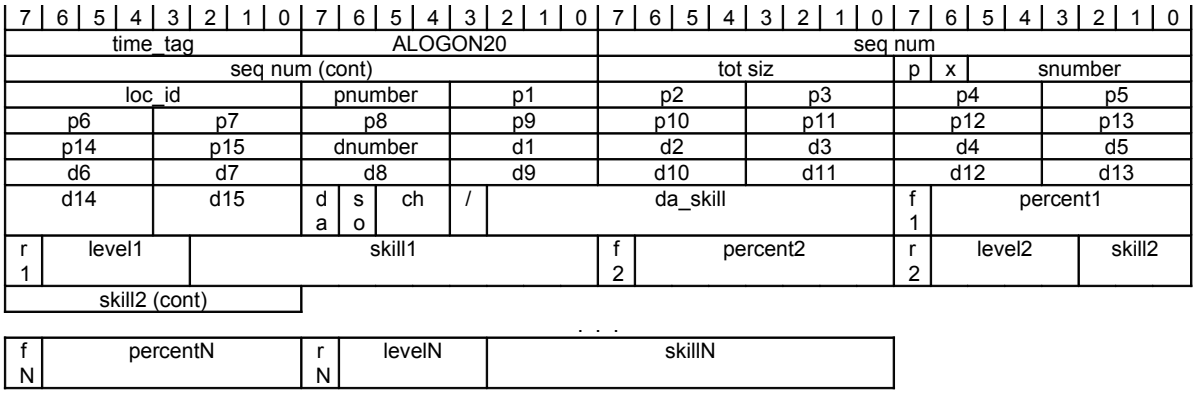
	Switch	Vect	Skill
First Release	CM3.1	y/n	y/n
Last Release	CM5.2		

3.4.2.1.1 **→ALOGON20, without EAS (New CM3.1)**

7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
time_tag								ALOGON20								seq num															
seq num (cont)																tot siz								p		x		f		s	////
loc_id								pnumber				pos																			
dnumber				d1				d2				d3				d4				d5				d6						d7	
d8				d9																											

time_tag	seconds past minute when event occurred, 0-59
opcode	ALOGON20 = 206 (0xce)
seq_num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
p	change pending; 0 = no changes pending, 1 = a change is pending for this agent. (If pending is due to **MVAGSTFD**, the **ALOGON** message is not sent until the agent ends call/ACW/DAC.)
x	translation; 0 = normal logon, 1 = translation pump-up status (agent logged in or AAS agent released from maintenance busy) or DME change.
f	0 = agent successfully logged into split, 1 = login failed (used with **MVAGSTFD**, not login from phone)
s	0 = non-EAS, no skills in message.
loc_id	location id number, 0-44 (G3V8-G3V11), 0-250 (MV2 and later), 0 is default for no location id
pnumber	number of digits in "pos", will always be 5 for non-EAS
pos	agent position, 1-10000.
dnumber	number of digits in agent logon ID, 1-9
d1-d9	digits of the logon ID without EAS, packed decimal format, left-justified where 0 is null and hex a is digit 0, zero fill digits to a byte boundary, (for example logid 4000 will be 4aaa0).

3.4.2.1.2 **→ALOGON20, with EAS (New CM3.1)**



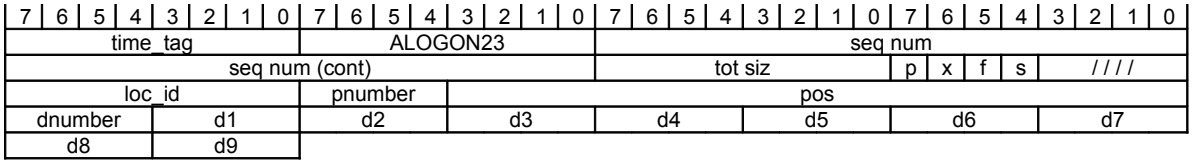
time_tag	seconds past minute when event occurred, 0-59
opcode	ALOGON20 = 206 (0xce)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
p	change pending; 0 = no changes pending, 1 = a change is pending for this agent.
x	translation; 0 = normal logon, 1 = translation pump-up status (agent logged in or AAS agent released from maintenance busy) or DME change.
snumber	number of skills in the message (1-60)
loc_id	location id number, 0-44 (G3V8-G3V11), 0-250 (MV2 and later), 0 is default for no location id
pnumber	number of digits in agent position, 1-15
p1-p15	agent position, with EAS (0-9999999) login ID in packed decimal format, left-justified where 0 is null and hex a is digit 0, zero fill to a byte boundary.
dnumber	number of digits in the physical extension, 1-15
d1-d15	the digits of the physical extension in packed decimal format, left-justified where 0 is null and hex a is digit 0, zero fill to a byte boundary.
da	flag indicating direct agent calls should be received first. Valid only when call handling preference is percent allocation.
so	service objective flag: 0=not in use, 1=in use. Valid only when call handling preference is skill level or greatest need.
ch	agent's call handling preference, 0 = skill level, 1 = greatest need, 2 = percent allocation
da_skill	agent's direct-agent skill (must be one of agent's 60 skills)
f 1-60	logon fail bits. For non-zero skills, look at this bit to see whether login to pending skill has failed. 0 = logon to pending skill succeeded, 1 = logon failed. It is possible to send all skills with fail bits set. f1=skill1, f2=skill2, etc.
percent 1-60	percent allocation for this skill, 0-100. Must be 0 when skill level is reserve (rN=1). Must be non-zero when call handling preference is percent allocation and skill level is not reserve (rN=0). Valid only when call handling preference is percent allocation.
r 1-60	flag indicating level is reserve level(r=1) rather than skill level(r=0).
level 1-60	skill level (0-15 representing levels 1-16) or reserve level (1-2).
skill 1-60	skills 1-999 (G3V8-G3V11), 1-2000 (MV2 and later), assigned to the agent in the order administered. Sending zero indicates no skill administered (or skill not measured by CMS).

3.4.2.2 →ALOGON23 (New CM6.0)

	Switch	Vect	Skill
First Release	CM6.0	y/n	y/n
Last Release	CM6.3		

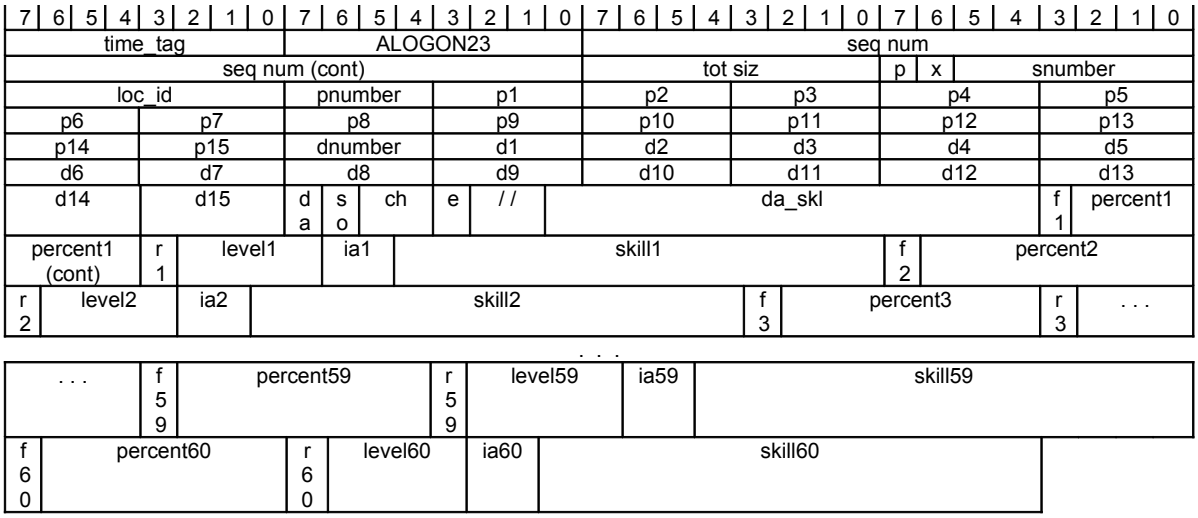
3.4.2.2.1 →ALOGON23, without EAS (New CM6.0)

The changes for **ALOGON23** are for the EAS version of **ALOGON** only. Since both the non-EAS version and the EAS version use the same opcode, the non-EAS version is also to use the new opcode for SPI language 23.



time_tag	seconds past minute when event occurred, 0-59
opcode	ALOGON23 = 5 (0x05)
seq_num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
p	change pending; 0 = no changes pending, 1 = a change is pending for this agent. (If pending is due to **MVAGSTFD**, the **ALOGON23** message is not sent until the agent ends call/ACW/DAC.)
x	translation; 0 = normal logon, 1 = translation pump-up status (agent logged in or AAS agent released from maintenance busy) or DME change.
f	0 = agent successfully logged into split, 1 = login failed (used with **MVAGSTFD**, not login from phone)
s	0 = non-EAS, no skills in message.
loc_id	location id number, 0-44 (G3V8-G3V11), 0-250 (MV2 and later), 0 is default for no location id
pnumber	number of digits in "pos", will always be 5 for non-EAS
pos	agent position, 1-10000.
dnumber	number of digits in agent logon ID, 1-9
d1-d9	digits of the logon ID without EAS, packed decimal format, left-justified where 0 is null and hex a is digit 0, zero fill digits to a byte boundary, (for example logid 4000 will be 4aaa0).

3.4.2.2.2 **→ALOGON23, with EAS (New CM6.0)**



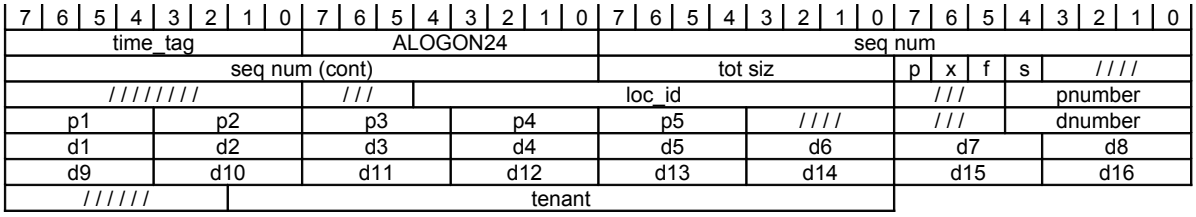
ALOGON23 (continued)

time_tag	seconds past minute when event occurred, 0-59
opcode	ALOGON23 = 5 (0x05)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
p	change pending; 0 = no changes pending, 1 = a change is pending for this agent.
x	translation; 0 = normal logon, 1 = translation pump-up status (agent logged in or AAS agent released from maintenance busy) or DME change.
snumber	number of skills in the message (1-60)
loc_id	location id number, 0-44 (G3V8-G3V11), 0-250 (MV2 and later), 0 is default for no location id
pnumber	number of digits in agent position, 1-15
p1-p15	agent position, with EAS login ID in packed decimal format, left-justified where 0 is null and hex a is digit 0, zero fill to a byte boundary.
dnumber	number of digits in the physical extension, 1-15
d1-d15	the digits of the physical extension in packed decimal format, left-justified where 0 is null and hex a is digit 0, zero fill to a byte boundary.
da	flag indicating direct agent calls should be received first. Valid only when call handling preference is percent allocation.
so	service objective flag: 0=not in use, 1=in use. Valid only when call handling preference is skill level or greatest need.
ch	agent's call handling preference, 0 = skill level, 1 = greatest need, 2 = percent allocation
e	end bit, set when this is the last message of the set needed to satisfy the request
da_skill	agent's direct-agent skill (must be one of agent's 120 skills)
f 1-60	logon fail bits. For non-zero skills, look at this bit to see whether login to pending skill has failed. 0 = logon to pending skill succeeded, 1 = logon failed. It is possible to send all skills with fail bits set. f1=skill1, f2=skill2, etc.
percent 1-60	percent allocation for this skill, 0-100. Must be 0 when skill level is reserve (rN=1). Must be non-zero when call handling preference is percent allocation and skill level is not reserve (rN=0). Valid only when call handling preference is percent allocation.
r 1-60	flag indicating level is reserve level(r=1) rather than skill level(r=0).
level 1-60	skill level (0-15 representing levels 1-16) or reserve level (1-2).
ia 1-60	field that indicates whether the corresponding skill is interruptible. This field is only valid when the corresponding “rx” (reserve) field is 0. 0 not an interruptible skill 1 auto-in-interrupt 2 manual-in-interrupt 3 notify-interrupt
skill 1-60	skills 1-8000 assigned to the agent in the order administered. Sending zero indicates no skill administered (or skill not measured by CMS).

3.4.2.3 **→ALOGON24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y/n	y/n
Last Release			

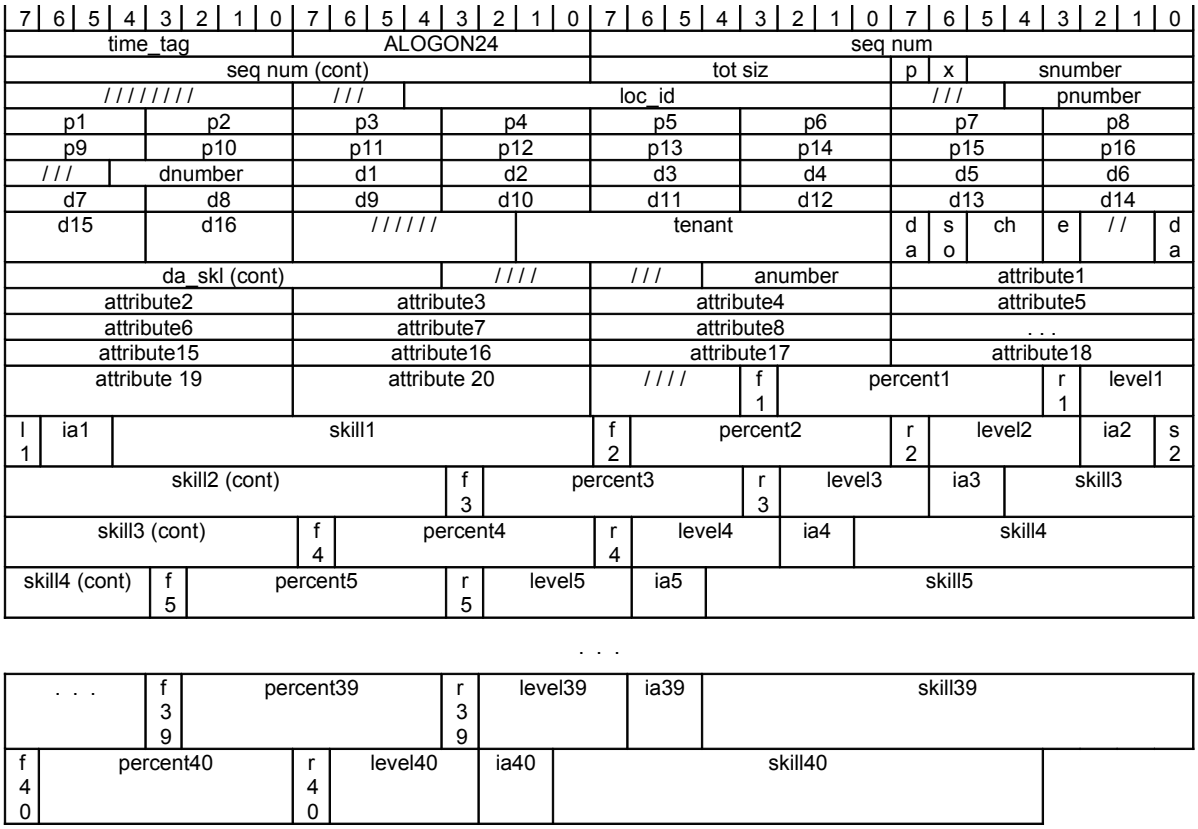
3.4.2.3.1 **→ALOGON24, without EAS (New CM7.0)**



time_tag	seconds past minute when event occurred, 0-59
opcode	ALOGON24 = 40 (0x28)
seq_num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
p	change pending; 0 = no changes pending, 1 = a change is pending for this agent. (If pending is due to **MVAGSTFD**, the **ALOGON24** message is not sent until the agent ends call/ACW/DAC.)
x	translation; 0 = normal logon, 1 = translation pump-up status (agent logged in or AAS agent released from maintenance busy) or DME change.
f	0 = agent successfully logged into split, 1 = login failed (used with **MVAGSTFD**, not login from phone)
s	0 = non-EAS, no skills in message.
loc_id	**location id number, 0-2000, 0 is default for no location id (13 bits for future expansion)**
pnumber	**number of digits in "pos", will always be 5 for non-EAS**
pos	agent position, 1-10000.
dnumber	**number of digits in agent logon ID, 1-16**
d1-d16	**digits of the logon ID without EAS, packed decimal format, left-justified where 0 is null and hex a is digit 0, zero fill digits to a byte boundary, (for example logid 4000 will be 4aaa0).**
tenant	**Station tenant number, 1-250**

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3.4.2.3.2 **→ALOGON24, with EAS (New CM7.0)**



. . .

ALOGON24 (continued)

time_tag	seconds past minute when event occurred, 0-59
opcode	ALOGON24 = 40 (0x28)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
p	change pending; 0 = no changes pending, 1 = a change is pending for this agent.
x	translation; 0 = normal logon, 1 = translation pump-up status (agent logged in or AAS agent released from maintenance busy) or DME change.
snumber	number of skills in the message (1-40)
loc_id	**location id number, 0-2000, 0 is default for no location id (13 bits for future expansion)**
pnumber	**number of digits in agent position, 1-16**
p1-p16	**agent position, with EAS login ID in packed decimal format, left-justified where 0 is null and hex a is digit 0, zero fill to a byte boundary.**
dnumber	**number of digits in the physical extension, 1-16**
d1-d16	**the digits of the physical extension in packed decimal format, left-justified where 0 is null and hex a is digit 0, zero fill to a byte boundary.**
da	flag indicating direct agent calls should be received first. Valid only when call handling preference is percent allocation.
so	service objective flag: 0=not in use, 1=in use. Valid only when call handling preference is skill level or greatest need.
ch	agent's call handling preference, 0 = skill level, 1 = greatest need, 2 = percent allocation
e	**end bit, set when this is the last message of the set needed to satisfy the request**
da_skill	agent's direct-agent skill (must be one of agent's 120 skills)
tenant	**Agent tenant number, 1-250**
anumber	**number of alphanumeric characters in attribute, 0-20**
attribute1-20	**Agent attribute, a 20 character alphanumeric string**
f **1-40**	logon fail bits. For non-zero skills, look at this bit to see whether login to pending skill has failed. 0 = logon to pending skill succeeded, 1 = logon failed. It is possible to send all skills with fail bits set. f1=skill1, f2=skill2, etc.
percent **1-40**	percent allocation for this skill, 0-100. Must be 0 when skill level is reserve (rN=1). Must be non-zero when call handling preference is percent allocation and skill level is not reserve (rN=0). Valid only when call handling preference is percent allocation.
r **1-40**	flag indicating level is reserve level(r=1) rather than skill level(r=0).
level **1-40**	skill level (0-15 representing levels 1-16) or reserve level (1-2).
ia **1-40**	field that indicates whether the corresponding skill is interruptible. This field is only valid when the corresponding “rx” (reserve) field is 0. 0 not an interruptible skill 1 auto-in-interrupt 2 manual-in-interrupt 3 notify-interrupt
skill **1-40**	skills 1-8000 assigned to the agent in the order administered. Sending zero indicates no skill administered (or skill not measured by the reporting adjunct).

3.4.3 → ALOGOFF20 and ALOGOFF24

An **Agent Logoff** message is sent by the switch whenever an agent logs off. The reporting adjunct will update the status of the agent position to reflect that the position is no longer staffed.

3.4.3.1 →ALOGOFF20 (New CM3.1)

	Switch	Vect	Skill
First Release	CM3.1	y/n	y/n
Last Release	CM6.3		

7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
time_tag								ALOGOFF20								seq num															
seq num (cont)																tot siz								pnumber				pos or p1-p15			
pos (non-EAS) or p1-p15 (EAS) (cont)																								/ / / /				reason			

time_tag seconds past minute when event occurred, 0-59
opcode ALOGOFF20 = 207 (0xcf)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
pnumber number of digits in agent position, 1-15, will always be 5 for non-EAS
pos/p1-p15 agent position - without EAS "pos" (1-10000), with EAS "p1-p15" (0-9999999) login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary
reason reason code (0-9); 0 is the default reason code.

3.4.3.2 →ALOGOFF24 (New CM7.0)

	Switch	Vect	Skill
First Release	CM7.0	y/n	y/n
Last Release			

An **Agent Logoff** message is sent by the switch whenever an agent logs off. The reporting adjunct will update the status of the agent position to reflect that the position is no longer staffed.

7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
time_tag								ALOGOFF24								seq num															
seq num (cont)																tot siz								/ / /				pnumber			
pos (non-EAS) or p1-p16 (EAS)																				/ / / /				/ / / /				reason			

time_tag seconds past minute when event occurred, 0-59
opcode ALOGOFF24 = 57 (0x39)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
pnumber **number of digits in agent position, 1-16, will always be 5 for non-EAS**
pos/p1-p16 **agent position - without EAS "pos" (1-10000), with EAS "p1-p16" (0-9999999) login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary**
reason reason code (0-9); 0 is the default reason code.

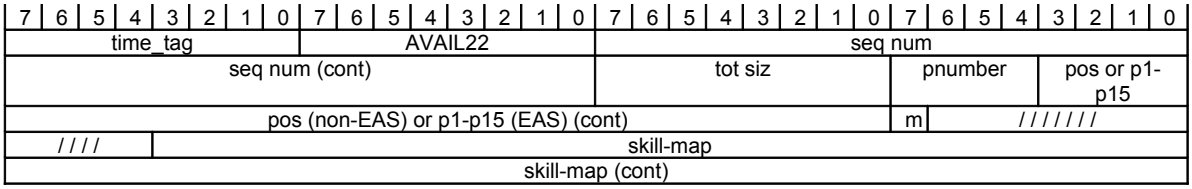
3.4.4 → AVAIL22, AVAIL23 and AVAIL24

The **Agent State Change** message is sent by the switch when an agent position changes state. The agent position will change states when the agent's work mode or call state changes.

This is a variable length message based on "pnumber" and whether or not EAS is active. Prior to **AVAIL23**, with EAS, this message will contain 8 additional bytes for the "skill-map". Starting with **AVAIL23** and the expansion to 120 skills, with EAS, this message will contain 16 additional bytes for the expanded “skill-map”.

3.4.4.1 →AVAIL22 (New CM5.2)

	Switch	Vect	Skill
First Release	CM5.2	y/n	y/n
Last Release	CM5.2		

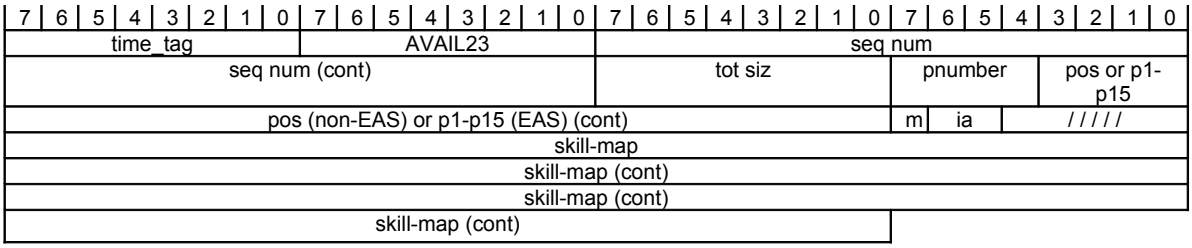


time_tag seconds past minute when event occurred, 0-59
opcode AVAIL22 = 1 (0x01)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
pnumber number of digits in agent position, 1-15, will always be 5 for non-EAS
pos/p1-p15 agent position - without EAS "pos" (1-10000), with EAS "p1-p15" (0-9999999) login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary
m flag to indicate the agent returned to available because of Multiple Call Handling. This flag notifies IQ tracking that this astate change is not for a call release.
skill-map one bit for each staffed skill. The skills are mapped left to right and correspond to the order of the skills in the **ALOGON** message.

- 1 means agent is available for this skill
- 0 means agent is not available for this skill or the skill is not assigned/measured; The reporting adjunct will report the agent activity for this skill as OTHER.

3.4.4.2 **→AVAIL23 (New CM6.0)**

	Switch	Vect	Skill
First Release	CM6.0	y/n	y/n
Last Release	CM6.3		



time_tag seconds past minute when event occurred, 0-59
opcode AVAIL23 = 14 (0x0e)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
pnumber number of digits in agent position, 1-15, will always be 5 for non-EAS
pos/p1-p15 agent position - without EAS "pos" (1-10000), with EAS "p1-p15" login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary
m flag to indicate the agent returned to available because of Multiple Call Handling. This flag notifies IQ tracking that this astate change is not for a call release.
ia flag indicating agent became available due to being interrupted while in Aux with an Interruptible Aux reason code.

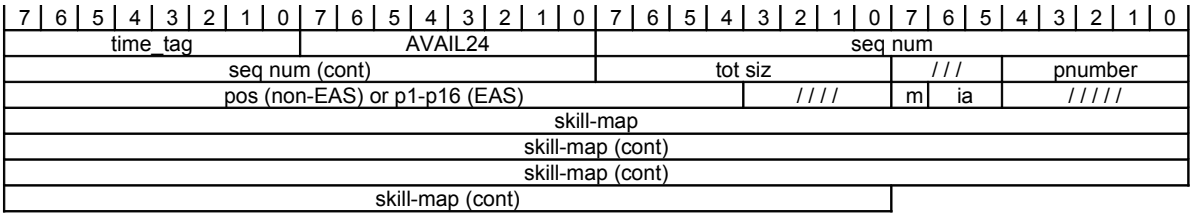
0 N/A (agent was not interrupted while in Aux)
1 auto-in-interrupt
2 manual-in-interrupt
3 notify-interrupt

skill-map one bit for each staffed skill (1-120). The skills are mapped left to right and correspond to the order of the skills in the ALOGON message.

- 1 means agent is available for this skill
- 0 means agent is not available for this skill or the skill is not assigned/measured; The reporting adjunct will report the agent activity for this skill as OTHER.

3.4.4.3 **→AVAIL24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y/n	y/n
Last Release			



time_tag seconds past minute when event occurred, 0-59
opcode AVAIL24 = 58 (0x3a)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
pnumber number of digits in agent position, 1-16, will always be 5 for non-EAS
pos/p1-p16 agent position - without EAS "pos" (1-10000), with EAS "p1-p16" login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary
m flag to indicate the agent returned to available because of Multiple Call Handling. This flag notifies IQ tracking that this astate change is not for a call release.
ia flag indicating agent became available due to being interrupted while in Aux with an Interruptible Aux reason code.

0 N/A (agent was not interrupted while in Aux)
1 auto-in-interrupt
2 manual-in-interrupt
3 notify-interrupt

skill-map (EAS) one bit for each staffed skill (1-120). The skills are mapped left to right and correspond to the order of the skills in the ALOGON message.

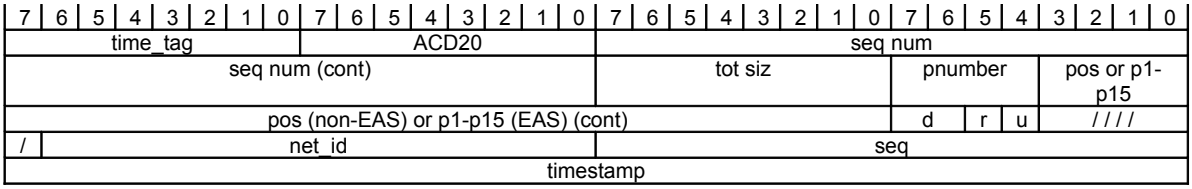
- 1 means agent is available for this skill
- 0 means agent is not available for this skill or the skill is not assigned/measured; The reporting adjunct will report the agent activity for this skill as OTHER.

3.4.5 → ACD20 and ACD24

This is a variable length message based on "pnumber" and "u".

3.4.5.1 →ACD20 (New CM3.1)

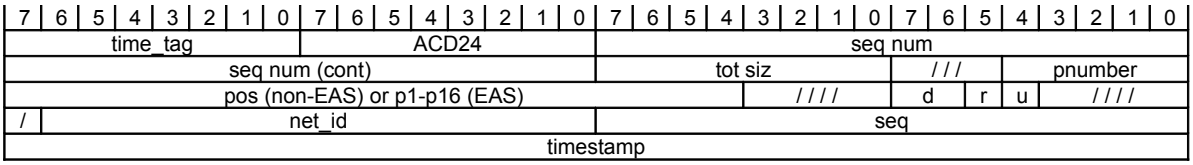
	Switch	Vect	Skill
First Release	CM3.1	y/n	y/n
Last Release	CM6.3		



time_tag	seconds past minute when event occurred, 0-59
opcode	ACD20 = 209 (0xd1)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
pnumber	number of digits in agent position, 1-15, will always be 5 for non-EAS
pos/p1-p15	agent position - without EAS "pos" (1-10000), with EAS "p1-p15" (0-9999999) login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary
d	direction of call 1 means call is outbound (ACDO) 2 means call is inbound (ACD, ACDI)
r	reconnect; 1 means the ACD call is being reconnected after being on hold.
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.5.2 **→ACD24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y/n	y/n
Last Release			



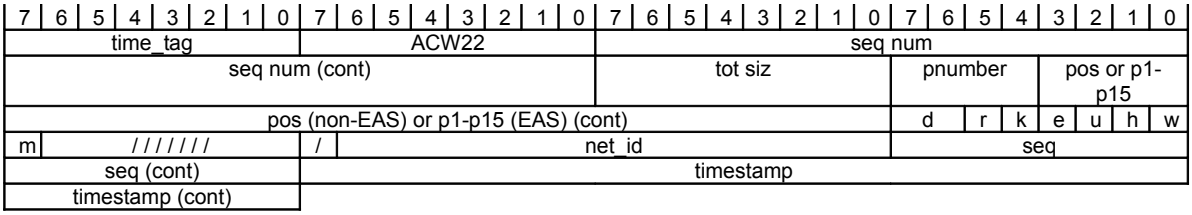
time_tag	seconds past minute when event occurred, 0-59
opcode	ACD24 = 59 (0x3b)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
pnumber	**number of digits in agent position, 1-16, will always be 5 for non-EAS**
pos/p1-p16	**agent position - without EAS "pos" (1-10000), with EAS "p1-p16" (0-9999999) login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary**
d	direction of call 1 means call is outbound (ACDO) 2 means call is inbound (ACD, ACDI)
r	reconnect; 1 means the ACD call is being reconnected after being on hold.
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.6 → ACW22 and ACW24

This is a variable length message based on "pnumber" and "u".

3.4.6.1 →ACW22 (New CM5.2)

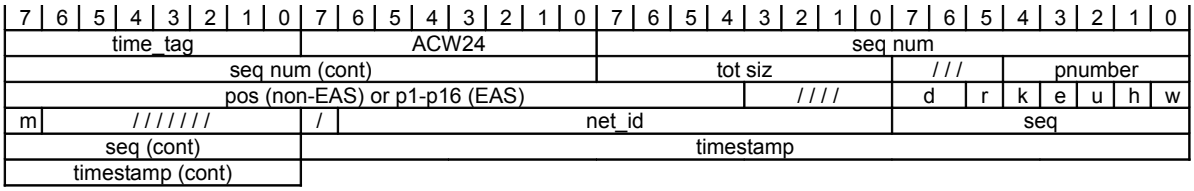
	Switch	Vect	Skill
First Release	CM5.2	y/n	y/n
Last Release	CM6.3		



time_tag	seconds past minute when event occurred, 0-59
opcode	ACW22 = 255 (0xff)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
pnumber	number of digits in agent position, 1-15, will always be 5 for non-EAS
pos/p1-p15	agent position - without EAS "pos" (1-10000), with EAS "p1-p15" (0-9999999) login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary
d	direction of call – 0 means no call (ACW) – 1 means call is outbound (ACWO) – 2 means call is inbound (ACWI)
r	reconnect; 1 means the ACWO/ACWI call is being reconnected after being on hold.
k	keyboard dialed; 1 means the ACWO call was placed by an adjunct on behalf of the agent.
e	external call; 1 = external off-premises call dialed by agent or on behalf of agent. DCS calls are considered off-premises.
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
h	flag to indicate the agent has just put a call on hold.
w	flag to indicate the observer is waiting for another call to observe.
m	flag to indicate the agent called a measured internal destination. This flag will only be set if the "e" flag is not set, that is, the call is not external.
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.6.2 **→ ACW24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y/n	y/n
Last Release			



time_tag seconds past minute when event occurred, 0-59
opcode ACW24 = 60 (0x3c)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
pnumber number of digits in agent position, 1-16, will always be 5 for non-EAS
pos/p1-p16 agent position - without EAS "pos" (1-10000), with EAS "p1-p16" (0-9999999) login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary
d direction of call
- 0 means no call (ACW)
- 1 means call is outbound (ACWO)
- 2 means call is inbound (ACWI)

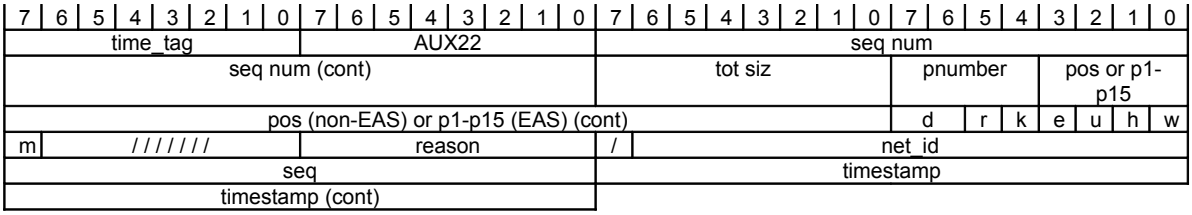
r reconnect; 1 means the ACWO/ACWI call is being reconnected after being on hold.
k keyboard dialed; 1 means the ACWO call was placed by an adjunct on behalf of the agent.
e external call; 1 = external off-premises call dialed by agent or on behalf of agent. DCS calls are considered off-premises.
u flag indicating that the UCID fields (net_id, seq and timestamp) are present
h flag to indicate the agent has just put a call on hold.
w flag to indicate the observer is waiting for another call to observe.
m flag to indicate the agent called a measured internal destination. This flag will only be set if the "e" flag is not set, that is, the call is not external.
net_id network entity identifier, 1-32767
seq call sequence number, 0-65535
timestamp unix style (seconds since 1/1/70) time call assigned a UCID

3.4.7 → AUX22, AUX23 and AUX24

This is a variable length message based on "pnumber" and "u".

3.4.7.1 →AUX22 (New CM5.2)

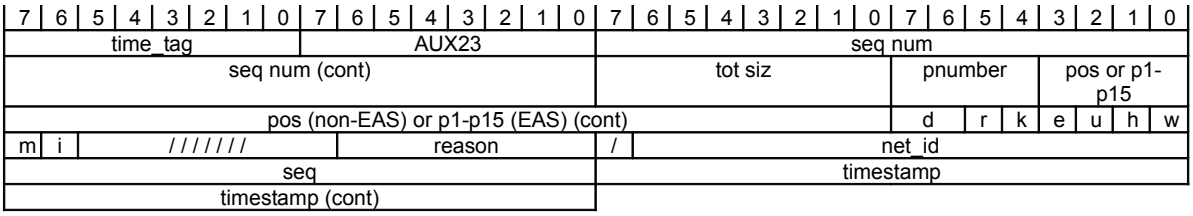
	Switch	Vect	Skill
First Release	CM5.2	y/n	y/n
Last Release	CM5.2		



time_tag	seconds past minute when event occurred, 0-59
opcode	AUX22 = 3 (0x03)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
pnumber	number of digits in agent position, 1-15, will always be 5 for non-EAS
pos/p1-p15	agent position - without EAS "pos" (1-10000), with EAS "p1-p15" (0-9999999) login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary
d	direction of call: 0 no call (AUX) 1 call is outbound (AUXO) 2 call is inbound (AUXI)
r	reconnect; 1 means the AUXO/AUXI call is being reconnected after being on hold.
k	keyboard dialed; 1 means the AUXO call was placed by an adjunct on behalf of the agent.
e	external call; 1 = external off-premises call dialed by agent or on behalf of agent. DCS calls are considered off-premises.
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
h	flag to indicate the agent has put a call on hold.
w	flag to indicate the observer is waiting for another call to observe.
m	flag to indicate the agent called a measured internal destination. This flag will only be set if the "e" flag is not set, that is, the call is not external.
reason	reason code (0-99); 0 is the default reason code.
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.7.2 **→AUX23 (New CM6.0)**

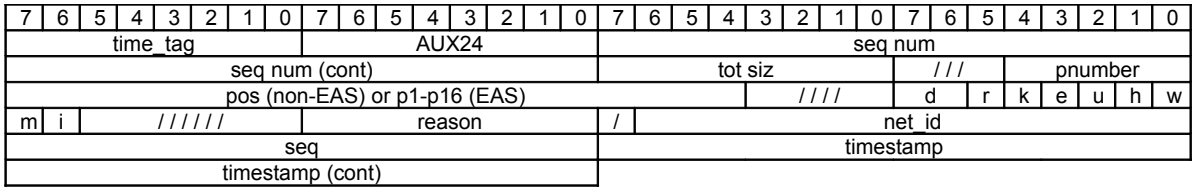
	Switch	Vect	Skill
First Release	CM6.0	y/n	y/n
Last Release	CM6.3		



time_tag	seconds past minute when event occurred, 0-59
opcode	AUX23 = 15 (0x0F)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
pnumber	number of digits in agent position, 1-15, will always be 5 for non-EAS
pos/p1-p15	agent position - without EAS "pos" (1-10000), with EAS "p1-p15" login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary
d	direction of call: 0 no call (AUX) 1 call is outbound (AUXO) 2 call is inbound (AUXI)
r	reconnect; 1 means the AUXO/AUXI call is being reconnected after being on hold.
k	keyboard dialed; 1 means the AUXO call was placed by an adjunct on behalf of the agent.
e	external call; 1 = external off-premises call dialed by agent or on behalf of agent. DCS calls are considered off-premises.
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
h	flag to indicate the agent has put a call on hold.
w	flag to indicate the observer is waiting for another call to observe.
m	flag to indicate the agent called a measured internal destination. This flag will only be set if the "e" flag is not set, that is, the call is not external.
i	flag that indicates that the AUX reason code entered is interruptible. The flag is to be set to 1 when the agent used an AUX reason code that has the Aux Work/Interruptible field on the Reason Code Names screen set to “y”.
reason	reason code (0-99); 0 is the default reason code.
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.7.3 **→AUX24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y/n	y/n
Last Release			



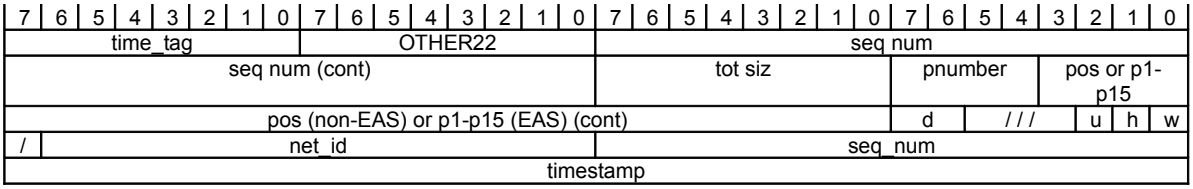
time_tag seconds past minute when event occurred, 0-59
opcode AUX24 = 61 (0x3d)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
pnumber number of digits in agent position, 1-16, will always be 5 for non-EAS
pos/p1-p16 agent position - without EAS "pos" (1-10000), with EAS "p1-p16" login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary
d direction of call:
0 no call (AUX)
1 call is outbound (AUXO)
2 call is inbound (AUXI)
r reconnect; 1 means the AUXO/AUXI call is being reconnected after being on hold.
k keyboard dialed; 1 means the AUXO call was placed by an adjunct on behalf of the agent.
e external call; 1 = external off-premises call dialed by agent or on behalf of agent. DCS calls are considered off-premises.
u flag indicating that the UCID fields (net_id, seq and timestamp) are present
h flag to indicate the agent has put a call on hold.
w flag to indicate the observer is waiting for another call to observe.
m flag to indicate the agent called a measured internal destination. This flag will only be set if the "e" flag is not set, that is, the call is not external.
i flag that indicates that the AUX reason code entered is interruptible. The flag is to be set to 1 when the agent used an AUX reason code that has the Aux Work/Interruptible field on the Reason Code Names screen set to “y”.
reason reason code (0-99); 0 is the default reason code.
net_id network entity identifier, 1-32767
seq call sequence number, 0-65535
timestamp unix style (seconds since 1/1/70) time call assigned a UCID

3.4.8 → OTHER22 and OTHER24

This is a variable length message based on "pnumber".

3.4.8.1 → OTHER22 (New CM5.2)

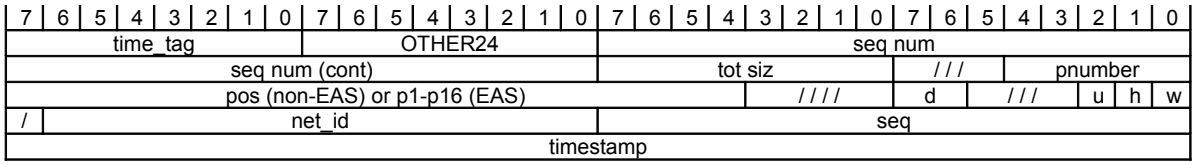
	Switch	Vect	Skill
First Release	CM5.2	y/n	y/n
Last Release	CM6.3		



time_tag	seconds past minute when event occurred, 0-59
opcode	OTHER22 = 250 (0xfa)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
pnumber	number of digits in agent position, 1-15, will always be 5 for non-EAS
pos/p1-p15	agent position - without EAS "pos" (1-10000), with EAS "p1-p15" (0-9999999) login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary
d	direction of call – 0 means no call – 1 means call is outbound – 2 means call is inbound
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present.
h	flag to indicate the agent has put a call on hold.
w	flag to indicate the observer is waiting for another call to observe.
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.8.2 **→OTHER24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y/n	y/n
Last Release			



time_tag seconds past minute when event occurred, 0-59
opcode OTHER24 = 62 (0x3e)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
pnumber number of digits in agent position, 1-16, will always be 5 for non-EAS
pos/p1-p16 agent position - without EAS "pos" (1-10000), with EAS "p1-p16" (0-9999999) login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary
d direction of call
- 0 means no call
- 1 means call is outbound
- 2 means call is inbound

u flag indicating that the UCID fields (net_id, seq and timestamp) are present.
h flag to indicate the agent has put a call on hold.
w flag to indicate the observer is waiting for another call to observe.
net_id network entity identifier, 1-32767
seq call sequence number, 0-65535
timestamp unix style (seconds since 1/1/70) time call assigned a UCID

3.4.9 → DIGITS20 and DIGITS24

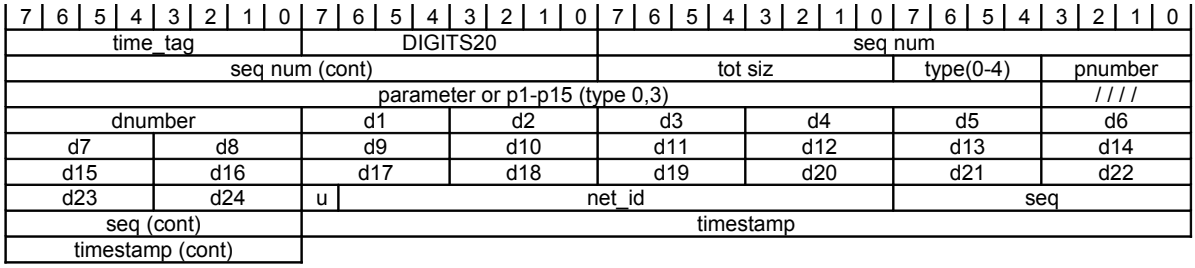
The **Digits Dialed** message is sent by the switch for outbound calls to tell the reporting adjunct what digits the agent dialed (trunk/feature access codes and authorization/account codes not included). This sends ISDN SID/ANI for inbound calls.

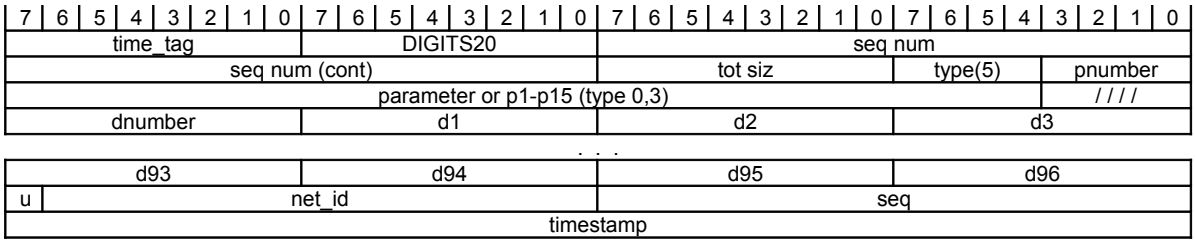
Call Work Codes are only sent when the agent is talking on an ACD call or is in ACW for a call.

These are variable-length messages based on the length field. The first twelve bytes will always be sent.

3.4.9.1 →DIGITS20 (New CM3.1)

	Switch	Vect	Skill
First Release	CM3.1	y/n	y/n
Last Release	CM6.3		





DIGITS20 (continued)

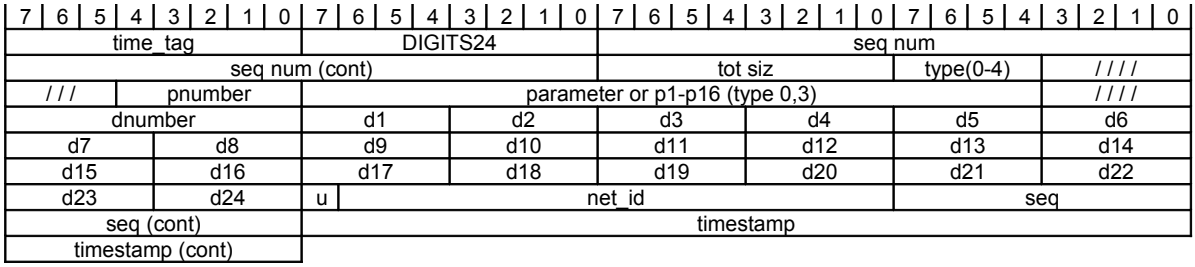
time_tag	seconds past minute when event occurred, 0-59
opcode	DIGITS20 = 213 (0xd5)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
type	type of digits

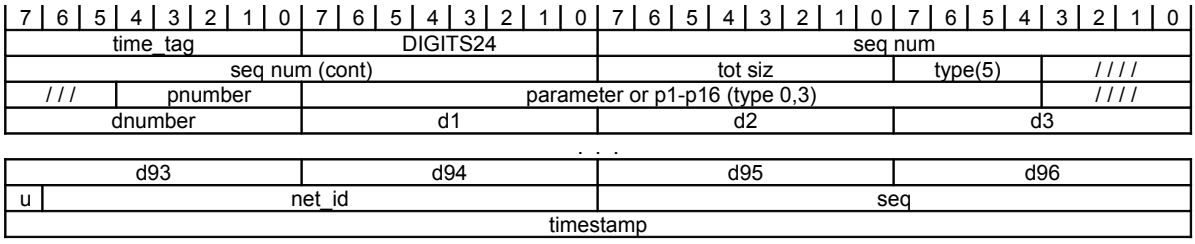
type	definition	parameter field contents
0	agent dialed digits (outbound call)	agent position
1	SID/ANI digits	internal trunk number(ITN)
2	Collected digits	internal trunk number(ITN)
3	Call work codes	agent position
4	ISDN II-digits	internal trunk number(ITN)
5	ASAI UUI (MV2 or later)	internal trunk number (ITN)

pnumber	number of digits in "parameter"/"p1-p15", 1-15, agent position for outbound calls and call work codes, will always be 5 for non-EAS and for "parameter" of internal trunk number (ITN).
parameter/p1-p15	agent position for outbound calls and call work codes, without EAS "parameter" (1-10000), with EAS "p1-p15" (0-9999999) login ID in packed decimal format, left-justified where 0 is null and hex a is digit 0, zero fill to a byte boundary. internal trunk number(ITN) "parameter" used to identify trunk facilities, 1-32766
dnumber	number of digits: 1-24 (type 0-4) and 1-96 (type 5)
d1-d24	for type=0-4, digits of destination number in packed decimal format, left-justified where 0 is null, hex a is digit 0, and hex c is # (for Collected digits). When type = 4 (II-digits), only d1-d2 will be populated, and leading zeros will be included.
d1-d96	for type=5, ASAI UUI (treated as binary data)
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.9.2 **→DIGITS24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y/n	y/n
Last Release			





DIGITS24 (continued)

time_tag	seconds past minute when event occurred, 0-59
opcode	DIGITS24 = 63 (0x3f)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
type	type of digits

type	definition	parameter field contents
0	agent dialed digits (outbound call)	agent position
1	SID/ANI digits	internal trunk number(ITN)
2	Collected digits	internal trunk number(ITN)
3	Call work codes	agent position
4	ISDN II-digits	internal trunk number(ITN)
5	ASAI UUI (MV2 or later)	internal trunk number (ITN)

pnumber	**number of digits in "parameter"/"p1-p16", 1-16, agent position for outbound calls and call work codes, will always be 5 for non-EAS and for "parameter" of internal trunk number (ITN).**
parameter/ p1-p16	**agent position for outbound calls and call work codes, without EAS "parameter" (1-10000), with EAS "p1-p16" (0-9999999) login ID in packed decimal format, left-justified where 0 is null and hex a is digit 0, zero fill to a byte boundary. internal trunk number(ITN) "parameter" used to identify trunk facilities, 1-65535**
dnumber	number of digits: 1-24 (type 0-4) and 1-96 (type 5)
d1-d24	for type=0-4, digits of destination number in packed decimal format, left-justified where 0 is null, hex a is digit 0, and hex c is # (for Collected digits). When type = 4 (II-digits), only d1-d2 will be populated, and leading zeros will be included.
d1-d96	for type=5, ASAI UUI (treated as binary data)
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

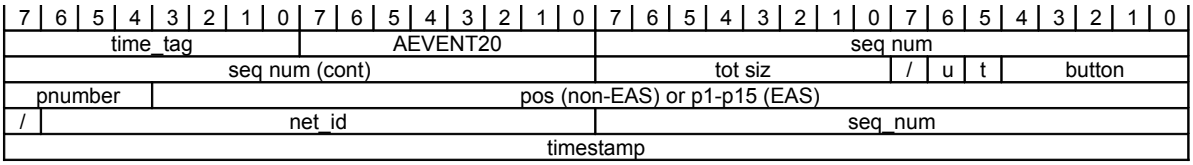
3.4.10 → AEVENT20 and AEVENT24

The **Agent Event** message is sent by the switch when an agent presses an event button or an event related to the agent has occurred. This includes stroke count, malicious call, supervisor assist, audio difficulty, RONA, ROIF, ROOF, "special" reason code buttons and agent notifications. The reporting adjunct will generate an immediate exception for malicious call, audio difficulty, RONA, ROIF or ROOF. The reporting adjunct will record the number of occurrences of supervisor assists and stroke counts on an agent and split/skill basis.

Assist and stroke count will only be pegged in the reporting adjunct if the agent is on an ACD call (split/skill or direct agent) or in ACW. Audio difficulty and MCT are pegged anytime the agent has any call. If all calls are on hold, The reporting adjunct pegs the event for the last call put on hold.

3.4.10.1 **→AEVENT20 (New CM3.1)**

	Switch	Vect	Skill
First Release	CM3.1	y/n	y/n
Last Release	CM6.3		

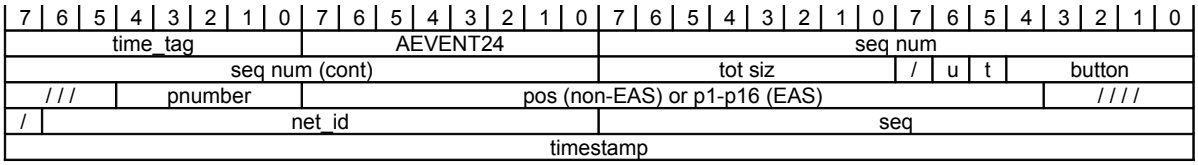


time_tag seconds past minute when event occurred, 0-59
opcode AEVENT20 = 214 (0xd6)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
u flag indicating that the UCID fields (net_id, seq and timestamp) are present
t, button type of button, button pressed
- t=0 miscellaneous button type, button = 0-7 as follows:
  - 0 supervisor assist or malicious call event completed (CLEAR)
  - 1 malicious call started
  - 2 supervisor assist call started
  - 3 call redirected because no answer while ringing at agent position: RONA (Redirect On No Answer), or not delivered due to connectivity failure: ROIF (Redirect on IP Failure, CM2.1 or later) or ROOF (Redirect on Optim Failure, CM5.0 or later).
  - 4 agent entered AUX reason code 0, *, # or timed out.
  - 5 agent entered logout reason code 0, *, # or timed out.
  - 6 agent has been notified for interruptible work while in AUX (CM6.0 and later).
  - 7 agent has been cleared from interruptible work while in AUX (CM6.0 and later).
- t=1 stroke count button type, button = 0-9 as follows: is a stroke button, 0-9
  - 0 audio difficulty
  - N stroke 1-9

pnumber number of digits in agent position, 1-15, will always be 5 for non-EAS
pos/p1-p15 agent position, without EAS "pos" (1-10000), with EAS "p1-p15" (0-9999999) login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary
net_id network entity identifier, 1-32767
seq call sequence number, 0-65535
timestamp unix style (seconds since 1/1/70) time call assigned a UCID

3.4.10.2 **→AEVENT24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y/n	y/n
Last Release			



time_tag seconds past minute when event occurred, 0-59
opcode AEVENT24 = 64 (0x40)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
u flag indicating that the UCID fields (net_id, seq and timestamp) are present
t, button type of button, button pressed

- t=0 miscellaneous button type, button = 0-7 as follows:
  - 0 supervisor assist or malicious call event completed (CLEAR)
  - 1 malicious call started
  - 2 supervisor assist call started
  - 3 call redirected because no answer while ringing at agent position: RONA (Redirect On No Answer), or not delivered due to connectivity failure: ROIF (Redirect on IP Failure, CM2.1 or later) or ROOF (Redirect on Optim Failure, CM5.0 or later).
  - 4 agent entered AUX reason code 0, *, # or timed out.
  - 5 agent entered logout reason code 0, *, # or timed out.
  - 6 agent has been notified for interruptible work while in AUX (CM6.0 and later).
  - 7 agent has been cleared from interruptible work while in AUX (CM6.0 and later).
- t=1 stroke count button type, button = 0-9 as follows: is a stroke button, 0-9
  - 0 audio difficulty
  - N stroke 1-9

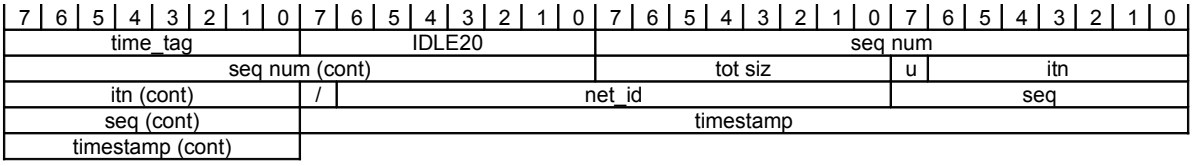
pnumber **number of digits in agent position, 1-16, will always be 5 for non-EAS**
pos/p1-p16 **agent position, without EAS "pos" (1-10000), with EAS "p1-p16" (0-9999999) login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary**
net_id network entity identifier, 1-32767
seq call sequence number, 0-65535
timestamp unix style (seconds since 1/1/70) time call assigned a UCID

3.4.11 → IDLE20, IDLE23 and IDLE24

The **IDLE** message is sent by the switch when a facility goes idle.

3.4.11.1 → IDLE20 (New CM3.1)

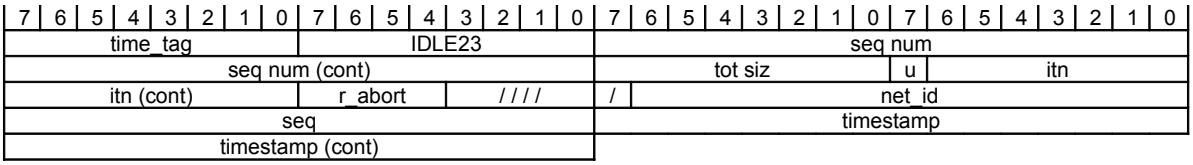
	Switch	Vect	Skill
First Release	CM3.1	y/n	y/n
Last Release	CM5.2		



time_tag	seconds past minute when event occurred, 0-59
opcode	IDLE20 = 215 (0xd7)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
itn	internal trunk number used to identify trunk facilities, 1-32766
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.11.2 **→IDLE23 (New CM6.0)**

	Switch	Vect	Skill
First Release	CM6.0	y/n	y/n
Last Release	CM6.3		



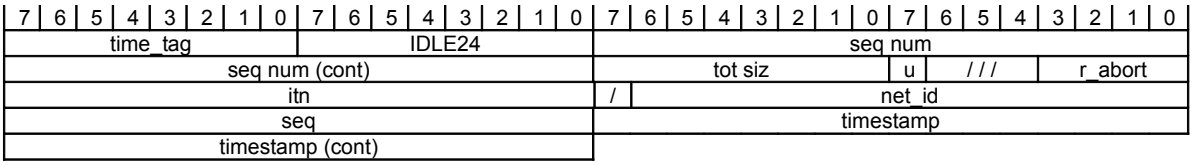
time_tag seconds past minute when event occurred, 0-59
opcode IDLE23 = 27 (0x1b)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
u flag indicating that the UCID fields (net_id, seq and timestamp) are present
itn internal trunk number used to identify trunk facilities, 1-32766
r_abort Reason for why call was aborted and may be requeued later with same ucid.

0 Default – call was not aborted for later requeue
1 Call could not be delivered because media or other resources were not available when agent was selected
2 Wait time increased drastically
3 Call was aborted after being delivered to the agent because the call was still receiving uninterruptible wait treatment
4 Recovering from a network failure - e.g., Call was cancelled because 182 Queued message was not received from CM within configured timeout
5 Recovering from a Voice Portal failure - Call was cancelled due to a VP application failure
6 Cancelled due to caller interaction - Caller has subsequently indicated a change to the disposition via a prompt from ICR

net_id network entity identifier, 1-32767
seq call sequence number, 0-65535
timestamp unix style (seconds since 1/1/70) time call assigned a UCID

3.4.11.3 **→IDLE24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y/n	y/n
Last Release			



time_tag seconds past minute when event occurred, 0-59
opcode IDLE24 = 65 (0x41)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
u flag indicating that the UCID fields (net_id, seq and timestamp) are present
r_abort Reason for why call was aborted and may be requeued later with same ucid.

- 0 Default – call was not aborted for later requeue
- 1 Call could not be delivered because media or other resources were not available when agent was selected
- 2 Wait time increased drastically
- 3 Call was aborted after being delivered to the agent because the call was still receiving uninterruptible wait treatment
- 4 Recovering from a network failure - e.g., Call was cancelled because 182 Queued message was not received from CM within configured timeout
- 5 Recovering from a Voice Portal failure - Call was cancelled due to a VP application failure
- 6 Cancelled due to caller interaction - Caller has subsequently indicated a change to the disposition via a prompt from ICR

itn **internal trunk number used to identify trunk facilities, 1-65535**
net_id network entity identifier, 1-32767
seq call sequence number, 0-65535
timestamp unix style (seconds since 1/1/70) time call assigned a UCID

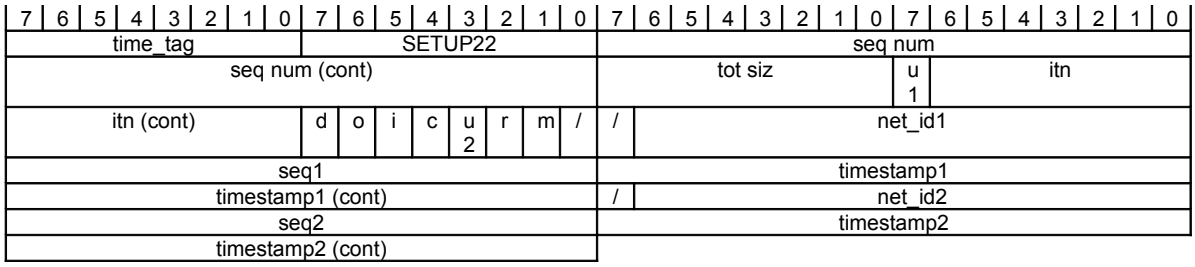
3.4.12 → SETUP22 and SETUP24

The **SETUP** message is sent by the switch when a trunk is seized or as the first message of a call entering the reporting adjunct measured domain. The universal call id (UCID) associated with a call must be sent at the beginning of the call (or when it enters the measured domain).

3.4.12.1 →SETUP22 (New CM5.2)

	Switch	Vect	Skill
First Release	CM5.2	y/n	y/n
Last Release	CM6.3		

This is a variable length message based on the "u1" and "u2" bits. The first ten bytes are always sent.

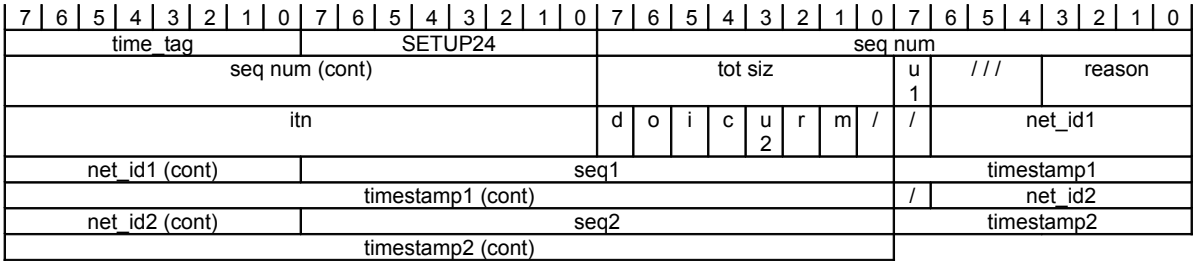


time_tag	seconds past minute when event occurred, 0-59
opcode	SETUP22 = 254 (0xfe)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
u1	flag indicating that the UCID fields (net_id1, seq1, and timestamp1) are present.
itn	internal trunk number used to identify trunk facilities, 1-32766
d	direction; 0 = outbound, 1 = inbound
o	OCM call; 0 = no, 1 = call distributed by OCM
i	ignore messsage indication for ProActive Contact support; 0 = accept, 1 = ignore
c	converse flag indicating SETUP sent because call being routed to a converse-on skill port.
u2	flag indicating that the UCID fields (net_id2, seq2, and time_stamp2) are present. This is the parent held call UCID provided with a call transferred from one CM to another.
r	re-sent call with the same UCID of a call that was aborted while still active and has returned. Default is 0.
m	media-added - The itn for the call has changed because media was added by a SIP INVITE with replaces for ICR support. This SETUP will be immediately followed by an XFER (rename) and an IDLE for the first itn. The UCID included in all the messages will be the same as the original SETUP. Default is 0.
net_id1	network entity identifier; 1-32767
seq1	call sequence number; 0-65535
timestamp1	unix style (seconds since 1/1/70) time call assigned a UCID
net_id2	network entity identifier; 1-32767
seq2	call sequence number; 0-6553.
timestamp2	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.12.2 **→SETUP24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y/n	y/n
Last Release			

This is a variable length message based on the "u1" and "u2" bits. The first ten bytes are always sent.



time_tag	seconds past minute when event occurred, 0-59
opcode	SETUP24 = 66 (0x42)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
u1	flag indicating that the UCID fields (net_id1, seq1, and timestamp1) are present.
reason	**reason the setup was sent:**
	0 New contact – the first message for a contact
	1 The SETUP is part of a RENAME
	2 Adding a trunk – an example is outflow
	3 LAR redirect of the call
	4 Converse command connection to the IVR port
	5 CTI phantom call
itn	**internal trunk number used to identify trunk facilities, 1-65535**
d	direction; 0 = outbound, 1 = inbound
o	OCM call; 0 = no, 1 = call distributed by OCM
i	ignore messsage indication for ProActive Contact support; 0 = accept, 1 = ignore
c	converse flag indicating SETUP sent because call being routed to a converse-on skill port.
u2	flag indicating that the UCID fields (net_id2, seq2, and time_stamp2) are present. This is the parent held call UCID provided with a call transferred from one CM to another.
r	re-sent call with the same UCID of a call that was aborted while still active and has returned. Default is 0.
m	media-added - The itn for the call has changed because media was added by a SIP INVITE with replaces for ICR support. This SETUP will be immediately followed by an XFER (rename) and an IDLE for the first itn. The UCID included in all the messages will be the same as the original SETUP. Default is 0.
net_id1	network entity identifier; 1-32767
seq1	call sequence number; 0-65535
timestamp1	unix style (seconds since 1/1/70) time call assigned a UCID
net_id2	network entity identifier; 1-32767
seq2	call sequence number; 0-6553.
timestamp2	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.13 → QUEUED20, QUEUED23 and QUEUED24

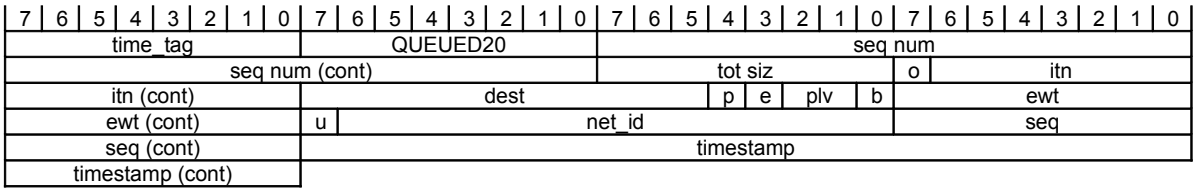
The **QUEUED** message is sent by the switch when a facility queues to a skill hunt group.

This is a variable length message based on the "e" bit and "u" bit. The message may optionally contain the expected wait time (EWT) for the split. EWT is calculated by the switch, but stored and reported by the reporting adjunct. The message may optionally contain the UCID in CM3.1 and later.

3.4.13.1 → QUEUED20 (New CM3.1)

	Switch	Vect	Skill
First Release	CM3.1	y/n	y/n
Last Release	CM5.2		

Twelve bytes (the first 11 bytes + the first byte of the UCID for the “u” flag) will always be sent.

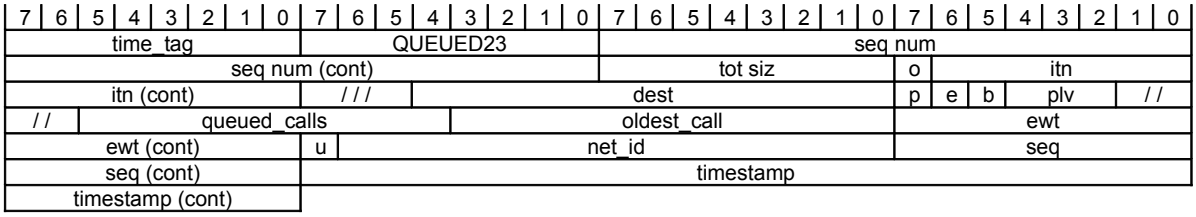


time_tag	seconds past minute when event occurred, 0-59
opcode	QUEUED20 = 217 (0xd9)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
o	OCM call; 0 = no, 1 = call distributed by OCM
itn	internal trunk number used to identify trunk facilities, 1-32766
dest	destination: split/skill number, 1-2000 (MV2 and later)
p	priority call (w/o vectoring enabled) 0 = none, 1 = priority
e	ewt flag indicating that ewt field is present If not, EWT for that priority is understood by the reporting adjunct to be null)
plv	priority level (with vectoring enabled); 0 = low, 1 = medium, 2 = high, 3 = top (Send 1 or 2 if vectoring active but non-vector controlled split.)
b	backup split/skill; 0 = main, 1 = backup
ewt	expected wait time for the priority level specified in the plv field, 1-32767 seconds (vectoring only)
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.13.2 **→QUEUED23 (New CM6.0)**

	Switch	Vect	Skill
First Release	CM6.0	y/n	y/n
Last Release	CM6.3		

Sixteen bytes (the first 15 bytes plus the first byte of UCID for the “u” flag) will always be sent.

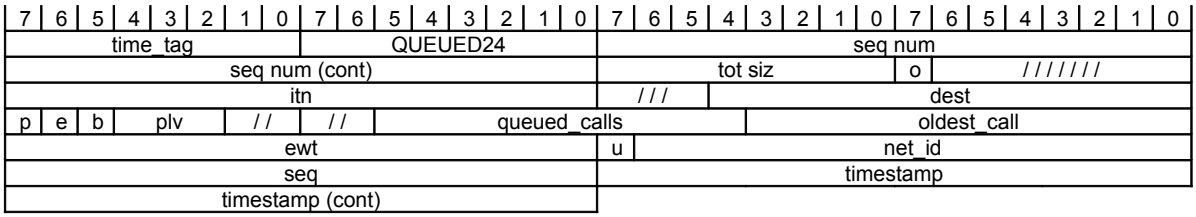


time_tag	seconds past minute when event occurred, 0-59
opcode	QUEUED23 = 28 (0x1c)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
o	OCM call; 0 = no, 1 = call distributed by OCM
itn	internal trunk number used to identify trunk facilities, 1-32766
dest	destination: split/skill number, 1-8000
p	priority call (w/o vectoring enabled) 0 = none, 1 = priority
e	ewt flag indicating that ewt field is present If not, EWT for that priority is understood by the reporting adjunct to be null)
b	backup split/skill; 0 = main, 1 = backup
plv	priority level (with vectoring enabled); 0 = low, 1 = medium, 2 = high, 3 = top, 4 = redirect (Send 1 or 2 if vectoring active but non-vector controlled split.)
queued_calls	number of calls in queue including this added one, 0-999 (Over 999 will be sent as 0x3ff)
oldest_call	times in secs of the oldest call in queue, 0-999 secs
ewt	expected wait time for the priority level specified in the plv field, 1-32767 seconds (vectoring only).
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.13.3 **→QUEUED24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y/n	y/n
Last Release			

Seventeen bytes (the first 16 bytes plus the first byte of UCID for the “u” flag) will always be sent.



time_tag	seconds past minute when event occurred, 0-59
opcode	QUEUED24 = 67 (0x43)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
o	OCM call; 0 = no, 1 = call distributed by OCM
itn	**internal trunk number used to identify trunk facilities, 1-65535**
dest	destination: split/skill number, 1-8000
p	priority call (w/o vectoring enabled) 0 = none, 1 = priority
e	ewt flag indicating that ewt field is present If not, EWT for that priority is understood by the reporting adjunct to be null)
b	backup split/skill; 0 = main, 1 = backup
plv	priority level (with vectoring enabled); 0 = low, 1 = medium, 2 = high, 3 = top, 4 = redirect (Send 1 or 2 if vectoring active but non-vector controlled split.)
queued_calls	number of calls in queue including this added one, 0-999 (Over 999 will be sent as 0x3ff)
oldest_call	times in secs of the oldest call in queue, 0-999 secs
ewt	expected wait time for the priority level specified in the plv field, 1-32767 seconds (vectoring only).
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

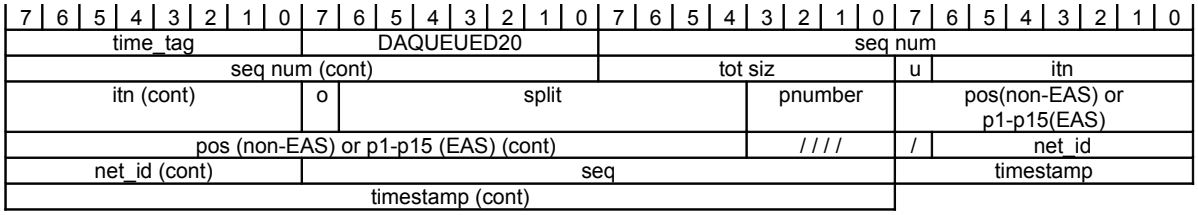
3.4.14 → DAQUEUED20, DAQUEUED23 and DAQUEUED24

The **DAQUEUED** message is sent by the switch when a facility is extended to an agent (direct agent call). **DAQUEUED** is sent independent of the call actually queuing.

This is a variable length message based on "pnumber" and "u".

3.4.14.1 → DAQUEUED20 (New CM3.1)

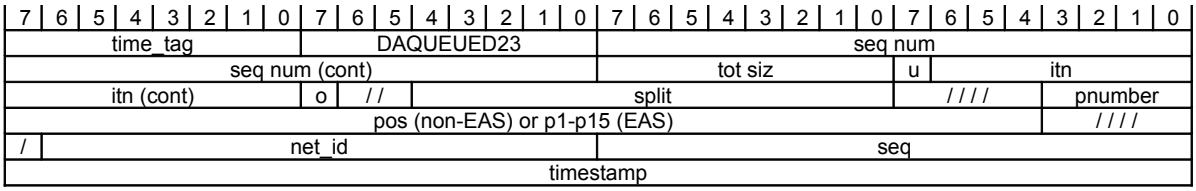
	Switch	Vect	Skill
First Release	CM3.1	y/n	y/n
Last Release	CM5.2		



time_tag	seconds past minute when event occurred, 0-59
opcode	DAQUEUED20 = 218 (0xda)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
itn	internal trunk number used to identify trunk facilities, 1-32766
o	OCM call; 0 = no, 1 = call distributed by OCM
split	split or skill, 1-999 (G3V8-G3V11), 1-2000 (MV2 and later)
pnumber	number of digits in agent position, 1-15, will always be 5 for non-EAS
pos/p1-p15	agent position - without EAS "pos" (1-5200 for G3V2-G3V4 or 1-10000 G3V5 and later), with EAS "p1-p15" (0-9999999) login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.14.2 **→DAQUEUED23 (New CM6.0)**

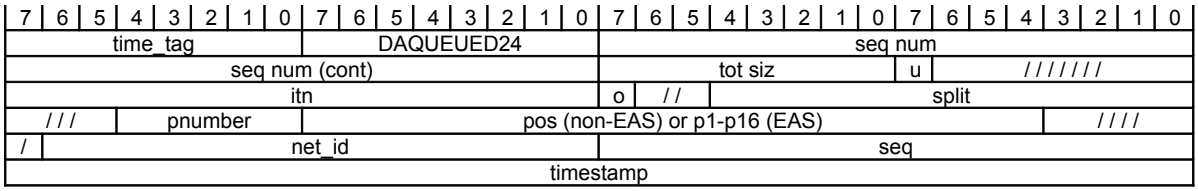
	Switch	Vect	Skill
First Release	CM6.0	y/n	y/n
Last Release	CM6.3		



time_tag	seconds past minute when event occurred, 0-59
opcode	DAQUEUED23= 29 (0x1d)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
itn	internal trunk number used to identify trunk facilities, 1-32766
o	OCM call; 0 = no, 1 = call distributed by OCM
split	split or skill, 1-8000
pnumber	number of digits in agent position, 1-15, will always be 5 for non-EAS
pos/p1-p15	agent position - without EAS "pos" (1-10000), with EAS "p1-p15" login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.14.3 **→DAQUEUED24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y/n	y/n
Last Release			



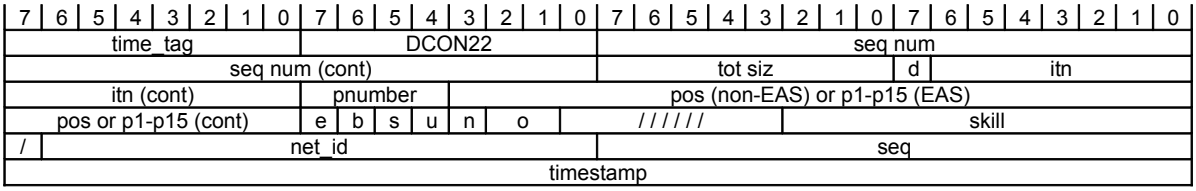
time_tag	seconds past minute when event occurred, 0-59
opcode	DAQUEUED24 = 68 (0x44)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
itn	**internal trunk number used to identify trunk facilities, 1-65535**
o	OCM call; 0 = no, 1 = call distributed by OCM
split	split or skill, 1-8000
pnumber	**number of digits in agent position, 1-16, will always be 5 for non-EAS**
pos/p1-p16	**agent position - without EAS "pos" (1-10000), with EAS "p1-p16" login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary**
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.15 → DCON22, DCON23 and DCON24

The **DCON** message is sent by the switch when a facility dequeues and connects to an agent. **DCON** is also sent to indicate someone such as a service observer has bridged on to the call ("b" bit is set). This is a variable length message based on "pnumber" and the "s" and "u" bits. The "s" bit is only set for EAS and only when ringing an agent with a call for a measured skill. The message may optionally contain the UCID in CM5.2 and later. When the "s" bit is set, the message will contain 2 additional bytes for the "skill" field.

3.4.15.1 →DCON22 (New CM5.2)

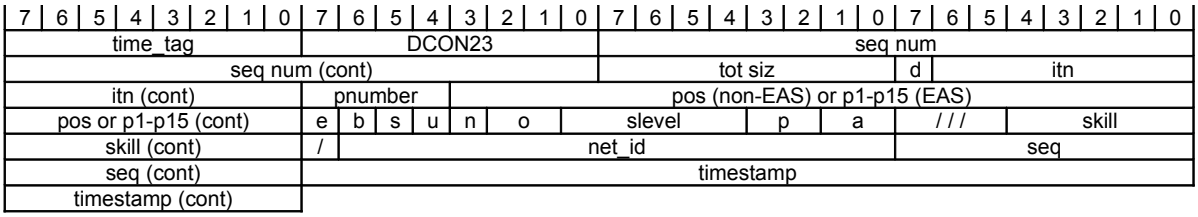
	Switch	Vect	Skill
First Release	CM5.2	y/n	y/n
Last Release	CM5.2		



time_tag	seconds past minute when event occurred, 0-59
opcode	DCON22 = 253 (0xfd)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
d	direction; 0=outbound, 1=inbound
itn	internal trunk number used to identify trunk facilities, 1-32766
pnumber	number of digits in agent position, 1-15, will always be 5 for non-EAS 0=an unmeasured endpoint received the call. In this case, no position is sent.
pos/p1-p15	agent position without EAS "pos" (1-10000), with EAS "p1-p15" (0-9999999) login ID in packed decimal format, left-justified where 0 is null and hex a is digit 0, zero fill to a byte boundary
e	call to extension; 0=no (ACD or direct agent), 1=extension call (Note: a call to the login ID with EAS that does not have the direct agent COR will be considered an extension call)
b	bridging; 0 = not bridging on to a call, 1 = bridging on to a call
s	0 = no skill in message, 1 = skill in message (ACD or Direct Agent call)
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
n	flag indicating that call was dequeued after multiple split queueing
o	type of observing when "b" bit is set due to service observing. Field for IQ 5.0 support starting with CM 5.2. 0 none (bridging case) 1 agent 2 VDN
skill	skill matched 1-999 (G3V8-G3V11), 1-2000 (MV2 and later)
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.15.2 **→DCON23 (New CM6.0)**

	Switch	Vect	Skill
First Release	CM6.0	y/n	y/n
Last Release	CM6.3		

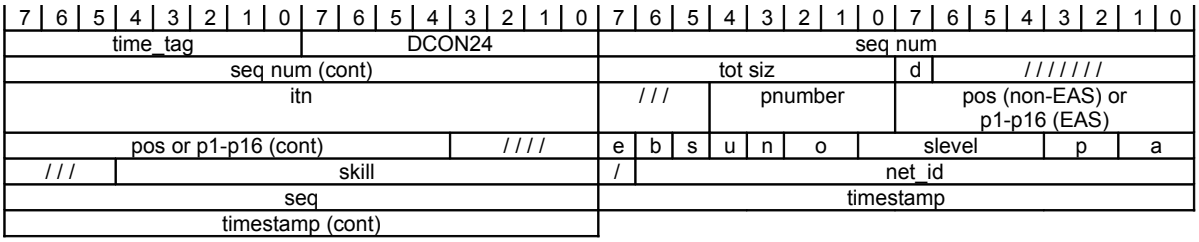


time_tag seconds past minute when event occurred, 0-59
opcode DCON23 = 11 (0x0b)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
d direction; 0=outbound, 1=inbound
itn internal trunk number used to identify trunk facilities, 1-32766
pnumber number of digits in agent position, 1-15, will always be 5 for non-EAS
0=an unmeasured endpoint received the call. In this case, no position is sent.
pos/p1-p15 agent position without EAS "pos" (1-10000), with EAS "p1-p15" login ID in packed decimal format, left-justified where 0 is null and hex a is digit 0, zero fill to a byte boundary
e call to extension; 0=no (ACD or direct agent), 1=extension call (Note: a call to the login ID with EAS that does not have the direct agent COR will be considered an extension call)
b bridging; 0 = not bridging on to a call, 1 = bridging on to a call
s 0 = no skill in message, 1 = skill in message (ACD or Direct Agent call)
u flag indicating that the UCID fields (net_id, seq and timestamp) are present
n flag indicating that call was dequeued after multiple split queueing
o type of observing when "b" bit is set due to service observing.
- 0 none (bridging case)
- 1 agent
- 2 VDN

slevel Level assigned to the agent for the skill that the call was delivered for, 0 (default) or level 1-16.
p
0 Call was not delivered via the preferred skill level check vector command or no skill preference was specified (default)
1 Agent's level for the skill does not match the level specified in the check vector command (either this was the only eligible agent or none of the eligible agent's assigned level for the skill matched)
2 Agent's level for the skill matches the preferred skill level (or range) specified by the check vector command.
a
0 for the not applicable case (default)
1 ACD call was routed to the agent after being queued (call surplus)
2 ACD call was routed to the agent without being queued (agent surplus)
skill skill matched 1-8000
net_id network entity identifier, 1-32767
seq call sequence number, 0-65535
timestamp unix style (seconds since 1/1/70) time call assigned a UCID

3.4.15.3 **→DCON24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y/n	y/n
Last Release			



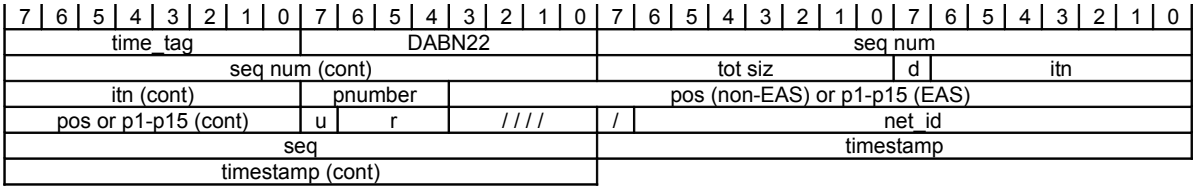
time_tag	seconds past minute when event occurred, 0-59
opcode	DCON24 = 69 (0x45)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
d	direction; 0=outbound, 1=inbound
itn	**internal trunk number used to identify trunk facilities, 1-65535**
pnumber	**number of digits in agent position, 1-16, will always be 5 for non-EAS 0=an unmeasured endpoint received the call. In this case, no pos/p1-p16 is sent.**
pos/p1-p16	**agent position without EAS "pos" (1-10000), with EAS "p1-p16" login ID in packed decimal format, left-justified where 0 is null and hex a is digit 0, zero fill to a byte boundary**
e	call to extension; 0=no (ACD or direct agent), 1=extension call (Note: a call to the login ID with EAS that does not have the direct agent COR will be considered an extension call)
b	bridging; 0 = not bridging on to a call, 1 = bridging on to a call
s	0 = no skill in message, 1 = skill in message (ACD or Direct Agent call)
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
n	flag indicating that call was dequeued after multiple split queueing
o	type of observing when "b" bit is set due to service observing. – 0 none (bridging case) – 1 agent – 2 VDN
slevel	Level assigned to the agent for the skill that the call was delivered for, 0 (default) or level 1-16.
p	0 Call was not delivered via the preferred skill level check vector command or no skill preference was specified (default) 1 Agent's level for the skill does not match the level specified in the check vector command (either this was the only eligible agent or none of the eligible agent's assigned level for the skill matched) 2 Agent's level for the skill matches the preferred skill level (or range) specified by the check vector command.
a	0 for the not applicable case (default) 1 ACD call was routed to the agent after being queued (call surplus) 2 ACD call was routed to the agent without being queued (agent surplus)
skill	skill matched 1-8000
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.16 → DABN22 and DABN24

The **DABN** message is sent by the switch when a facility abandons in processing, in queue, while ringing or when a caller (agent or other party) drops from hold.

3.4.16.1 → DABN22 (New CM5.2)

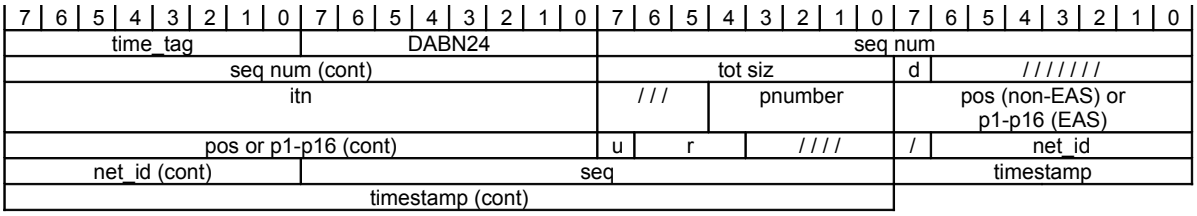
	Switch	Vect	Skill
First Release	CM5.2	y/n	y/n
Last Release	CM6.3		



time_tag	seconds past minute when event occurred, 0-59
opcode	DABN22 = 252 (0xfc)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
d	1 = agent dropped last conferenced party, 0 = doesn't apply
itn	internal trunk number used to identify trunk facilities, 1-32766
pnumber	number of digits in agent position, 1-15, will always be 5 for non-EAS
pos/p1-p15	agent position without EAS "pos" (1-10000), with EAS "p1-p15" (0-9999999) login ID in packed decimal format, left-justified where 0 is null and hex a is digit 0, zero fill to a byte boundary. The agent position is the identity of the agent who put the call on hold for the held call dropped case.
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
r	reason field. 0 normal call abandoned 1 held call dropped 2-7 for future use
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.16.2 **→DABN24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y/n	y/n
Last Release			



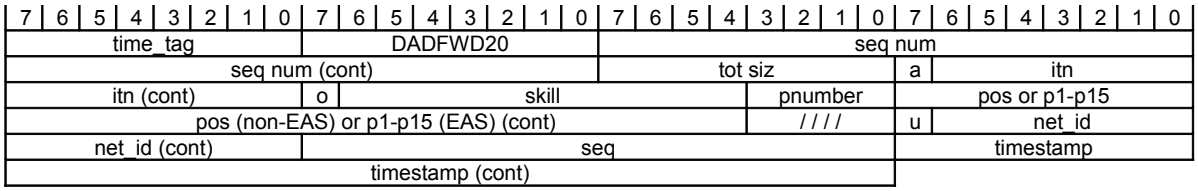
time_tag	seconds past minute when event occurred, 0-59
opcode	DABN24 = 70 (0x46)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
d	1 = agent dropped last conferenced party, 0 = doesn't apply
itn	**internal trunk number used to identify trunk facilities, 1-65535**
pnumber	**number of digits in agent position, 1-16, will always be 5 for non-EAS**
pos/p1-p16	**agent position without EAS "pos" (1-10000), with EAS "p1-p16" (0-9999999) login ID in packed decimal format, left-justified where 0 is null and hex a is digit 0, zero fill to a byte boundary. The agent position is the identity of the agent who put the call on hold for the held call dropped case.**
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
r	reason field. 0 normal call abandoned 1 held call dropped 2-7 for future use
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.17 → DADFWD20 and DADFWD24

The **DADFWD20** message is sent by the switch when a facility is forwarded to an agent via direct agent calling. This is a variable length message based on "pnumber" and "u".

3.4.17.1 → DADFWD20 (New CM3.1)

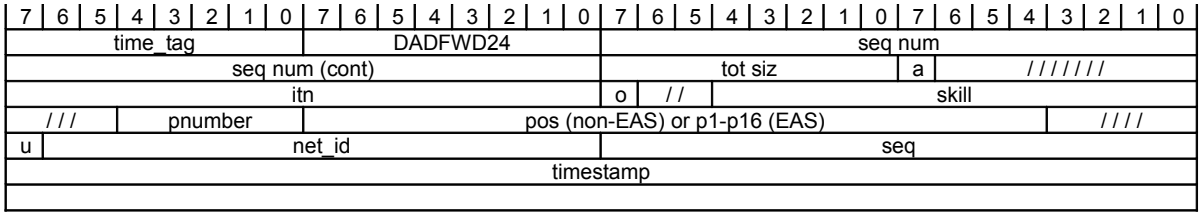
	Switch	Vect	Skill
First Release	CM3.1	y/n	y/n
Last Release	CM6.3		



time_tag	seconds past minute when event occurred, 0-59
opcode	DADFWD20 = 221 (0xdd)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
a	ASAI adjunct routing; 1 = adjunct (re)routed call
itn	internal trunk number used to identify trunk facilities, 1-32766
o	OCM call; 0 = no, 1 = call distributed by OCM
skill	skill number or split number call queued to, 1-999 (G3V8-G3V11), 1-2000 (MV2 and later)
pnumber	number of digits in agent position, 1-15, will always be 5 for non-EAS
pos/p1-p15	agent position - without EAS "pos" (1-10000), with EAS "p1-p15" (0-9999999) login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.17.2 **→DADFWD24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y/n	y/n
Last Release			



time_tag	seconds past minute when event occurred, 0-59
opcode	DADFWD24 = 42 (0x2a)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
a	ASAI adjunct routing; 1 = adjunct (re)routed call
itn	**internal trunk number used to identify trunk facilities, 1-65535**
o	OCM call; 0 = no, 1 = call distributed by OCM
skill	**skill number or split number call queued to, 1-8000**
pnumber	**number of digits in agent position, 1-16, will always be 5 for non-EAS**
pos/p1-p16	**agent position - without EAS "pos" (1-10000), with EAS "p1-p16" (0-9999999) login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary**
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.18 → DFWD20, DFWD23 and DFWD24

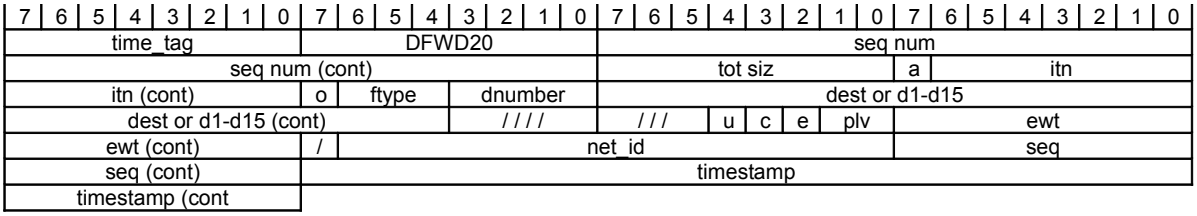
The **DFWD** message is sent by the switch when a facility is forwarded internally. This is a variable length message based on "dnumber" and the "e" and "u" bits. When the "e" bit is set the message will contain two additional bytes for the "ewt" field. If "dnumber" is zero then there will be NO "dest"/"d1-d15" field in message.

The message may optionally contain the expected wait time (EWT) for the split, which is based on the history of calls that have queued to this split (ftype = 4). EWT is calculated by the switch, and sent to CMS. The message may optionally contain the UCID in CM3.1 and later.

When forwarded to an agent position (ftype = 6) the "dest" will contain the login ID packed decimal format, left-justified with EAS.

3.4.18.1 → DFWD20 (New CM3.1)

	Switch	Vect	Skill
First Release	CM3.1	y/n	y/n
Last Release	CM5.2		



DFWD20 (continued)

time_tag	seconds past minute when event occurred, 0-59
opcode	DFWD20 = 222 (0xde)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
a	ASAI adjunct routing; 1 = adjunct (re)routed call
itn	internal trunk number used to identify trunk facilities, 1-32766
o	OCM call; 0 = no, 1 = call distributed by OCM
ftype	forwarding type

0	redirected (intraflow) to a VDN
1	interflow
2	lookahead interflow, BSR interflow, or Network Call Redirection
3	unknown destination
4	redirected (intraflow) to a split
5	not used
6	redirected (intraflow) to an extension
7	redirected (intraflow) to an attendant position

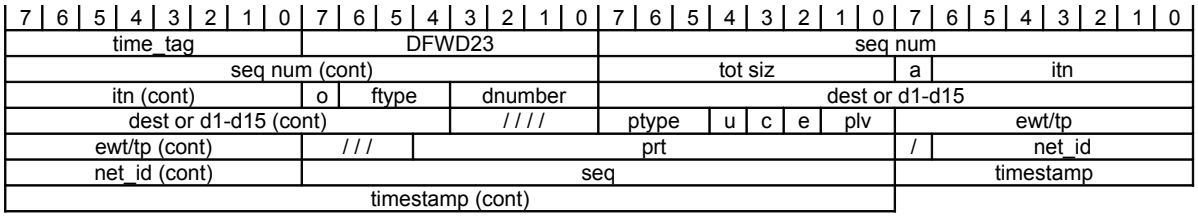
dnumber	number of digits in "dest"/"d1-d15", 5 for ftype=0 and 5 for ftype=4 and 1-15 for ftype=6(EAS) 0=call was forwarded to an unmeasured endpoint. In this case, no destination is sent.
dest/d1-d15	destination: with EAS "d1-d15" (0-9999999) login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary. If "dnumber" is zero then there will be NO "dest"/"d1-d15" field in message.

ftype	dest field represents
0 (intraflow to VDN)	vector number
4 (intraflow to split)	split
6 (intraflow to extension)	agent position

c	call deflection flag, valid only when ftype=2. When this bit is set it indicates that the BSR or route-to number interflow was successful by means of call redirection through the public network.
e	ewt flag (If set, last two bytes are part of the message, only if ftype is 4. If not set, EWT is understood by CMS to be null)
plv	priority (with vectoring): If ftype is 4, and the call is directed to a split via a messaging split, route to number, or adjunct routing, priorities pegged are: 1 = medium (no priority was sent), 2 = high (a priority was sent).
ewt	expected wait time for the priority level specified in the plv field, 0-32767 seconds (with vectoring)
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.18.2 **→DFWD23 (New CM6.0)**

	Switch	Vect	Skill
First Release	CM6.0	y/n	y/n
Last Release	CM6.3		



DFWD23 (continued)

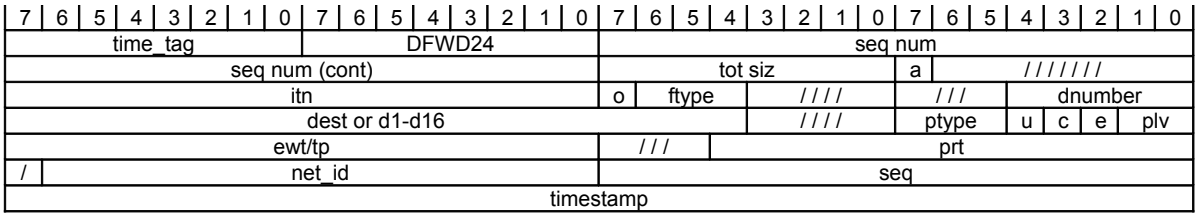
time_tag	seconds past minute when event occurred, 0-59
opcode	DFWD23 = 12 (0x0c)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
a	ASAI adjunct routing; 1 = adjunct (re)routed call
itn	internal trunk number used to identify trunk facilities, 1-32766
o	OCM call; 0 = no, 1 = call distributed by OCM
ftype	forwarding type 0 redirected (intraflow) to a VDN 1 interflow 2 lookahead interflow, BSR interflow, or Network Call Redirection 3 unknown destination 4 redirected (intraflow) to a split 5 redirected to another VDN by a Policy Routing Table (PRT) – the iln of the VDN the call is routed to included in the dest field 6 redirected (intraflow) to an extension 7 redirected (intraflow) to an attendant position
dnumber	number of digits in "dest"/"d1-d15", 5 for ftype=0 and 5 for ftype=4 and 1-15 for ftype=6(EAS) 0=call was forwarded to an unmeasured endpoint. In this case, no destination is sent.
dest/d1-d15	destination: with EAS "d1-d15" in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary. If "dnumber" is zero then there will be NO "dest"/"d1-d15" field in message.

ftype	dest field represents
0 (intraflow to VDN)	vector number or zero if routing to PRT
4 (intraflow to split)	split
5 (PRT)	VDN we are routing to
6 (intraflow to extension)	agent position

ptype	0 No PRT number included (i.e., ftype is not 5) 1 Percentage Allocation type PRT 2-7 Reserved for future use
c	call deflection flag, valid only when ftype=2. When this bit is set it indicates that the BSR or route-to number interflow was successful by means of call redirection through the public network.
e	ewt/tp flag If set, ewt/tp two byte field is part of the message. If set when the ftype = 4. the field contains ewt. If set when the ftype = 5, the field contains the target percentage (tp).
plv	priority (with vectoring): If ftype is 4, and the call is directed to a split via a messaging split, route to number, or adjunct routing, priorities pegged are: 1 = medium (no priority was sent), 2 = high (a priority was sent).
ewt/tp	With ftype = 4, contains the expected wait time for the priority level specified in the plv field, 0-32767 seconds. With ftype = 5, contains the tp (target percentage) 1-100 assigned in the PRT table (in prt field) to the VDN in the dest field, R6 and later.
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
prt	The number of the Policy Routing Table used to route the call to this subsequent VDN range 1-8000 – set to all 0's if no PRT number included
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.18.3 **→DFWD24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y/n	y/n
Last Release			



DFWD24 (continued)

time_tag	seconds past minute when event occurred, 0-59
opcode	DFWD24 = 71 (0x47)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
a	ASAI adjunct routing; 1 = adjunct (re)routed call
itn	**internal trunk number used to identify trunk facilities, 1-65535**
o	OCM call; 0 = no, 1 = call distributed by OCM
ftype	forwarding type

0 redirected (intraflow) to a VDN
1 interflow
2 lookahead interflow, BSR interflow, or Network Call Redirection
3 unknown destination
4 redirected (intraflow) to a split
5 redirected to another VDN by a Policy Routing Table (PRT) – the iln of the VDN the call is routed to included in the dest field
6 redirected (intraflow) to an extension
7 redirected (intraflow) to an attendant position

dnumber	**number of digits in "dest"/"d1-d16", 5 for ftype=0 and 5 for ftype=4 and 1-24 for ftype=6(EAS)** **0=call was forwarded to an unmeasured endpoint. In this case, no destination is sent.**
dest/d1-d16	**destination: with EAS "d1-d16" in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary. If "dnumber" is zero then there will be NO "dest"/"d1-d16" field in message.**

ftype	dest field represents
0 (intraflow to VDN)	vector number or zero if routing to PRT
4 (intraflow to split)	split
5 (PRT)	VDN we are routing to
6 (intraflow to extension)	agent position

ptype
0 No PRT number included (i.e., ftype is not 5)
1 Percentage Allocation type PRT
2-7 Reserved for future use

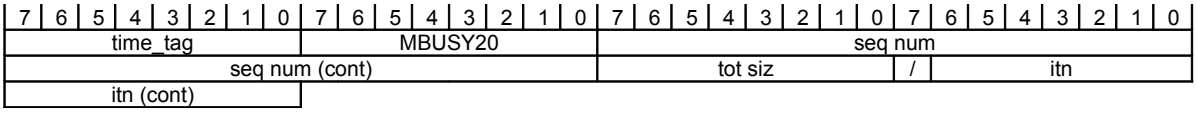
c	call deflection flag, valid only when ftype=2. When this bit is set it indicates that the BSR or route-to number interflow was successful by means of call redirection through the public network.
e	ewt/tp flag If set, ewt/tp two byte field is part of the message. If set when the ftype = 4. the field contains ewt. If set when the ftype = 5, the field contains the target percentage (tp).
plv	priority (with vectoring): If ftype is 4, and the call is directed to a split via a messaging split, route to number, or adjunct routing, priorities pegged are: 1 = medium (no priority was sent), 2 = high (a priority was sent).
ewt/tp	With ftype = 4, contains the expected wait time for the priority level specified in the plv field, 0-32767 seconds. With ftype = 5, contains the tp (target percentage) 1-100 assigned in the PRT table (in prt field) to the VDN in the dest field, R6 and later.
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
prt	The number of the Policy Routing Table used to route the call to this subsequent VDN range 1-8000 – set to all 0's if no PRT number included
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.19 → MBUSY20 and MBUSY24

The **MBUSY** message is sent by the switch when a trunk is busied out.

3.4.19.1 → MBUSY20 (New CM3.1)

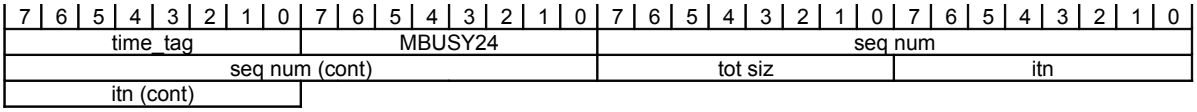
	Switch	Vect	Skill
First Release	CM3.1	y/n	y/n
Last Release	CM6.3		



time_tag	seconds past minute when event occurred, 0-59
opcode	MBUSY20 = 223 (0xdf)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
itn	internal trunk number used to identify trunk facilities, 1-32766

3.4.19.2 → MBUSY24 (New CM7.0)

	Switch	Vect	Skill
First Release	CM3.1	y/n	y/n
Last Release			



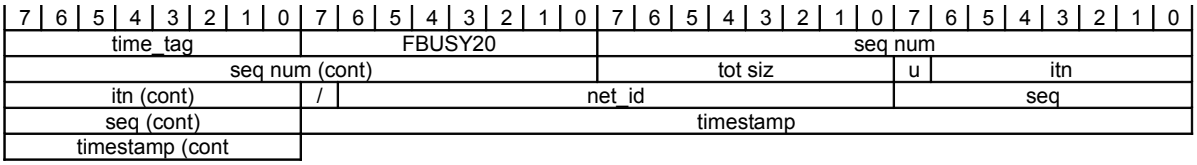
time_tag	seconds past minute when event occurred, 0-59
opcode	MBUSY24 = 72 (0x48)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
itn	**internal trunk number used to identify trunk facilities, 1-65535**

3.4.20 →FBUSY20 and FBUSY24

The **FBUSY** message is sent by the switch when a facility receives forced busy.

3.4.20.1 →FBUSY20 (New CM3.1)

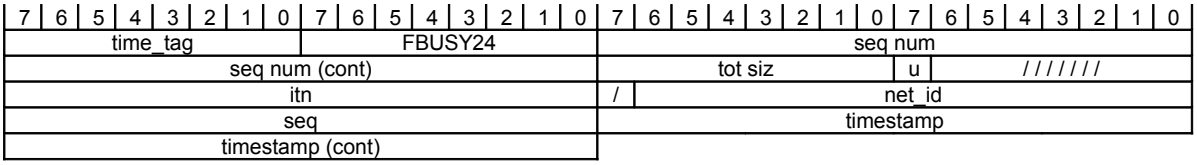
	Switch	Vect	Skill
First Release	CM3.1	y/n	y/n
Last Release	CM6.3		



time_tag	seconds past minute when event occurred, 0-59
opcode	FBUSY20 = 224 (0xe0)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
itn	internal trunk number used to identify trunk facilities, 1-32766
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.20.2 →FBUSY24 (New CM7.0)

	Switch	Vect	Skill
First Release	CM7.0	y/n	y/n
Last Release			



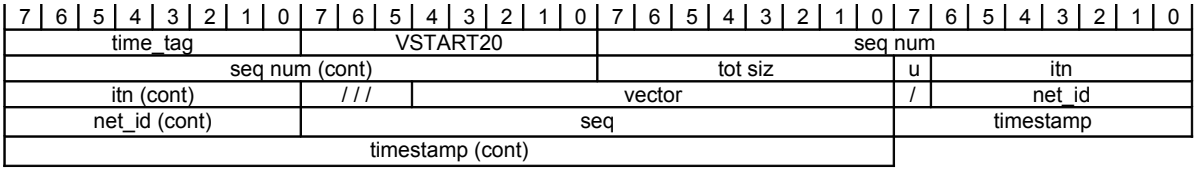
time_tag	seconds past minute when event occurred, 0-59
opcode	FBUSY24 = 73 (0x49)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
itn	**internal trunk number used to identify trunk facilities, 1-65535**
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.21 →VSTART20 and VSTART24

The **VSTART** message is sent by the switch when a facility starts vector processing.

3.4.21.1 →VSTART20 (New CM3.1)

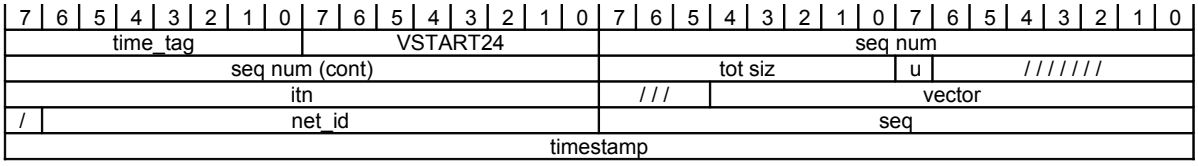
	Switch	Vect	Skill
First Release	CM3.1	y	y/n
Last Release	CM6.3		



time_tag	seconds past minute when event occurred, 0-59
opcode	VSTART20 = 225 (0xe1)
seq	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
itn	internal trunk number used to identify trunk facilities, 1-32766
vector	vector number, 1-999 (G3V8-CM2), 1-2000 (CM3-CM5.2), 1-8000 (CM6.0 and later)
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.21.2 →VSTART24 (New CM7.0)

	Switch	Vect	Skill
First Release	CM7.0	y	y/n
Last Release			



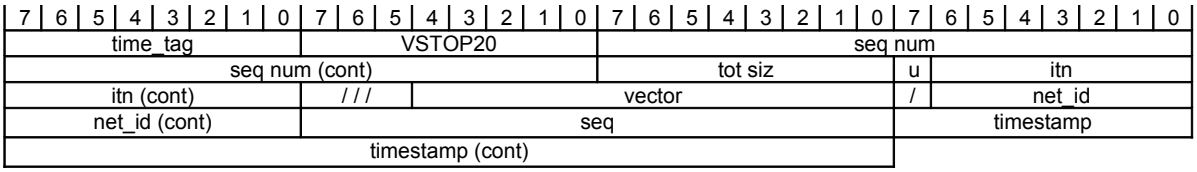
time_tag	seconds past minute when event occurred, 0-59
opcode	VSTART24 = 74 (0x4a)
seq	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
itn	**internal trunk number used to identify trunk facilities, 1-65535**
vector	vector number, 1-999 (G3V8-CM2), 1-2000 (CM3-CM5.2), 1-8000 (CM6.0 and later)
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.22 →VSTOP20 and VSTOP24

The **VSTOP** message is sent by the switch when a facility stops vector processing.

3.4.22.1 →VSTOP20 (New CM3.1)

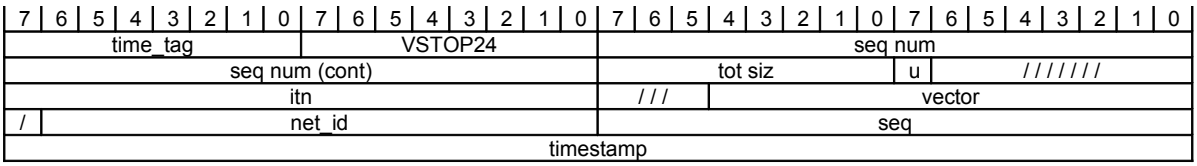
	Switch	Vect	Skill
First Release	CM3.1	y	y/n
Last Release	CM6.3		



time_tag	seconds past minute when event occurred, 0-59
opcode	VSTOP20 = 226 (0xe2)
seq_num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
itn	internal trunk number used to identify trunk facilities, 1-32766
vector	vector number, 1-999 (G3V8-CM2), 1-2000 (CM3-CM5.2), 1-8000 (CM6.0 and later)
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.22.2 →VSTOP24 (New CM7.0)

	Switch	Vect	Skill
First Release	CM7.0	y	y/n
Last Release			



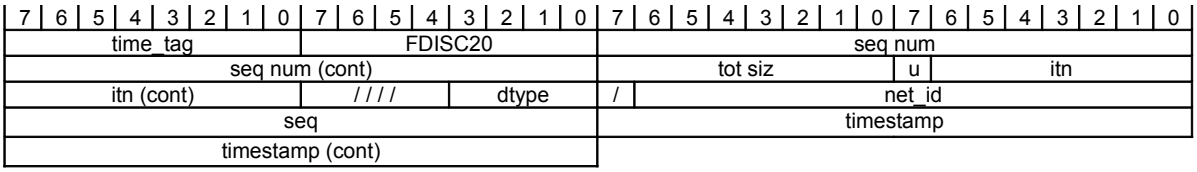
time_tag	seconds past minute when event occurred, 0-59
opcode	VSTOP24 = 75 (0x4b)
seq_num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
itn	**internal trunk number used to identify trunk facilities, 1-65535**
vector	vector number, 1-999 (G3V8-CM2), 1-2000 (CM3-CM5.2), 1-8000 (CM6.0 and later)
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.23 → FDISC20 and FDISC24

The **FDISC20** message is sent by the switch when a facility receives a forced disconnect. Forced disconnects may happen as a result of a "fdisc" vector step, a vector timeout event, or a "reply best" vector step. This message provides the reporting adjunct the ability to associate a vector, VDN, and trunk by the itn in this message and the DNEVENT and VSTART messages.

3.4.23.1 → FDISC20 (New CM3.1)

	Switch	Vect	Skill
First Release	CM3.1	y	y/n
Last Release	CM6.3		



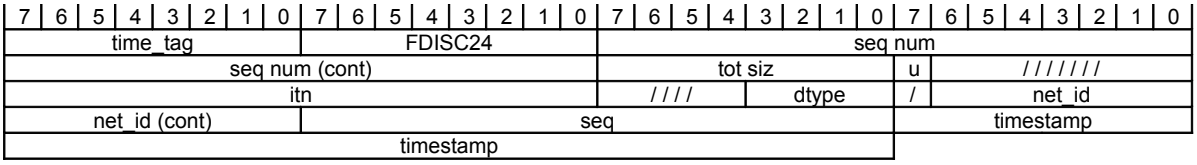
time_tag	seconds past minute when event occurred, 0-59
opcode	FDISC20 = 227 (0xe3)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
itn	internal trunk number used to identify trunk facilities, 1-32766
dtype	type of disconnect

- 0 disconnect vector step executed
- 1 vector timeout
- 2 reply best vector step executed

net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.23.2 **→FDISC24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y	y/n
Last Release			



time_tag seconds past minute when event occurred, 0-59
opcode FDISC24 = 76 (0x4c)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
u flag indicating that the UCID fields (net_id, seq and timestamp) are present
itn **internal trunk number used to identify trunk facilities, 1-65535**
dtype type of disconnect

0 disconnect vector step executed
1 vector timeout
2 reply best vector step executed

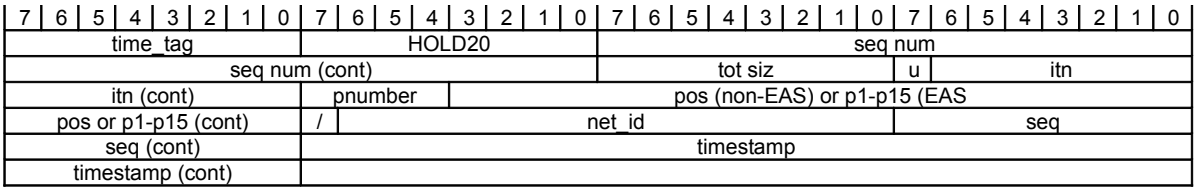
net_id network entity identifier, 1-32767
seq call sequence number, 0-65535
timestamp unix style (seconds since 1/1/70) time call assigned a UCID

3.4.24 → HOLD20 and HOLD24

The **HOLD** message is sent by the switch when a facility is put on "hold" by an agent. This is a variable length message based on "pnumber".

3.4.24.1 → HOLD20 (New CM3.1)

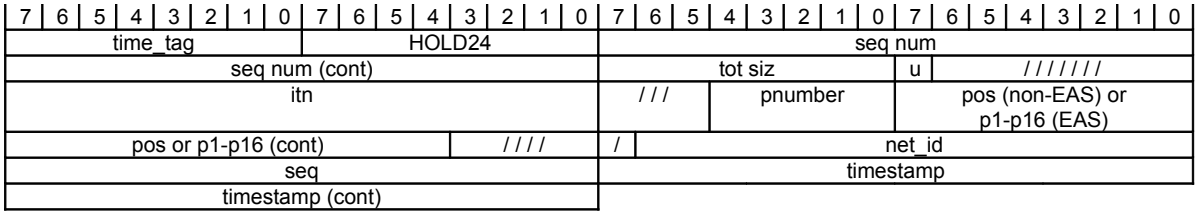
	Switch	Vect	Skill
First Release	CM3.1	y/n	y/n
Last Release	CM6.3		



time_tag	seconds past minute when event occurred, 0-59
opcode	HOLD20 = 228 (0xe4)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
itn	internal trunk number used to identify trunk facilities, 1-32766
pnumber	number of digits in agent position, 1-15, will always be 5 for non-EAS
pos/p1-p15	agent position - without EAS "pos" (1-10000), with EAS "p1-p15" (0-9999999) login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.24.2 **→HOLD24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y/n	y/n
Last Release			



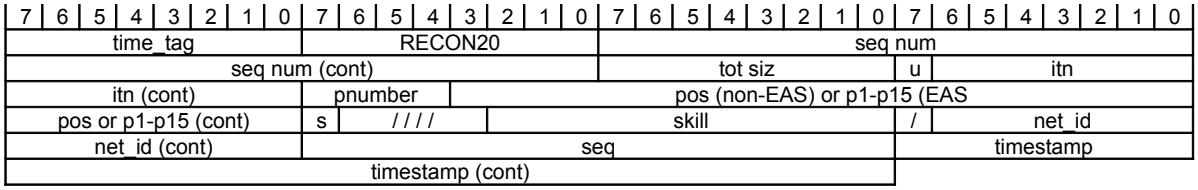
time_tag seconds past minute when event occurred, 0-59
opcode HOLD24 = 77 (0x4d)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
u flag indicating that the UCID fields (net_id, seq and timestamp) are present
itn internal trunk number used to identify trunk facilities, 1-65535
pnumber number of digits in agent position, 1-16, will always be 5 for non-EAS
pos/p1-p16 agent position - without EAS "pos" (1-10000), with EAS "p1-p16" (0-9999999) login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary
net_id network entity identifier, 1-32767
seq call sequence number, 0-65535
timestamp unix style (seconds since 1/1/70) time call assigned a UCID

3.4.25 → RECON20, RECON23 and RECON24

The **RECON** message is sent by the switch when a facility is reconnected after being put on "hold" by an agent. This is a variable length message based on "pnumber" and the "s" and "u" bits. The "s" bit is only set for EAS and only when reconnecting to a call for a CMS measured skill. When the "s" bit is set, the message will contain one additional byte for the "skill" field. Without EAS the first 2 bits of the "skill" field are set to 0. The message may optionally contain the UCID.

3.4.25.1 → RECON20 (New CM3.1)

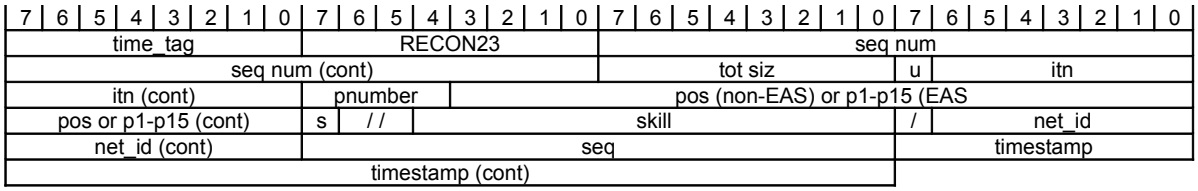
	Switch	Vect	Skill
First Release	CM3.1	y/n	y/n
Last Release	CM5.2		



time_tag	seconds past minute when event occurred, 0-59
opcode	RECON20 = 229 (0xe5)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
itn	internal trunk number used to identify trunk facilities, 1-32766
pnumber	number of digits in agent position, 1-15, will always be 5 for non-EAS
pos/p1-p15	agent position - without EAS "pos" (1-10000), with EAS "p1-p15" (0-9999999) login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary
s	0 = no skill in message, 1 = skill in message
skill	skill matched 1-999 (G3V8-G3V11), 1-2000 (MV2 and later)
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.25.2 **→RECON23 (New CM6.0)**

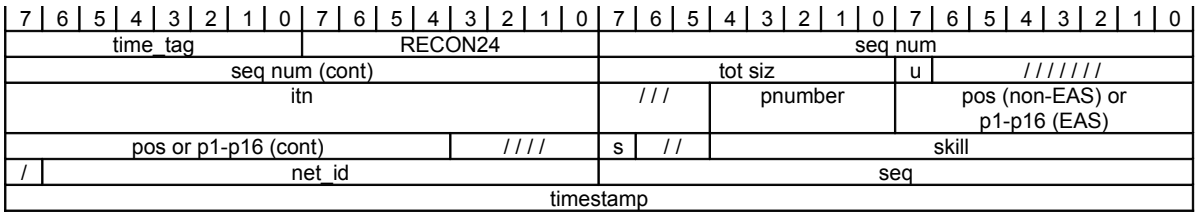
	Switch	Vect	Skill
First Release	CM6.0	y/n	y/n
Last Release	CM6.3		



time_tag	seconds past minute when event occurred, 0-59
opcode	RECON23 = 30 (0x1e)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
itn	internal trunk number used to identify trunk facilities, 1-32766
pnumber	number of digits in agent position, 1-15, will always be 5 for non-EAS
pos/p1-p15	agent position - without EAS "pos" (1-10000), with EAS "p1-p15" login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary
s	0 = no skill in message, 1 = skill in message
skill	skill matched 1-8000
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.25.3 **→RECON24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y/n	y/n
Last Release			



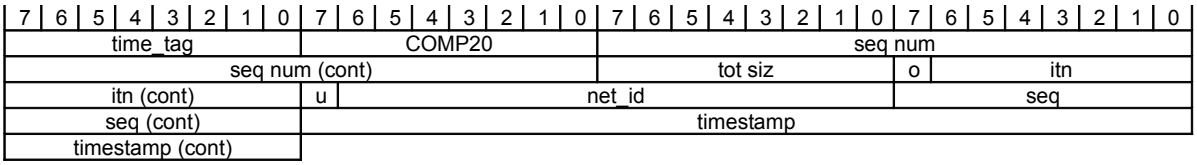
time_tag	seconds past minute when event occurred, 0-59
opcode	RECON24 = 78 (0x4e)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
itn	**internal trunk number used to identify trunk facilities, 1-65535**
pnumber	**number of digits in agent position, 1-16, will always be 5 for non-EAS**
pos/p1-p16	**agent position - without EAS "pos" (1-10000), with EAS "p1-p16" login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary**
s	0 = no skill in message, 1 = skill in message
skill	skill matched 1-8000
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.26 → COMP20 and COMP24

The **COMP** message is sent by the switch when a trunk's far end has answered (outbound call has been completed).

3.4.26.1 → COMP20 (New CM3.1)

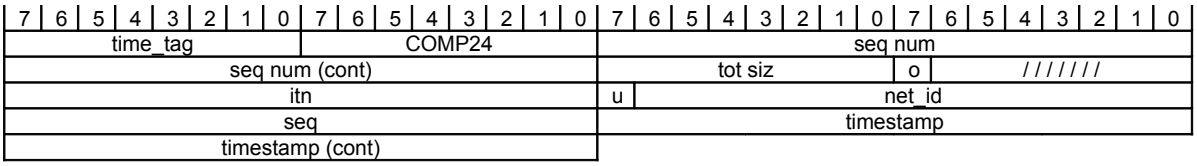
	Switch	Vect	Skill
First Release	CM3.1	y/n	y/n
Last Release	CM6.3		



time_tag	seconds past minute when event occurred, 0-59
opcode	COMP20 = 230 (0xe6)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
o	OCM call; 0 = no, 1 = call distributed by ocm
itn	internal trunk number used to identify trunk facilities, 1-32766
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.26.2 → COMP24 (New CM7.0)

	Switch	Vect	Skill
First Release	CM7.0	y/n	y/n
Last Release			



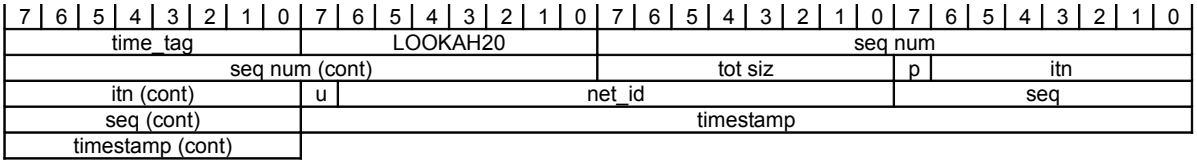
time_tag	seconds past minute when event occurred, 0-59
opcode	COMP24 = 79 (0x4f)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
o	OCM call; 0 = no, 1 = call distributed by ocm
itn	**internal trunk number used to identify trunk facilities, 1-65535**
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.27 →LOOKAH20 and LOOKAH24

The **LOOKAH** message is sent by the switch when a facility attempts to lookahead interflow (LAI), BSR poll (p bit is set for poll), BSR interflow or NCR (NCD or ENCT) is invoked. This is followed by a DFWD if the interflow (LAI or BSR) or NCR network redirection is successful.

3.4.27.1 →LOOKAH20 (New CM3.1)

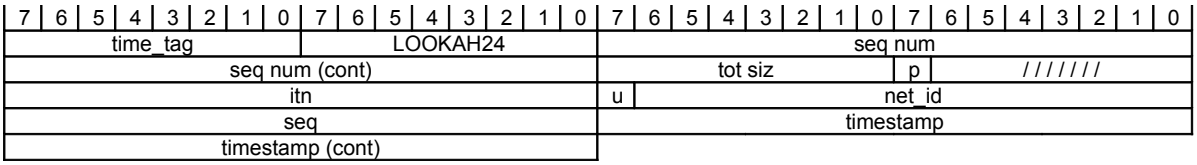
	Switch	Vect	Skill
First Release	CM3.1	y	y/n
Last Release			



time_tag	seconds past minute when event occurred, 0-59
opcode	LOOKAH20 = 231 (0xe7)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
p	poll bit indicating lookahead is for BSR polling
itn	internal trunk number used to identify trunk facilities, 1-32766
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.27.2 →LOOKAH24 (New CM7.0)

	Switch	Vect	Skill
First Release	CM3.1	y	y/n
Last Release			



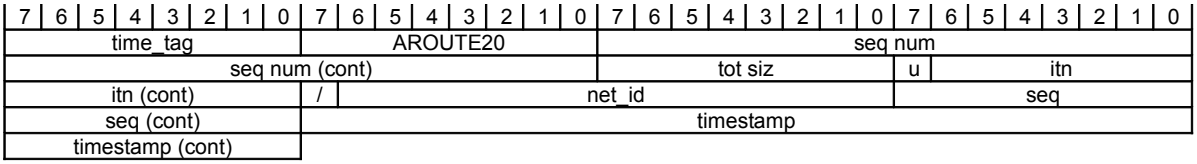
time_tag	seconds past minute when event occurred, 0-59
opcode	LOOKAH24 = 80 (0x50)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
p	poll bit indicating lookahead is for BSR polling
itn	**internal trunk number used to identify trunk facilities, 1-65535**
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.28 → AROUTE20 and AROUTE24

The **AROUTE20** message is sent by the switch when a facility requests adjunct routing.

3.4.28.1 → AROUTE20 (New CM3.1)

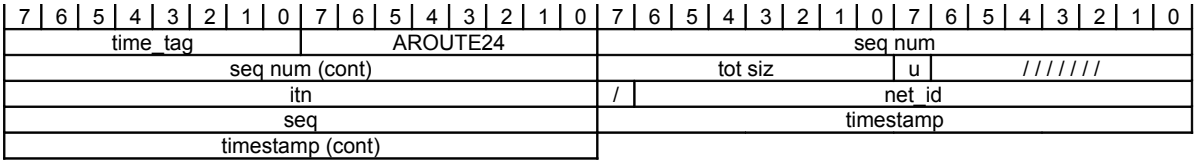
	Switch	Vect	Skill
First Release	CM3.1	y	y/n
Last Release	CM6.3		



time_tag	seconds past minute when event occurred, 0-59
opcode	AROUTE20 = 232 (0xe8)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
itn	internal trunk number used to identify trunk facilities, 1-32766
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.28.2 → AROUTE24 (New CM7.0)

	Switch	Vect	Skill
First Release	CM7.0	y	y/n
Last Release			



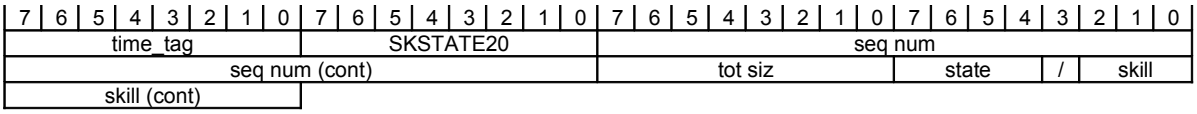
time_tag	seconds past minute when event occurred, 0-59
opcode	AROUTE24 = 81 (0x51)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
itn	**internal trunk number used to identify trunk facilities, 1-65535**
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.29 → SKSTATE20 and SKSTATE23

The **Skill State** message is sent by the switch when a skill changes state with respect to its Service Level Supervisor threshold levels. This message is also sent during a full translation pumpup and a logon/status pumpup for any measured skills. Both forms of pumpup include this status information.

3.4.29.1 → SKSTATE20 (New CM3.1)

	Switch	Vect	Skill
First Release	CM3.1	y	y
Last Release	CM5.2		



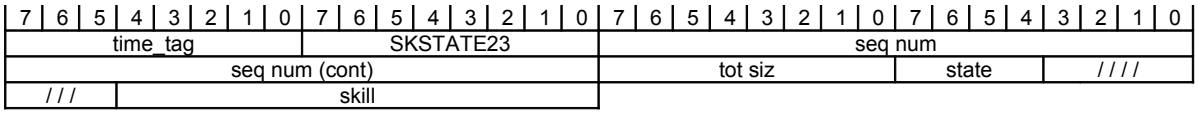
time_tag seconds past minute when message was formatted, 0-59
opcode SKSTATE20 = 233 (0xe9)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
state skill service level state (0 used by CMS to mean UNKNOWN)

State	Meaning
1	normal
2	over threshold one (overload1)
3	over threshold two (overload2)
4	behind target percent (MV2 and later)
5	auto reserving (MV2 and later)

skill skill number 1-999 (G3V8-G3V11), 1-2000 (MV2 and later)

3.4.29.2 **→SKSTATE23 (New CM6.0)**

	Switch	Vect	Skill
First Release	CM6.0	y	y
Last Release			



time_tag seconds past minute when message was formatted, 0-59
opcode SKSTATE23 = 31 (0x1f)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
state skill service level state (0 used by CMS to mean UNKNOWN)

State	Meaning
1	normal
2	over threshold one (overload1)
3	over threshold two (overload2)
4	behind target percent (MV2 and later)
5	auto reserving (MV2 and later)

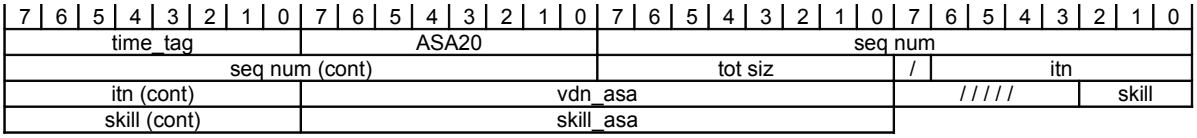
skill skill number 1-8000

3.4.30 → ASA20 and ASA23

This message is sent by the switch whenever the average speed of answer value changes, and is used to update the switch-based rolling average-speed-of-answer (ASA) status information that the reporting adjunct stores for each split/skill and VDN.

3.4.30.1 → ASA20 (New CM3.1)

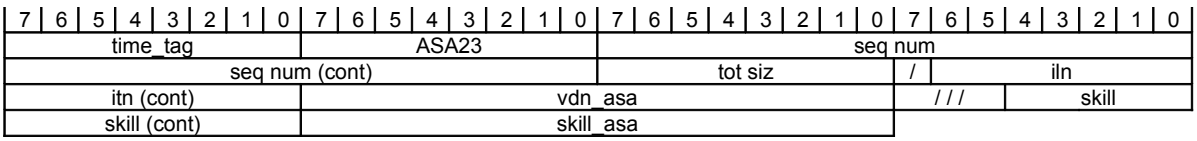
	Switch	Vect	Skill
First Release	CM3.1	y/n	y/n
Last Release	CM5.2		



time_tag seconds past minute when event occurred, 0-59
opcode ASA20 = 234 (0xea)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
iln internal line number for VDN, 1-32766; 0 means no vdn_asa included
vdn_asa average speed of answer for calls to the VDN, 0-65535 seconds
skill skill number, 1-999 (G3V8-G3V11), 1-2000 (MV2 and later), 0 means no skill_asa value is set
skill_asa average speed of answer for calls to the skill, 0-65535 seconds

3.4.30.2 → ASA23 (New CM6.0)

	Switch	Vect	Skill
First Release	CM6.0	y/n	y/n
Last Release	CM6.3		



time_tag seconds past minute when event occurred, 0-59
opcode ASA23 = 32 (0x20)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
iln internal line number for VDN, 1-32766; 0 means no vdn_asa included
vdn_asa average speed of answer for calls to the VDN, 0-65535 seconds
skill skill number, 1-8000 (CM6.0 and later), 0 means no skill_asa value is set
skill_asa average speed of answer for calls to the skill, 0-65535 seconds

3.4.31 → DNEVENT20 and DNEVENT24

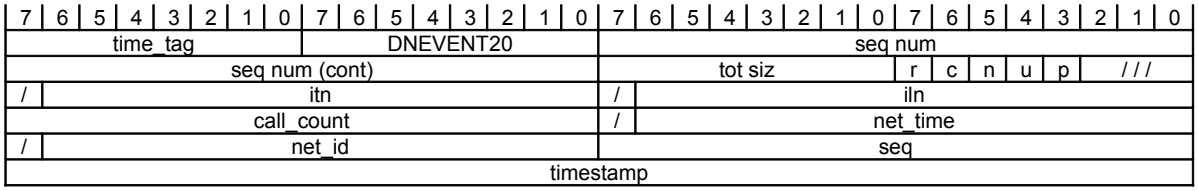
The **VDN Event** message is sent by the switch when a trunk is used to process a VDN call. It reports the association between the dialed VDN and the trunk that is carrying the call. For both measured and unmeasured trunks, this message will be preceded by a **SETUP**. The next message expected for the trunk will be a start vector processing message (**VSTART**).

This is a variable length message based on the "c", "n" and "u" bits. The first twelve bytes will always be sent. The message may optionally contain the VDN call count, which represents the current number of active calls that are associated with the VDN. The VDN call count is incremented and decremented by the switch, but is stored and reported by the reporting adjunct. The message may optionally contain the time spent by the call in VDN processing elsewhere in the network. The message may also optionally contain the UCID.

The message may contain any one or more: call_count, net_time and UCID. Regardless of which of these fields are present, they will always be in the following order: call_count, net_time, UCID.

3.4.31.1 → DNEVENT20 (New CM3.1)

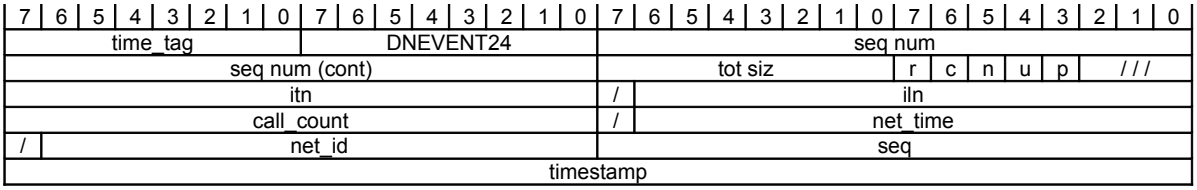
	Switch	Vect	Skill
First Release	CM3.1	y	y/n
Last Release	CM6.3		



time_tag	seconds past minute when event occurred, 0-59
opcode	DNEVENT20 = 235 (0xeb)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
r	return to VDN, 1=yes, 0=no
c	call count flag - indicates active calls present for this VDN
n	net time flag - indicates call has been interflowed from another switch in the network
u	UCID flag - indicates that UCID fields (net_id, seq and timestamp) are present
p	POM (Proactive Outreach Manager) VDN flag, 1=yes, 0=no (CM6.2 and later)
itn	internal trunk number, 1-32766
iln	internal line number (VDN), 1-32766
call_count	active calls associated with this VDN, 0-7712
net_time	time spent in VDN processing at prior sending switch(es), 0-32766 (times greater than 32766 seconds are truncated to 32766)
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.31.2 **→DNEVENT24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y	y/n
Last Release			



time_tag	seconds past minute when event occurred, 0-59
opcode	DNEVENT24 = 83 (0x53)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
r	return to VDN, 1=yes, 0=no
c	call count flag - indicates active calls present for this VDN
n	net time flag - indicates call has been interflowed from another switch in the network
u	UCID flag - indicates that UCID fields (net_id, seq and timestamp) are present
p	POM (Proactive Outreach Manager) VDN flag, 1=yes, 0=no (CM6.2 and later)
itn	**internal trunk number, 1-65535**
iln	internal line number (VDN), 1-32766
call_count	active calls associated with this VDN, 0-7712
net_time	time spent in VDN processing at prior sending switch(es), 0-32766 (times greater than 32766 seconds are truncated to 32766)
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

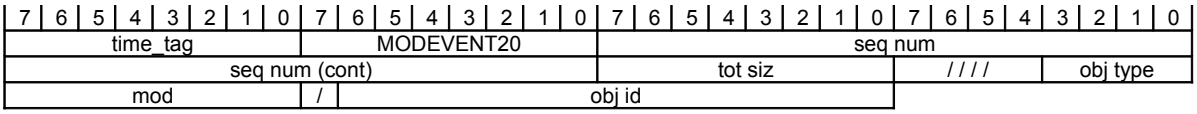
3.4.32 → MODEVENT20 (New CM3.1)

	Switch	Vect	Skill
First Release	CM3.1	y	y/n
Last Release			

The **Modification Event** message is sent by the switch when an entity on the switch has had its administration changed and its modification number incremented. Entities included in this protocol are measured VDNs, vectors, and BSR application plans.

The modification number for a measured VDN is incremented and the **MODEVENT20** message is sent whenever the assigned vector or the assigned skills change.

The modification number for a vector is incremented and the **MODEVENT20** message is sent whenever the vector is modified. If a vector is modified, this means that something in one or more steps has changed. This includes a successful submission of the vector form on the SAT, and a successful processing of the **CHGVEC** message from the reporting adjunct.



time_tag	seconds past minute when event occurred, 0-59
opcode	MODEVENT20 = 236 (0xec)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
obj type	type of object which changed: 0=vector, 1=VDN
mod	modification tag number, 0-255
obj id	identifier for object

obj type	entity	obj id values
0	vector	1-999 (G3V8-CM2), 1-2000 (CM3-CM5.2), 1-8000 (CM6.0 and later)
1	VDN (iln)	1-32767

3.4.33 ←RAUDIT and RAUDIT24

Whenever the reporting adjunct receives a progress mark or state change (**FSTATE**) for an unmeasured trunk, it sets an AT4 timer (5 minutes). If this timer expires, the reporting adjunct will send an **RAUDIT** message to the switch and reset the AT4 timer. The expected response from the switch is an *AUDIT/TAUDIT* message for the requested trunk facility (internal trunk number).

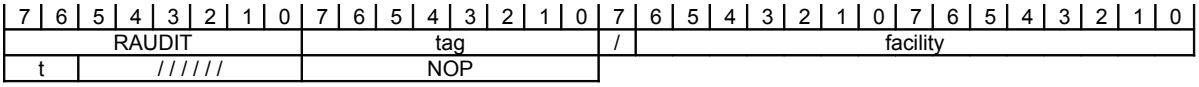
Whenever the reporting adjunct receives a DNEVENT message with a VDN count, it sets an AT6 timer (default 2 minutes) for the VDN. If this timer expires, the reporting adjunct will send an **RAUDIT** message to the switch and reset the timer. The expected response from the switch is a *VAUDIT* message for the requested VDN (internal line number).

The reporting adjunct checks the EWT status of a split/skill based on an AT5 (default 2 minute) timer. If any non-NULL EWT values for that split/skill have not been updated by a **QUEUED** or **DFWD**-to-split message since the timer was last reset and the EWT values cannot be inferred, then an **RAUDIT** message with t=1 is sent to the switch. The reporting adjunct will expect an **EWTAUDIT** message with the updated expected wait times for the requested split/skill.

The EWT for a particular skill will be updated via an **EWTAUDIT** to the reporting adjunct when a call comes out of queue (calculated using the top position, CM5.2.1 and later) or when the last agent logs out of the skill (CM6.2 and later).

3.4.33.1 ←RAUDIT

	Switch	Vect	Skill
First Release	G3V1	y/n	y/n
Last Release	CM6.3		

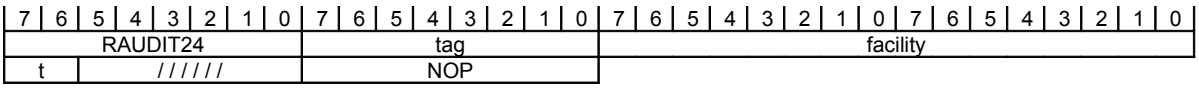


opcode RAUDIT = 8 (0x08)
tag message tag, 0-255 (it will be returned in the switch response message)
facility t=1 split/skill number
t=2 internal trunk number, 1-32766
t=3 internal line number, 1-32766
t audit type

1 EWT (Expected Wait Time) (facility = split/skill)
2 unmeasured trunk facility (facility = itn)
3 VDN counted calls (facility = iln)

3.4.33.2 **← RAUDIT24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y/n	y/n
Last Release			



opcode RAUDIT24 = 66 (0x42)
tag message tag, 0-255 (it will be returned in the switch response message)
facility
- t=1 split/skill number
- **t=2 internal trunk number, 1-65535**
- t=3 internal line number, 1-32766

t audit type

1. EWT (Expected Wait Time) (facility = split/skill)
2. unmeasured trunk facility (facility = itn)
3. VDN counted calls (facility = iln)

3.4.34 → TAUDIT20 and TAUDIT24

(Replacements for AAUDIT, EWTAUDIT and VAUDIT)

NOTE: IQ can receive an **EWTAUDIT** or **VAUDIT**. These items will be used to update the database for expected wait time or VDN calls-counted (active calls). IQ does not currently process any other audit messages.

An **Audit** message is sent by the switch as part of a periodic task or at the request of the reporting adjunct. The audit is designed to ensure that the reporting adjunct database reflects the current status of the switch database.

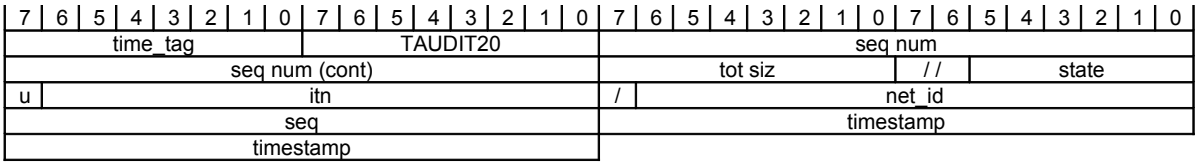
The switch audits the measured agents and facilities every 10 minutes and sends **AAUDIT** equivalent or **TAUDIT** messages.

When the reporting adjunct receives an **AAUDIT** equivalent or **TAUDIT** message, it compares the status of the agent or facility with that in the reporting adjunct real-time database. If a discrepancy is found, the reporting adjunct will log an error, abort call recording, initialize the agent position or trunk associated with the call, and then wait for the next initializing event before continuing to track the agent or facility.

If the reporting adjunct receives an audit message inconsistent with the current state of the trunk or agent, the reporting adjunct changes the state to OTHER for agent inconsistencies and UNKNOWN for trunk inconsistencies. When the state changes or another audit is received, the reporting adjunct updates the agent/trunk to the switch state.

3.4.34.1 **→TAUDIT20 (New CM3.1)**

	Switch	Vect	Skill
First Release	CM3.1	y/n	y/n
Last Release	CM6.3		



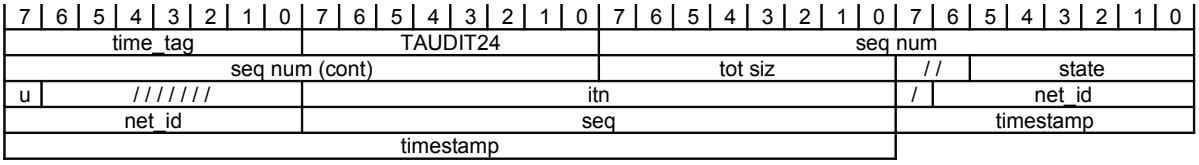
time_tag seconds past minute when event occurred, 0-59
opcode TAUDIT20 = 238 (0xee)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
state current state

0	IDLE	trunk is available for calls (in idle pool)
2	QUEUED	trunk is queued to a split
3	DCON	trunk is connected to an agent position
6	MBUSY	trunk is maintenance-busy
9	OTHER	trunk is SEIZED, DABN, DFWD, FBUSY or FDISC actual state can be sent, but OTHER is acceptable
15	HOLD	trunk is on-hold at an agent position

u flag indicating that the UCID fields (net_id, seq and timestamp) are present
itn internal trunk number, 1-32766
net_id network entity identifier, 1-32767
seq call sequence number, 0-65535
timestamp unix style (seconds since 1/1/70) time call assigned a UCID

3.4.34.2 **→TAUDIT24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y/n	y/n
Last Release			



time_tag seconds past minute when event occurred, 0-59
opcode TAUDIT24 = 84 (0x54)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
state current state

0	IDLE	trunk is available for calls (in idle pool)
2	QUEUED	trunk is queued to a split
3	DCON	trunk is connected to an agent position
6	MBUSY	trunk is maintenance-busy
9	OTHER	trunk is SEIZED, DABN, DFWD, FBUSY or FDISC actual state can be sent, but OTHER is acceptable
15	HOLD	trunk is on-hold at an agent position

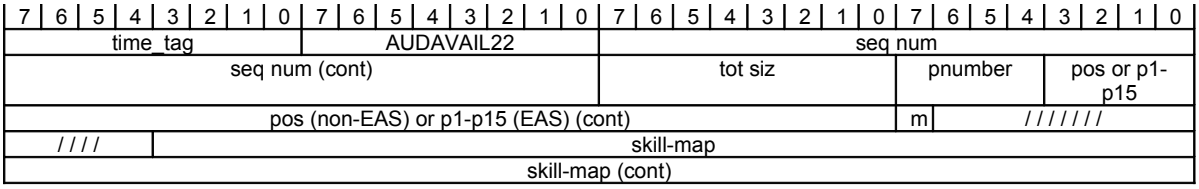
u flag indicating that the UCID fields (net_id, seq and timestamp) are present
itn **internal trunk number, 1-65535**
net_id network entity identifier, 1-32767
seq call sequence number, 0-65535
timestamp unix style (seconds since 1/1/70) time call assigned a UCID

3.4.35 → AUDAVAIL22, AUDAVAIL23 and AUDAVAIL24

This is a variable length message based on "pnumber" and whether or not EAS is active. Prior to **AUDAVAIL23**, with EAS, this message will contain 8 additional bytes for the "skill-map". Starting with **AUDAVAIL23** and the expansion to 120 skills, with EAS, this message will contain 16 additional bytes for the expanded “skill-map”.

3.4.35.1 →AUDAVAIL22 (New CM5.2)

	Switch	Vect	Skill
First Release	CM5.2	y/n	y/n
Last Release	CM5.2		



time_tag — seconds past minute when event occurred, 0-59

opcode — AUDAVAIL22 = 2 (0x02)

seq num — sequence number, 0-0xffff ffff

tot siz — total size in bytes, 1-255

pnumber — number of digits in agent position, 1-15, will always be 5 for non-EAS

pos/p1-p15 — agent position - without EAS "pos" (1-10000), with EAS "p1-p15" (0-9999999) login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary

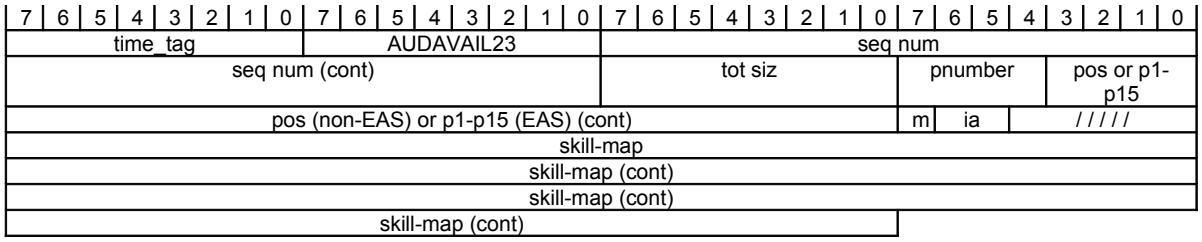
m — flag to indicate the agent returned to available because of Multiple Call Handling. This is for IQ 5.0 support starting with CM 5.2.

skill-map — one bit for each staffed skill. The skills are mapped left to right and correspond to the order of the skills in the **ALOGON** message.

- 1 means agent state applies for this skill
- 0 means agent state does not apply for this skill or the skill is not assigned/measured; CMS will report the agent activity for this skill as OTHER.

3.4.35.2 **→AUDAVAIL23 (New CM6.0)**

	Switch	Vect	Skill
First Release	CM6.0	y/n	y/n
Last Release	CM6.3		



time_tag seconds past minute when event occurred, 0-59

opcode AUDAVAIL23 = 8 (0x08)

seq num sequence number, 0-0xffff ffff

tot siz total size in bytes, 1-255

pnumber number of digits in agent position, 1-15, will always be 5 for non-EAS

pos/p1-p15 agent position - without EAS "pos" (1-10000), with EAS "p1-p15" login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary

m flag to indicate the agent returned to available because of Multiple Call Handling. This is for IQ 5.0 support starting with CM 5.2.

ia flag indicating agent became available due to being interrupted while in Aux with an Interruptible Aux reason code.

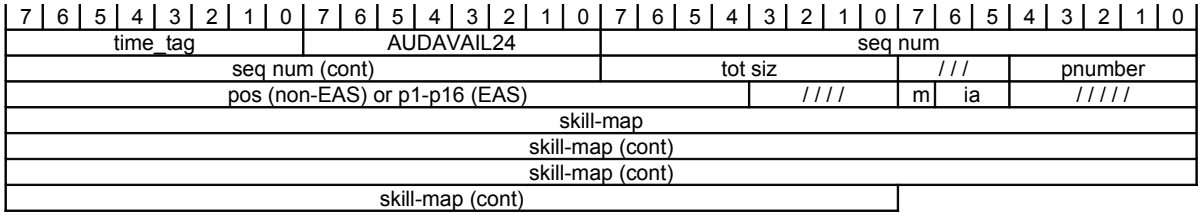
0 N/A (agent was not interrupted while in Aux)
1 auto-in-interrupt
2 manual-in-interrupt
3 notify-interrupt

skill-map one bit for each staffed skill (1-120). The skills are mapped left to right and correspond to the order of the skills in the **ALOGON** message.

- 1 means agent state applies for this skill
- 0 means agent state does not apply for this skill or the skill is not assigned/measured; CMS will report the agent activity for this skill as OTHER.

3.4.35.3 **→AUDAVAIL24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y/n	y/n
Last Release			



time_tag seconds past minute when event occurred, 0-59
opcode AUDAVAIL24 = 85 (0x55)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
pnumber number of digits in agent position, 1-16, will always be 5 for non-EAS
pos/p1-p16 agent position - without EAS "pos" (1-10000), with EAS "p1-p16" login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary
m flag to indicate the agent returned to available because of Multiple Call Handling. This is for IQ 5.0 support starting with CM 5.2.
ia flag indicating agent became available due to being interrupted while in Aux with an Interruptible Aux reason code.

0 N/A (agent was not interrupted while in Aux)
1 auto-in-interrupt
2 manual-in-interrupt
3 notify-interrupt

skill-map one bit for each staffed skill (1-120). The skills are mapped left to right and correspond to the order of the skills in the **ALOGON** message.

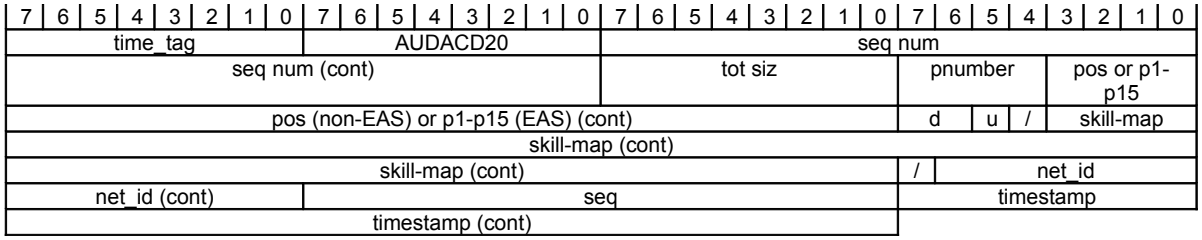
- 1 means agent state applies for this skill
- 0 means agent state does not apply for this skill or the skill is not assigned/measured; CMS will report the agent activity for this skill as OTHER.

3.4.36 → AUDACD20, AUDACD23 and AUDACD24

This is a variable length message based on "pnumber" and whether or not EAS is active. Prior to **AUDACD23**, with EAS, this message will contain 7 additional bytes for the "skill-map" and without EAS since the d/u flag byte still exists, the first 4 bits of the "skill-map" are set to 0. Starting with **AUDACD23** and the expansion to 120 skills, with EAS, this message will contain 15 additional bytes for the expanded "skill-map".

3.4.36.1 → AUDACD20 (New CM3.1)

	Switch	Vect	Skill
First Release	CM3.1	y/n	y/n
Last Release	CM5.2		



time_tag — seconds past minute when event occurred, 0-59

opcode — AUDACD20 = 240 (0xf0)

seq num — sequence number, 0-0xffff ffff

tot siz — total size in bytes, 1-255

pnumber — number of digits in agent position, 1-15, will always be 5 for non-EAS

pos/p1-p15 — agent position - without EAS "pos" (1-10000), with EAS "p1-p15" (0-9999999) login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary

d — direction of call
- 1 means call is outbound (ACDO)
- 2 means call is inbound (ACDI)

Note: this information is not available with AAUDIT.

u — flag indicating that the UCID fields (net_id, seq and timestamp) are present

skill-map — one bit for each staffed skill. The skills are mapped left to right and correspond to the order of the skills in the **ALOGON** message.
- 1 means agent state applies for this skill
- 0 means agent state does not apply for this skill or the skill is not assigned/measured; the reporting adjunct will report the agent activity for this skill as OTHER.

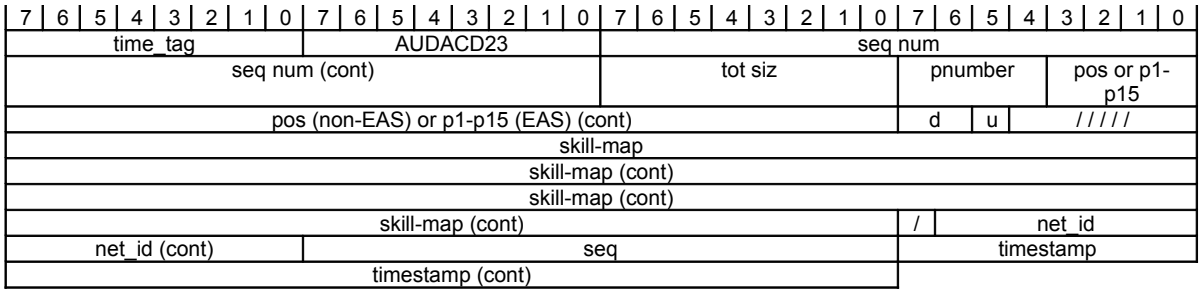
net_id — network entity identifier, 1-32767

seq — call sequence number, 0-65535

timestamp — unix style (seconds since 1/1/70) time call assigned a UCID

3.4.36.2 **→AUDACD23 (New CM6.0)**

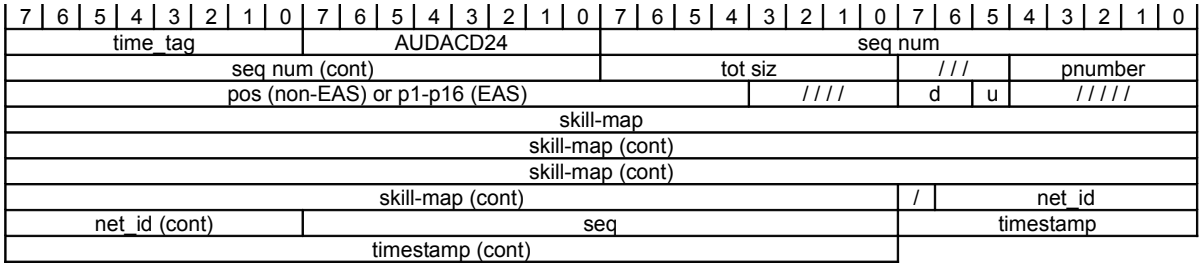
	Switch	Vect	Skill
First Release	CM6.0	y/n	y/n
Last Release	CM6.3		



time_tag	seconds past minute when event occurred, 0-59
opcode	AUDACD23 = 6 (0x06)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
pnumber	number of digits in agent position, 1-15, will always be 5 for non-EAS
pos/p1-p15	agent position - without EAS "pos" (1-10000), with EAS "p1-p15" login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary
d	direction of call – 1 means call is outbound (ACDO) – 2 means call is inbound (ACDI) Note: this information is not available with AAUDIT.
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
skill-map	one bit for each staffed skill (1-120). The skills are mapped left to right and correspond to the order of the skills in the ALOGON message. – 1 means agent state applies for this skill – 0 means agent state does not apply for this skill or the skill is not assigned/measured; the reporting adjunct will report the agent activity for this skill as OTHER.
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.36.3 **→AUDACD24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y/n	y/n
Last Release			



time_tag — seconds past minute when event occurred, 0-59
opcode — AUDACD24 = 86 (0x56)
seq num — sequence number, 0-0xffff ffff
tot siz — total size in bytes, 1-255
pnumber — number of digits in agent position, 1-16, will always be 5 for non-EAS
pos/p1-p16 — agent position - without EAS "pos" (1-10000), with EAS "p1-p16" login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary
d — direction of call
- 1 means call is outbound (ACDO)
- 2 means call is inbound (ACDI)

Note: this information is not available with AAUDIT.

u — flag indicating that the UCID fields (net_id, seq and timestamp) are present
skill-map — one bit for each staffed skill (1-120). The skills are mapped left to right and correspond to the order of the skills in the ALOGON message.
- 1 means agent state applies for this skill
- 0 means agent state does not apply for this skill or the skill is not assigned/measured; the reporting adjunct will report the agent activity for this skill as OTHER.

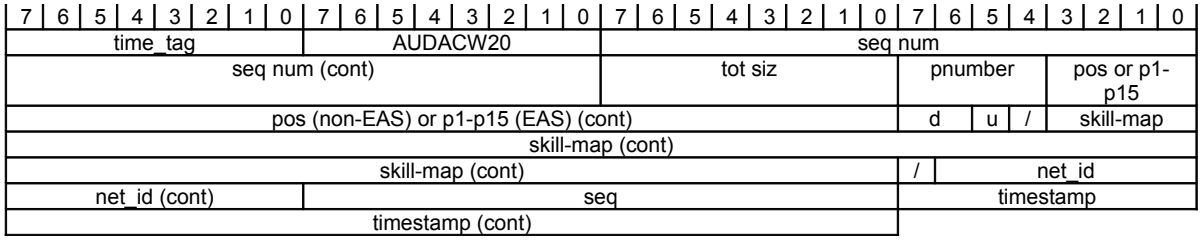
net_id — network entity identifier, 1-32767
seq — call sequence number, 0-65535
timestamp — unix style (seconds since 1/1/70) time call assigned a UCID

3.4.37 → AUDACW20, AUDACW23 and AUDACW24

This is a variable length message based on "pnumber" and whether or not EAS is active. Prior to **AUDACW23**, with EAS, this message will contain 7 additional bytes for the "skill-map" and without EAS (since the d/u flag byte still exists) the first 4 bits of the "skill-map" are set to 0. Starting with **AUDACD23** and the expansion to 120 skills, with EAS, this message will contain 15 additional bytes for the expanded "skill-map".

3.4.37.1 → AUDACW20 (New CM3.1)

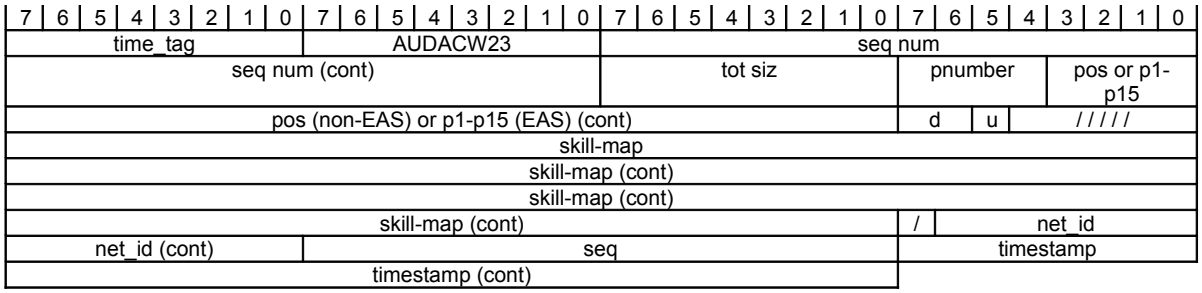
	Switch	Vect	Skill
First Release	CM3.1	y/n	y/n
Last Release	CM5.2		



time_tag	seconds past minute when event occurred, 0-59
opcode	AUDACW20 = 241 (0xf1)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
pnumber	number of digits in agent position, 1-15, will always be 5 for non-EAS
pos/p1-p15	agent position - without EAS "pos" (1-10000), with EAS "p1-p15" (0-9999999) login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary
d	direction of call – 0 means no call (ACW) – 1 means call is outbound (ACWO) – 2 means call is inbound (ACWI) Note: this information is not available with AAUDIT.
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
skill-map	one bit for each staffed skill. The skills are mapped left to right and correspond to the order of the skills in the **ALOGON** message. – 1 means agent state applies for this skill – 0 means agent state does not apply for this skill or the skill is not assigned/measured; the reporting adjunct will report the agent activity for this skill as OTHER.
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.37.2 **→AUDACW23 (New CM6.0)**

	Switch	Vect	Skill
First Release	CM6.0	y/n	y/n
Last Release	CM6.3		



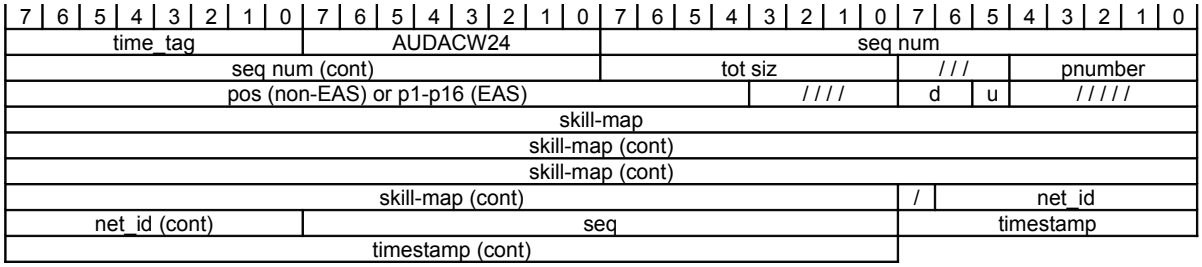
time_tag seconds past minute when event occurred, 0-59
opcode AUDACW23 = 7 (0x07)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
pnumber number of digits in agent position, 1-15, will always be 5 for non-EAS
pos/p1-p15 agent position - without EAS "pos" (1-10000), with EAS "p1-p15" login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary
d direction of call
- 0 means no call (ACW)
- 1 means call is outbound (ACWO)
- 2 means call is inbound (ACWI)

Note: this information is not available with AAUDIT.
u flag indicating that the UCID fields (net_id, seq and timestamp) are present
skill-map one bit for each staffed skill (1-120). The skills are mapped left to right and correspond to the order of the skills in the **ALOGON** message.
- 1 means agent state applies for this skill
- 0 means agent state does not apply for this skill or the skill is not assigned/measured; the reporting adjunct will report the agent activity for this skill as OTHER.

net_id network entity identifier, 1-32767
seq call sequence number, 0-65535
timestamp unix style (seconds since 1/1/70) time call assigned a UCID

3.4.37.3 **→AUDACW24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y/n	y/n
Last Release			



time_tag seconds past minute when event occurred, 0-59
opcode AUDACW24 = 87 (0x57)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
pnumber number of digits in agent position, 1-16, will always be 5 for non-EAS
pos/p1-p16 agent position - without EAS "pos" (1-10000), with EAS "p1-p16" login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary
d direction of call
- 0 means no call (ACW)
- 1 means call is outbound (ACWO)
- 2 means call is inbound (ACWI)

Note: this information is not available with AAUDIT.

u flag indicating that the UCID fields (net_id, seq and timestamp) are present
skill-map one bit for each staffed skill (1-120). The skills are mapped left to right and correspond to the order of the skills in the **ALOGON** message.
- 1 means agent state applies for this skill
- 0 means agent state does not apply for this skill or the skill is not assigned/measured; the reporting adjunct will report the agent activity for this skill as OTHER.

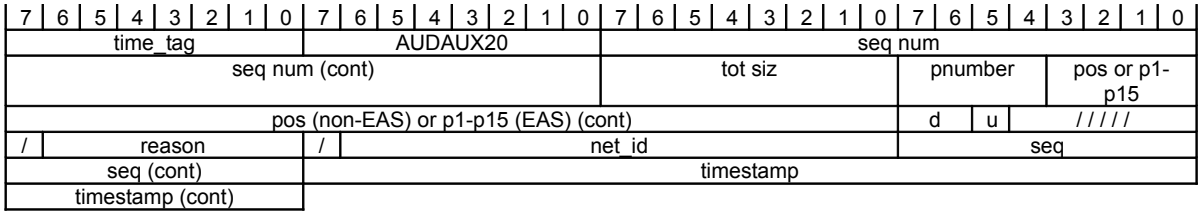
net_id network entity identifier, 1-32767
seq call sequence number, 0-65535
timestamp unix style (seconds since 1/1/70) time call assigned a UCID

3.4.38 → AUDAUX20, AUDAUX23 and AUDAUX24

This is a variable length message based on "pnumber".

3.4.38.1 → AUDAUX20 (New CM3.1)

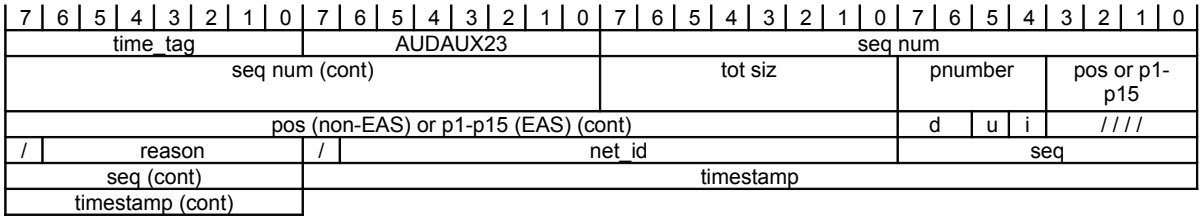
	Switch	Vect	Skill
First Release	CM3.1	y/n	y/n
Last Release	CM5.2		



time_tag	seconds past minute when event occurred, 0-59
opcode	AUDAUX20 = 242 (0xf2)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
pnumber	number of digits in agent position, 1-15, will always be 5 for non-EAS
pos/p1-p15	agent position - witout EAS "pos" (1-10000), with EAS "p1-p15" (0-9999999) login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary
d	direction of call – 0 means no call (AUX) – 1 means call is outbound (AUXO) – 2 means call is inbound (AUXI) Note: this information is not available with AAUDIT.
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
reason	reason code (0-99); 0 is the default reason code.
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.38.2 **→AUDAUX23 (New CM6.0)**

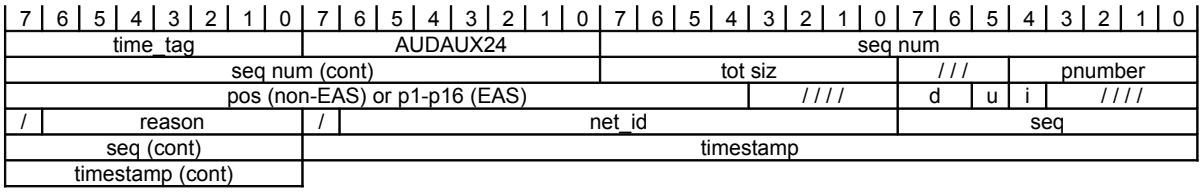
	Switch	Vect	Skill
First Release	CM6.0	y/n	y/n
Last Release	CM6.3		



time_tag	seconds past minute when event occurred, 0-59
opcode	AUDAUX23 = 16 (0x10)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
pnumber	number of digits in agent position, 1-15, will always be 5 for non-EAS
pos/p1-p15	agent position - witout EAS "pos" (1-10000), with EAS "p1-p15" (0-9999999) login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary
d	direction of call – 0 means no call (AUX) – 1 means call is outbound (AUXO) – 2 means call is inbound (AUXI) Note: this information is not available with AAUDIT.
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present
reason	reason code (0-99); 0 is the default reason code.
i	flag that indicates that the AUX reason code entered is interruptible. The flag is to be set to 1 when the agent used and AUX reason code that has the Aux Work/Interruptible field on the Reason Code Names screen set to “y”.
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.38.3 **→AUDAUX24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y/n	y/n
Last Release			



time_tag seconds past minute when event occurred, 0-59
opcode AUDAUX24 = 88 (0x58)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
pnumber number of digits in agent position, 1-16, will always be 5 for non-EAS
pos/p1-p16 agent position - witout EAS "pos" (1-10000), with EAS "p1-p16" (0-9999999) login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary
d direction of call
- 0 means no call (AUX)
- 1 means call is outbound (AUXO)
- 2 means call is inbound (AUXI)

Note: this information is not available with AAUDIT.

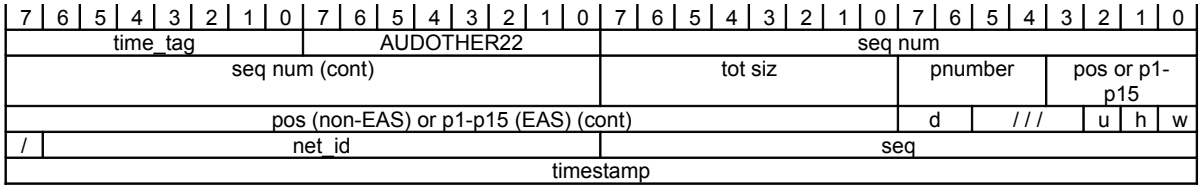
u flag indicating that the UCID fields (net_id, seq and timestamp) are present
reason reason code (0-99); 0 is the default reason code.
i flag that indicates that the AUX reason code entered is interruptible. The flag is to be set to 1 when the agent used and AUX reason code that has the Aux Work/Interruptible field on the Reason Code Names screen set to “y”.
net_id network entity identifier, 1-32767
seq call sequence number, 0-65535
timestamp unix style (seconds since 1/1/70) time call assigned a UCID

3.4.39 → AUDOTHER22 and AUDOTHER24

This is a variable length message based on "pnumber".

3.4.39.1 →AUDOTHER22 (New CM5.2)

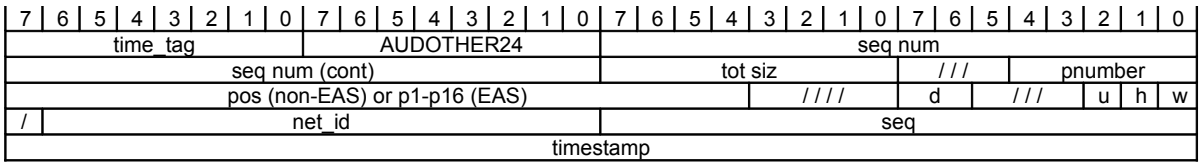
	Switch	Vect	Skill
First Release	CM5.2	y/n	y/n
Last Release	CM6.3		



time_tag	seconds past minute when event occurred, 0-59
opcode	AUDOTHER22 = 251 (0xfb)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
pnumber	number of digits in agent position, 1-15, will always be 5 for non-EAS
pos/p1-p15	agent position - without EAS "pos" (1-10000), with EAS "p1-p15" (0-9999999) login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary
d	direction of call – 0 means no call – 1 means call is outbound – 2 means call is inbound
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present.
h	flag to indicate the agent has just put a call on hold.
w	flag to indicate the observer is waiting fo another call to observe.
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.39.2 **→AUDOTHER24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y/n	y/n
Last Release			



time_tag	seconds past minute when event occurred, 0-59
opcode	AUDOTHER24 = 89 (0x59)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
pnumber	**number of digits in agent position, 1-16, will always be 5 for non-EAS**
pos/p1-p16	**agent position - without EAS "pos" (1-10000), with EAS "p1-p16" (0-9999999) login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary**
d	direction of call – 0 means no call – 1 means call is outbound – 2 means call is inbound
u	flag indicating that the UCID fields (net_id, seq and timestamp) are present.
h	flag to indicate the agent has just put a call on hold.
w	flag to indicate the observer is waiting fo another call to observe.
net_id	network entity identifier, 1-32767
seq	call sequence number, 0-65535
timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

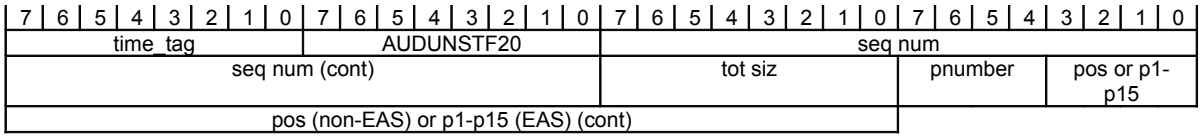
3.4.40 → AUDUNSTF20 and AUDUNSTF24

This message is non-EAS only.

This is a variable length message based on "pnumber".

3.4.40.1 → AUDUNSTF20 (New CM3.1)

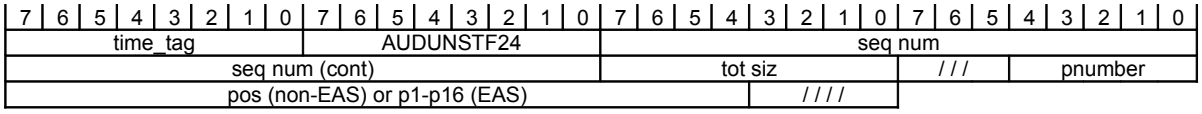
	Switch	Vect	Skill
First Release	CM3.1	y/n	y/n
Last Release	CM6.3		



time_tag seconds past minute when event occurred, 0-59
opcode AUDUNSTF20 = 244 (0xf4)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
pnumber number of digits in agent position, 1-15, will always be 5 for non-EAS
pos/p1-p15 agent position - without EAS "pos" (1-10000), with EAS "p1-p15" (0-9999999) login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary

3.4.40.2 → AUDUNSTF24 (New CM7.0)

	Switch	Vect	Skill
First Release	CM7.0	y/n	y/n
Last Release			



time_tag seconds past minute when event occurred, 0-59
opcode AUDUNSTF24 = 90 (0x5a)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
pnumber number of digits in agent position, 1-16, will always be 5 for non-EAS
pos/p1-p16 agent position - without EAS "pos" (1-10000), with EAS "p1-p16" (0-9999999) login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary

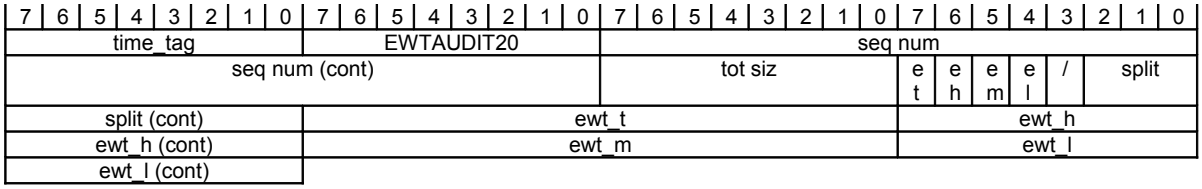
3.4.41 → EWTAUDIT20 and EWTAUDIT23

The **EWTAUDIT** message is sent by the switch when it receives an **RAUDIT** message from the reporting adjunct, indicating that it has not received a **QUEUED** or **DFWD** message for a particular split/skill within the expected timeframe.

The **EWTAUDIT** will contain the updated Expected Wait Times for all four priorities for a split/skill.

3.4.41.1 → EWTAUDIT20 (New CM3.1)

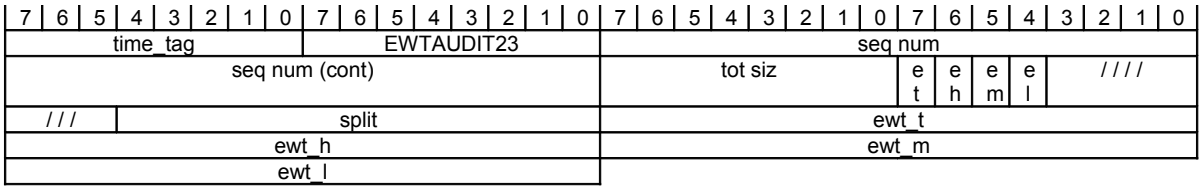
	Switch	Vect	Skill
First Release	CM3.1	y	y/n
Last Release	CM5.2		



time_tag seconds past minute when event occurred, 0-59
opcode EWTAUDIT20 = 245 (0xf5)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
et flag indicating whether ewt_t contains an EWT (when 0, ewt_t is null; when 1, ewt_t will contain a numeric value in seconds)
eh flag indicating whether ewt_h contains an EWT (when 0, ewt_h is null; when 1, ewt_h will contain a numeric value in seconds)
em flag indicating whether ewt_m contains an EWT (when 0, ewt_m is null; when 1, ewt_m will contain a numeric value in seconds)
el flag indicating whether ewt_l contains an EWT (when 0, ewt_l is null; when 1, ewt_l will contain a numeric value in seconds)
split split or skill number for which CMS needs an update for expected wait time. 1-999 (G3V8-G3V11), 1-2000 (MV2 and later)
ewt_t expected wait time for split/skill at top priority, 0-9999 seconds
ewt_h expected wait time for split/skill at high priority, 0-9999 seconds
ewt_m expected wait time for split/skill at medium priority, 0-9999 seconds
ewt_l expected wait time for split/skill at low priority, 0-9999 seconds

3.4.41.2 **→EWTAUDIT23 (New CM6.0)**

	Switch	Vect	Skill
First Release	CM6.0	y	y/n
Last Release			

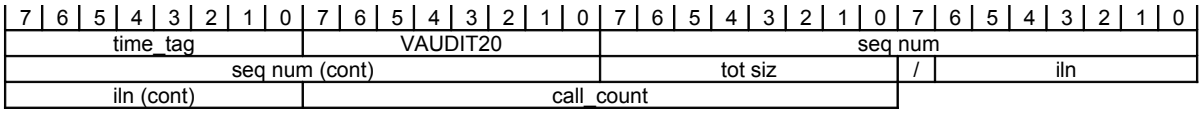


time_tag seconds past minute when event occurred, 0-59
opcode EWTAUDIT23 = 34 (0x22)
seq num sequence number, 0-0xffff ffff
tot siz total size in bytes, 1-255
et flag indicating whether ewt_t contains an EWT (when 0, ewt_t is null; when 1, ewt_t will contain a numeric value in seconds)
eh flag indicating whether ewt_h contains an EWT (when 0, ewt_h is null; when 1, ewt_h will contain a numeric value in seconds)
em flag indicating whether ewt_m contains an EWT (when 0, ewt_m is null; when 1, ewt_m will contain a numeric value in seconds)
el flag indicating whether ewt_l contains an EWT (when 0, ewt_l is null; when 1, ewt_l will contain a numeric value in seconds)
split split or skill number for which CMS needs an update for expected wait time. 1-8000
ewt_t expected wait time for split/skill at top priority, 0-9999 seconds
ewt_h expected wait time for split/skill at high priority, 0-9999 seconds
ewt_m expected wait time for split/skill at medium priority, 0-9999 seconds
ewt_l expected wait time for split/skill at low priority, 0-9999 seconds

3.4.42 → VAUDIT20 (New CM3.1)

	Switch	Vect	Skill
First Release	CM3.1	y	y/n
Last Release			

This message is sent by the switch to the reporting adjunct in response to an RAUDIT. It is used by the reporting adjunct to update the VDN call count for a VDN that has not had any new calls arrive and thus no recent DNEVENT messages have been received.



time_tag	seconds past minute when event occurred, 0-59
opcode	VAUDIT20 = 246 (0xf6)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
iln	internal line number, 1-32766
call_count	active calls associated with this VDN, 0-7712

3.4.43 → XFER20 and XFER24

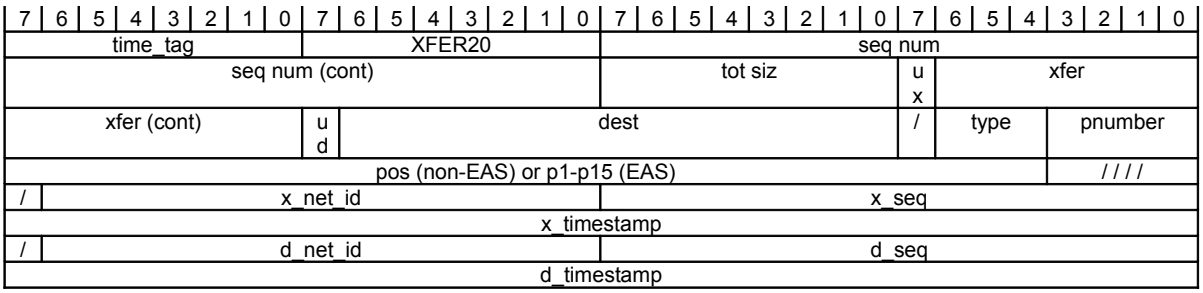
The **Transfer** message is sent by the switch when a call is merged, conferenced, transferred or path replacement done to another destination. It is an instruction to the reporting adjunct to copy the status of the transfer trunk (*xfer*) to the destination trunk (*dest*) and then track the call as if it arrived on the *dest* trunk. If the conference field (*c*) is set, the reporting adjunct will track both the *xfer* and *dest* trunks until one or both of the call drops from the conference.

Any of the trunks involved may be measured or unmeasured, currently being tracked or not (either the trunk or agent must be measured).From a reporting adjunct perspective, internal calls use internal unmeasured trunks.In all of these cases, the reporting adjunct will copy the status of the *xfer* trunk to the *dest* trunk and attempt to track the call(s).

For example, an internal call placed to the split and connected to a position will use an unmeasured trunk (the reporting adjunct will be tracking this trunk). If the agent transfers the call to another destination, the second call will also involve an unmeasured trunk (which may or may not be tracked by the reporting adjunct) and the **XFER20** message will instruct the reporting adjunct to transfer the status of the second call (if not being tracked the status is SEIZED) to the original call.

3.4.43.1 → XFER20 (New CM3.1)

	Switch	Vect	Skill
First Release	CM3.1	y/n	y/n
Last Release	CM6.3		

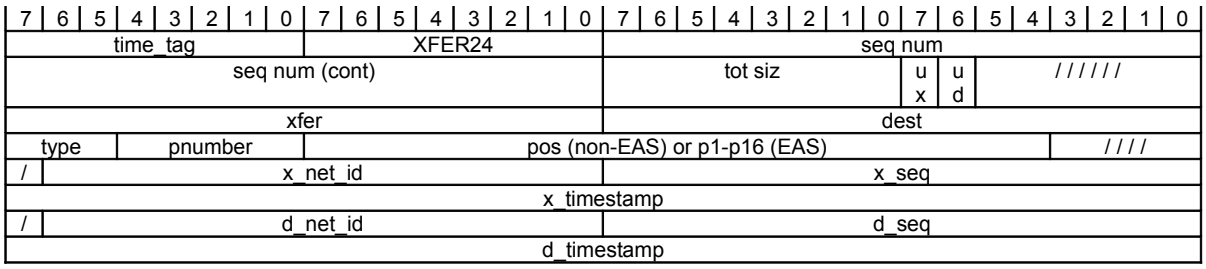


XFER20 (continued)

time_tag	seconds past minute when event occurred, 0-59
opcode	XFER20 = 247 (0xf7)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
ux	flag indicating that the UCID fields (net_id, seq and timestamp) are present for xfer itn
xfer	internal trunk number of transfer trunk, 1-32677
ud	flag indicating that the UCID fields (net_id, seq and timestamp) are present for dest itn
dest	internal trunk number of destination trunk, 1-32677
type	0=transfer, 1=conference, 2=special link (G2), 3=rename, 4=path replacement (G3V10 and later) Special link is used for transfers that involve two physical trunks and neither of the trunks drops. Rename is used when two or more facilities drop when a transfer occurs, the call continues on the "dest" facility.
pnumber	number of digits in agent position, 1-15, will always be 5 for non-EAS 0=an unmeasured endpoint transferred the call. In this case, no position is sent.
pos-p1-p15	agent position that initiates the transfer/conference, without EAS "pos" (1-10000), with EAS "p1-p15" (0-9999999) login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary flag indicating that the UCID fields (net_id, seq and timestamp) are present
x_net_id	network entity identifier, 1-32767
x_seq	call sequence number, 0-65535
x_timestamp	unix style (seconds since 1/1/70) time call assigned a UCID
d_net_id	network entity identifier, 1-32767
d_seq	call sequence number, 0-65535
d_timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.43.2 **→XFER24 (New CM7.0)**

	Switch	Vect	Skill
First Release	CM7.0	y/n	y/n
Last Release			



XFER24 (continued)

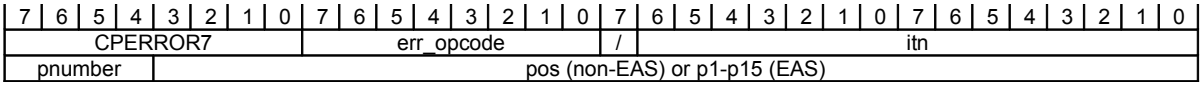
time_tag	seconds past minute when event occurred, 0-59
opcode	XFER24 = 91 (0x5b)
seq num	sequence number, 0-0xffff ffff
tot siz	total size in bytes, 1-255
ux	flag indicating that the UCID fields (net_id, seq and timestamp) are present for xfer itn
xfer	**internal trunk number of transfer trunk, 1-65535**
ud	flag indicating that the UCID fields (net_id, seq and timestamp) are present for dest itn
dest	**internal trunk number of destination trunk, 1-65535**
type	0=transfer, 1=conference, 2=special link (G2), 3=rename, 4=path replacement (G3V10 and later) Special link is used for transfers that involve two physical trunks and neither of the trunks drops. Rename is used when two or more facilities drop when a transfer occurs, the call continues on the "dest" facility.
pnumber	**number of digits in agent position, 1-16, will always be 5 for non-EAS 0=an unmeasured endpoint transferred the call. In this case, no position is sent.**
pos-p1-p16	**agent position that initiates the transfer/conference, without EAS "pos" (1-10000), with EAS "p1-p16" (0-9999999) login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary**
	flag indicating that the UCID fields (net_id, seq and timestamp) are present
x_net_id	network entity identifier, 1-32767
x_seq	call sequence number, 0-65535
x_timestamp	unix style (seconds since 1/1/70) time call assigned a UCID
d_net_id	network entity identifier, 1-32767
d_seq	call sequence number, 0-65535
d_timestamp	unix style (seconds since 1/1/70) time call assigned a UCID

3.4.44 ← CPERROR7 and CPERROR24

The **Call Processing Error** message will be sent by the reporting adjunct to the switch when it detects a protocol violation in call processing messaging. This is a variable length message based on "pnumber".

3.4.44.1 ←CPERROR7

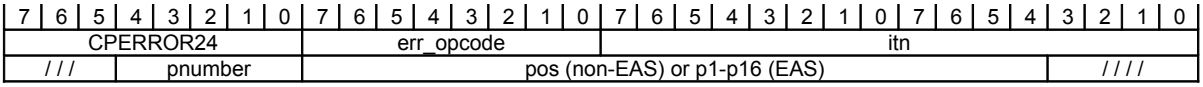
	Switch	Vect	Skill
First Release	G3V11	y/n	y/n
Last Release	CM6.3		



opcode CPERROR7 = 42 (0x2a)
err_opcode opcode of the message from the switch that violates the protocol
itn internal trunk number used to identify trunk facilities, 1-32766
pnumber number of digits in agent position, 1-15, will always be 5 for non-EAS
pos/p1-p15 agent position - without EAS "pos" (1-10000), with EAS "p1-p15" (0-9999999) login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary

3.4.44.2 ←CPERROR24 (New CM7.0)

	Switch	Vect	Skill
First Release	CM7.0	y/n	y/n
Last Release			



opcode CPERROR24 = 67 (0x43)
err_opcode opcode of the message from the switch that violates the protocol
itn internal trunk number used to identify trunk facilities, 1-65535
pnumber number of digits in agent position, 1-16, will always be 5 for non-EAS
pos/p1-p16 agent position - without EAS "pos" (1-10000), with EAS "p1-p16" (0-9999999) login ID in packed decimal format, left-justified where hex a is digit 0 and 0 in this field is null, zero fill to a byte boundary

3.5 **Miscellaneous**

3.5.1 ←**NOP**

	Switch	Vect	Skill
First Release	G3V1	y/n	y/n
Last Release			

The **No Operation** message may be sent by the reporting adjunct and ignored by the switch. Starting with R3V2, CMS will send this as a two-byte message.

7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
NOP								NOP							

opcode NOP = 11 (0xb)
opcode NOP = 11 (0xb) (sent twice)

4. Message Index

4.1 Alphabetical List

Table 1 Reporting Adjunct to Switch Messages – Alphabetical List

←Message	Code	Section In
APARMS7	x28	3.3.11.1
APARMS24	**x3f**	**3.3.11.2**
CHGNAM20	x32	3.3.28.1
CHGNAM24	**x41**	**3.3.28.2**
CHGVDNSK8	x2e	3.3.18.1
CHGVDNSK23	x3a	3.3.18.2
CHGVEC20	x33	3.3.22.1
CHGVEC24	**x44**	**3.3.22.2**
CPERROR7	x2a	3.4.44.1
CPERROR24	**x43**	**3.4.44.2**
MVAGSTFD21	x35	3.3.10.1
MVAGSTFD23	x38	3.3.10.2
MVAGSTFD24	**x3e**	**3.3.10.3**
MVVDN9	x2f	3.3.17.1
MVVDN23	x3c	3.3.17.2
NOP	x0b	3.5.1
RAUDIT	x08	3.4.33.1
RAUDIT24	**x42**	**3.4.33.2**
RCODENAM20	x31	3.3.27
REQLIST5	x23	3.3.23
REQNAM20	x30	3.3.25
RFTB	x06	3.2.7
RLTB	x0c	3.2.8
RTCS21	x34	3.2.1.1
RTCS23	x37	3.2.1.2
RTCS24	**x3d**	3.2.1.3
SKILLADD8	x2d	3.3.13.1
SKILLADD23	x39	3.3.13.2
SKILLADD24	**x40**	**3.3.13.3**
USERID21	x36	3.3.32
VPARMS	x15	3.3.19

Table 2 Switch to Reporting Adjunct Messages – Alphabetical List

→ Message	Code	Section In
ACD20	xd1	3.4.5.1
ACD24	**x3b**	**3.4.5.2**
ACW22	xff	3.4.6.1
ACW24	**x3c**	**3.4.6.2**
ADMLIST20	xc8	3.3.24
ADMNAM20	xc9	3.3.26.1
ADMNAM24	**x2b**	**3.3.26.2**
AEVENT20	xd6	3.4.10.1
AEVENT24	**x40**	**3.4.10.2**
AGENTSK20	xc2	3.3.12.1
AGENTSK23	x04	3.3.12.2
AGENTSK24	**x37**	**3.3.12.3**
AGMOVE20	xf8	3.3.9
ALOGOFF20	xcf	3.4.3.1
ALOGOFF24	**x39**	**3.4.3.2**
ALOGON20	xce	3.4.2.1
ALOGON23	x05	3.4.2.2
ALOGON24	**x28**	3.4.2.3
ARAGENT20	xbd	3.3.4.1
ARAGENT24	**x33**	**3.3.4.2**
AROUTE20	xe8	3.4.28.1
AROUTE24	**x51**	**3.4.28.2**
ARSPLIT20	xbe	3.3.5.1
ARSPLIT23	x13	3.3.5.2
ARSPLIT24	**x34**	**3.3.5.3**
ARTGVDN20	xc0	3.3.7.1
ARTGVDN24	**x36**	**3.3.7.2**
ARTKGRP20	xbf	3.3.6.1
ARTKGRP23	x14	3.3.6.2
ARTKGRP24	**x35**	**3.3.6.3**
ARTRUNK20	xc1	3.3.8.1
ARTRUNK24	**x26**	3.3.8.2
ARVDN20	xbc	3.3.3.1
ARVDN23	x12	3.3.3.2
ARVDN24	**x32**	**3.3.3.3**
ASA20	xea	3.4.30.1
ASA23	x20	3.4.30.2
AUDACD20	xf0	3.4.36.1
AUDACD23	x06	3.4.36.2
AUDACD24	**x56**	**3.4.36.3**
AUDACW20	xf1	3.4.37.1
AUDACW23	x07	3.4.37.2
AUDACW24	**x57**	**3.4.37.3**
AUDAUX20	xf2	3.4.38.1
AUDAUX23	x10	3.4.38.2
AUDAUX24	**x58**	**3.4.38.3**
AUDAVAIL22	x02	3.4.35.1
AUDAVAIL23	x08	3.4.35.2
AUDAVAIL24	**x55**	**3.4.35.3**
AUDOTHER22	xfb	3.4.39.1
AUDOTHER24	**x59**	**3.4.39.2**
AUDUNSTF20	xf4	3.4.40.1
AUDUNSTF24	**x5a**	**3.4.40.2**
AUX22	x03	3.4.7.1
AUX23	x0f	3.4.7.2
AUX24	**x3d**	**3.4.7.3**
AVAIL22	x01	3.4.4.1
AVAIL23	x0e	3.4.4.2
AVAIL24	**x3a**	**3.4.4.3**
BUSYOUT20	xba	3.3.1
COMP20	xe6	3.4.26.1
COMP24	**x4f**	**3.4.26.2**
CRCODE20	xcc	3.3.31
DABN22	xfc	3.4.16.1
DABN24	**x46**	**3.4.16.2**
DADFWD20	xdd	3.4.17.1
DADFWD24	**x2a**	**3.4.17.2**
DAQUEUED20	xda	3.4.14.1
DAQUEUED23	x1d	3.4.14.2
DAQUEUED24	**x44**	**3.4.14.3**
DCON22	xfd	3.4.15.1
DCON23	x0b	3.4.15.2
DCON24	**x45**	**3.4.15.3**
DFWD20	xde	3.4.18.1
DFWD23	x0c	3.4.18.2
DFWD24	**x47**	**3.4.18.3**
DIGITS20	xd5	3.4.9.1
DIGITS24	**x3f**	**3.4.9.2**
DNEVENT20	xeb	3.4.31.1
DNEVENT24	**x53**	**3.4.31.2**
EWTAUDIT20	xf5	3.4.41.1
EWTAUDIT23	x22	3.4.41.2
FBUSY20	xe0	3.4.20.1
FBUSY24	**x49**	**3.4.20.2**
FDISC20	xe3	3.4.23.1
FDISC24	**x4c**	**3.4.23.2**
HOLD20	xe4	3.4.24.1
HOLD24	**x4d**	**3.4.24.2**
IDLE20	xd7	3.4.11.1
IDLE23	x1b	3.4.11.2
IDLE24	**x41**	**3.4.11.3**

LOOKAH20	xe7	3.4.27.1

LOOKAH24	**x50**	**3.4.27.2**

MBUSY20	xdf	3.4.19.1
MBUSY24	**x48**	**3.4.19.2**
MODEVENT20	xec	3.4.32
MVRCODE20	xca	3.3.29
OTHER22	xfa	3.4.8.1
OTHER24	**x3e**	**3.4.8.2**
QUEUED20	xd9	3.4.13.1
QUEUED23	x1c	3.4.13.2
QUEUED24	**x43**	**3.4.13.3**
RCODE20	xcb	3.3.30
RECON20	xe5	3.4.25.1
RECON23	x1e	3.4.25.2
RECON24	**x4e**	**3.4.25.3**
RELBUSY20	xbb	3.3.2
SETUP22	xfe	3.4.12.1
SETUP24	**x42**	**3.4.12.2**
SKSTATE20	xe9	3.4.29.1
SKSTATE23	x1f	3.4.29.2
TAUDIT20	xee	3.4.34.1
TAUDIT24	**x54**	**3.4.34.2**
TGMOVE20	xc3	3.3.14.1
TGMOVE23	x16	3.3.14.2
TIME20	xcd	3.4.1
VAUDIT20	xf6	3.4.42
VDNMOVE20	xc5	3.3.16.1
VDNMOVE23	x18	3.3.16.2
VDNMOVE24	**x38**	**3.3.16.3**
VECTOR20	xc6	3.3.20.1
VECTOR24	**x52**	**3.3.20.2**
VSTART20	xe1	3.4.21.1
VSTART24	**x4a**	**3.4.21.2**
VSTOP20	xe2	3.4.22.1
VSTOP24	**x4b**	**3.4.22.2**
VTGMOVE20	xc4	3.3.15
XAGENT20	xb6	3.2.13.1
XAGENT24	**x2f**	**3.2.13.2**
XCLOSE20	xb3	3.2.10
XFAIL21	xf9	3.2.3.1
XFAIL23	x0d	3.2.3.2
XFAIL24	**x29**	3.2.3.3
XFER20	xf7	3.4.43.1
XFER24	**x5b**	**3.4.43.2**
XOPEN20	xb2	3.2.9
XSPLIT20	xb5	3.2.12.1
XSPLIT24	**x2e**	**3.2.12.2**
XSTAT20	xb1	3.2.5.1
XSTAT24	**x2c**	**3.2.5.2**
XTGVDN20	xb8	3.2.15.1
XTGVDN24	**x31**	**3.2.15.2**
XTKGRP20	xb7	3.2.14.1
XTKGRP24	**x30**	**3.2.14.2**
XTRUNK20	xb9	3.2.16.1
XTRUNK24	**x25**	3.2.16.2
XVDN20	xb4	3.2.11.1
XVDN23	x11	3.2.11.2
XVDN24	**x2d**	**3.2.11.3**

Table 3 Reporting Adjunct to Switch Messages – Opcode List

Message	Op Code	Description
RFTB	x06	Request Full Translations
RAUDIT	x08	Request Audit
NOP	x0b	No Operation (Filler)
RLTB	x0c	Request Logon Translations
VPARMS	x15	Request Vector Parameters
REQLIST5	x23	Request List of VDNs or Vectors
APARMS7	x28	Request Agent's Skills (SPI language 7)
CPERROR7	x2a	Call Processing Protocol Error (SPI language 7)
SKILLADD8	x2d	Add Skill to Agent (SPI language 8)
CHGVDNSK8	x2e	Change VDN's Skills (SPI language 8)
MVVDN9	x2f	Move VDNs (SPI language 9)
REQNAM20	x30	Request Name - IQ Names Admin (SPI langauge 20)
RCODENAM20	x31	Return Code - IQ Names Admin (SPI language 20)
CHGNAM20	x32	Change Name - IQ Names Admin (SPI language 20)
CHGVEC20	x33	Change Vector Parameters II (SPI language 20)
RTCS21	x34	Request Translation Change Status (SPI language 21)
MVAGSTFD21	x35	Move Agents While Staffed (SPI language 21)
USERID21	x36	User Identity (SPI language 21)
RTCS23	x37	Request Translation Change Status (SPI language 23)
MVAGSTFD23	x38	Move Agents While Staffed (SPI language 23)
SKILLADD23	x39	Add Skill to Agent (SPI language 23)
CHGVDNSK23	x3a	Change VDN's Skills (SPI language 23)
MVVDN23	x3c	Move VDNs (SPI language 23)
RTCS24	**x3d**	**Request Translation Change Status (SPI language 24)**
MVAGSTFD24	**x3e**	**Move Agents While Staffed (SPI language 24)**
APARMS24	**x3f**	**Request Agent's Skills (SPI language 24)**
SKILLADD24	**x40**	**Add Skill to Agent (SPI language 24)**
CHGNAM24	**x41**	**Change Name - IQ Names Admin (SPI language 24)**
RAUDIT24	**x42**	**Request Audit (SPI language 24)**
CPERROR24	**x43**	**Call Processing Protocol Error (SPI language 24)**
CHGVEC24	**x44**	**Change Vector Parameters (SPI language 24)**

Table 4 Switch to Reporting Adjunct Messages – Opcode List

Message	Code	Description
AVAIL22	x01	Agent State
AUDAVAIL22	x02	Agent State Audit
AUX22	x03	Agent State AUX
AGENTSK23	x04	Agent's Skills (SPI language 23)
ALOGON23	x05	Agent Logon (SPI language 23)
AUDACD23	x06	Agent State Audit (SPI language 23)
AUDACW23	x07	Agent State Audit (SPI language 23)
AUDAVAIL23	x08	Agent State Audit (SPI language 23)
DCON23	x0b	Trunk State Connected (SPI language 23)
DFWD23	x0c	Trunk State Forward (SPI language 23)
XFAIL23	x0d	Translation Pumpup Failed (SPI language 23)
AVAIL23	x0e	Agent State AVAIL (SPI language 23)
AUX23	x0f	Agent State AUX (SPI language 23)
AUDAUX23	x10	Agent State Audit (SPI language 23)
XVDN23	x11	VDN-Vector Translations (SPI language 23)
ARVDN23	x12	Add/Remove VDN (SPI language 23)
ARSPLIT23	x13	Add/Remove Split (SPI language 23)
ARTKGRP23	x14	Add/Remove Trunk Group (SPI language 23)
TGMOVE23	x16	Trunk Group Move (SPI language 23)
VDNMOVE23	x18	VDN-Vector Move (SPI language 23)
IDLE23	x1b	Trunk State Idle (SPI language 23)
QUEUED23	x1c	Trunk State Queued (SPI language 23)
DAQUEUED23	x1d	Trunk State Direct Agent Queued (SPI language 23)
RECON23	x1e	Trunk State Reconnect (SPI language 23)
SKSTATE23	x1f	Skill State (SPI language 23)
ASA23	x20	Average Speed of Answer (SPI language 23)
EWTAUDIT23	x22	Expected Wait Time (SPI language 23)
XTRUNK24	**x25**	**Trunk Translations (SPI language 24)**
ARTRUNK24	**x26**	**Add/Remove Trunk (SPI language 24)**
ALOGON24	**x28**	**Agent Logon (SPI language 24)**
XFAIL24	**x29**	**Translation Pumpup Failed (SPI language 24)**
DADFWD24	**x2a**	**Trunk State Direct Agent Forward (SPI language 24)**
ADMNAM24	**x2b**	**Names Admin from CMS (SPI language 24)**
XSTAT24	**x2c**	**Translation Status (SPI language 24)**
XVDN24	**x2d**	**VDN-Vector Translations (SPI language 24)**
XSPLIT24	**x2e**	**Split Translations (SPI language 24)**
XAGENT24	**x2f**	**Agent Position Translations (SPI language 24)**
XTKGRP24	**x30**	**TG to Split Translations (SPI language 24)**
XTGVDN24	**x31**	**TG to VDN Translations (SPI language 24)**
ARVDN24	**x32**	**Add/Remove VDN (SPI language 24)**
ARAGENT24	**x33**	**Add/Remove Agent (SPI language 24)**
ARSPLIT24	**x34**	**Add/Remove Split (SPI language 24)**
ARTKGRP24	**x35**	**Add/Remove Trunk Group (SPI language 24)**
ARTGVDN24	**x36**	**Add/Remove Trunk Group to VDN (SPI language 24)**
AGENTSK24	**x37**	**Agent's Skills (SPI language 24)**
VDNMOVE24	**x38**	**VDN-Vector Move (SPI language 24)**

ALOGOFF24	**x39**	**Agent Logoff (SPI language 24)**
AVAIL24	**x3a**	**Agent State AVAIL (SPI language 24)**
ACD24	**x3b**	**Agent State ACD (SPI language 24)**
ACW24	**x3c**	**Agent State ACW (SPI language 24)**
AUX24	**x3d**	**Agent State AUX (SPI language 24)**
OTHER24	**x3e**	**Agent State OTHER (SPI language 24)**
DIGITS24	**x3f**	**Digits (SPI language 24)**
AEVENT24	**x40**	**Agent Event (SPI language 24)**
IDLE24	**x41**	**Trunk State Idle (SPI language 24)**
SETUP24	**x42**	**Trunk State Seized (SPI language 24)**
QUEUED24	**x43**	**Trunk State Queued (SPI language 24)**
DAQUEUED24	**x44**	**Trunk State Direct Agent Queued (SPI language 24)**
DCON24	**x45**	**Trunk State Connected (SPI language 24)**
DABN24	**x46**	**Trunk State Abandoned SPI language 24)**
DFWD24	**x47**	**Trunk State Forward (SPI language 24)**
MBUSY24	**x48**	**Trunk State Make Busy (SPI language 24)**
FBUSY24	**x49**	**Trunk State Forced Busy (SPI language 24)**
VSTART24	**x4a**	**Trunk State Vector Start (SPI language 24)**
VSTOP24	**x4b**	**Trunk State Vector Stop (SPI language 24)**
FDISC24	**x4c**	**Trunk State Forced Disconnect (SPI language 24)**
HOLD24	**x4d**	**Trunk State Hold (SPI language 24)**
RECON24	**x4e**	**Trunk State Reconnect (SPI language 24)**
COMP24	**x4f**	**Trunk State Completed (SPI language 24)**
LOOKAH24	**x50**	**Trunk State Lookahead Attempted (SPI language 24)**
AROUTE24	**x51**	**Trunk State Adjunct Routing Attempted (SPI language 24)**
VECTOR24	**x52**	**Vector Parameters (SPI language 24)**
DNEVENT24	**x53**	**VDN Dialed Event (SPI language 24)**
TAUDIT24	**x54**	**Trunk Audit (SPI language 24)**
AUDAVAIL24	**x55**	**Agent State Audit (SPI language 24)**
AUDACD24	**x56**	**Agent State Audit (SPI language 24)**
AUDACW24	**x57**	**Agent State Audit (SPI language 24)**
AUDAUX24	**x58**	**Agent State Audit (SPI language 24)**
AUDOTHER24	**x59**	**Agent State Audit (SPI language 24)**
AUDUNSTF24	**x5a**	**Agent State Audit (SPI language 24)**
XFER24	**x5b**	**Transfer (SPI language 24)**
XSTAT20	xb1	Translation Status
XOPEN20	xb2	Open Translation Block
XCLOSE20	xb3	Close Translation Block
XVDN20	xb4	VDN-Vector Translations
XSPLIT20	xb5	Split Translations
XAGENT20	xb6	Agent Position Translations
XTKGRP20	xb7	TG to Split Translations
XTGVDN20	xb8	TG to VDN Translations
XTRUNK20	xb9	Trunk Translations
BUSYOUT20	xba	Busy-Out CMS
RELBUSY20	xbb	Release Busy CMS
ARVDN20	xbc	Add/Remove VDN
ARAGENT20	xbd	Add/Remove Agent
ARSPLIT20	xbe	Add/Remove Split
ARTKGRP20	xbf	Add/Remove Trunk Group

ARTGVDN20	xc0	Add/Remove Trunk Group to VDN
ARTRUNK20	xc1	Add/Remove Trunk
AGENTSK20	xc2	Agent's Skills
TGMOVE20	xc3	Trunk Group Move
VTGMOVE20	xc4	Trunk Group to VDN Move
VDNMOVE20	xc5	VDN-Vector Move
VECTOR20	xc6	Vector Parameters
ADMLIST20	xc8	List VDNs or Vectors
ADMNAM20	xc9	Names Admin from CMS
MVRCODE20	xca	Return Code from Move Request
RCODE20	xcb	Return Code
CRCODE20	xcc	Count of Return Codes
TIME20	xcd	Time of Day
ALOGON20	xce	Agent Logon
ALOGOFF20	xcf	Agent Logoff
ACD20	xd1	Agent State ACD
DIGITS20	xd5	Digits
AEVENT20	xd6	Agent Event
IDLE20	xd7	Trunk State Idle
QUEUED20	xd9	Trunk State Queued
DAQUEUED20	xda	Trunk State Direct Agent Queued
DADFWD20	xdd	Trunk State Direct Agent Forward
DFWD20	xde	Trunk State Forward
MBUSY20	xdf	Trunk State Make Busy
FBUSY20	xe0	Trunk State Forced Busy
VSTART20	xe1	Trunk State Vector Start
VSTOP20	xe2	Trunk State Vector Stop
FDISC20	xe3	Trunk State Forced Disconnect
HOLD20	xe4	Trunk State Hold
RECON20	xe5	Trunk State Reconnect
COMP20	xe6	Trunk State Completed
LOOKAH20	xe7	Trunk State Lookahead Attempted
AROUTE20	xe8	Trunk State Adjunct Routing Attempted
SKSTATE20	xe9	Skill State
ASA20	xea	Average Speed of Answer
DNEVENT20	xeb	VDN Dialed Event
MODEVENT20	xec	Modification Event
TAUDIT20	xee	Trunk Audit
AUDACD20	xf0	Agent State Audit
AUDACW20	xf1	Agent State Audit
AUDAUX20	xf2	Agent State Audit
AUDUNSTF20	xf4	Agent State Audit
EWTAUDIT20	xf5	Expected Wait Time
VAUDIT20	xf6	VDN Counted-Calls
XFER20	xf7	Transfer
AGMOVE20	xf8	Agent Move
XFAIL21	xf9	Translation Pumpup Failed
OTHER22	xfa	Agent State OTHER
AUDOTHER22	xfb	Agent State Audit
DABN22	xfc	Trunk State Abandoned

DCON22	xfd	Trunk State Connected
SETUP22	xfe	Trunk State Seized
ACW22	xff	Agent State ACW

5. **Virtual trunk mapping**

cabinet = switch representation / 12000
slot = (switch representation % 12000) / 500
circuit = (switch representation % 12000) % 500

% = modulus

Table 5 Virtual Trunk Mapping (examples)

Switch representation (logical ports)	As sent in SPI message			
	cabinet	carrier	slot	circuit
T00001	000	0	00	001
T00499	000	0	00	499
T00500	000	0	01	000
T00999	000	0	01	499
T01000	000	0	02	000
T01500	000	0	03	000
T02000	000	0	04	000
T02500	000	0	05	000
T03000	000	0	06	000
T35000	000	0	07	000
T04000	000	0	08	000
T04500	000	0	09	000
T05000	000	0	10	000
T05500	000	0	11	000
T06000	000	0	12	000
T06500	000	0	13	000
T07000	000	0	14	000
T07500	000	0	15	000
T08000	000	0	16	000
T08500	000	0	17	000
T09000	000	0	18	000
T09500	000	0	19	000
T10000	000	0	20	000
T10500	000	0	21	000
T11000	000	0	22	000
T11500	000	0	23	000
T12000	001	0	0	000
T12500	001	0	1	000
T13000	001	0	2	000
T13500	001	0	3	000
T14000	001	0	4	000
T14500	001	0	5	000
T15000	001	0	6	000
T15500	001	0	7	000
T16000	001	0	8	000
T16500	001	0	9	000
T17000	001	0	10	000
T17500	001	0	11	000

T18000	001	0	12	000
T18500	001	0	13	000
T19000	001	0	14	000
T19500	001	0	15	000
T20000	001	0	16	000
T20500	001	0	17	000
T21000	001	0	18	000
T21500	001	0	19	000
T22000	001	0	20	000
T22500	001	0	21	000
T23000	001	0	22	000
T23500	001	0	23	000
T24000	002	0	0	000

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